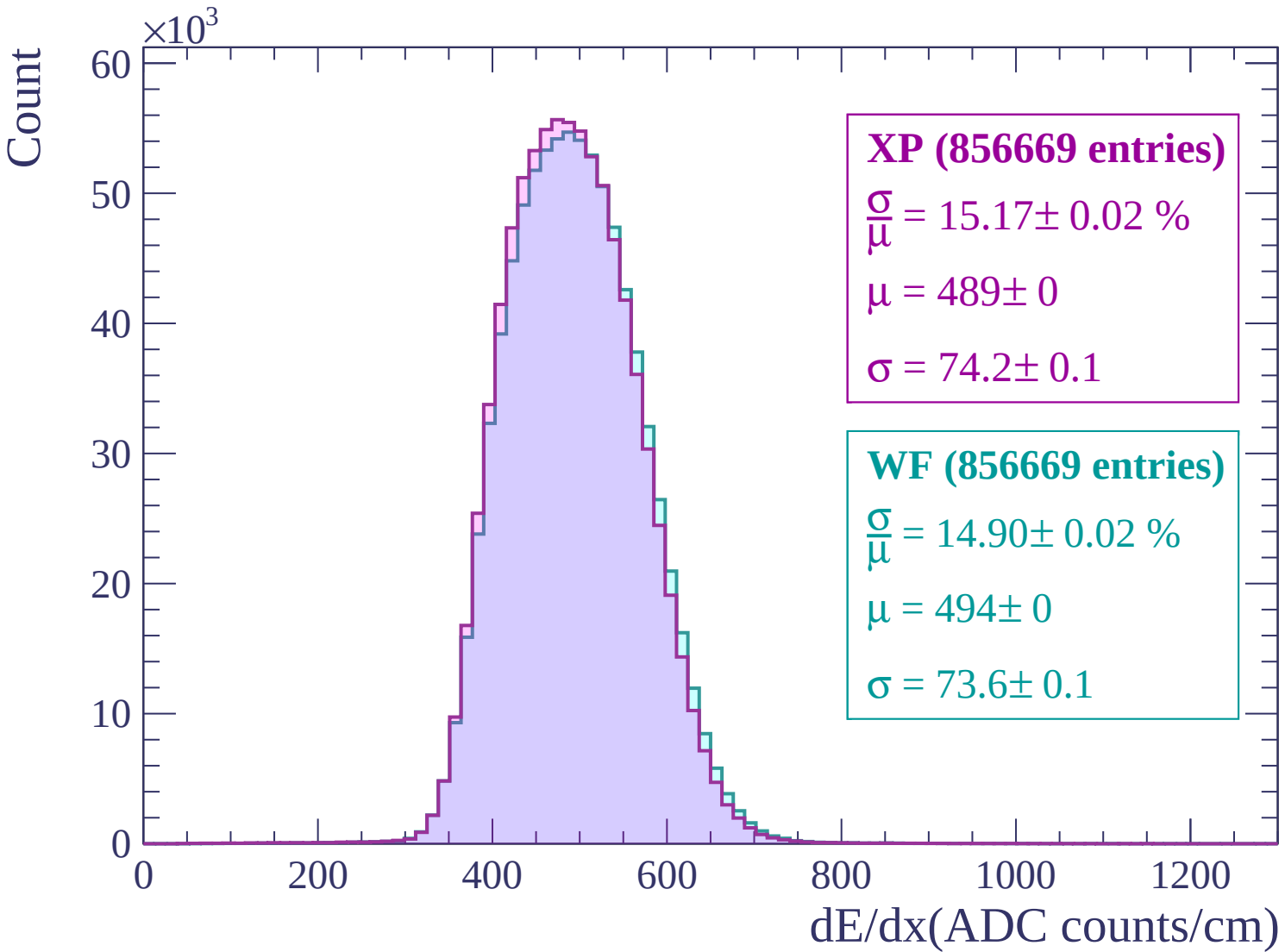
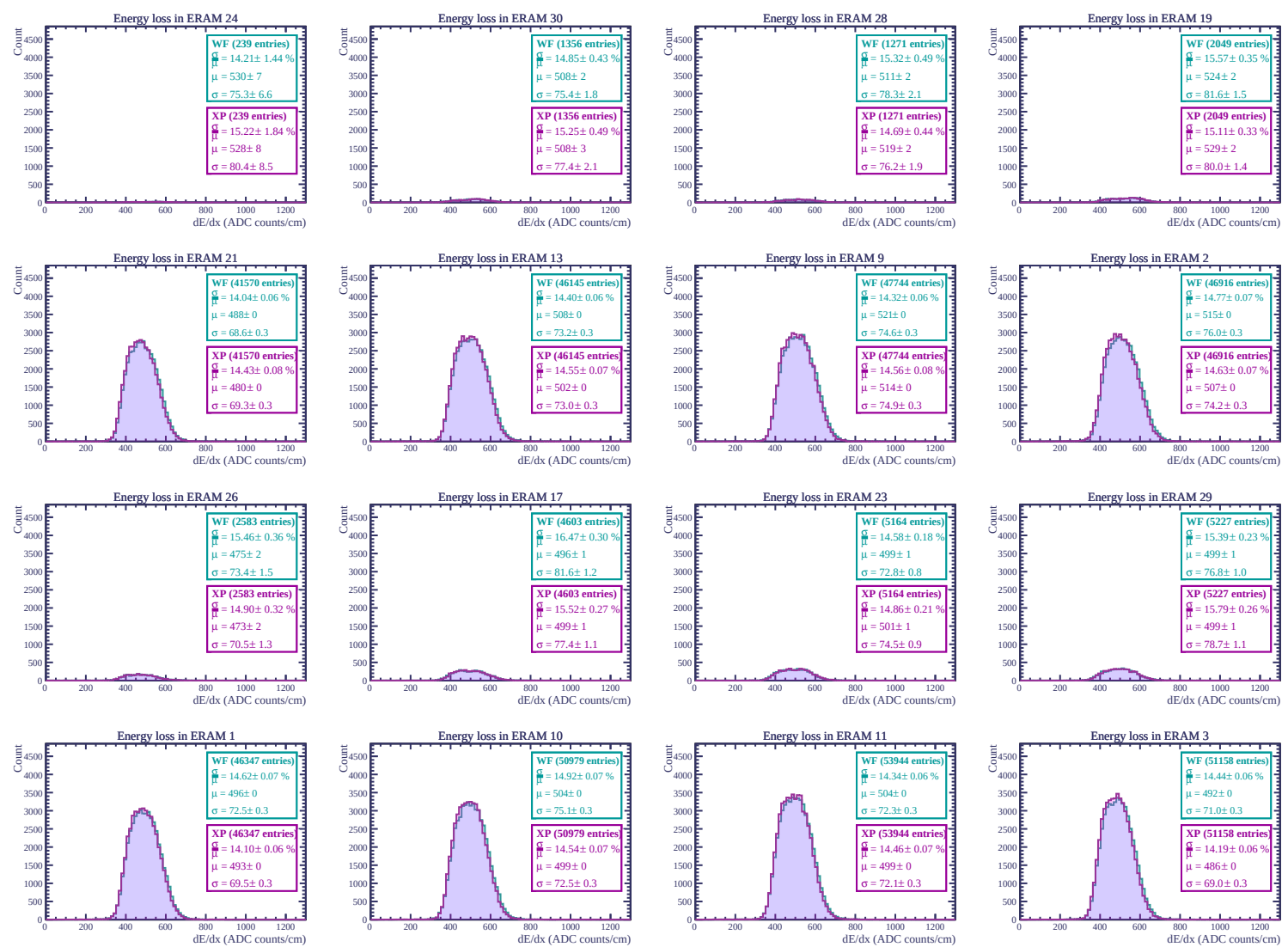
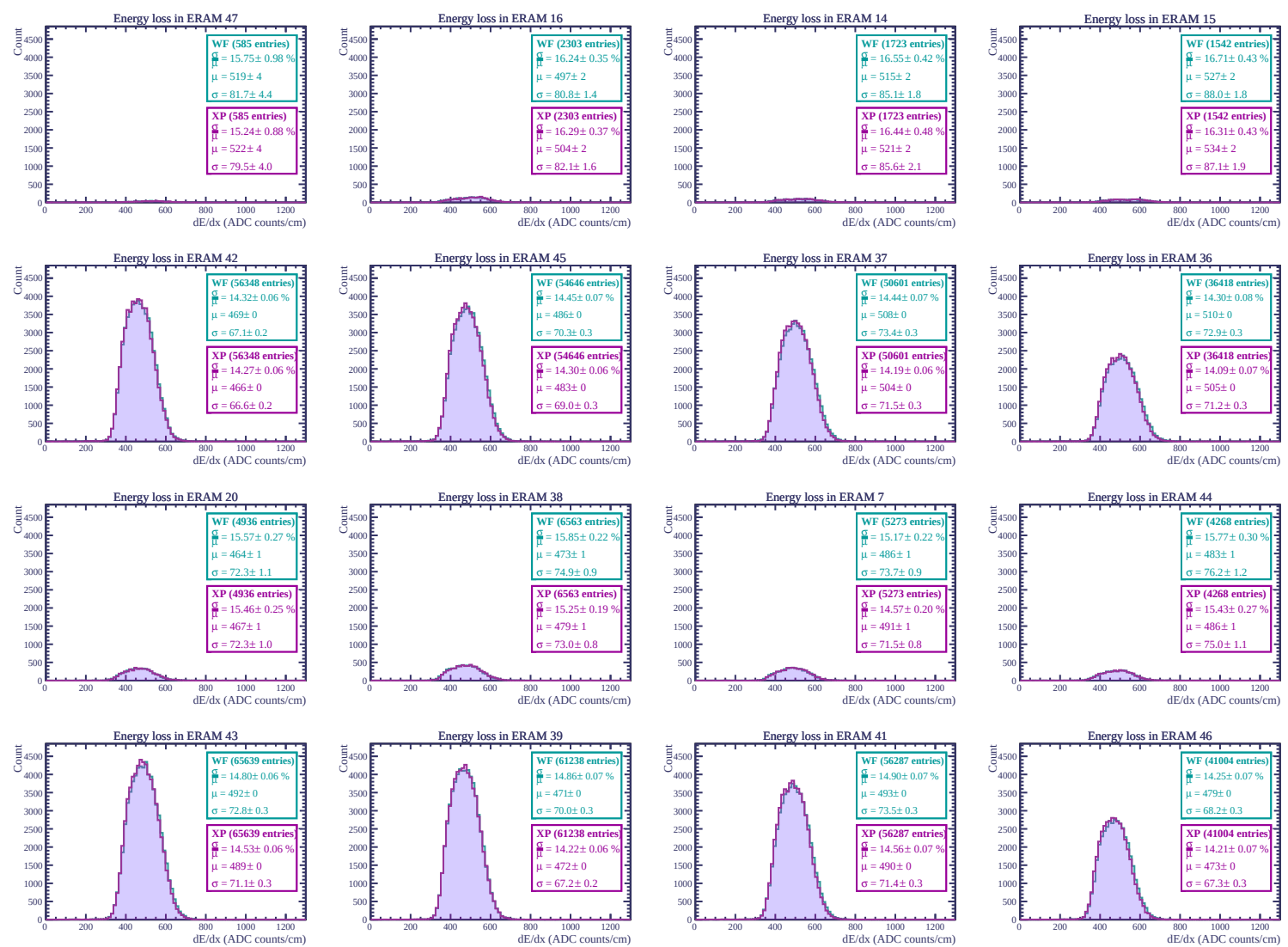
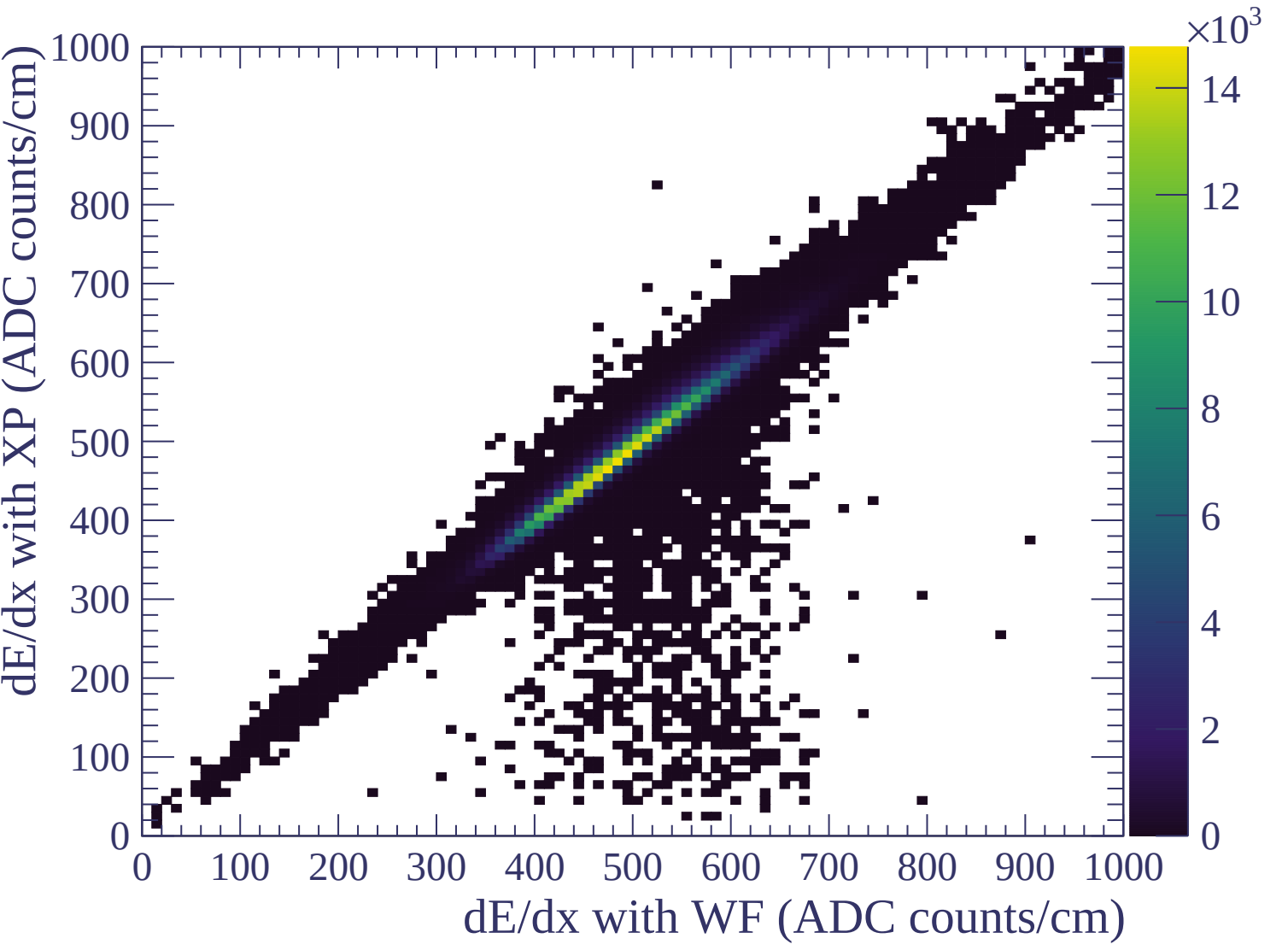
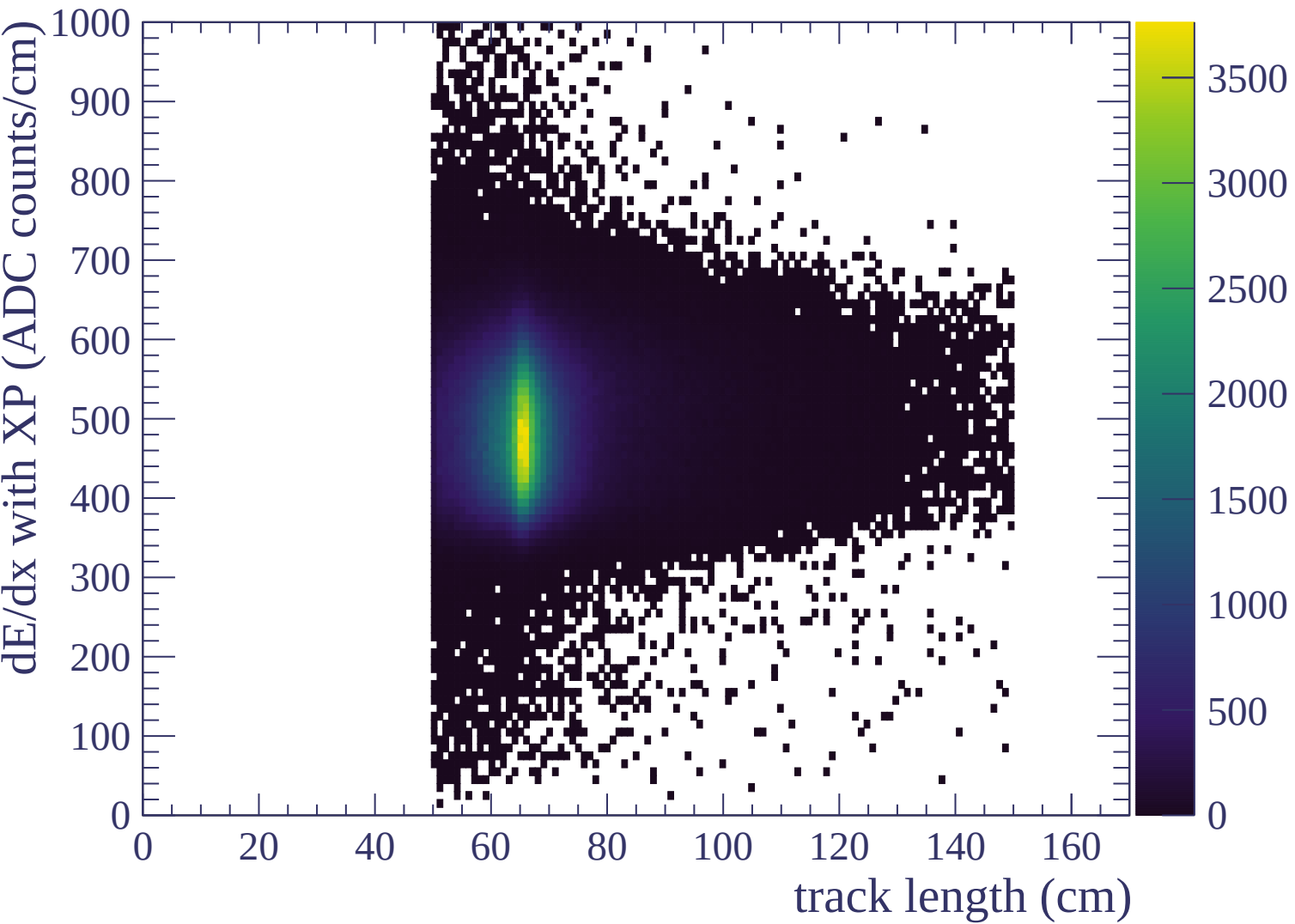


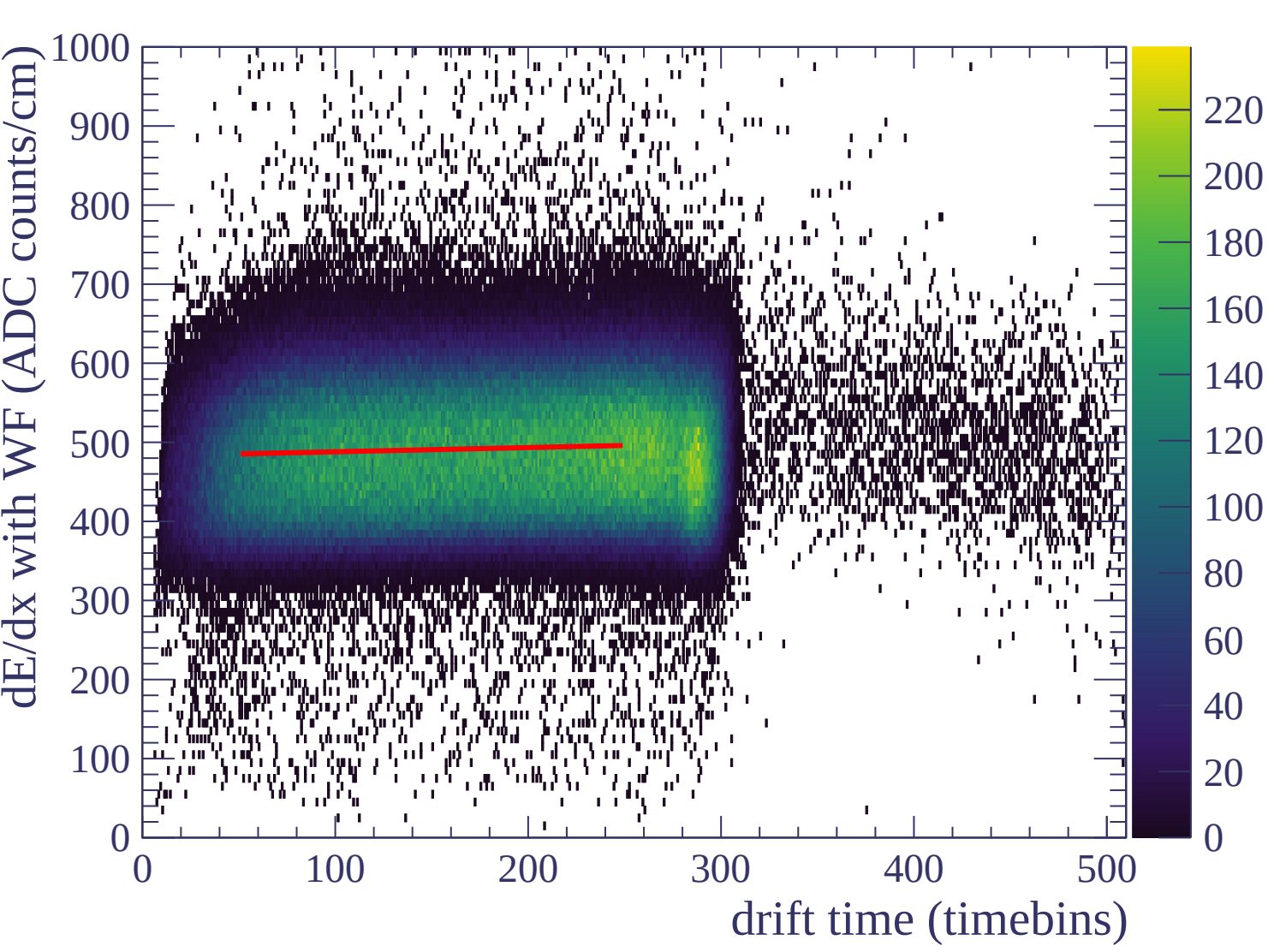


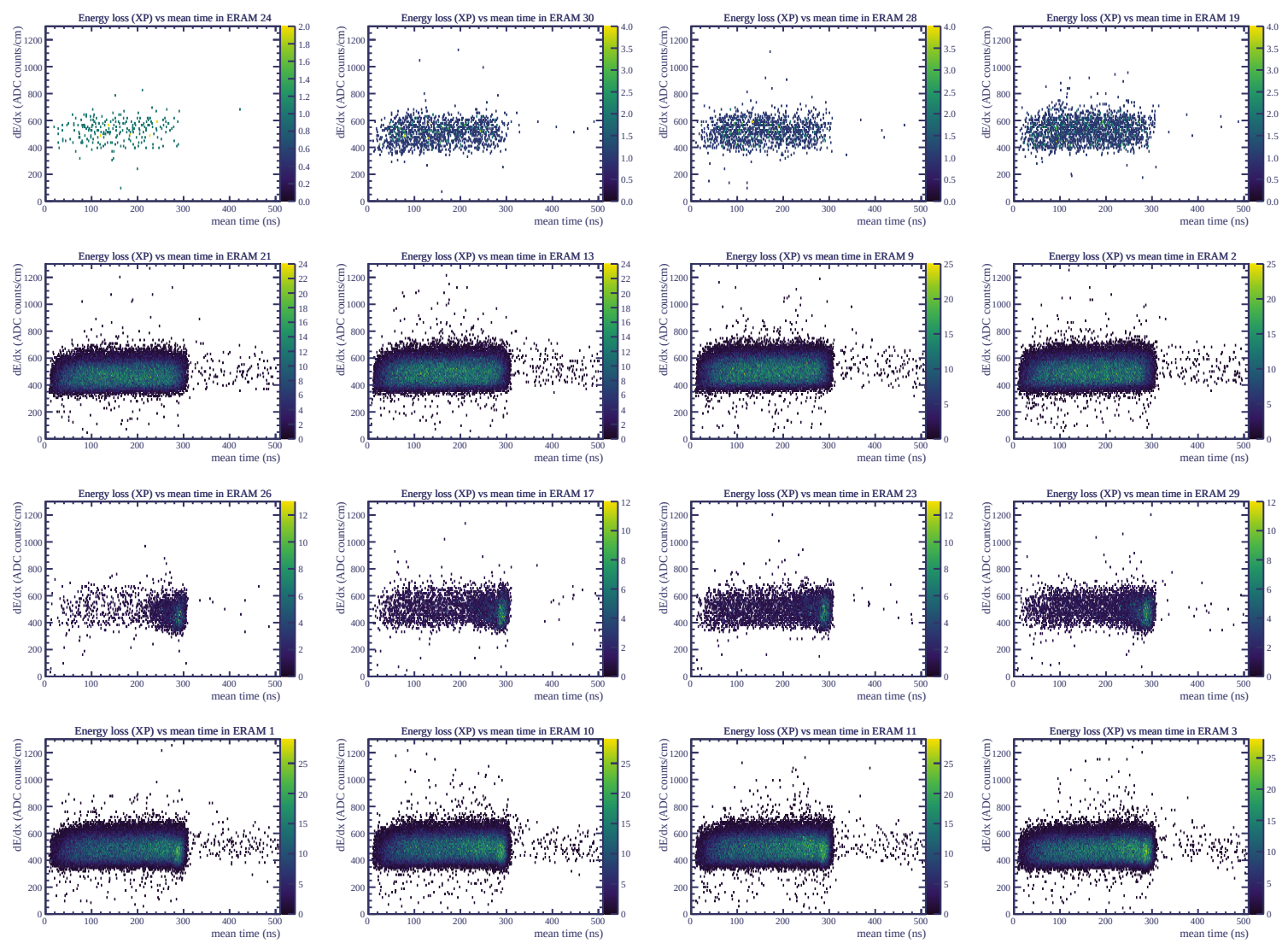


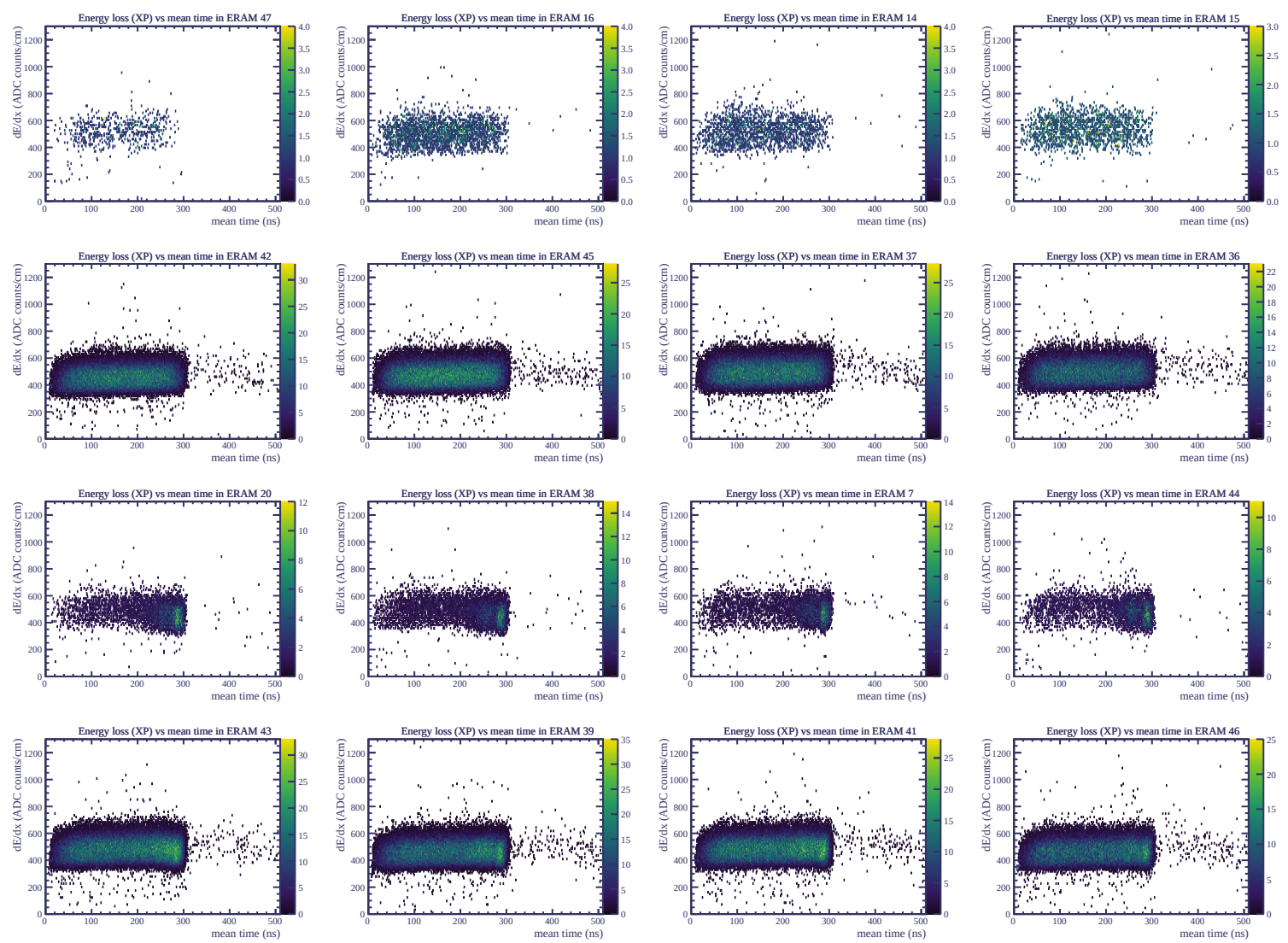




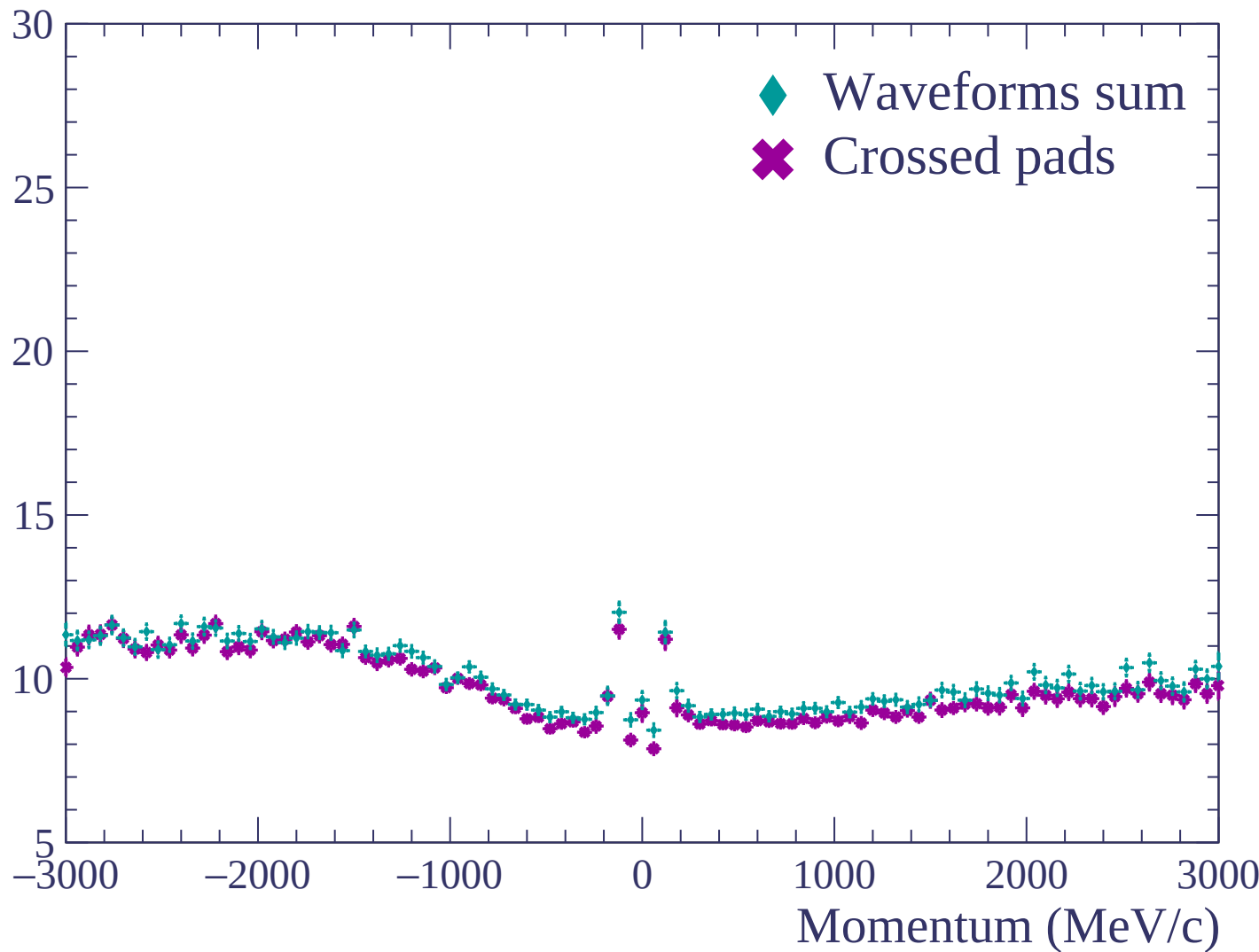


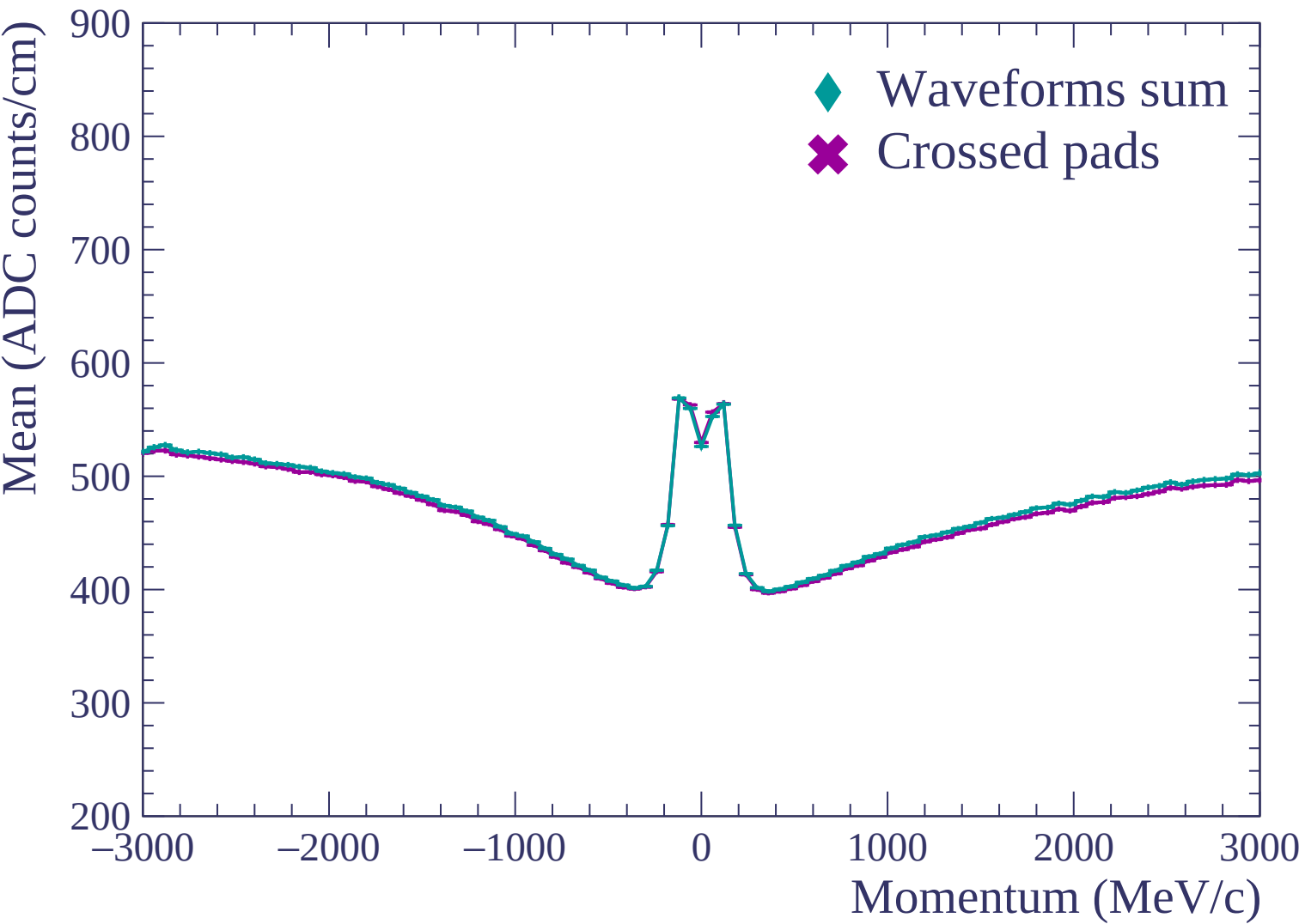


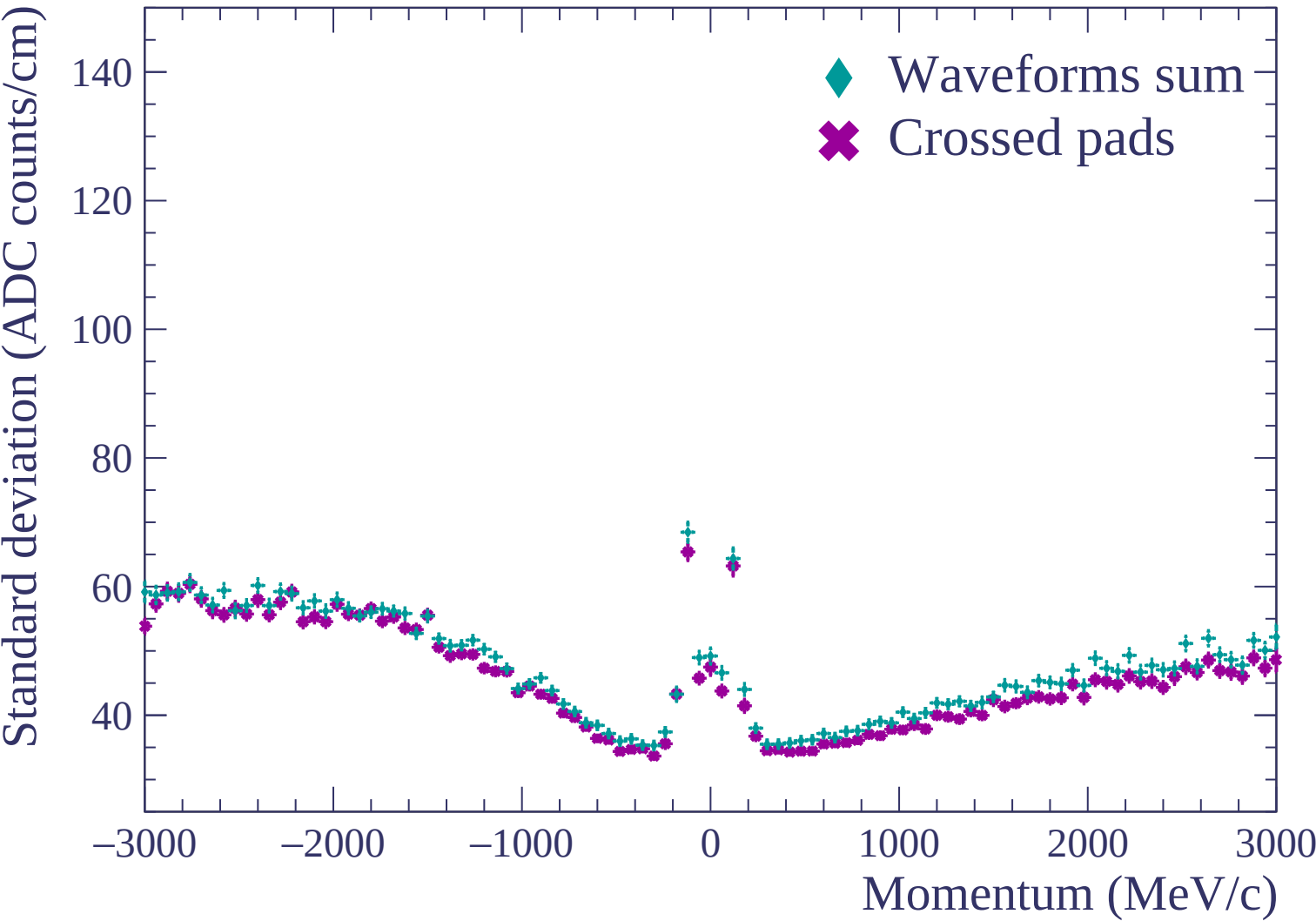


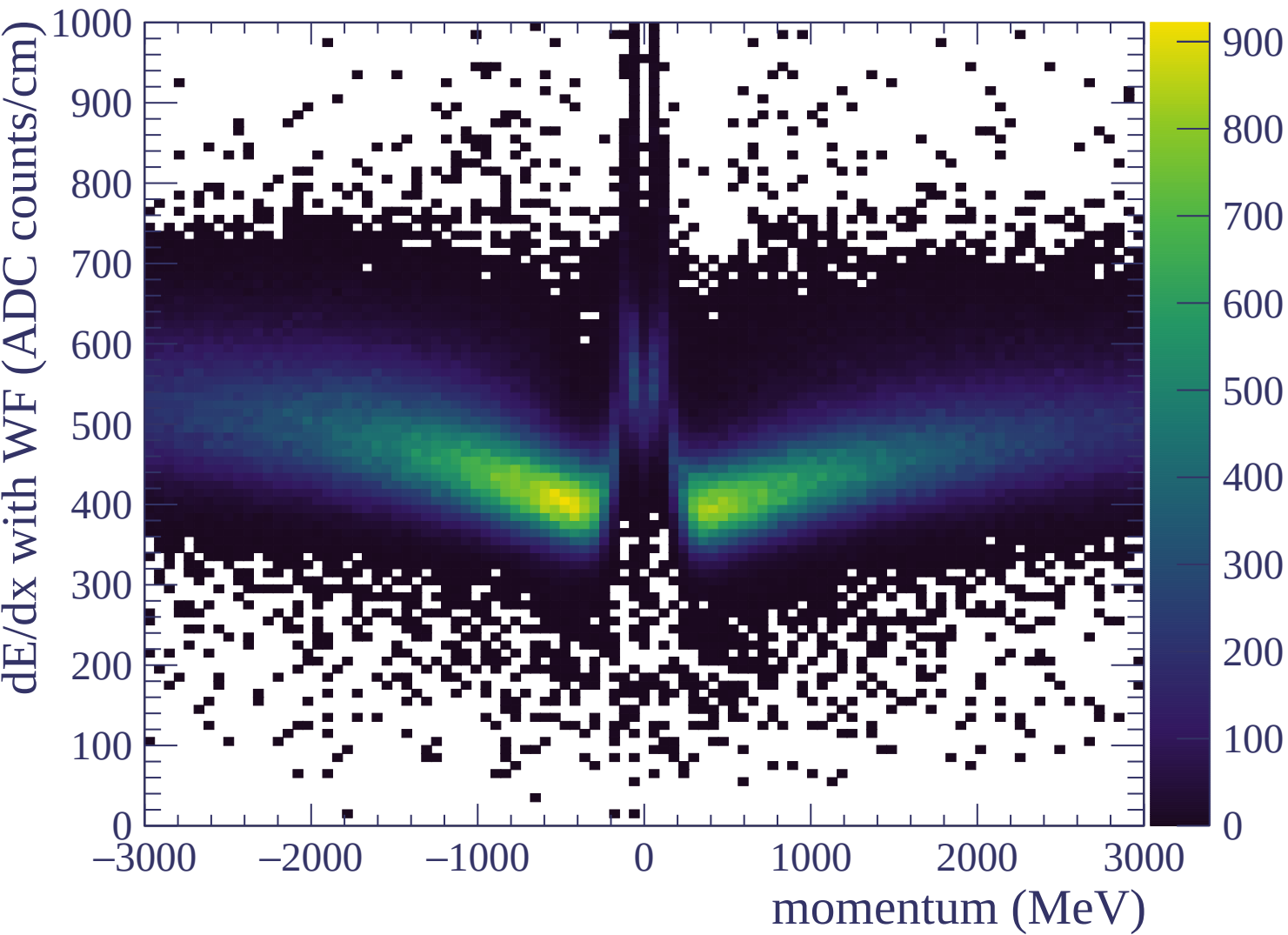


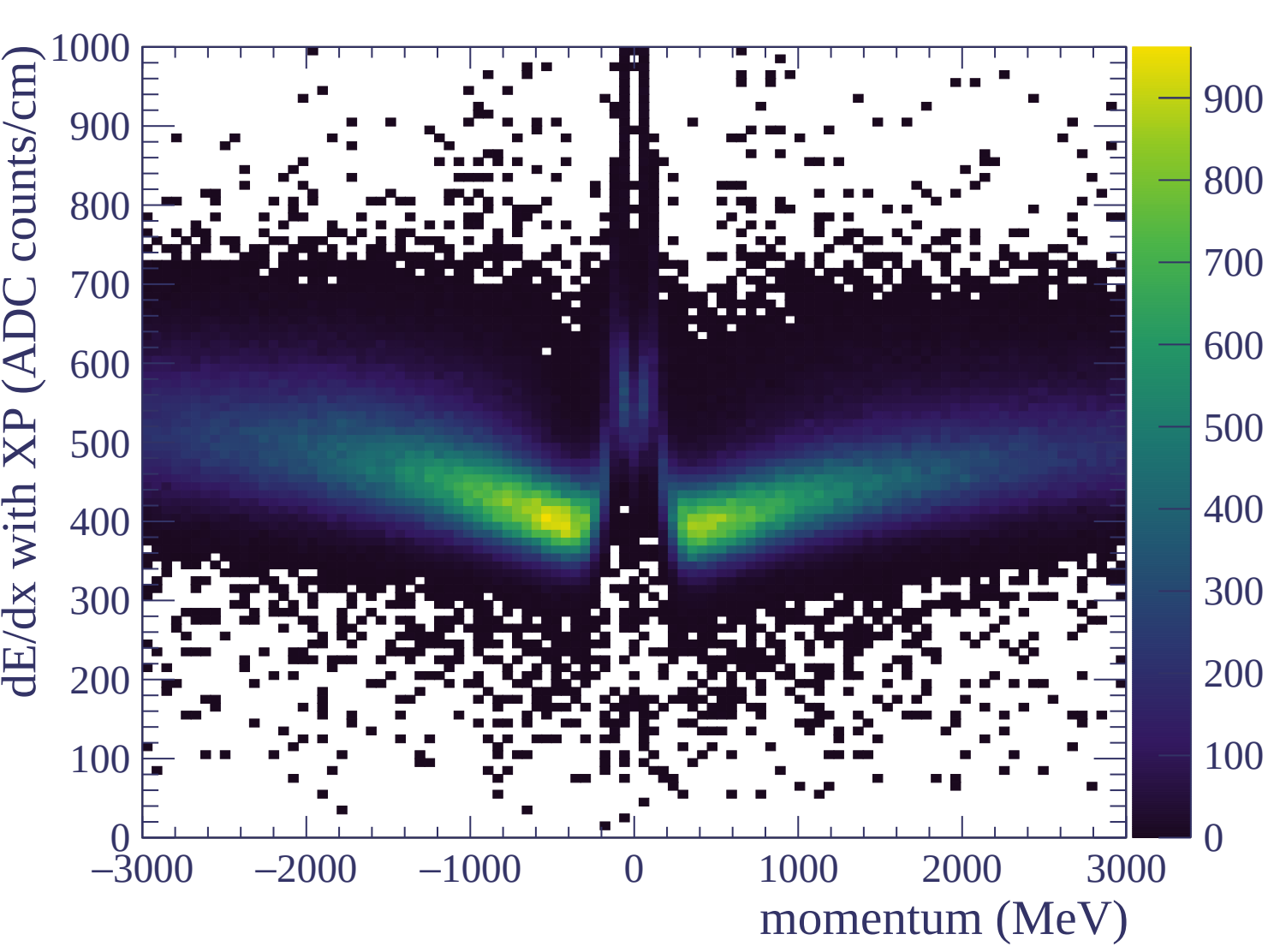
Resolution (%)

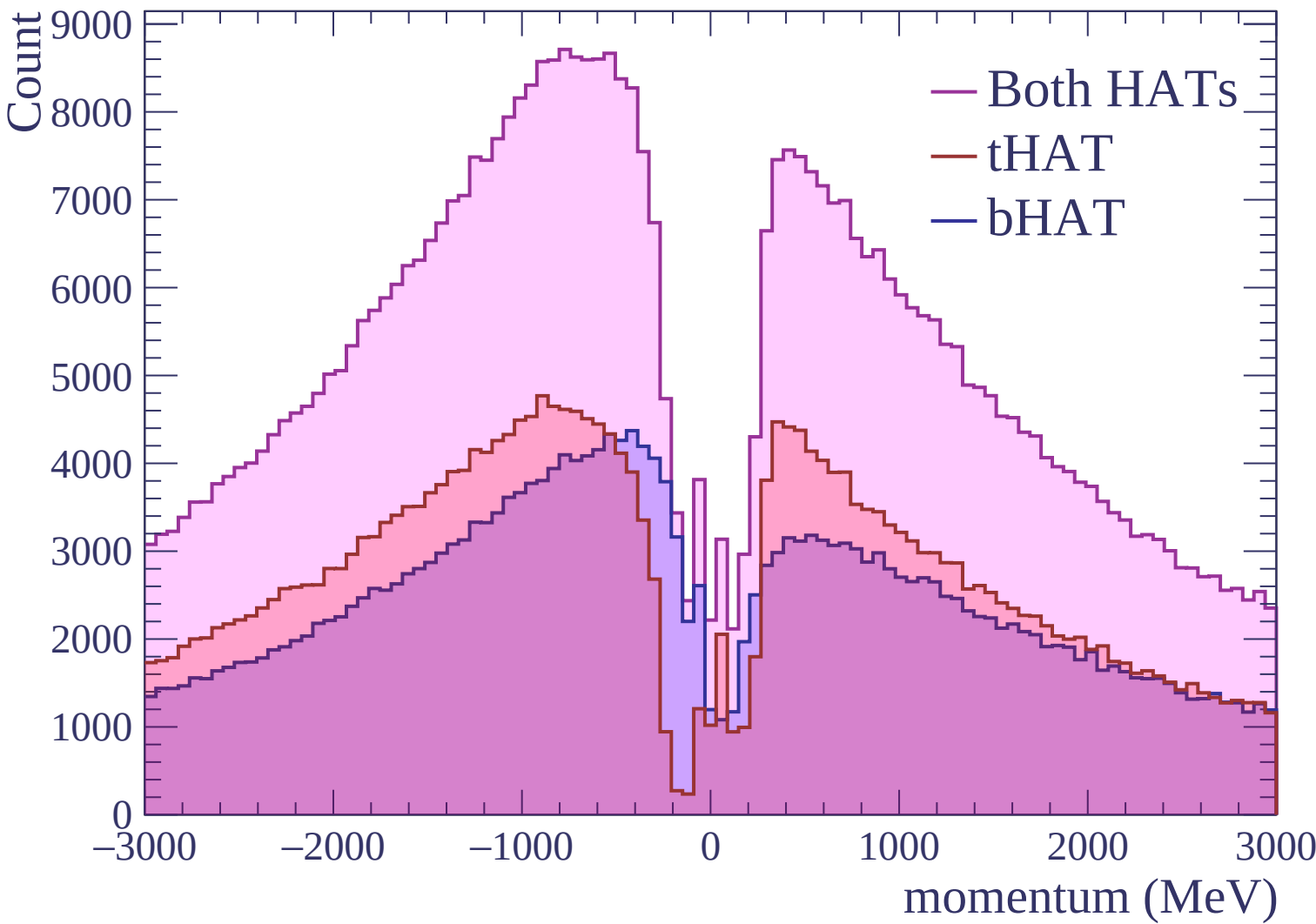






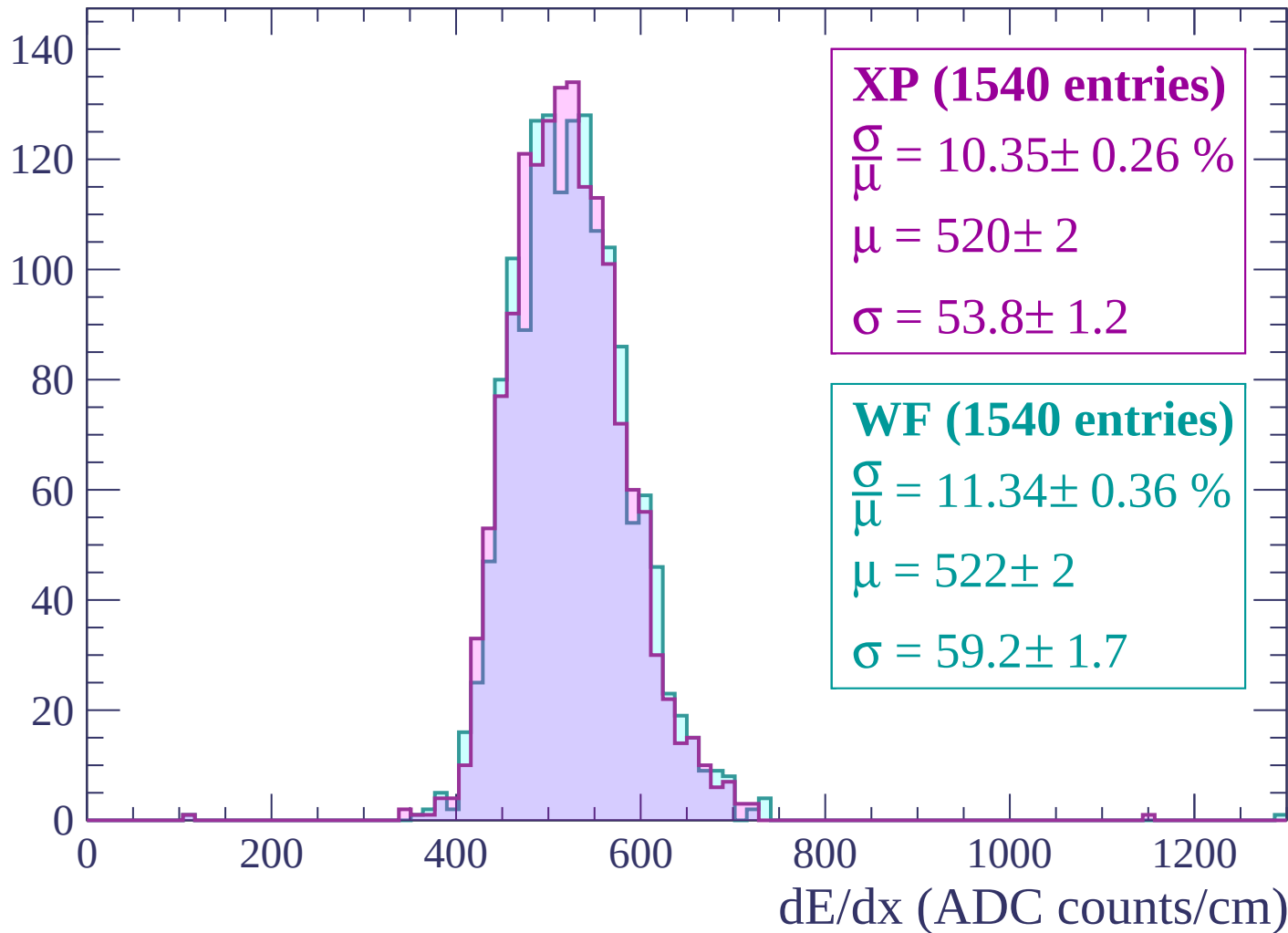






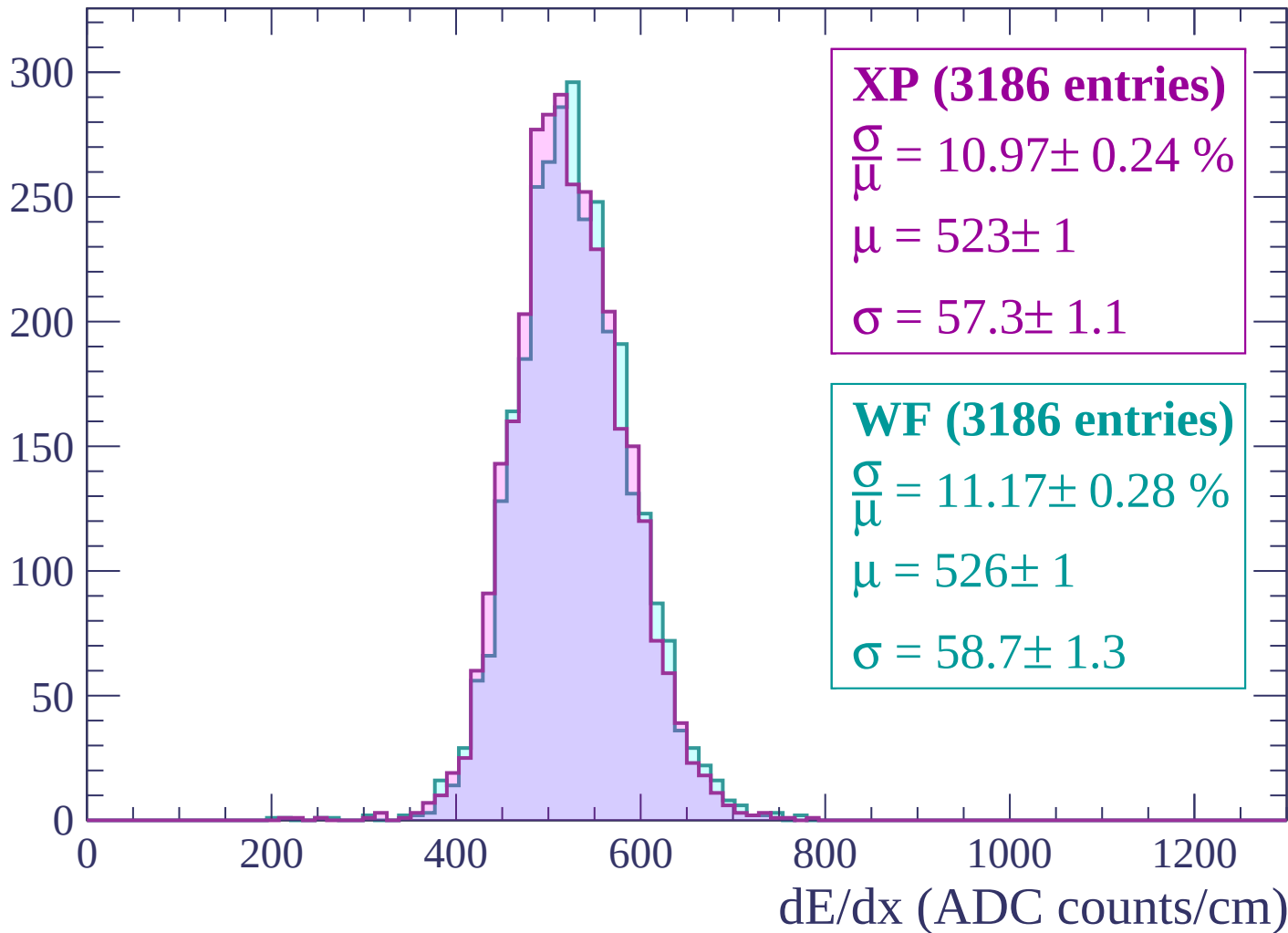
Energy loss | -3000 < p < -2940

Count



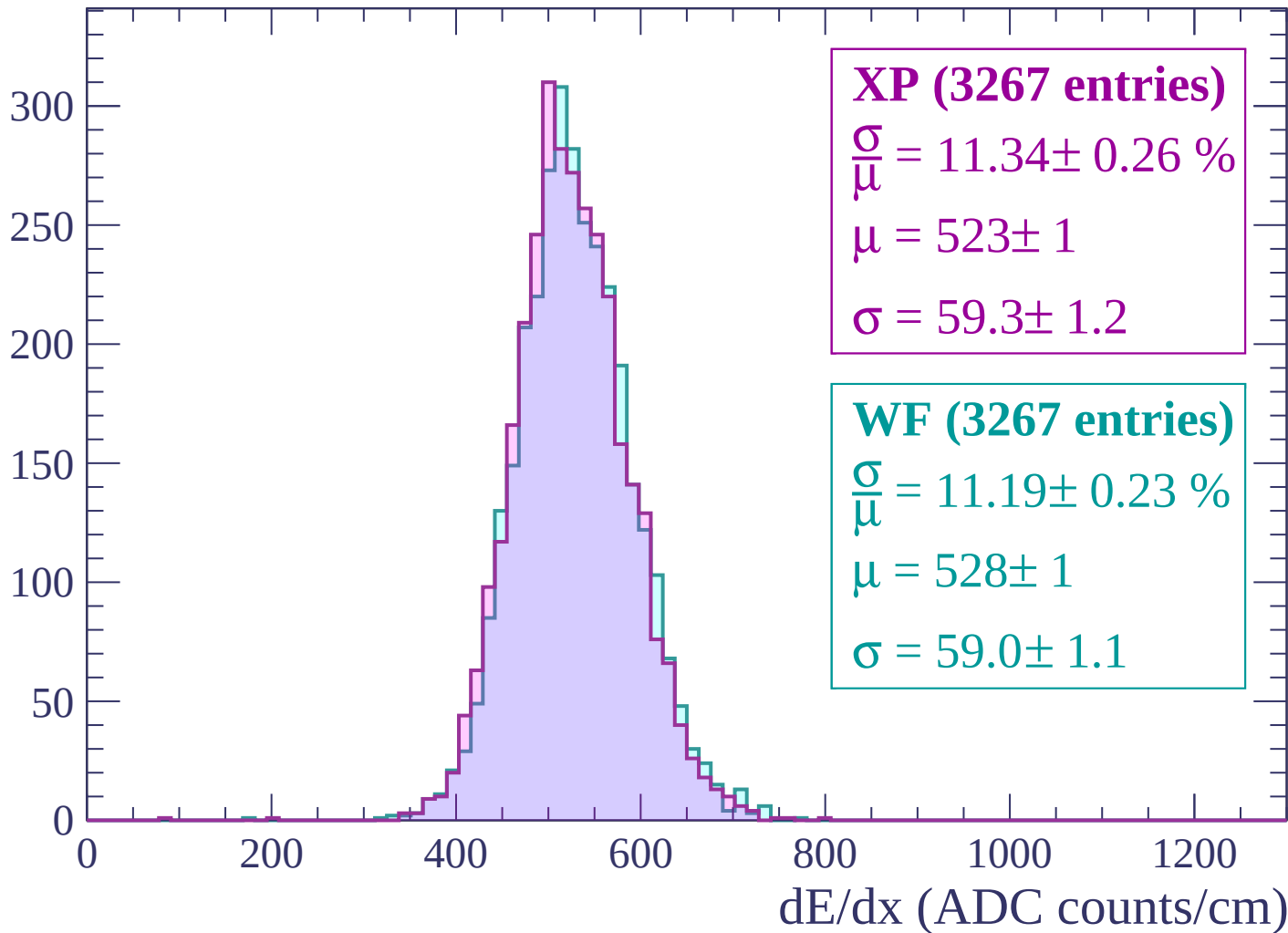
Energy loss | $-2940 < p < -2880$

Count



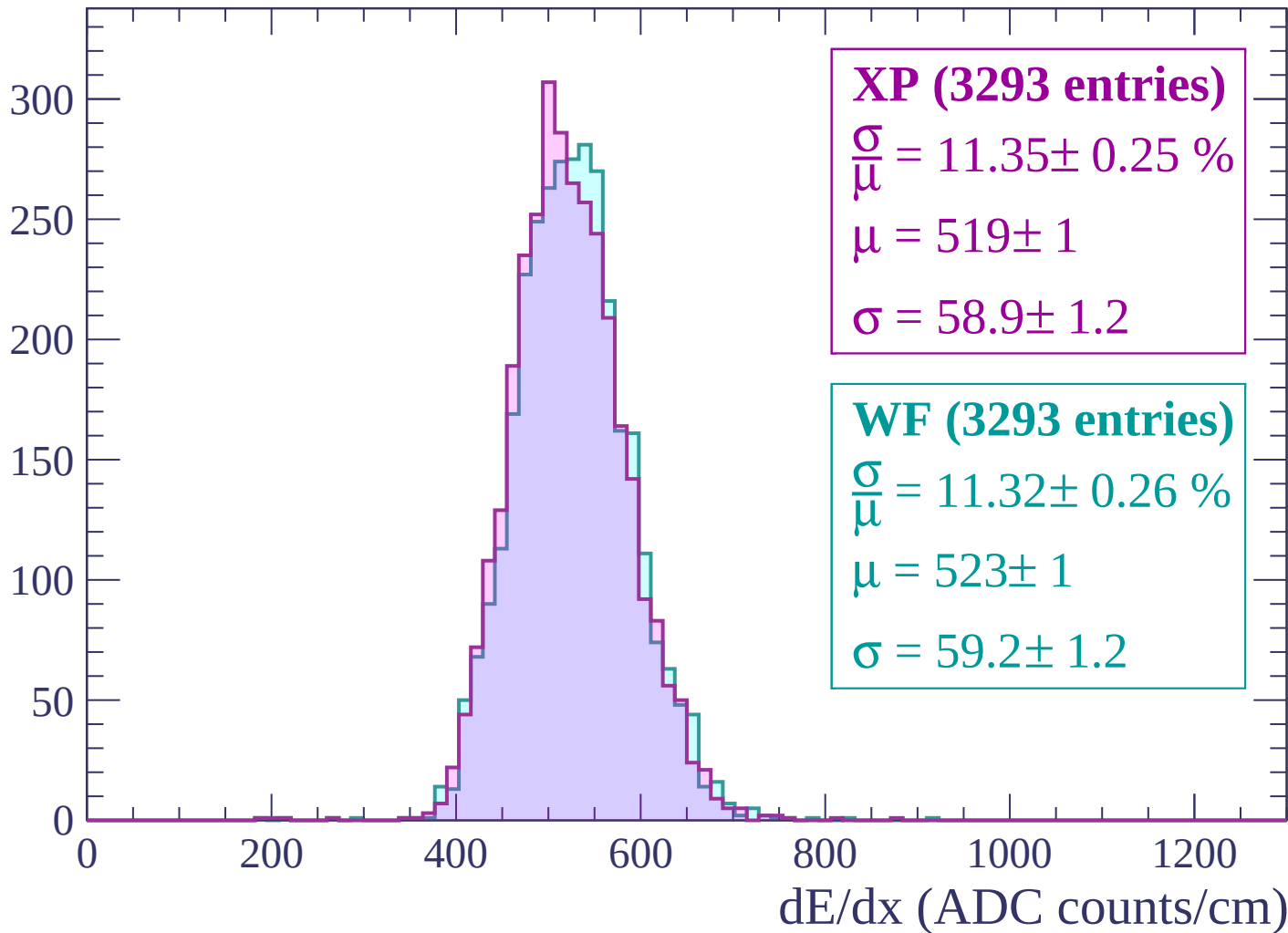
Energy loss | $-2880 < p < -2820$

Count



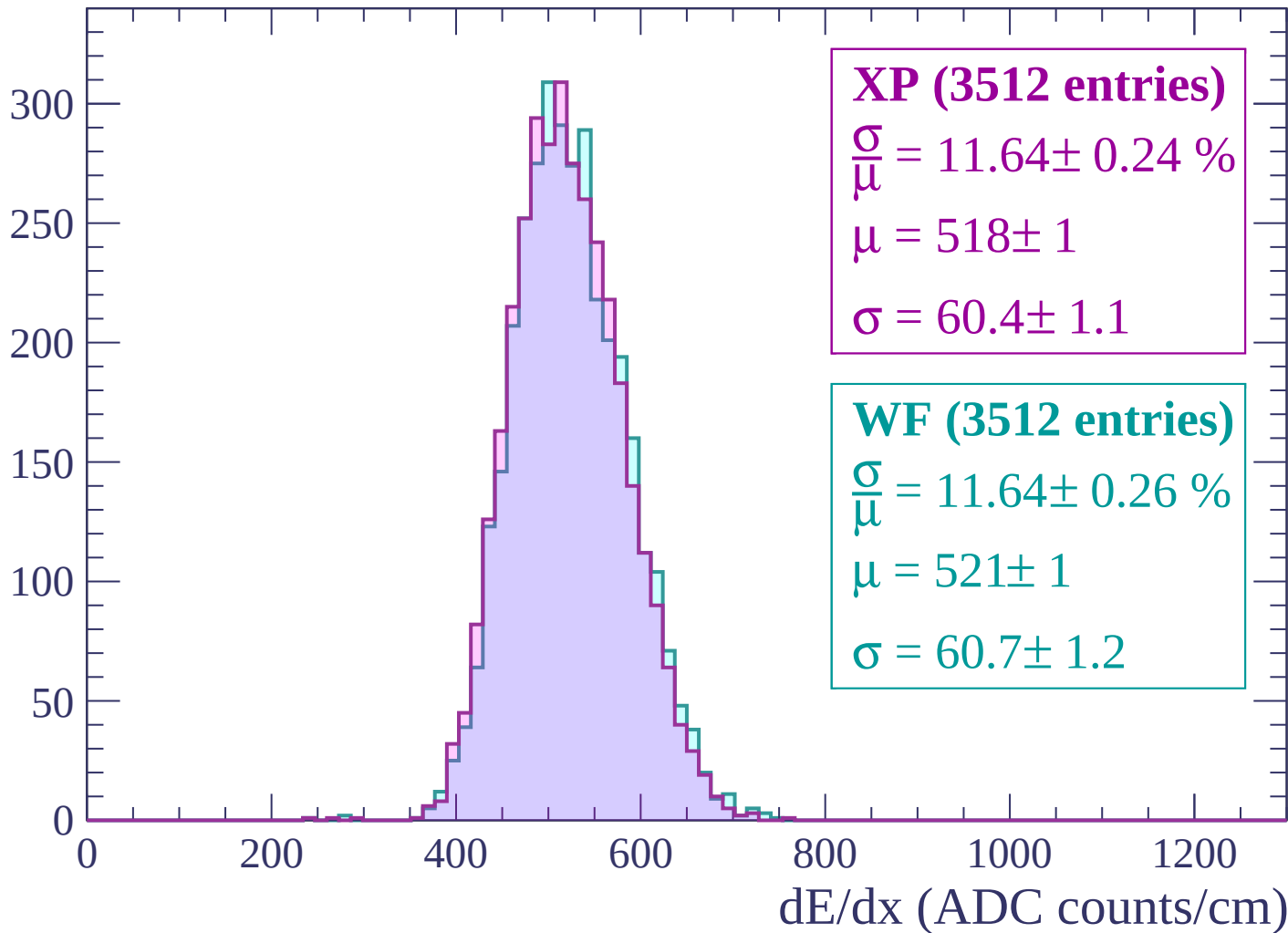
Energy loss | $-2820 < p < -2760$

Count



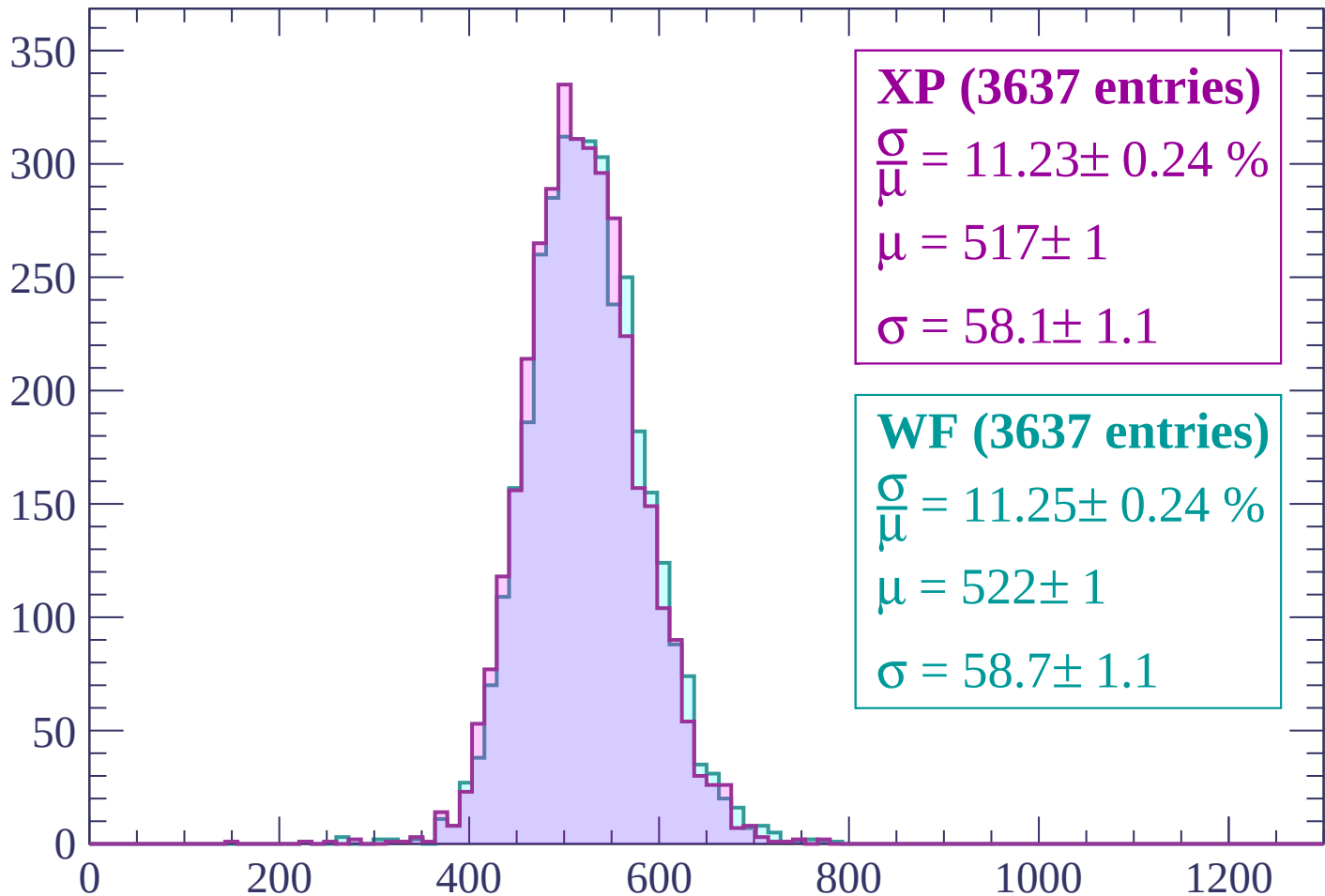
Energy loss | -2760 < p < -2700

Count



Energy loss | -2700 < p < -2640

Count



XP (3637 entries)

$$\frac{\sigma}{\mu} = 11.23 \pm 0.24 \%$$

$$\mu = 517 \pm 1$$

$$\sigma = 58.1 \pm 1.1$$

WF (3637 entries)

$$\frac{\sigma}{\mu} = 11.25 \pm 0.24 \%$$

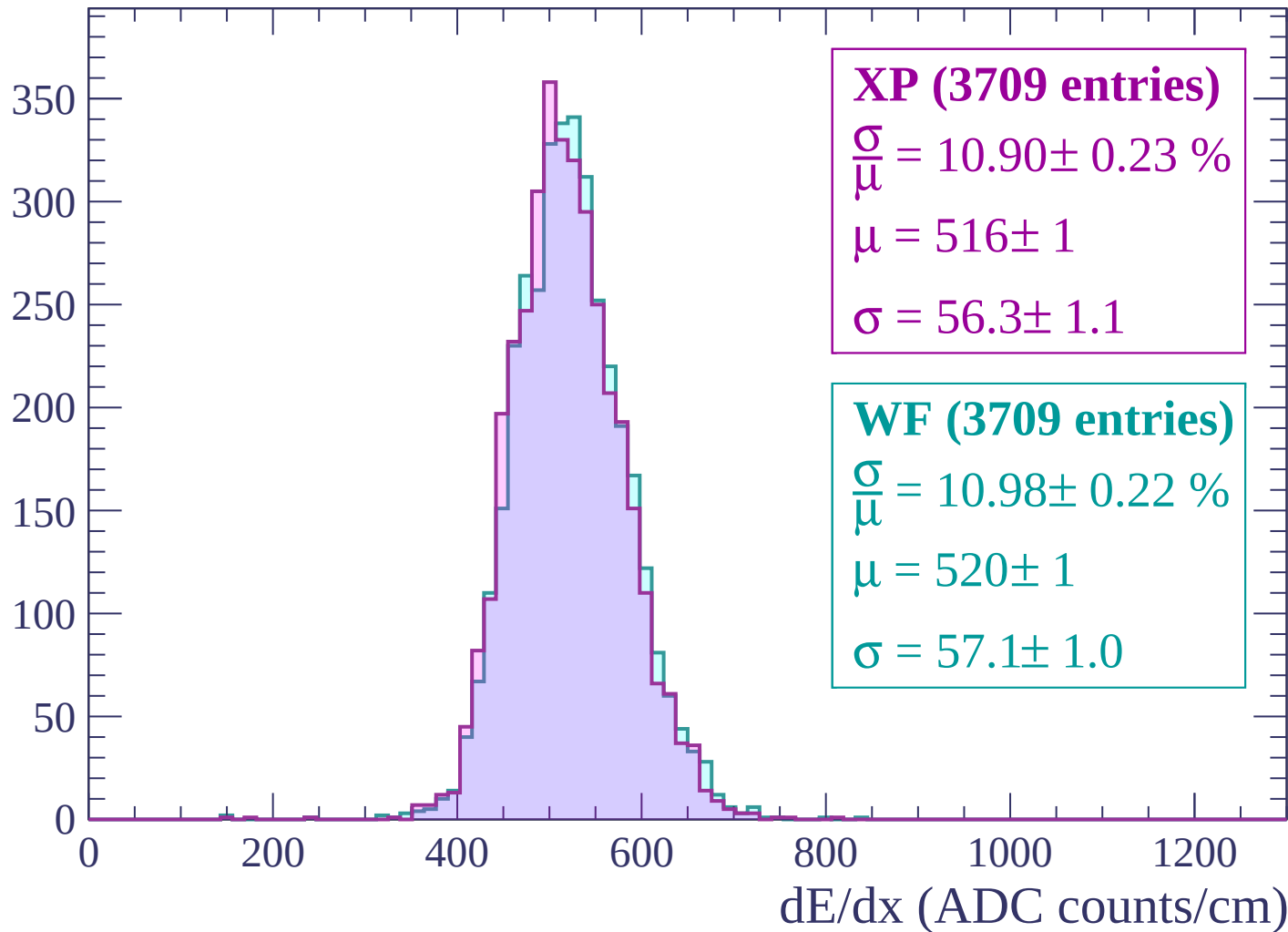
$$\mu = 522 \pm 1$$

$$\sigma = 58.7 \pm 1.1$$

dE/dx (ADC counts/cm)

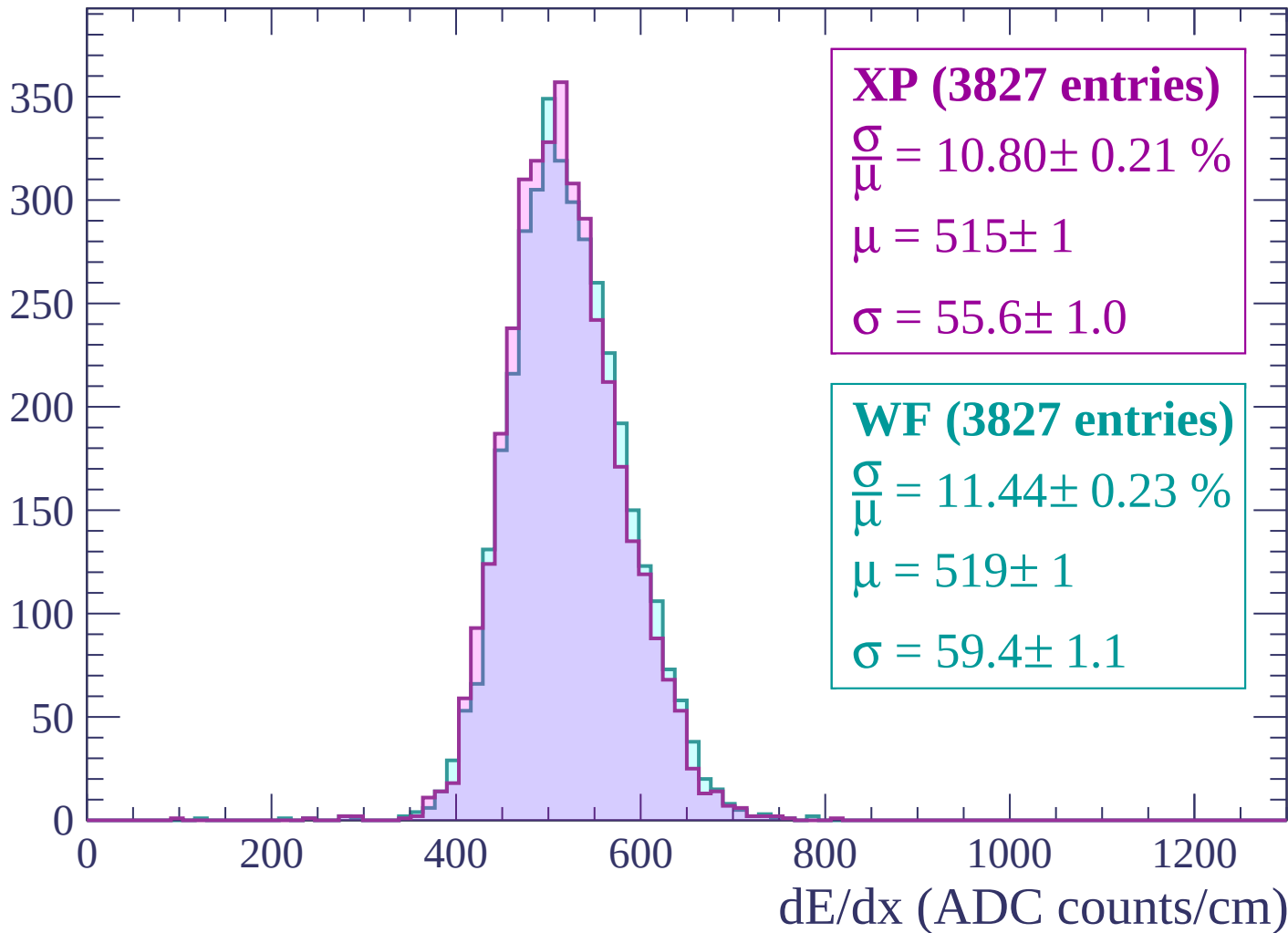
Energy loss | -2640 < p < -2580

Count



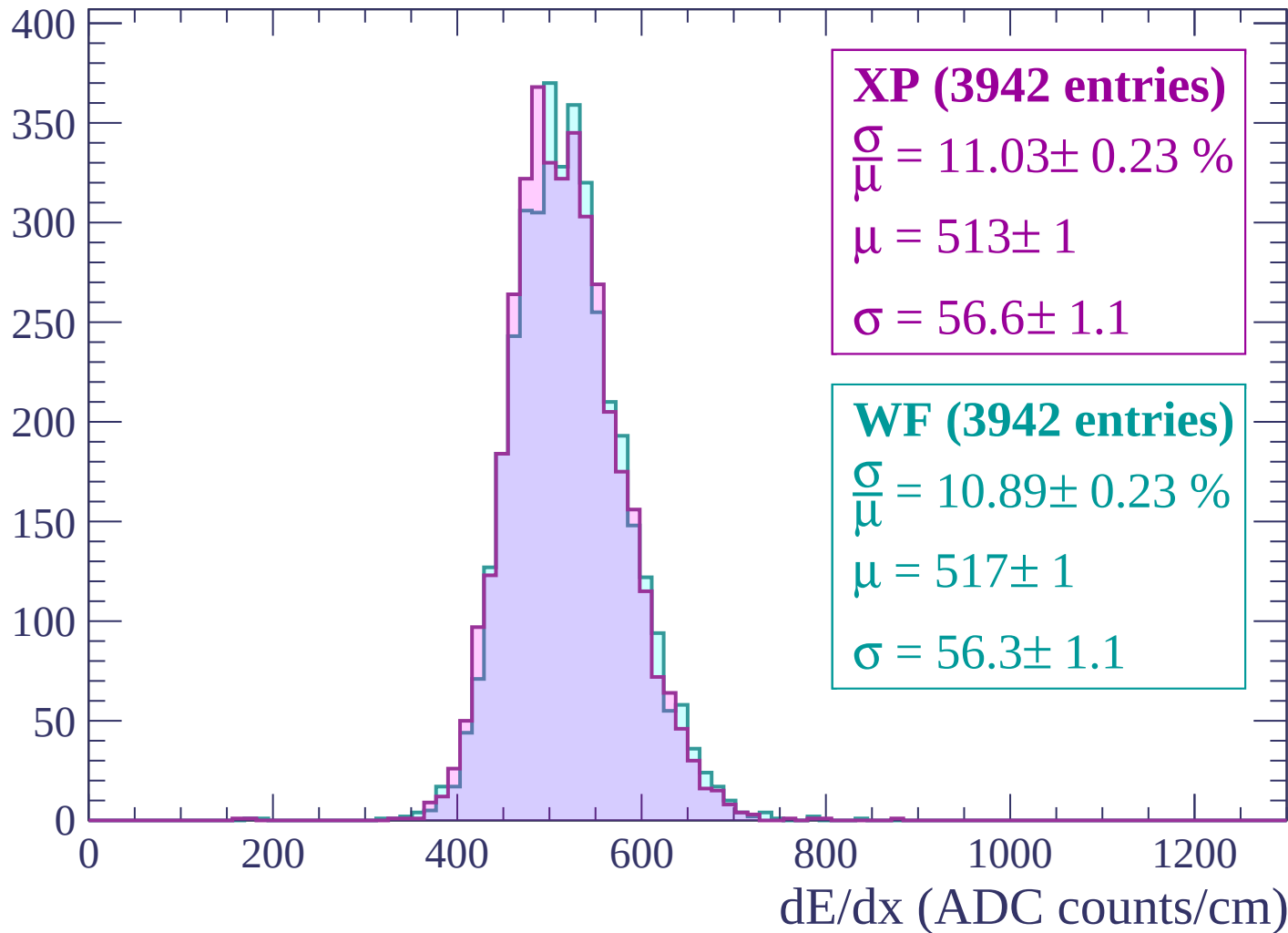
Energy loss | -2580 < p < -2520

Count



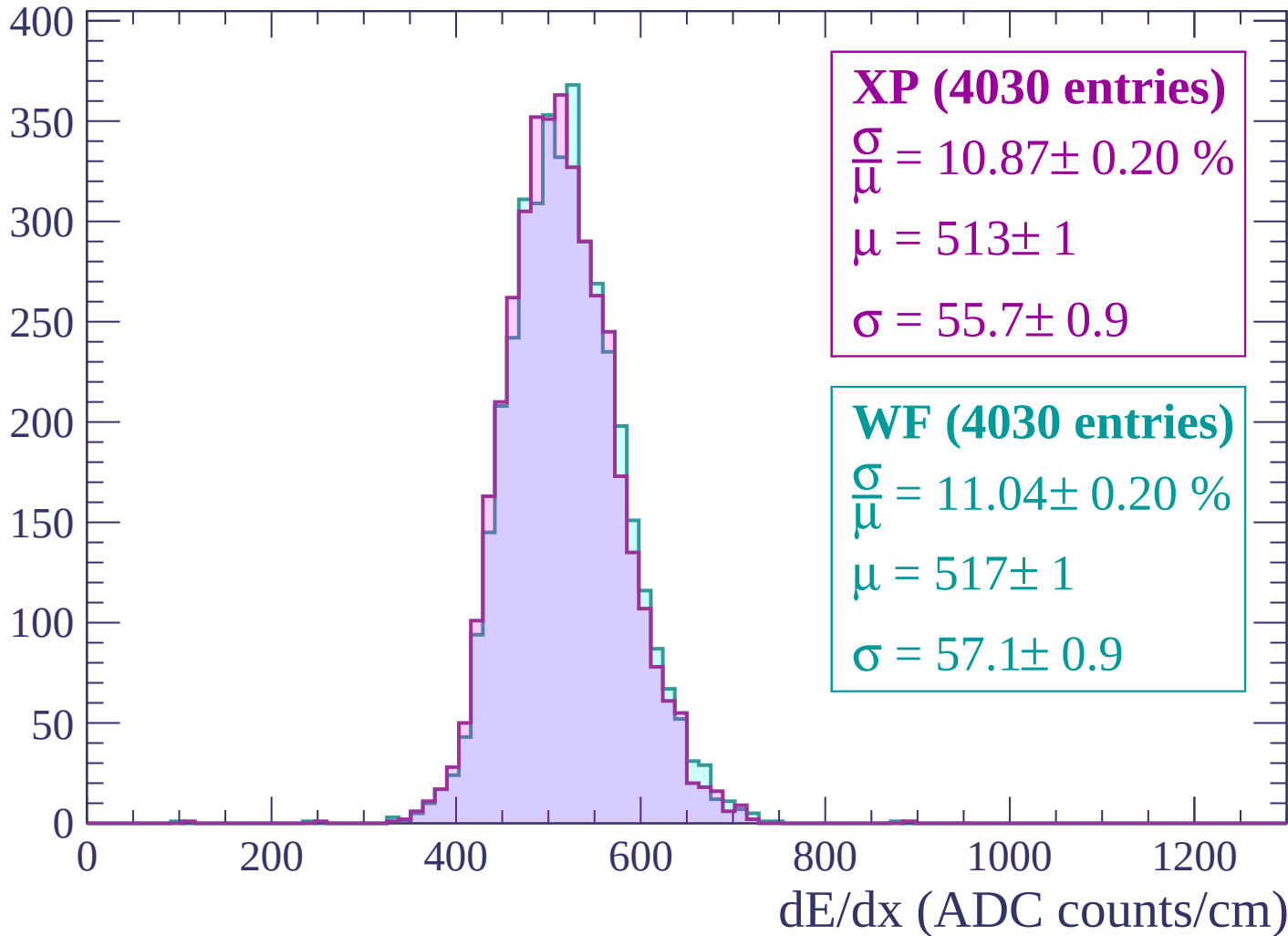
Energy loss | -2520 < p < -2460

Count



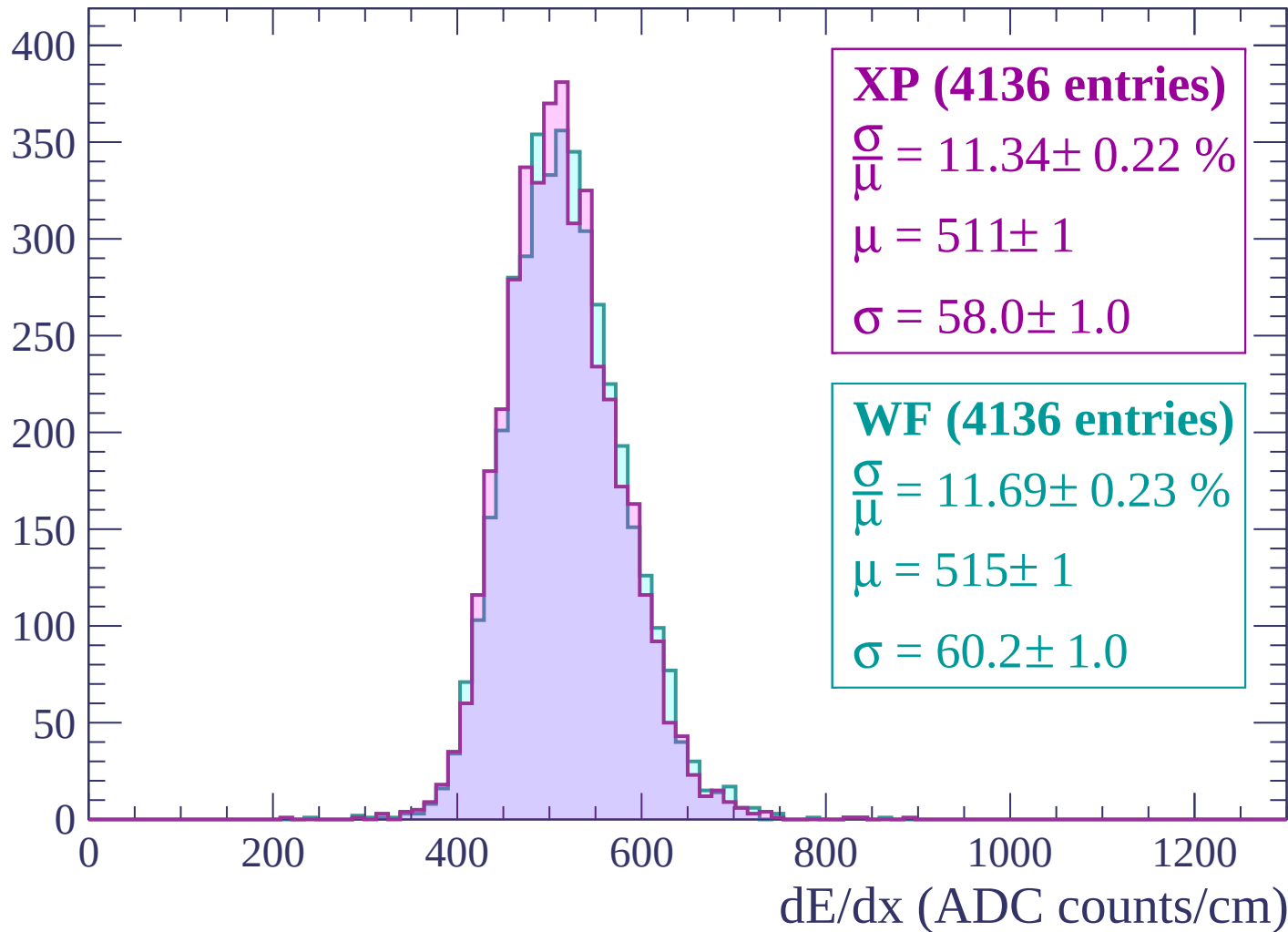
Energy loss | -2460 < p < -2400

Count



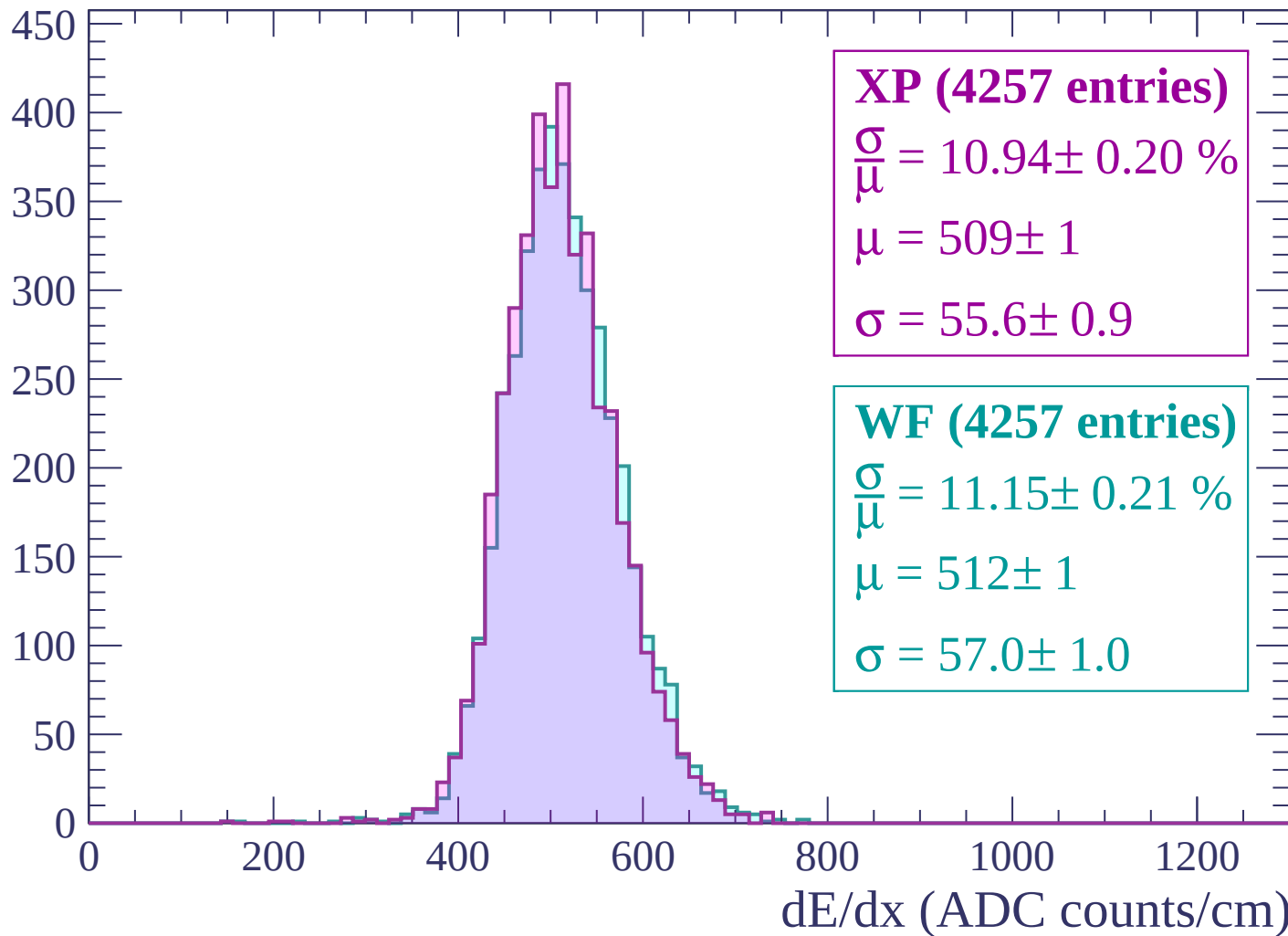
Energy loss | $-2400 < p < -2340$

Count



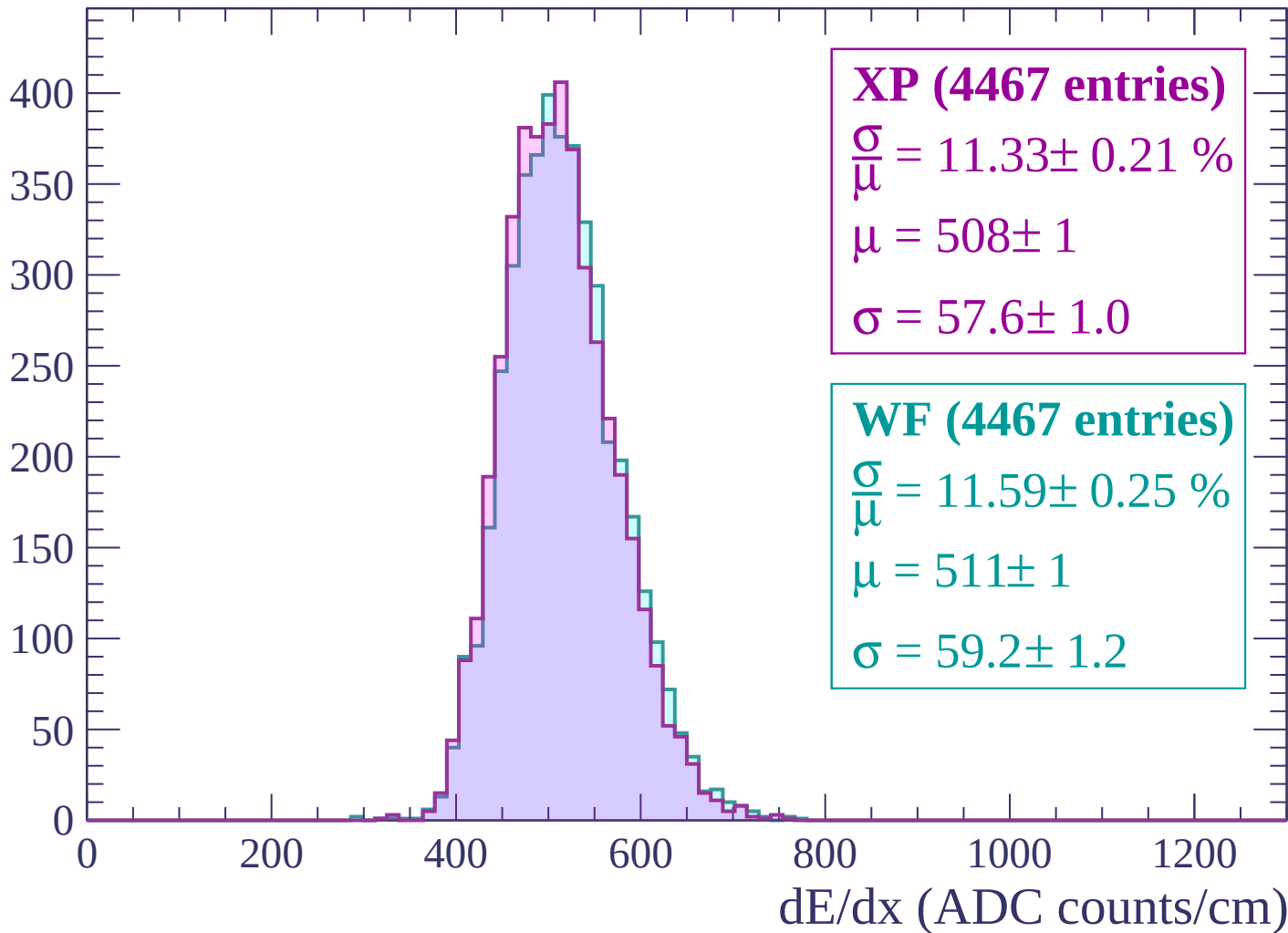
Energy loss | $-2340 < p < -2280$

Count



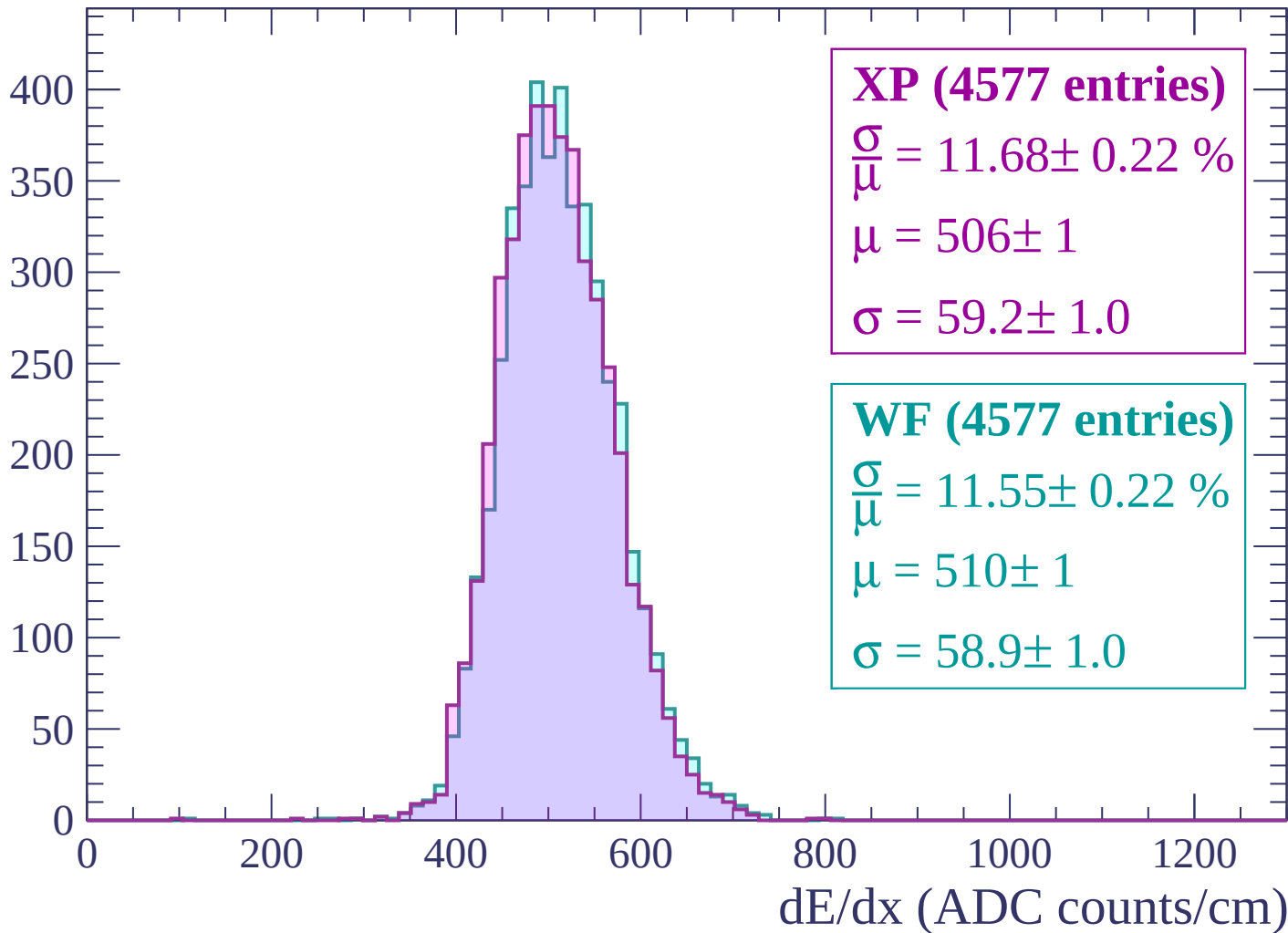
Energy loss | $-2280 < p < -2220$

Count



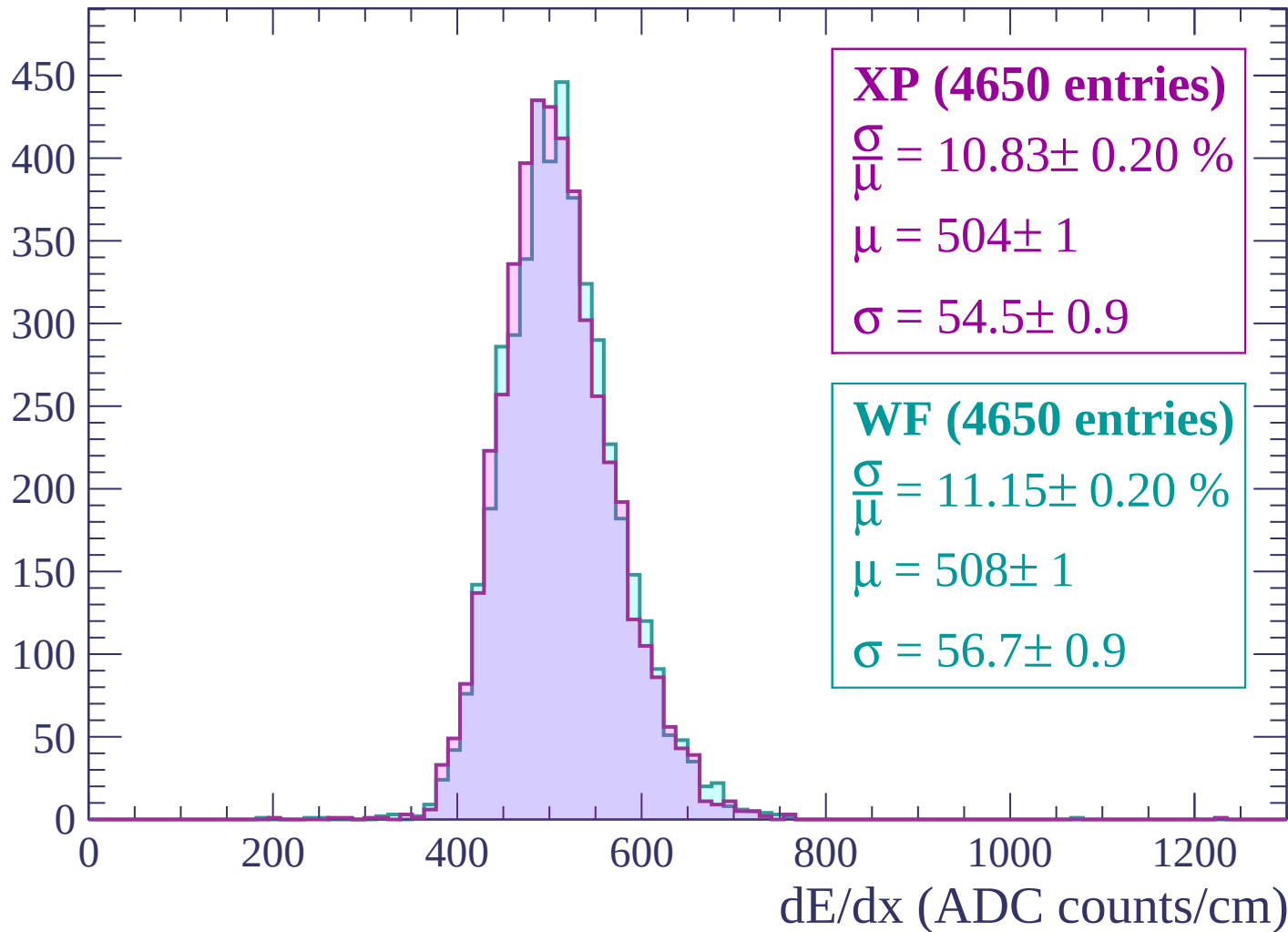
Energy loss | -2220 < p < -2160

Count



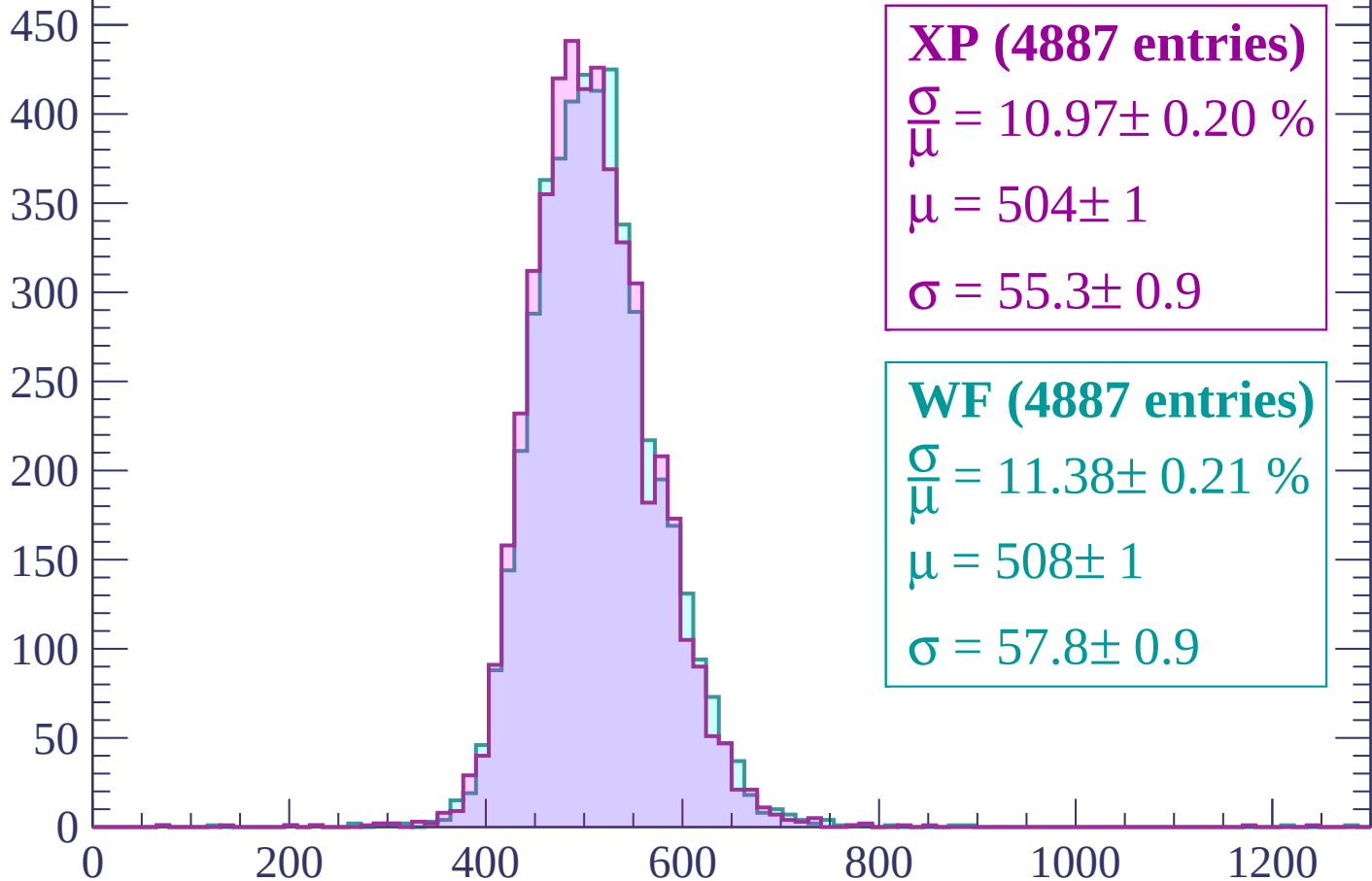
Energy loss | $-2160 < p < -2100$

Count



Energy loss | $-2100 < p < -2040$

Count



XP (4887 entries)

$$\frac{\sigma}{\mu} = 10.97 \pm 0.20 \%$$

$$\mu = 504 \pm 1$$

$$\sigma = 55.3 \pm 0.9$$

WF (4887 entries)

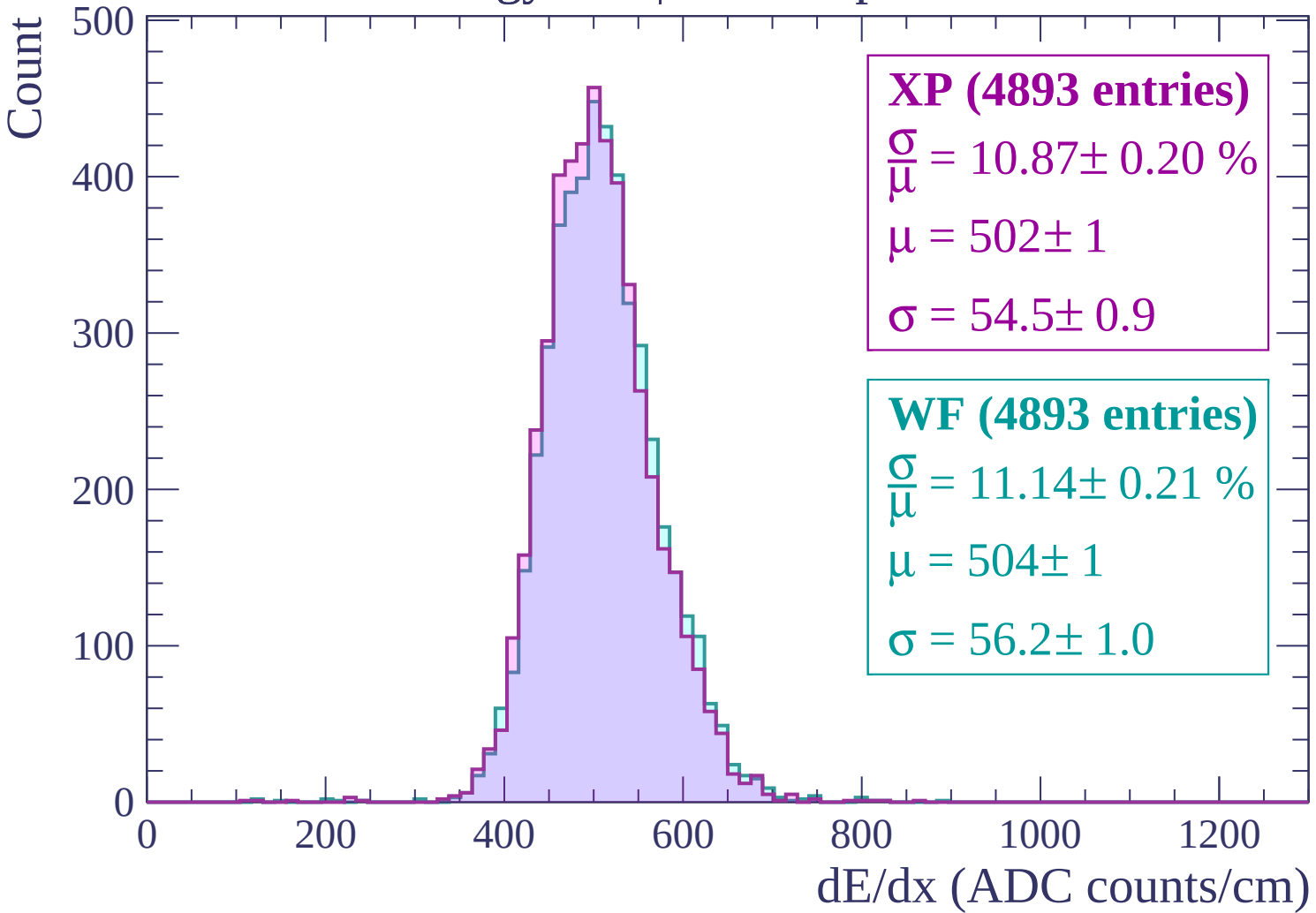
$$\frac{\sigma}{\mu} = 11.38 \pm 0.21 \%$$

$$\mu = 508 \pm 1$$

$$\sigma = 57.8 \pm 0.9$$

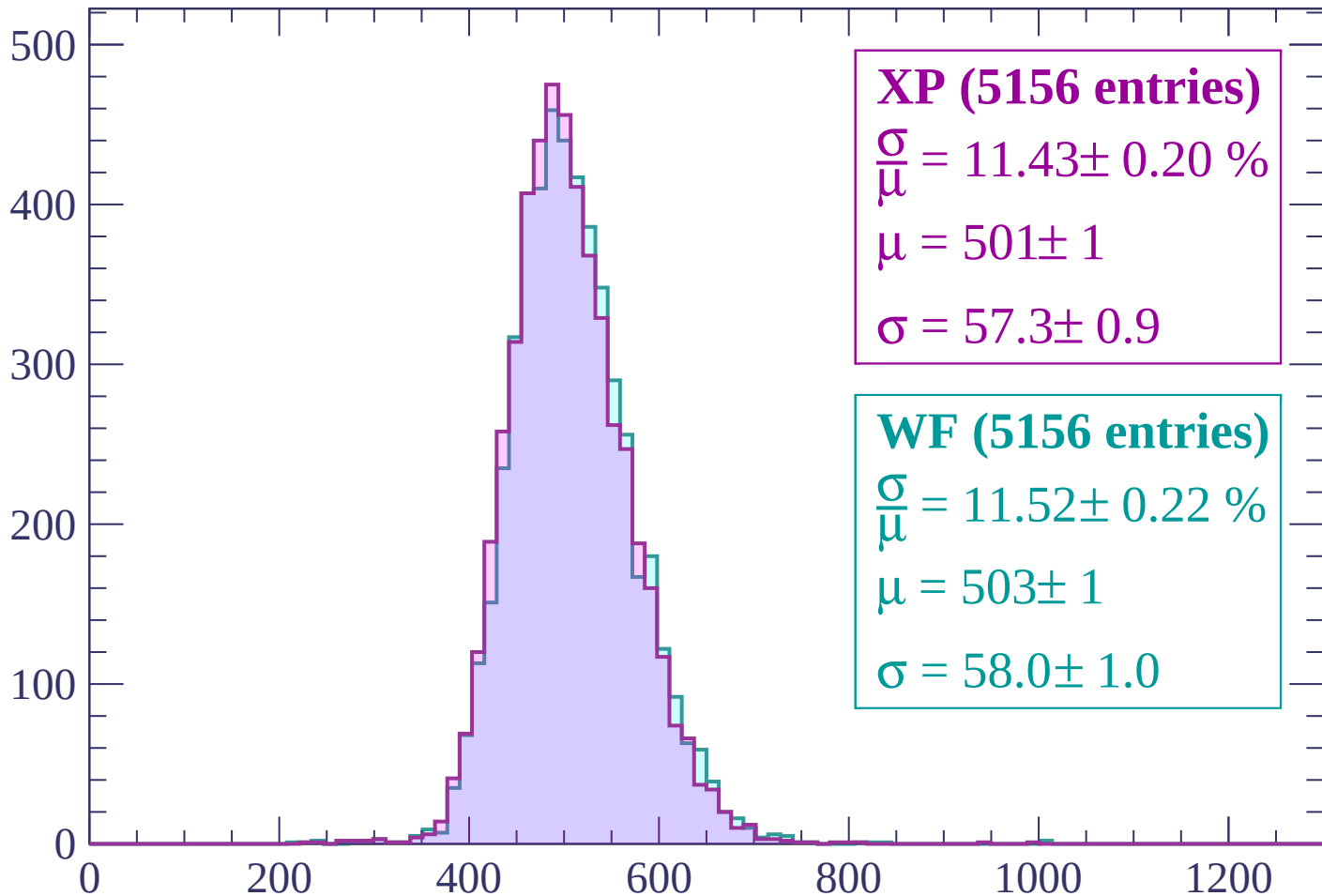
dE/dx (ADC counts/cm)

Energy loss | $-2040 < p < -1980$



Energy loss | -1980 < p < -1920

Count



XP (5156 entries)

$\frac{\sigma}{\mu} = 11.43 \pm 0.20 \%$

$\mu = 501 \pm 1$

$\sigma = 57.3 \pm 0.9$

WF (5156 entries)

$\frac{\sigma}{\mu} = 11.52 \pm 0.22 \%$

$\mu = 503 \pm 1$

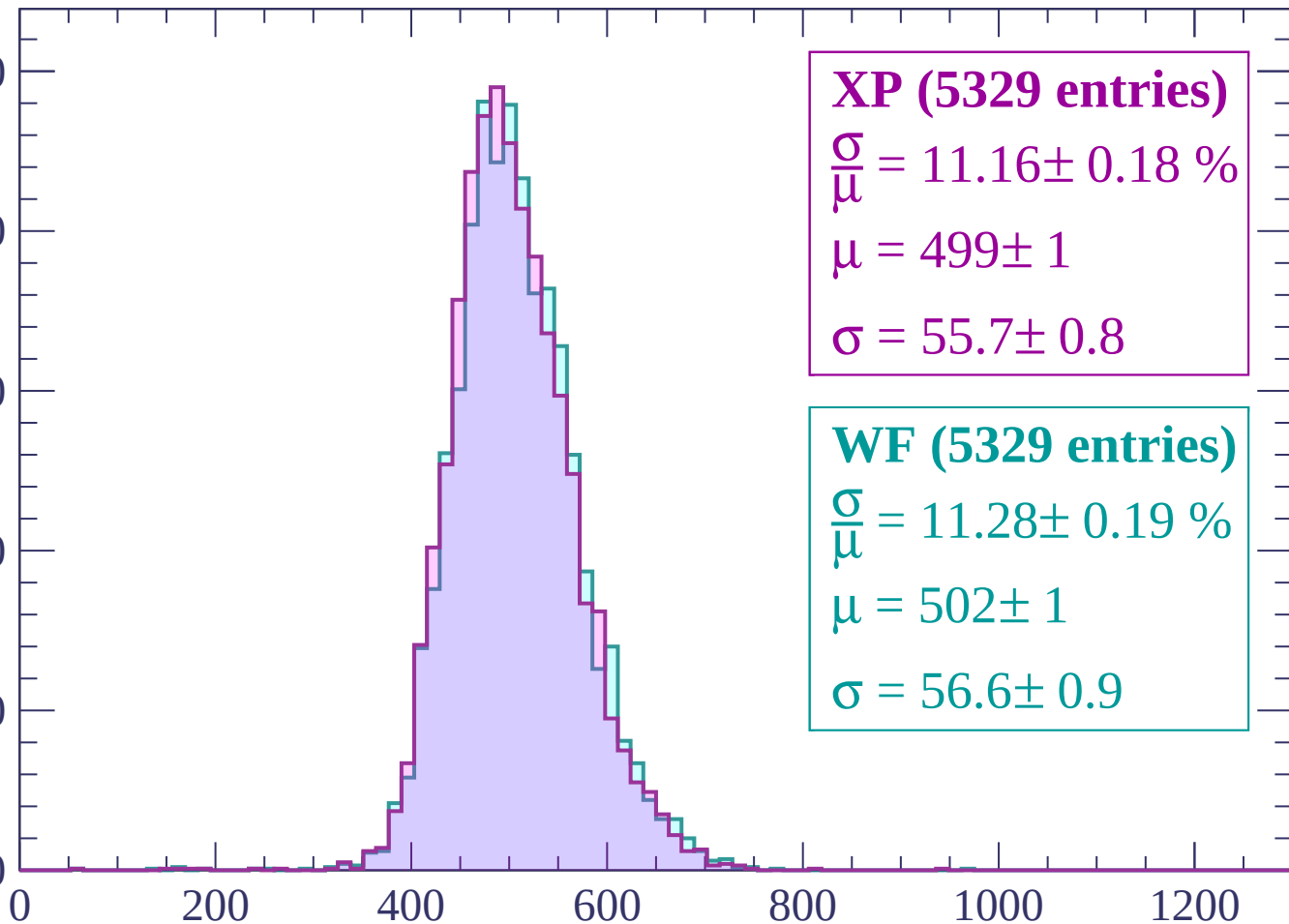
$\sigma = 58.0 \pm 1.0$

dE/dx (ADC counts/cm)

Energy loss | -1920 < p < -1860

Count

500
400
300
200
100
0



XP (5329 entries)

$\frac{\sigma}{\mu} = 11.16 \pm 0.18 \%$

$\mu = 499 \pm 1$

$\sigma = 55.7 \pm 0.8$

WF (5329 entries)

$\frac{\sigma}{\mu} = 11.28 \pm 0.19 \%$

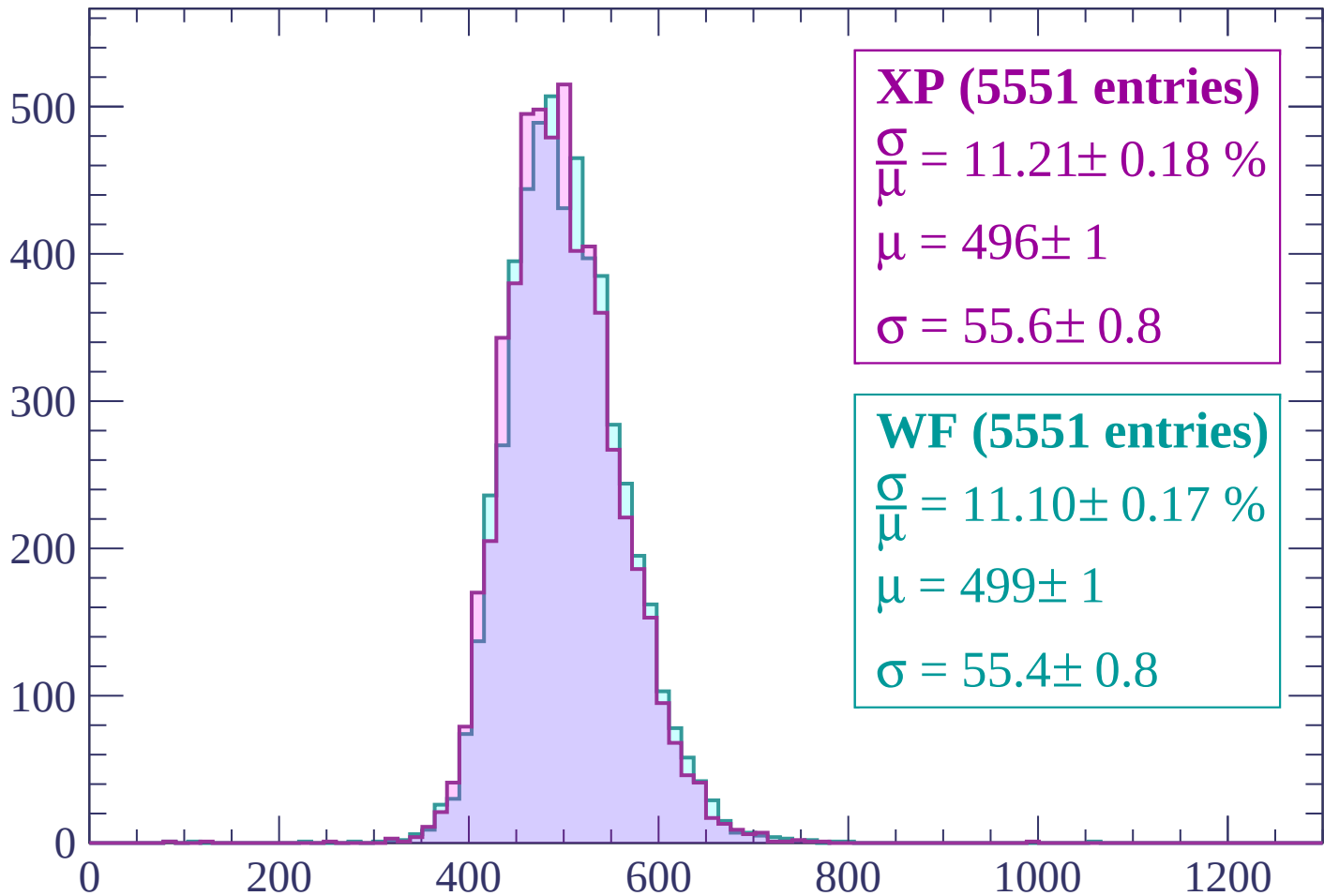
$\mu = 502 \pm 1$

$\sigma = 56.6 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss | -1860 < p < -1800

Count



XP (5551 entries)

$$\frac{\sigma}{\mu} = 11.21 \pm 0.18 \%$$

$$\mu = 496 \pm 1$$

$$\sigma = 55.6 \pm 0.8$$

WF (5551 entries)

$$\frac{\sigma}{\mu} = 11.10 \pm 0.17 \%$$

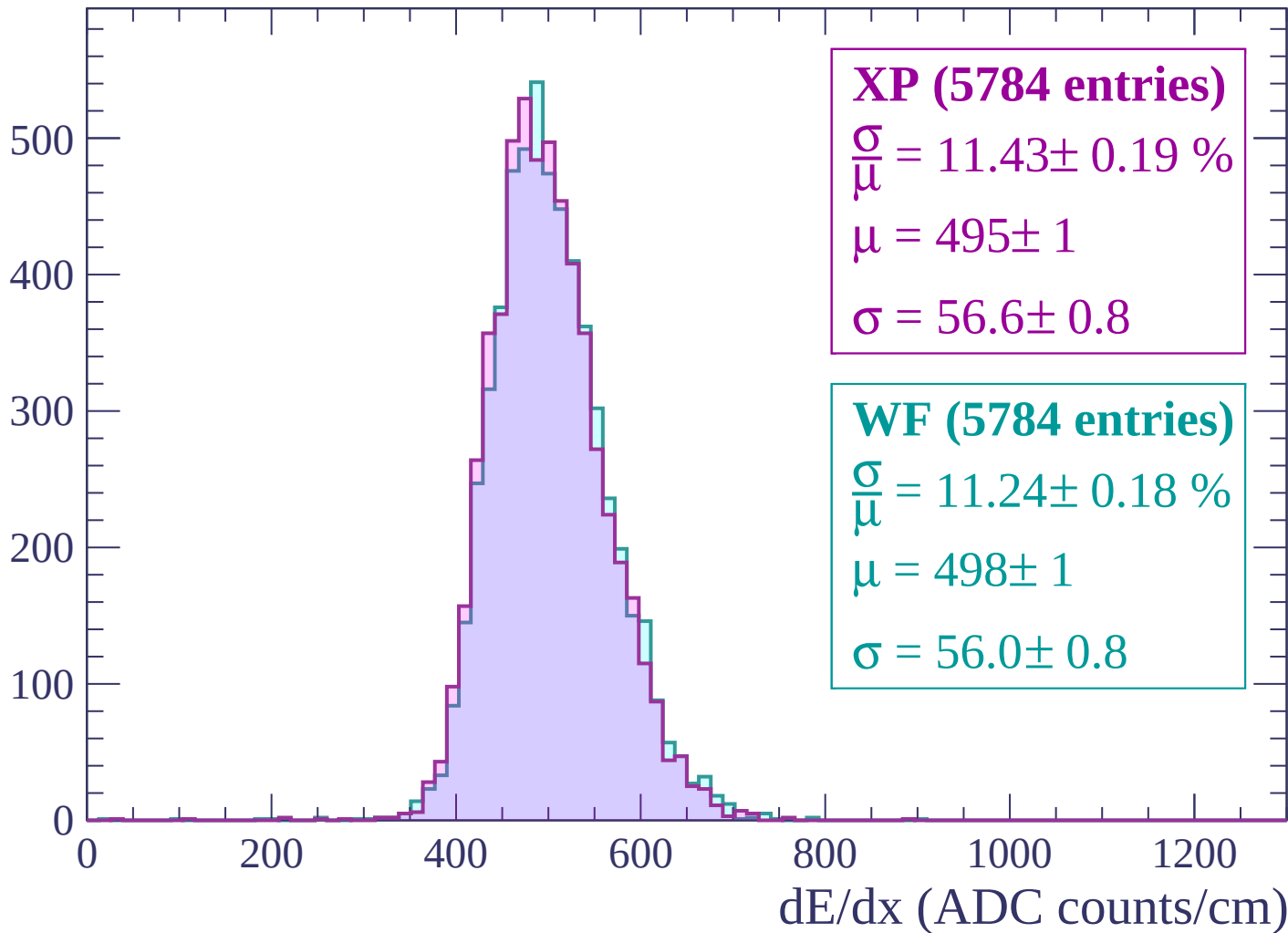
$$\mu = 499 \pm 1$$

$$\sigma = 55.4 \pm 0.8$$

dE/dx (ADC counts/cm)

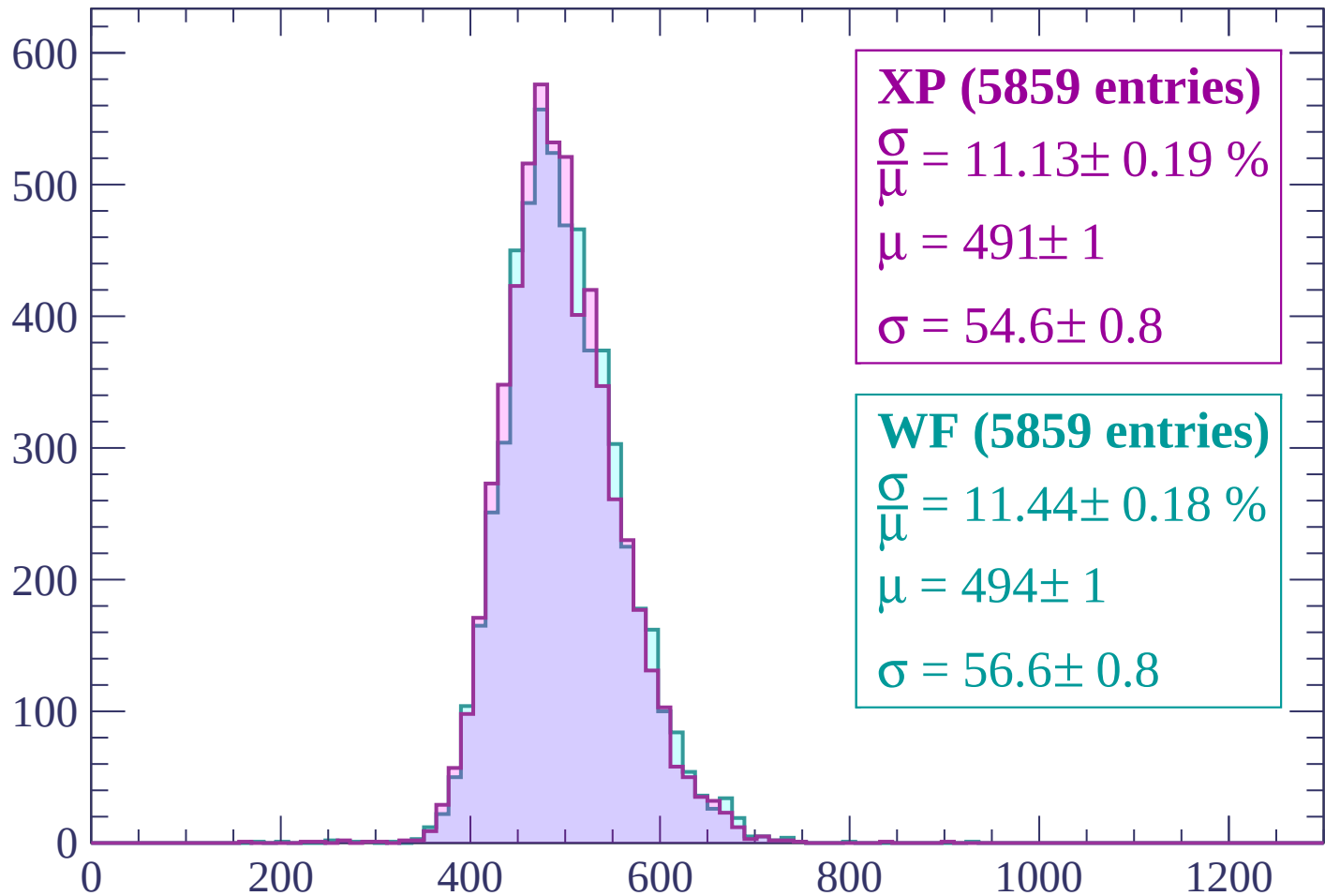
Energy loss | $-1800 < p < -1740$

Count



Energy loss | $-1740 < p < -1680$

Count



XP (5859 entries)

$\frac{\sigma}{\mu} = 11.13 \pm 0.19 \%$

$\mu = 491 \pm 1$

$\sigma = 54.6 \pm 0.8$

WF (5859 entries)

$\frac{\sigma}{\mu} = 11.44 \pm 0.18 \%$

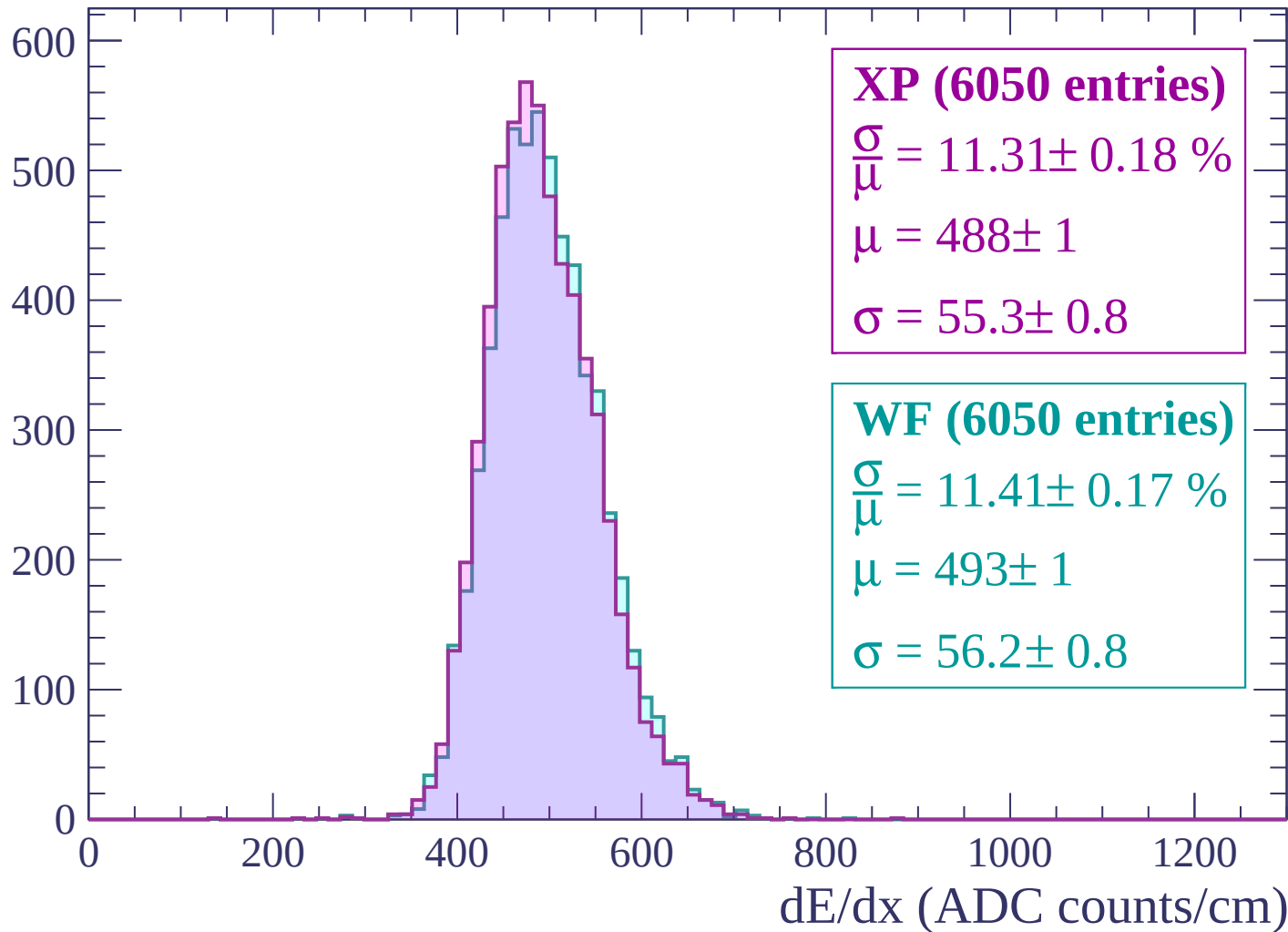
$\mu = 494 \pm 1$

$\sigma = 56.6 \pm 0.8$

dE/dx (ADC counts/cm)

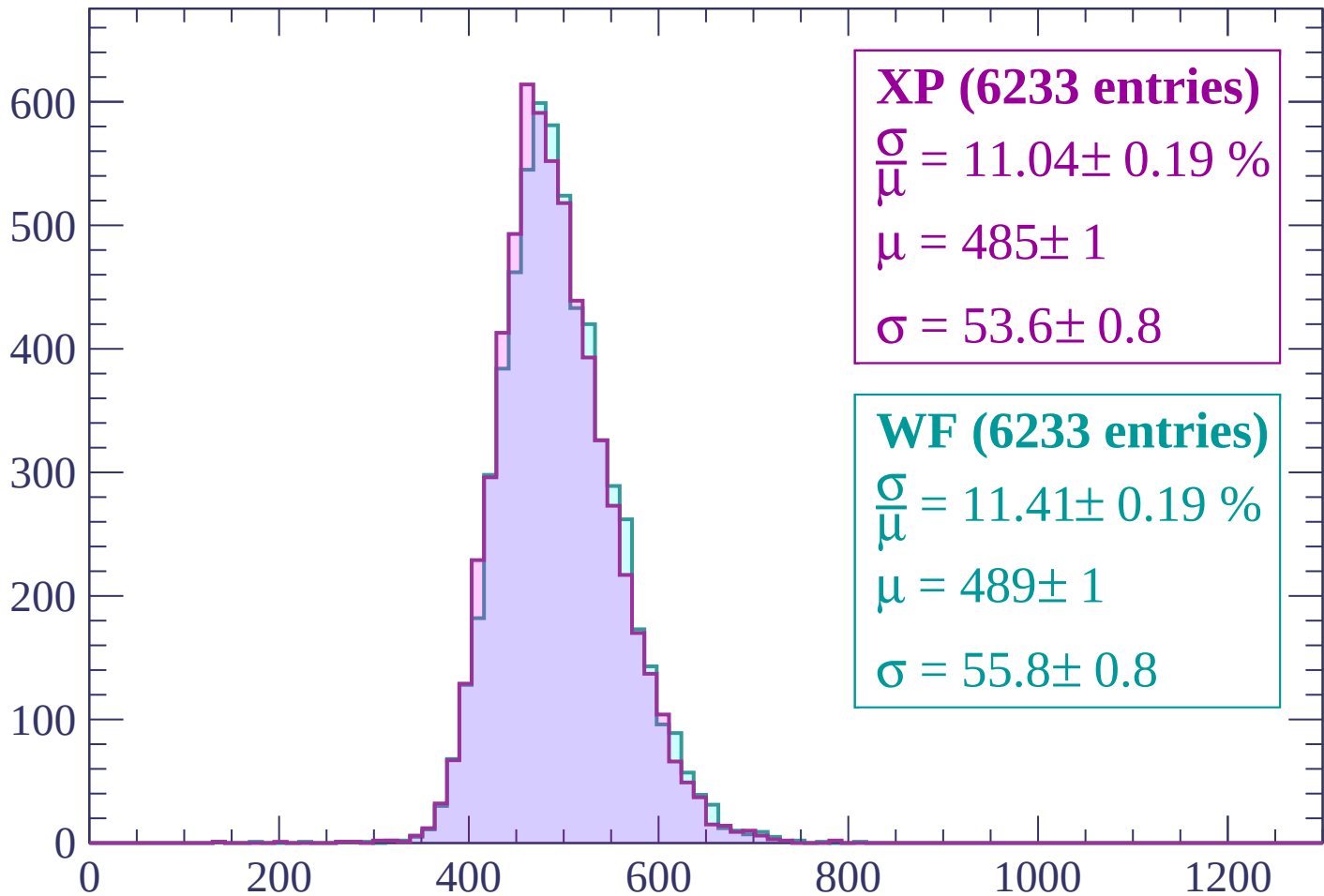
Energy loss | $-1680 < p < -1620$

Count



Energy loss | $-1620 < p < -1560$

Count



XP (6233 entries)

$$\frac{\sigma}{\mu} = 11.04 \pm 0.19 \%$$

$$\mu = 485 \pm 1$$

$$\sigma = 53.6 \pm 0.8$$

WF (6233 entries)

$$\frac{\sigma}{\mu} = 11.41 \pm 0.19 \%$$

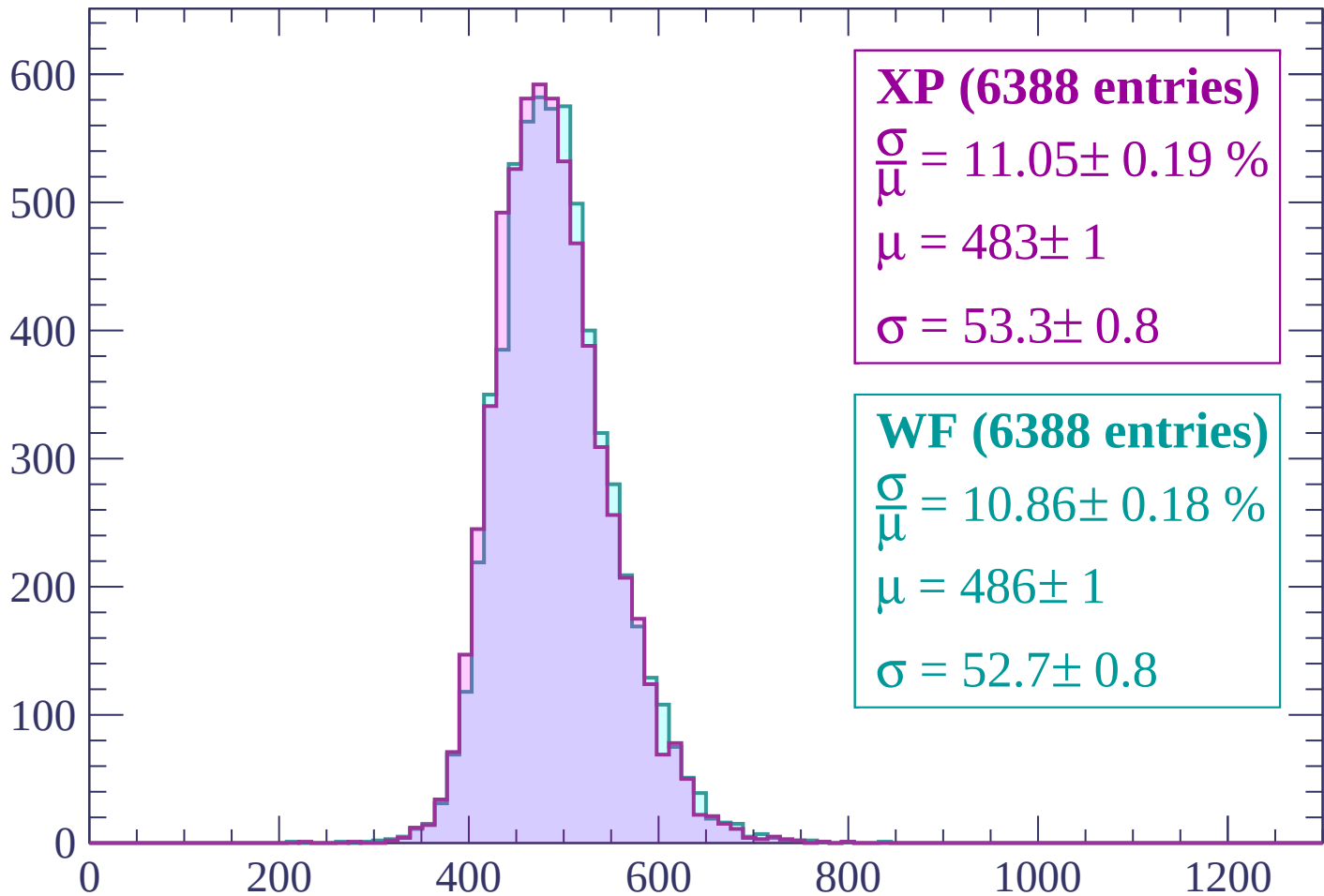
$$\mu = 489 \pm 1$$

$$\sigma = 55.8 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | -1560 < p < -1500

Count



XP (6388 entries)

$$\frac{\sigma}{\mu} = 11.05 \pm 0.19 \%$$

$$\mu = 483 \pm 1$$

$$\sigma = 53.3 \pm 0.8$$

WF (6388 entries)

$$\frac{\sigma}{\mu} = 10.86 \pm 0.18 \%$$

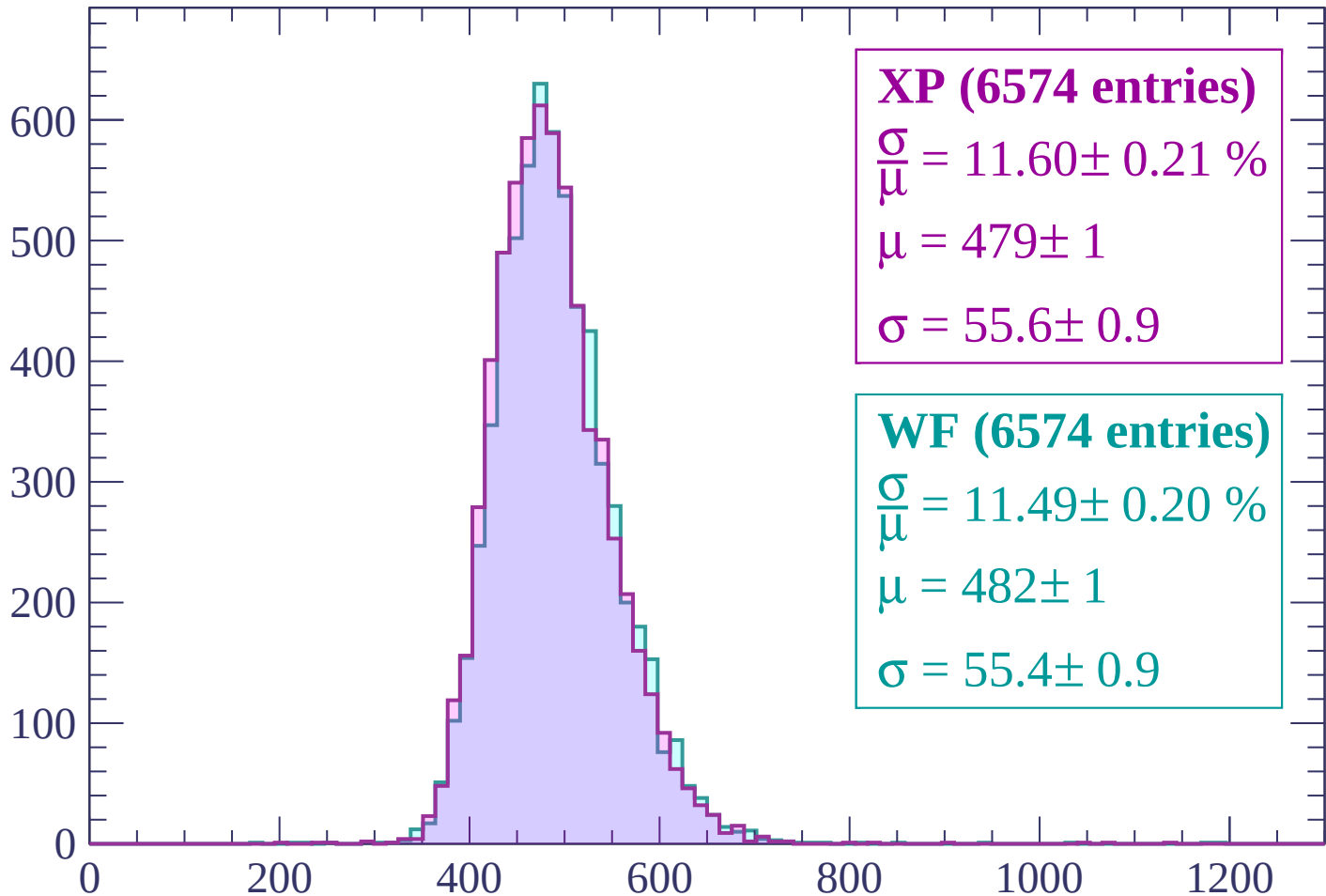
$$\mu = 486 \pm 1$$

$$\sigma = 52.7 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | -1500 < p < -1440

Count



XP (6574 entries)

$$\frac{\sigma}{\mu} = 11.60 \pm 0.21 \%$$

$$\mu = 479 \pm 1$$

$$\sigma = 55.6 \pm 0.9$$

WF (6574 entries)

$$\frac{\sigma}{\mu} = 11.49 \pm 0.20 \%$$

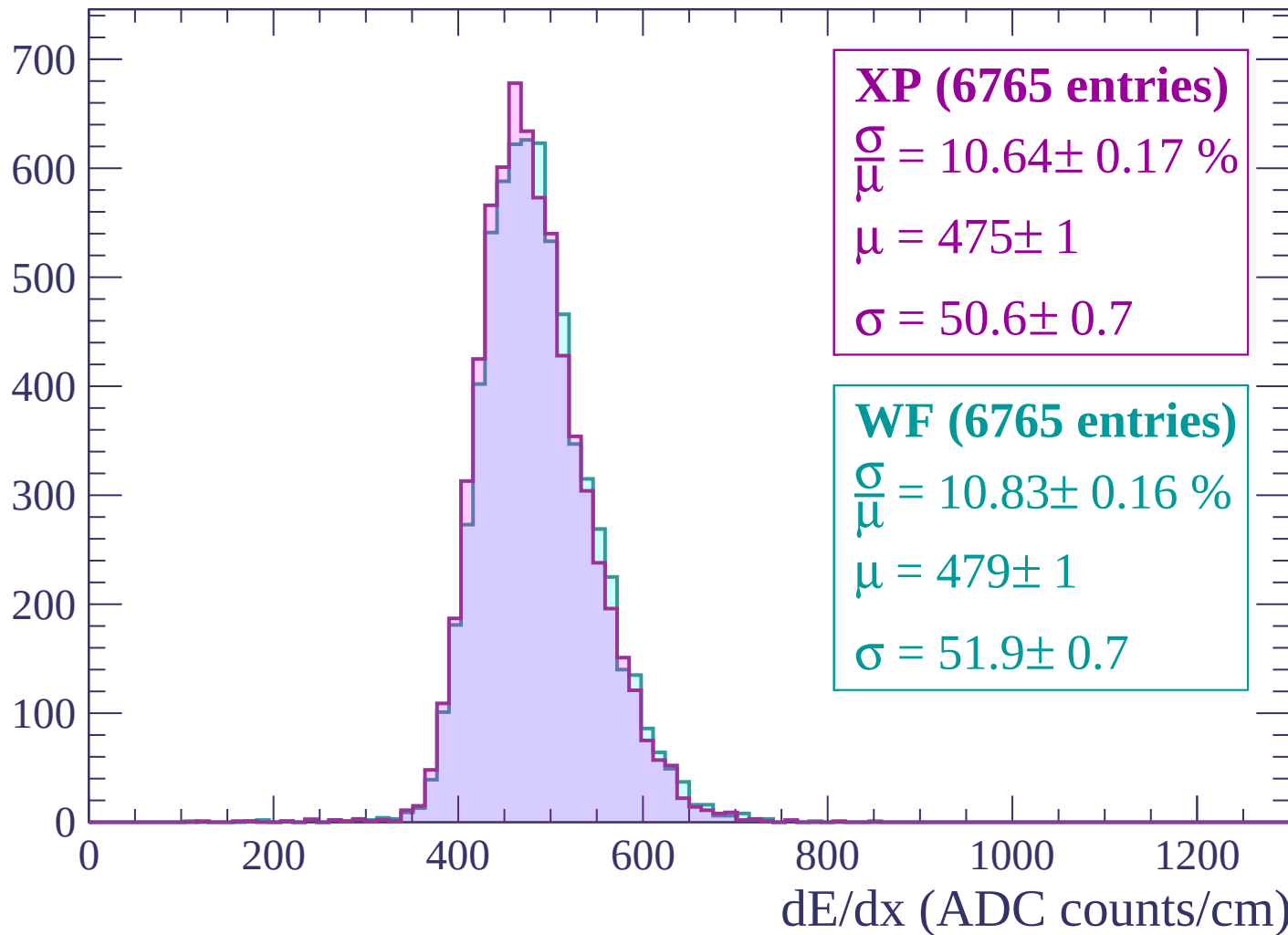
$$\mu = 482 \pm 1$$

$$\sigma = 55.4 \pm 0.9$$

dE/dx (ADC counts/cm)

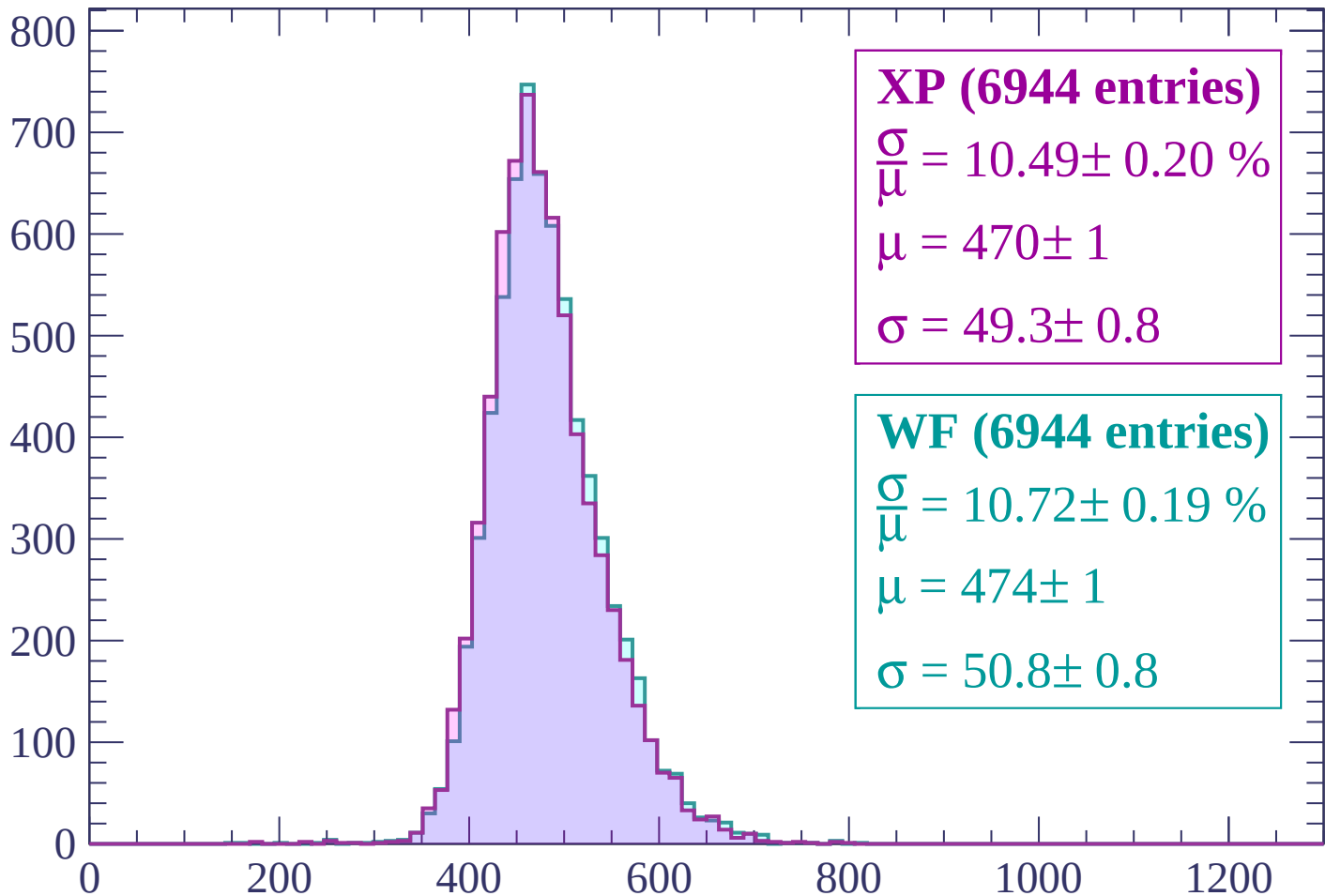
Energy loss | $-1440 < p < -1380$

Count



Energy loss | -1380 < p < -1320

Count



XP (6944 entries)

$\frac{\sigma}{\mu} = 10.49 \pm 0.20 \%$

$\mu = 470 \pm 1$

$\sigma = 49.3 \pm 0.8$

WF (6944 entries)

$\frac{\sigma}{\mu} = 10.72 \pm 0.19 \%$

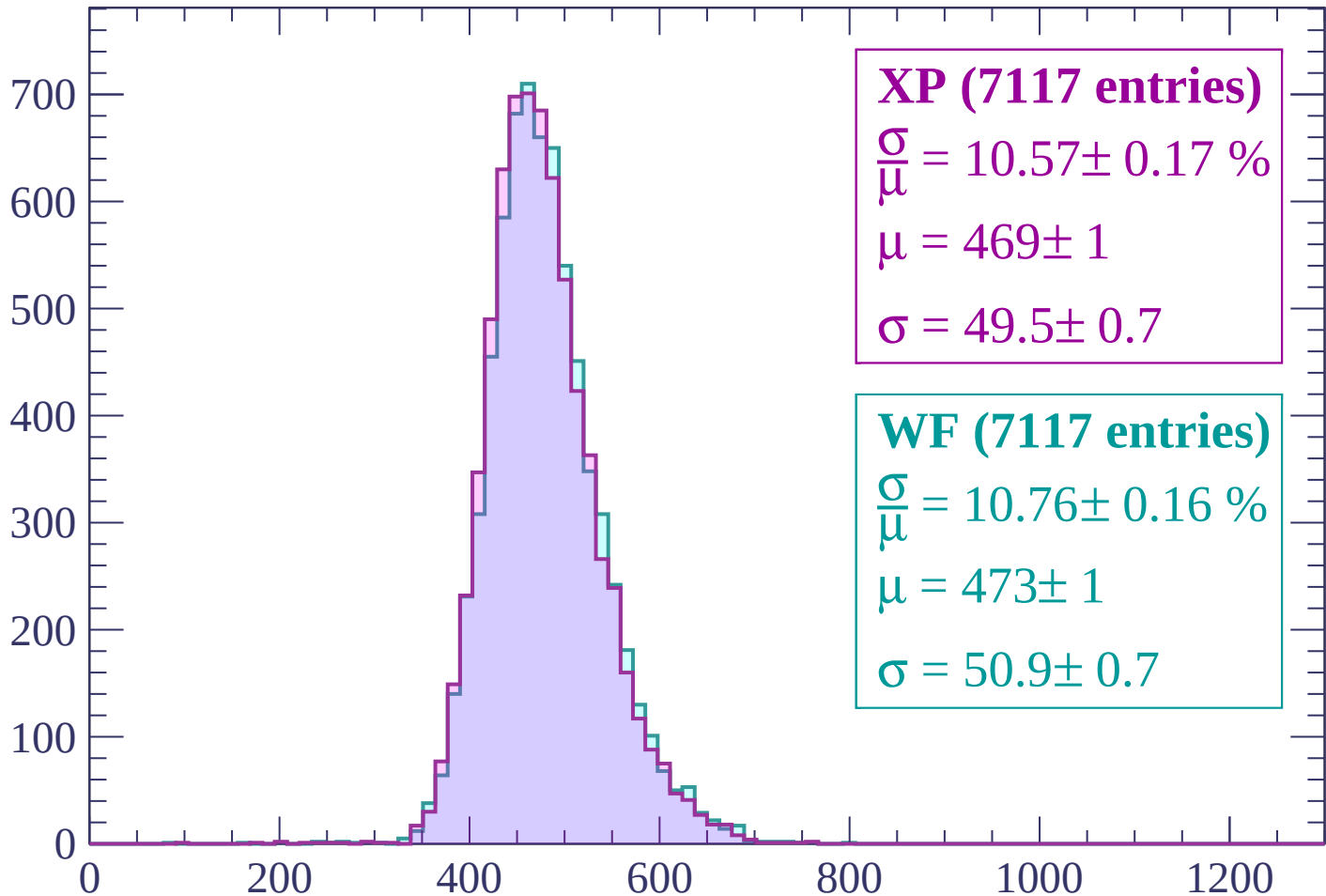
$\mu = 474 \pm 1$

$\sigma = 50.8 \pm 0.8$

dE/dx (ADC counts/cm)

Energy loss | $-1320 < p < -1260$

Count



XP (7117 entries)

$\frac{\sigma}{\mu} = 10.57 \pm 0.17 \%$

$\mu = 469 \pm 1$

$\sigma = 49.5 \pm 0.7$

WF (7117 entries)

$\frac{\sigma}{\mu} = 10.76 \pm 0.16 \%$

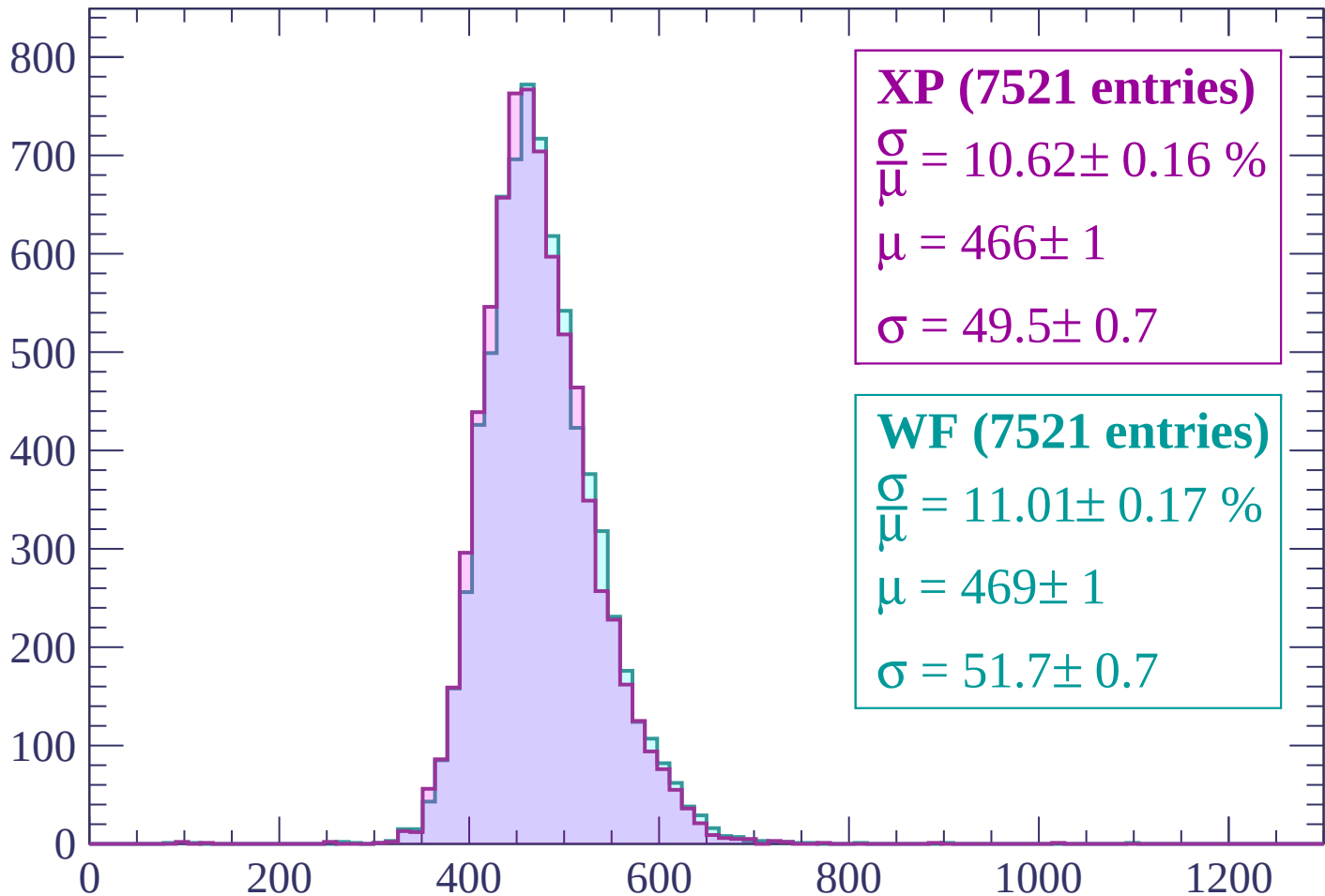
$\mu = 473 \pm 1$

$\sigma = 50.9 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $-1260 < p < -1200$

Count



XP (7521 entries)

$\frac{\sigma}{\mu} = 10.62 \pm 0.16 \%$

$\mu = 466 \pm 1$

$\sigma = 49.5 \pm 0.7$

WF (7521 entries)

$\frac{\sigma}{\mu} = 11.01 \pm 0.17 \%$

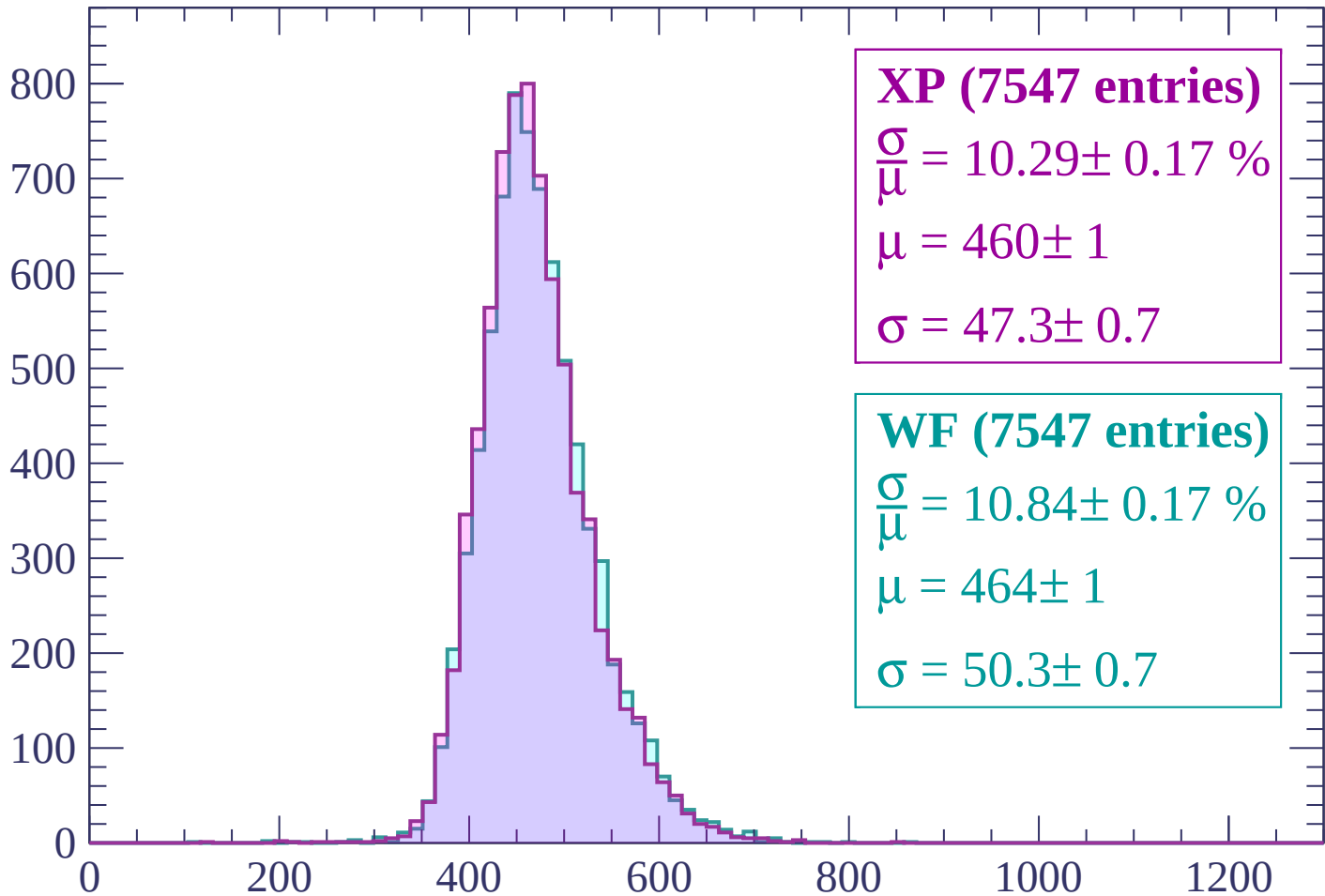
$\mu = 469 \pm 1$

$\sigma = 51.7 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $-1200 < p < -1140$

Count



XP (7547 entries)

$\frac{\sigma}{\mu} = 10.29 \pm 0.17 \%$

$\mu = 460 \pm 1$

$\sigma = 47.3 \pm 0.7$

WF (7547 entries)

$\frac{\sigma}{\mu} = 10.84 \pm 0.17 \%$

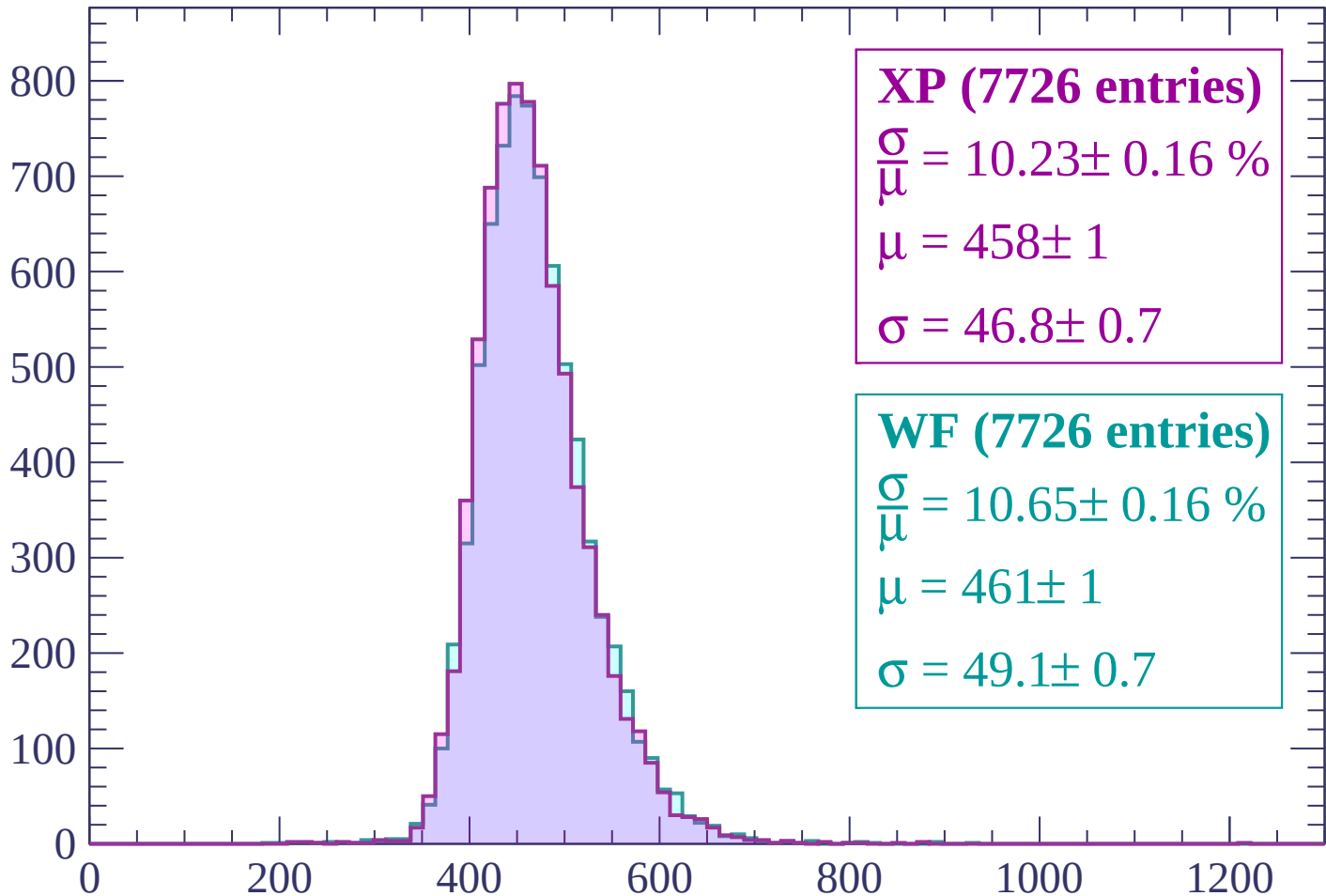
$\mu = 464 \pm 1$

$\sigma = 50.3 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $-1140 < p < -1080$

Count



XP (7726 entries)

$\frac{\sigma}{\mu} = 10.23 \pm 0.16 \%$

$\mu = 458 \pm 1$

$\sigma = 46.8 \pm 0.7$

WF (7726 entries)

$\frac{\sigma}{\mu} = 10.65 \pm 0.16 \%$

$\mu = 461 \pm 1$

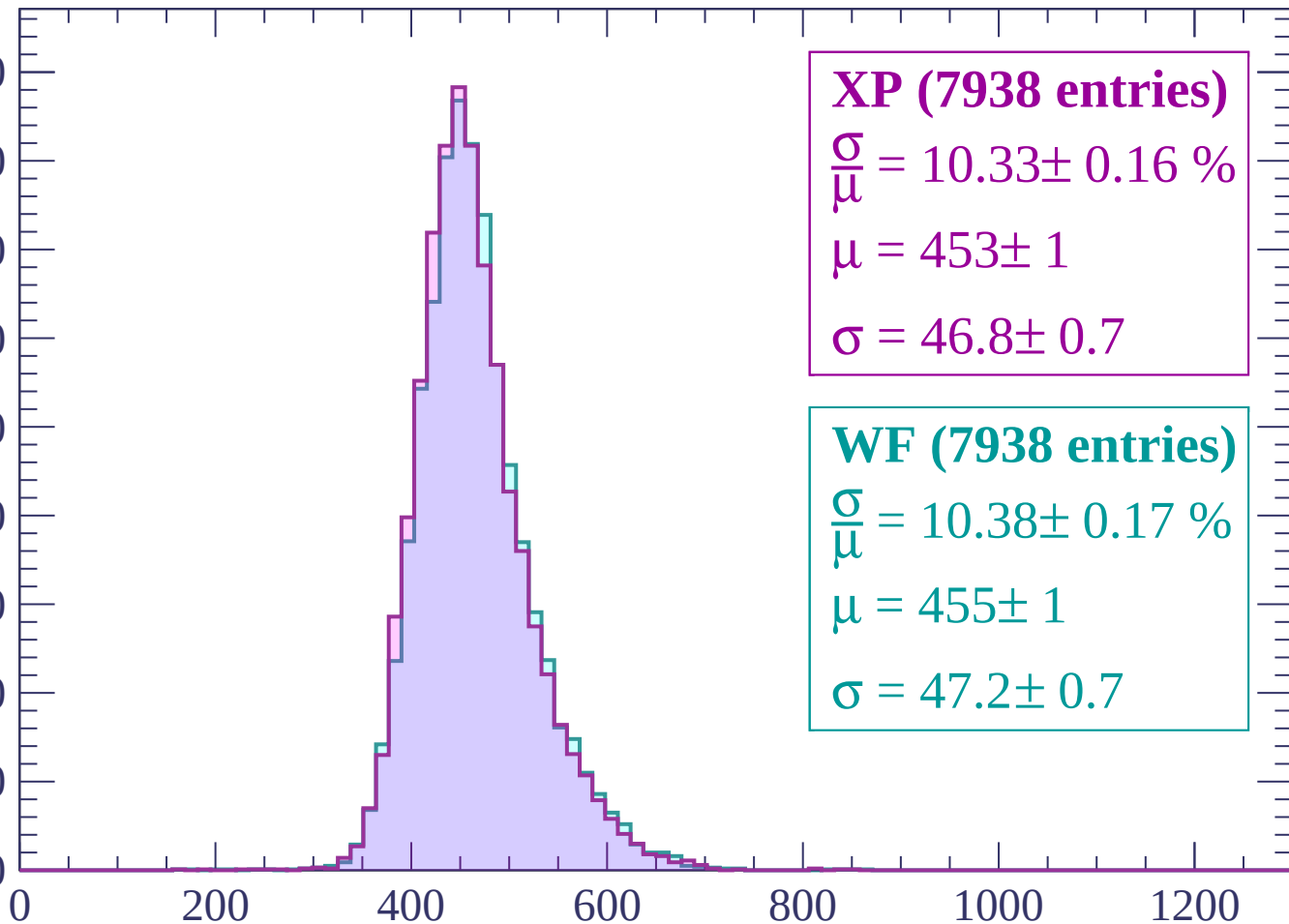
$\sigma = 49.1 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $-1080 < p < -1020$

Count

900
800
700
600
500
400
300
200
100
0



XP (7938 entries)

$\frac{\sigma}{\mu} = 10.33 \pm 0.16 \%$

$\mu = 453 \pm 1$

$\sigma = 46.8 \pm 0.7$

WF (7938 entries)

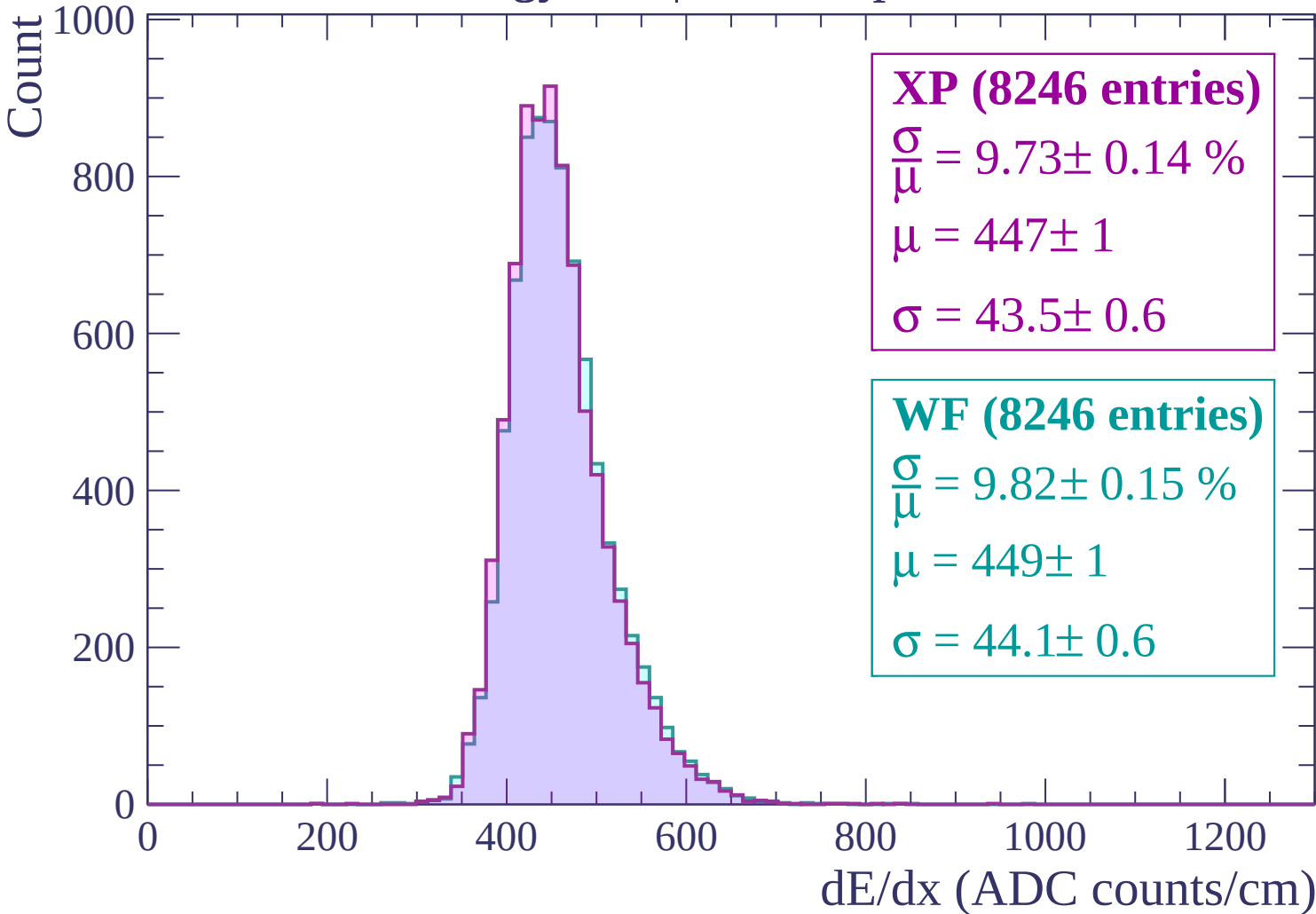
$\frac{\sigma}{\mu} = 10.38 \pm 0.17 \%$

$\mu = 455 \pm 1$

$\sigma = 47.2 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $-1020 < p < -960$



Energy loss | $-960 < p < -900$

Count

1000

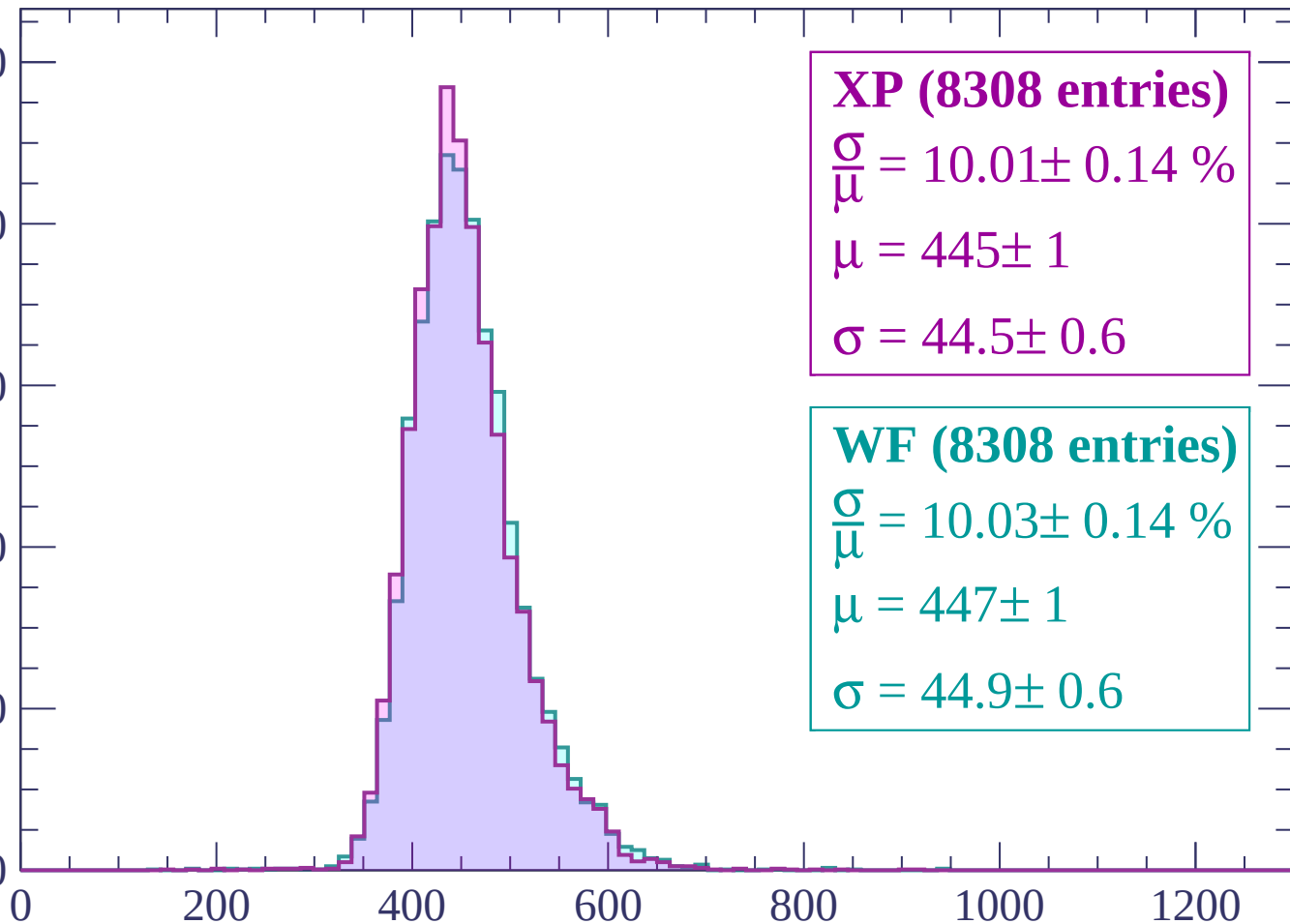
800

600

400

200

0



XP (8308 entries)

$\frac{\sigma}{\mu} = 10.01 \pm 0.14 \%$

$\mu = 445 \pm 1$

$\sigma = 44.5 \pm 0.6$

WF (8308 entries)

$\frac{\sigma}{\mu} = 10.03 \pm 0.14 \%$

$\mu = 447 \pm 1$

$\sigma = 44.9 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $-900 < p < -840$

Count

1000

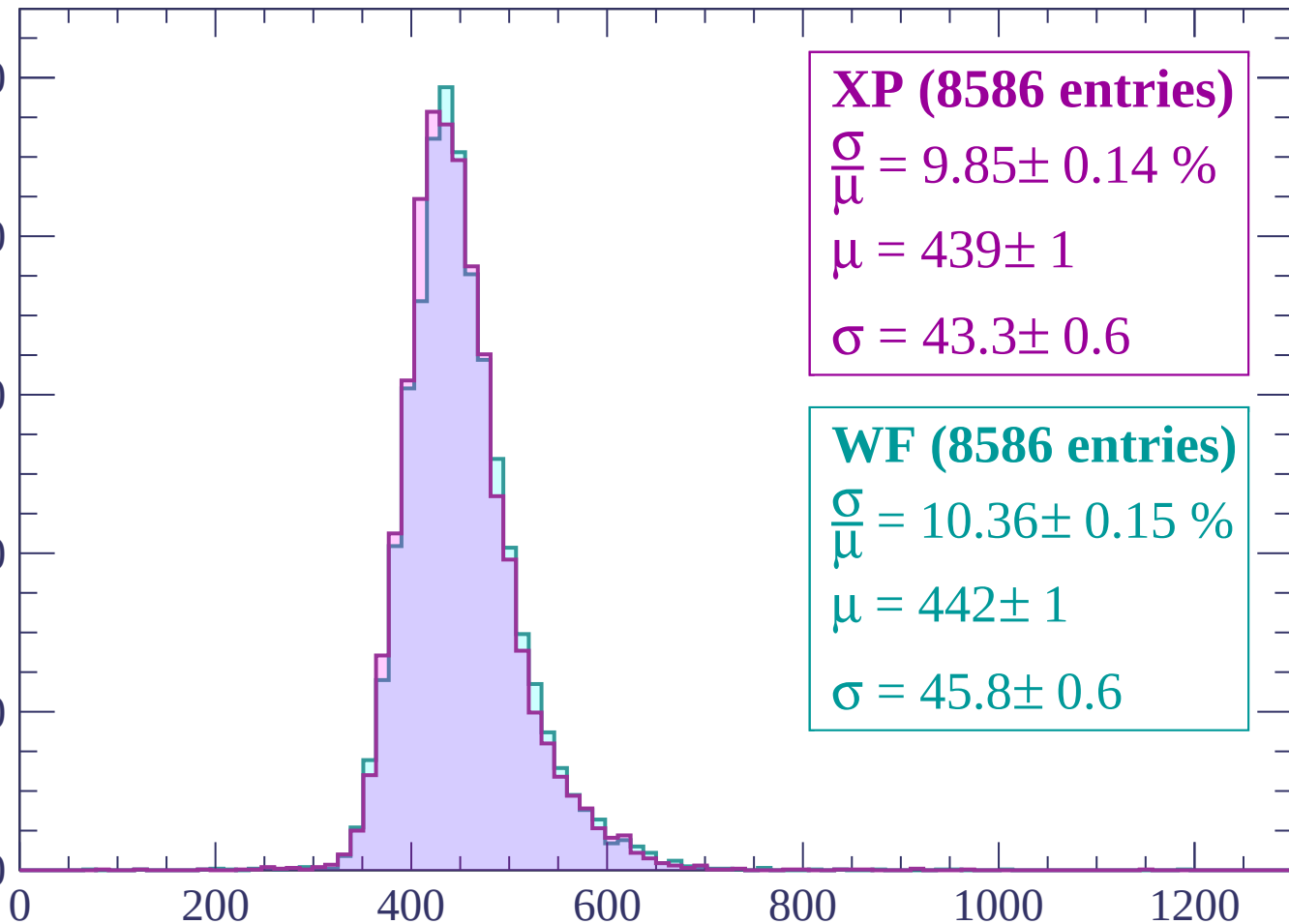
800

600

400

200

0



XP (8586 entries)

$\frac{\sigma}{\mu} = 9.85 \pm 0.14 \%$

$\mu = 439 \pm 1$

$\sigma = 43.3 \pm 0.6$

WF (8586 entries)

$\frac{\sigma}{\mu} = 10.36 \pm 0.15 \%$

$\mu = 442 \pm 1$

$\sigma = 45.8 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -780$

Count

1000

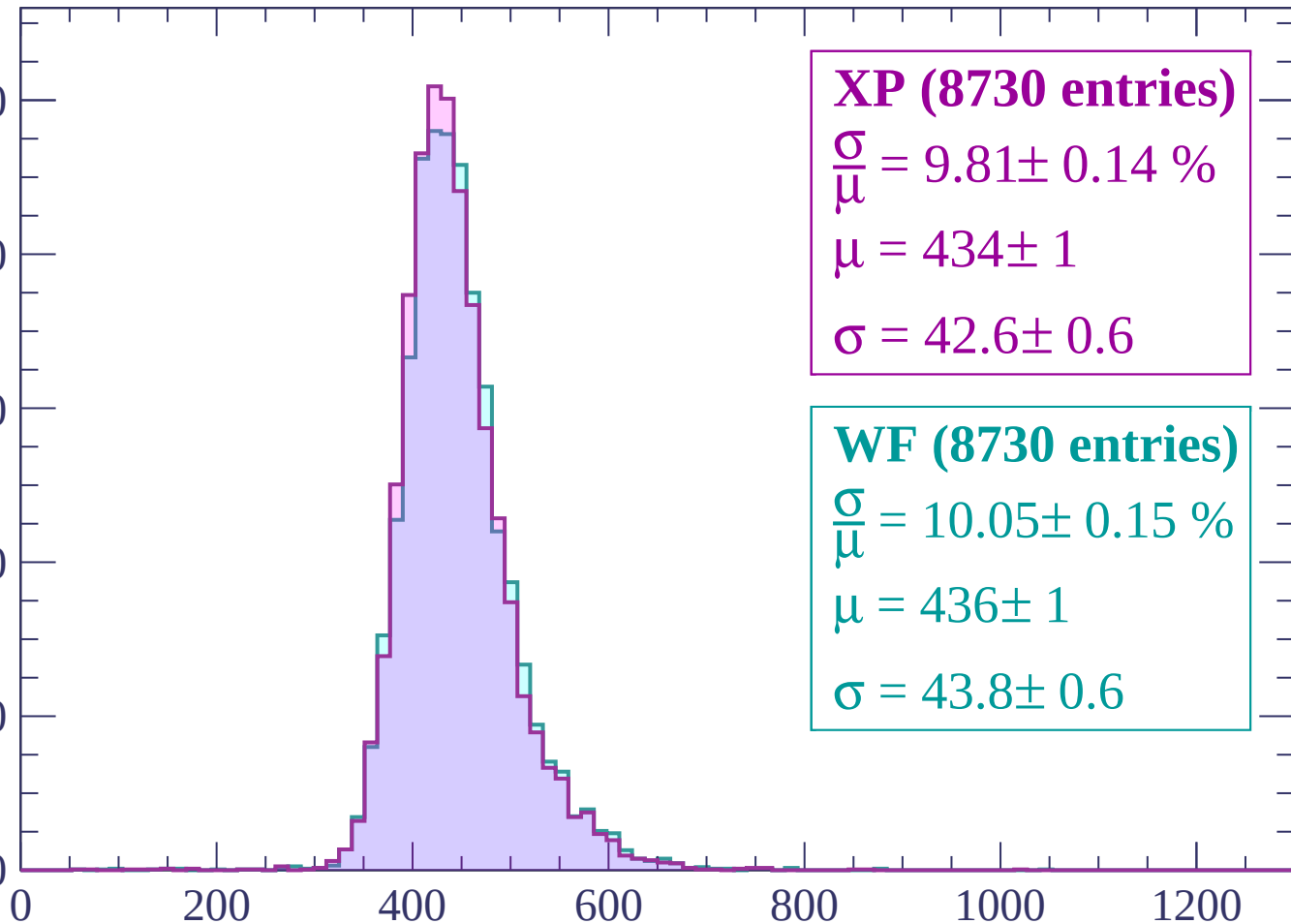
800

600

400

200

0



XP (8730 entries)

$\frac{\sigma}{\mu} = 9.81 \pm 0.14 \%$

$\mu = 434 \pm 1$

$\sigma = 42.6 \pm 0.6$

WF (8730 entries)

$\frac{\sigma}{\mu} = 10.05 \pm 0.15 \%$

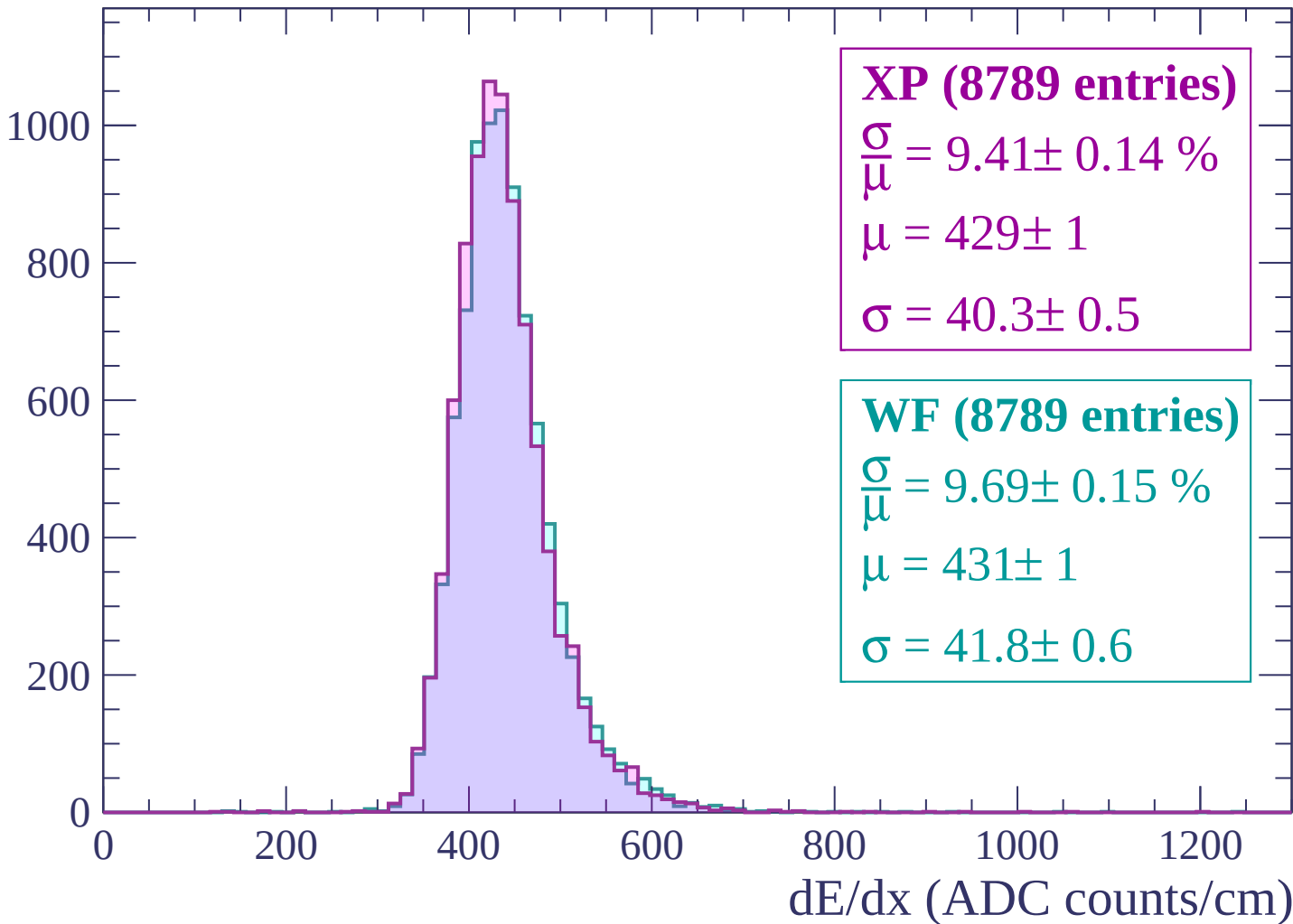
$\mu = 436 \pm 1$

$\sigma = 43.8 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $-780 < p < -720$

Count



Energy loss | -720 < p < -660

Count

1000

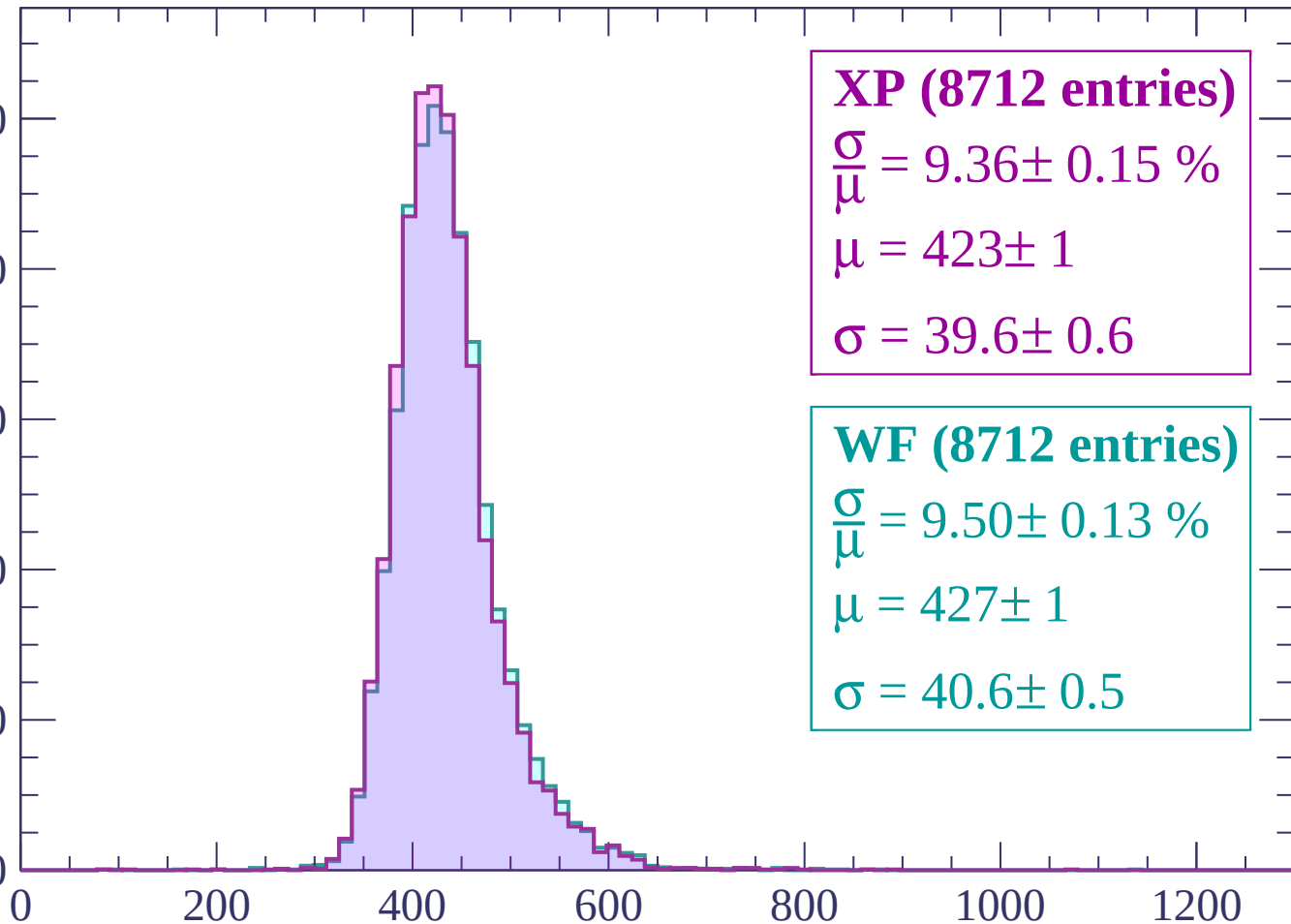
800

600

400

200

0



XP (8712 entries)

$\frac{\sigma}{\mu} = 9.36 \pm 0.15 \%$

$\mu = 423 \pm 1$

$\sigma = 39.6 \pm 0.6$

WF (8712 entries)

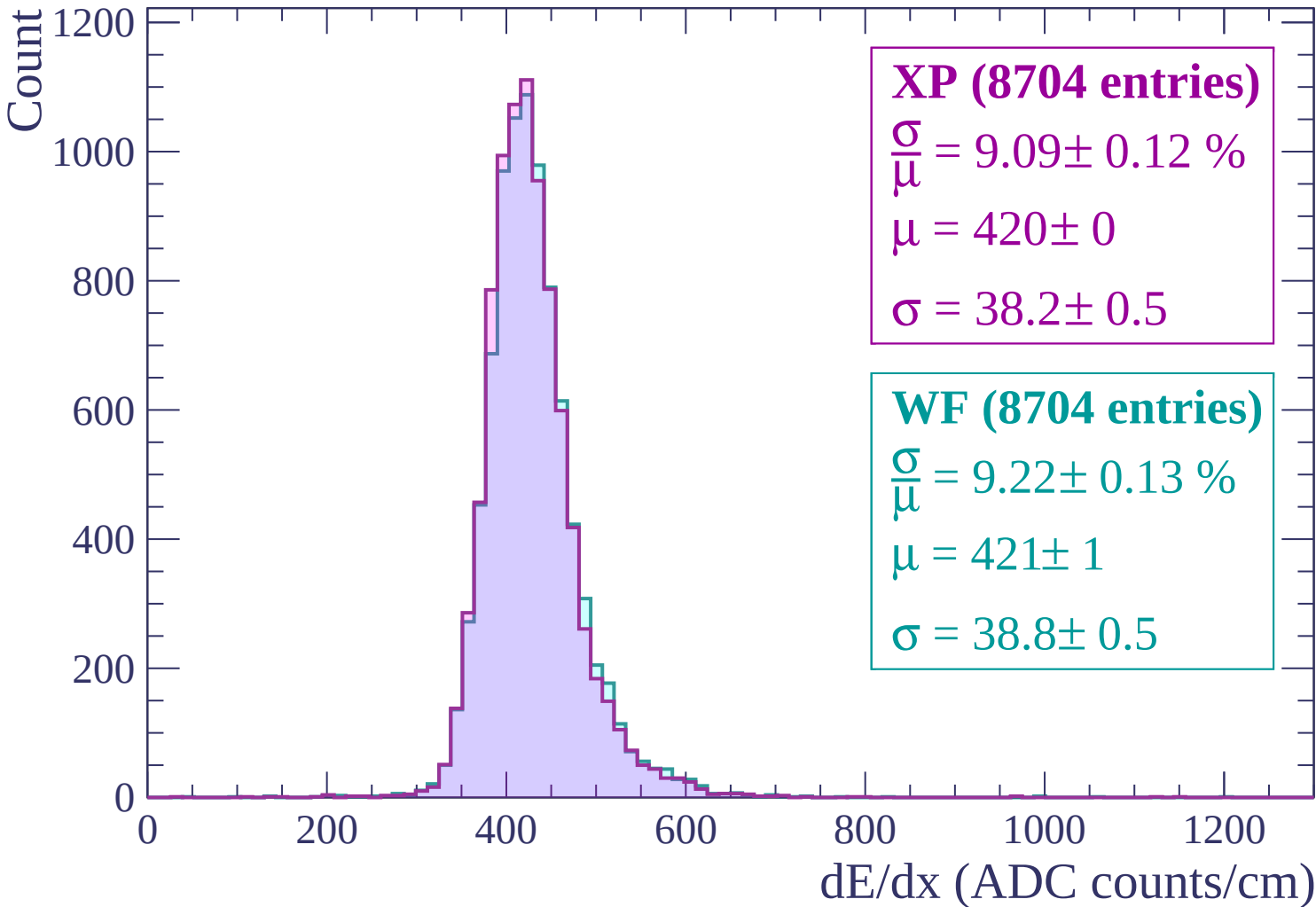
$\frac{\sigma}{\mu} = 9.50 \pm 0.13 \%$

$\mu = 427 \pm 1$

$\sigma = 40.6 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $-660 < p < -600$



Energy loss | $-600 < p < -540$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (8642 entries)

$\frac{\sigma}{\mu} = 8.78 \pm 0.13 \%$

$\mu = 415 \pm 0$

$\sigma = 36.4 \pm 0.5$

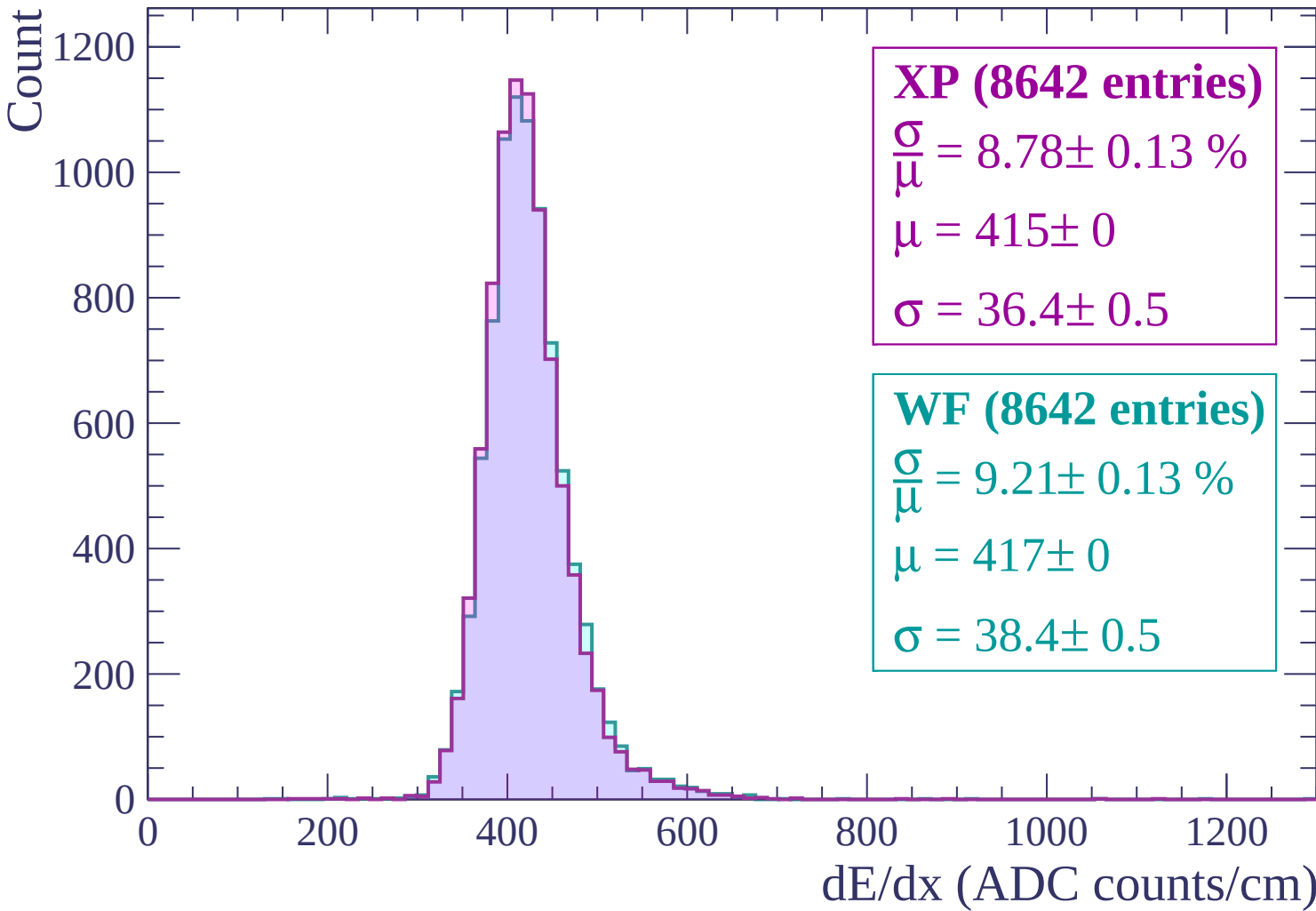
WF (8642 entries)

$\frac{\sigma}{\mu} = 9.21 \pm 0.13 \%$

$\mu = 417 \pm 0$

$\sigma = 38.4 \pm 0.5$

dE/dx (ADC counts/cm)



Energy loss | $-540 < p < -480$

Count

1200
1000
800
600
400
200
0

XP (8738 entries)

$$\frac{\sigma}{\mu} = 8.82 \pm 0.12 \%$$

$$\mu = 410 \pm 0$$

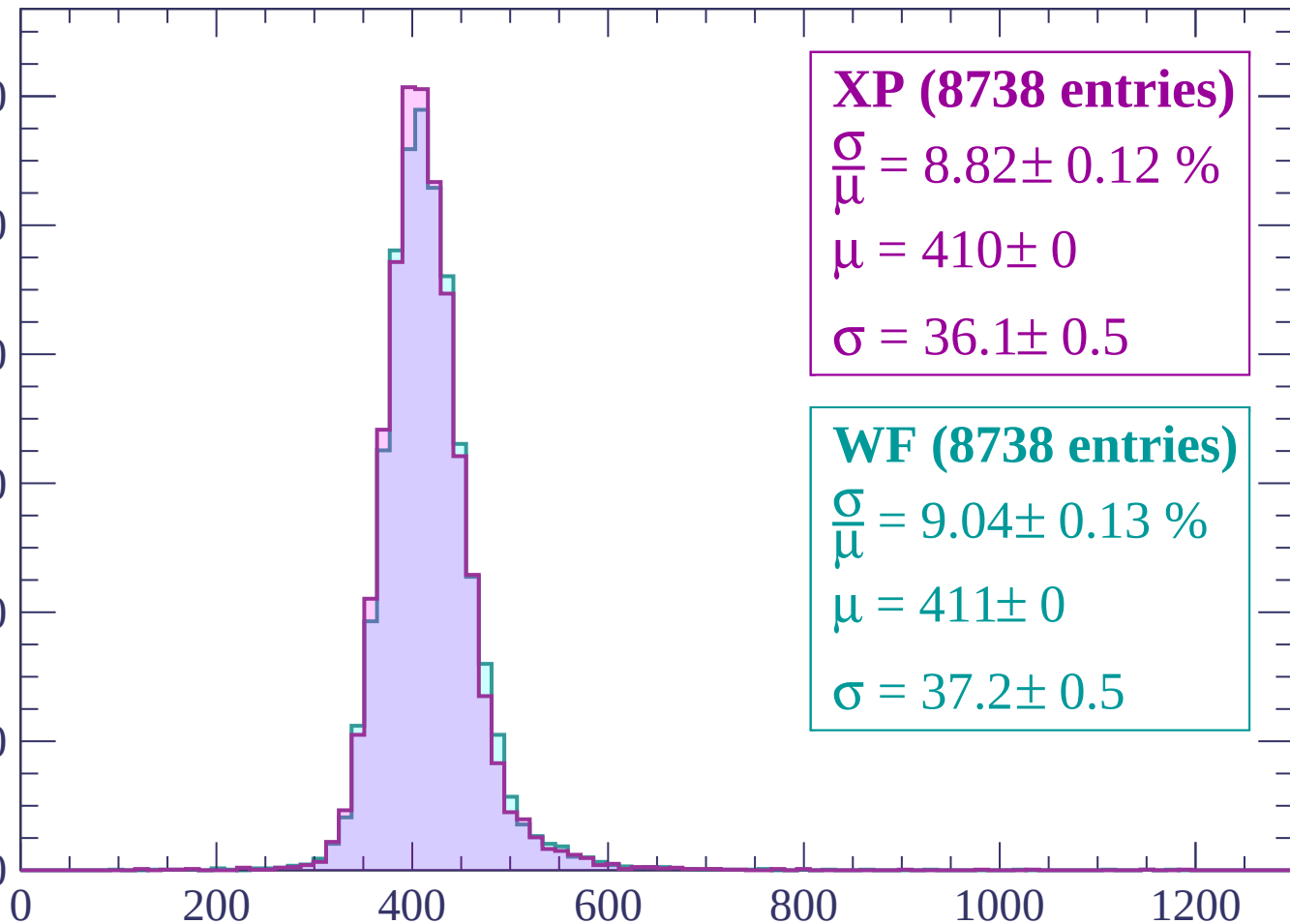
$$\sigma = 36.1 \pm 0.5$$

WF (8738 entries)

$$\frac{\sigma}{\mu} = 9.04 \pm 0.13 \%$$

$$\mu = 411 \pm 0$$

$$\sigma = 37.2 \pm 0.5$$



dE/dx (ADC counts/cm)

Energy loss | $-480 < p < -420$

Count

1200
1000
800
600
400
200
0

XP (8500 entries)

$$\frac{\sigma}{\mu} = 8.47 \pm 0.13 \%$$

$$\mu = 406 \pm 0$$

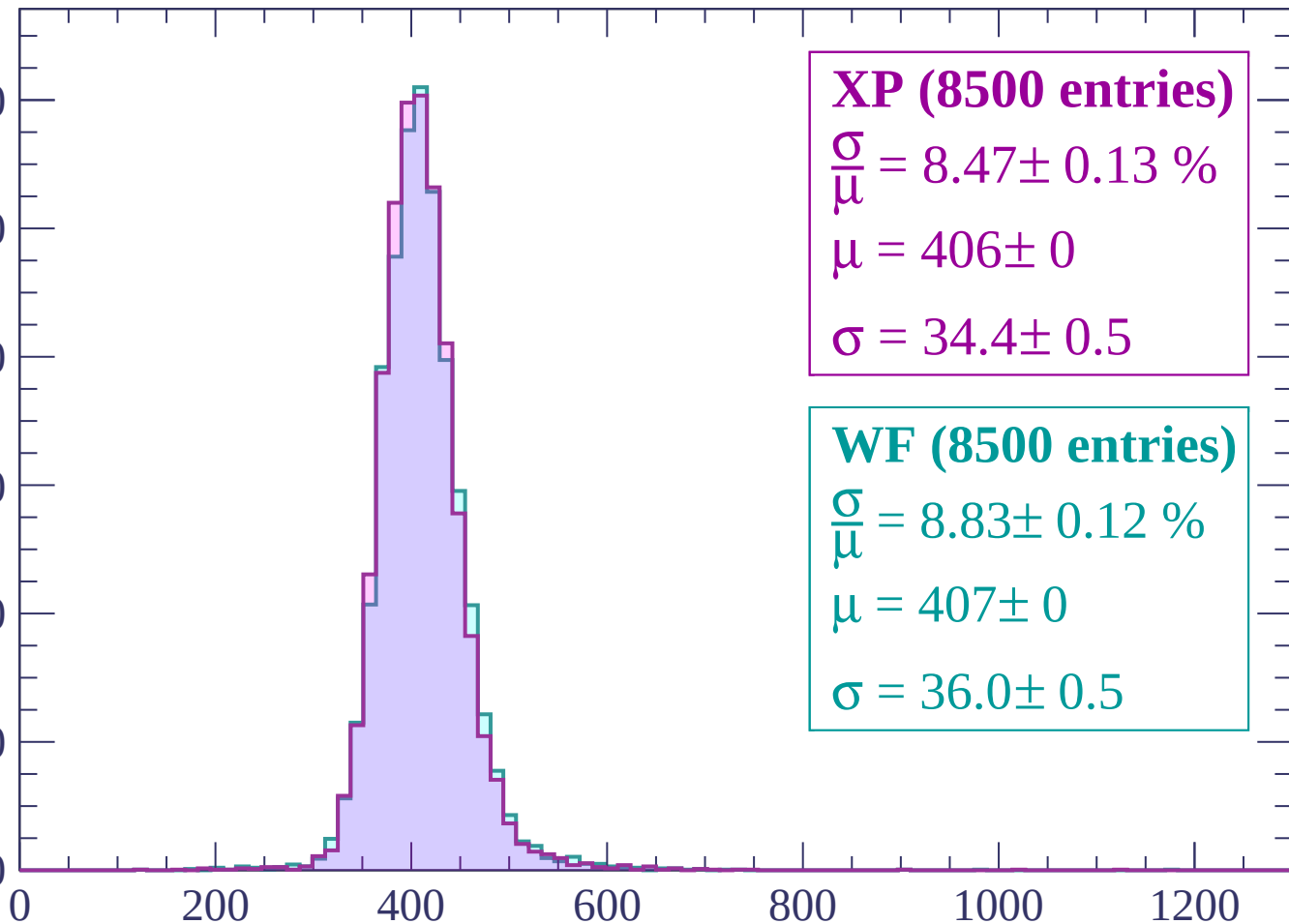
$$\sigma = 34.4 \pm 0.5$$

WF (8500 entries)

$$\frac{\sigma}{\mu} = 8.83 \pm 0.12 \%$$

$$\mu = 407 \pm 0$$

$$\sigma = 36.0 \pm 0.5$$



dE/dx (ADC counts/cm)

Energy loss | $-420 < p < -360$

Count

1200
1000
800
600
400
200
0

XP (8409 entries)

$$\frac{\sigma}{\mu} = 8.63 \pm 0.13 \%$$

$$\mu = 402 \pm 0$$

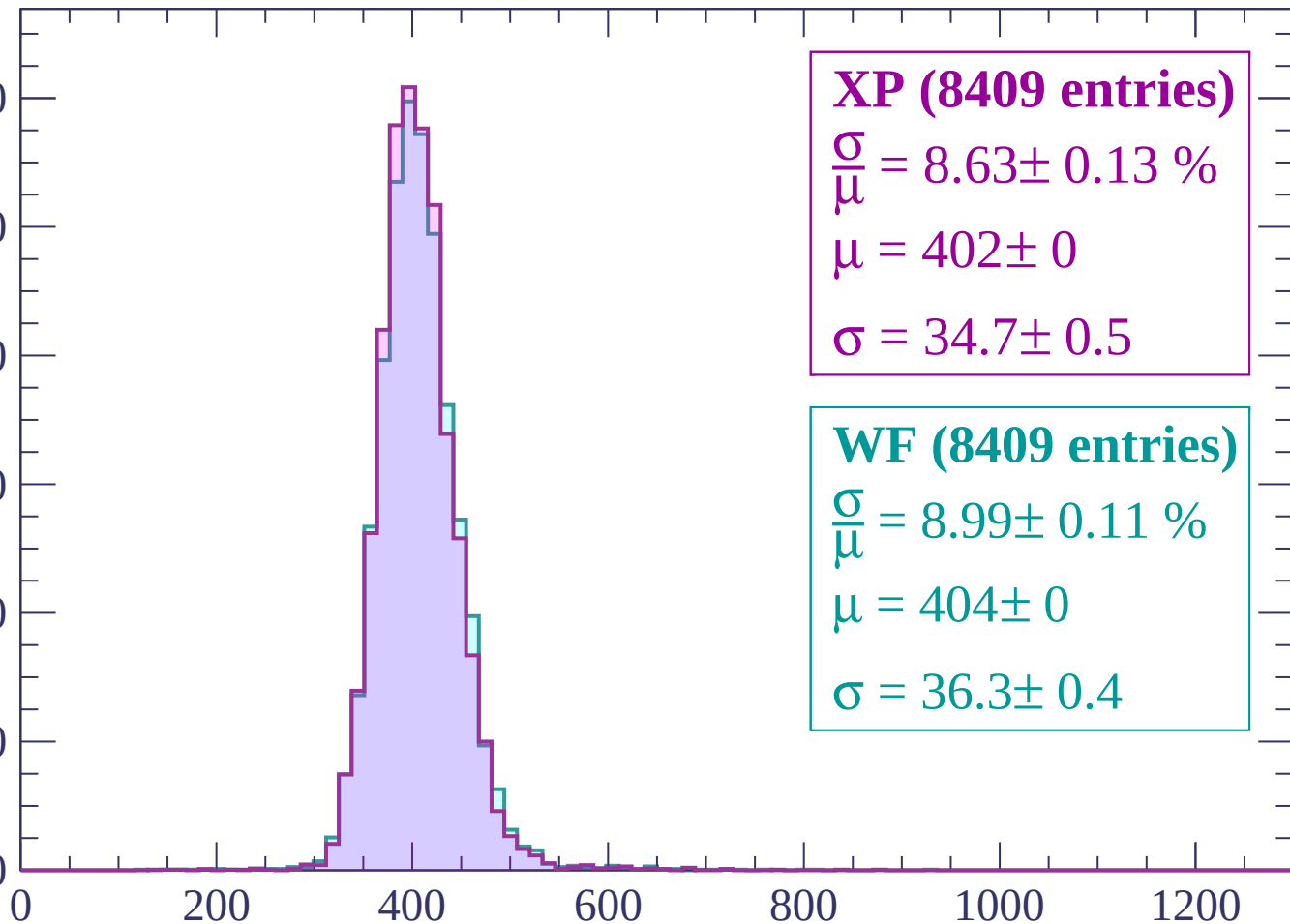
$$\sigma = 34.7 \pm 0.5$$

WF (8409 entries)

$$\frac{\sigma}{\mu} = 8.99 \pm 0.11 \%$$

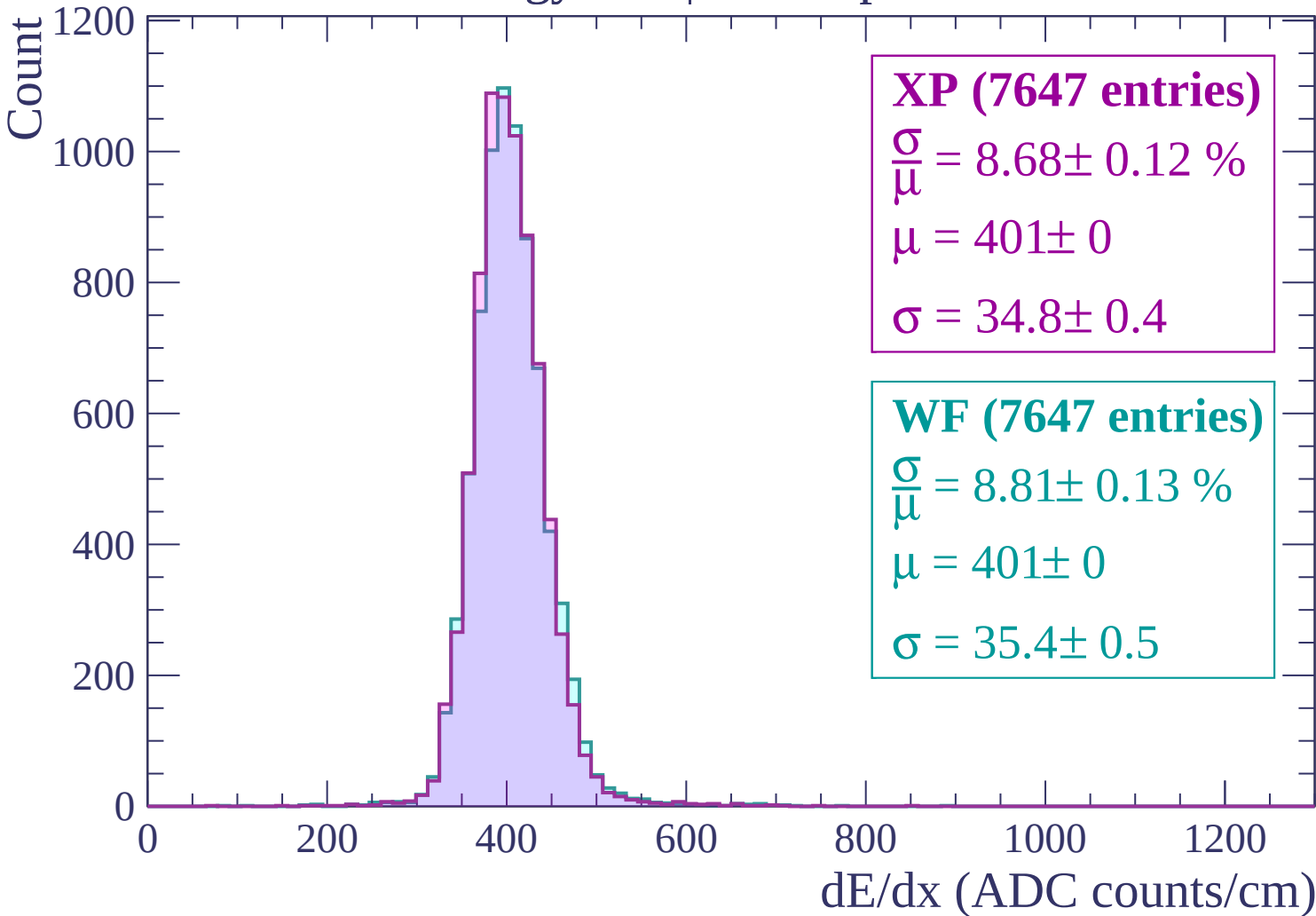
$$\mu = 404 \pm 0$$

$$\sigma = 36.3 \pm 0.4$$



dE/dx (ADC counts/cm)

Energy loss | $-360 < p < -300$



Energy loss | $-300 < p < -240$

Count

1000

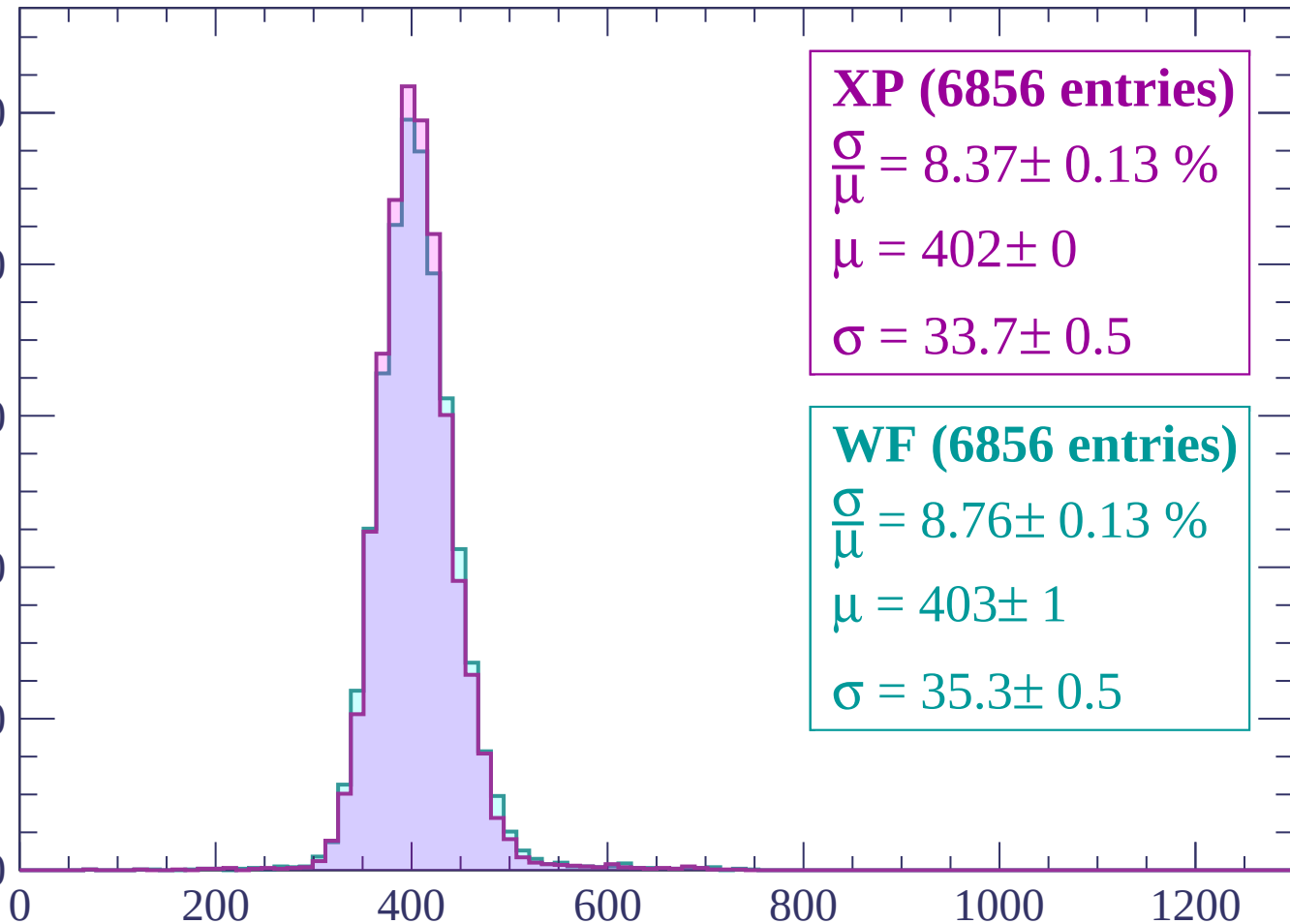
800

600

400

200

0



XP (6856 entries)

$\frac{\sigma}{\mu} = 8.37 \pm 0.13 \%$

$\mu = 402 \pm 0$

$\sigma = 33.7 \pm 0.5$

WF (6856 entries)

$\frac{\sigma}{\mu} = 8.76 \pm 0.13 \%$

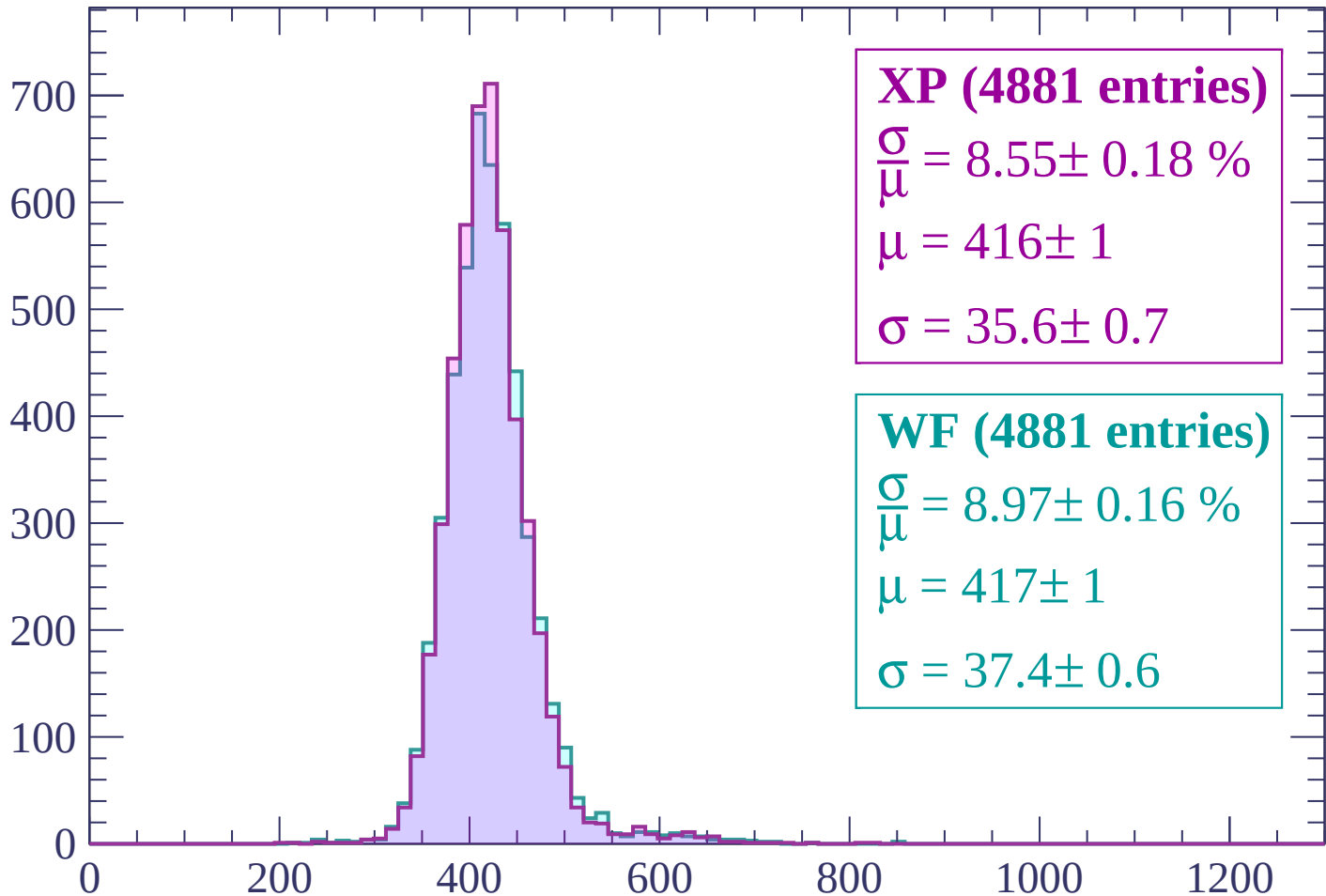
$\mu = 403 \pm 1$

$\sigma = 35.3 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -180$

Count



XP (4881 entries)

$$\frac{\sigma}{\mu} = 8.55 \pm 0.18 \%$$

$$\mu = 416 \pm 1$$

$$\sigma = 35.6 \pm 0.7$$

WF (4881 entries)

$$\frac{\sigma}{\mu} = 8.97 \pm 0.16 \%$$

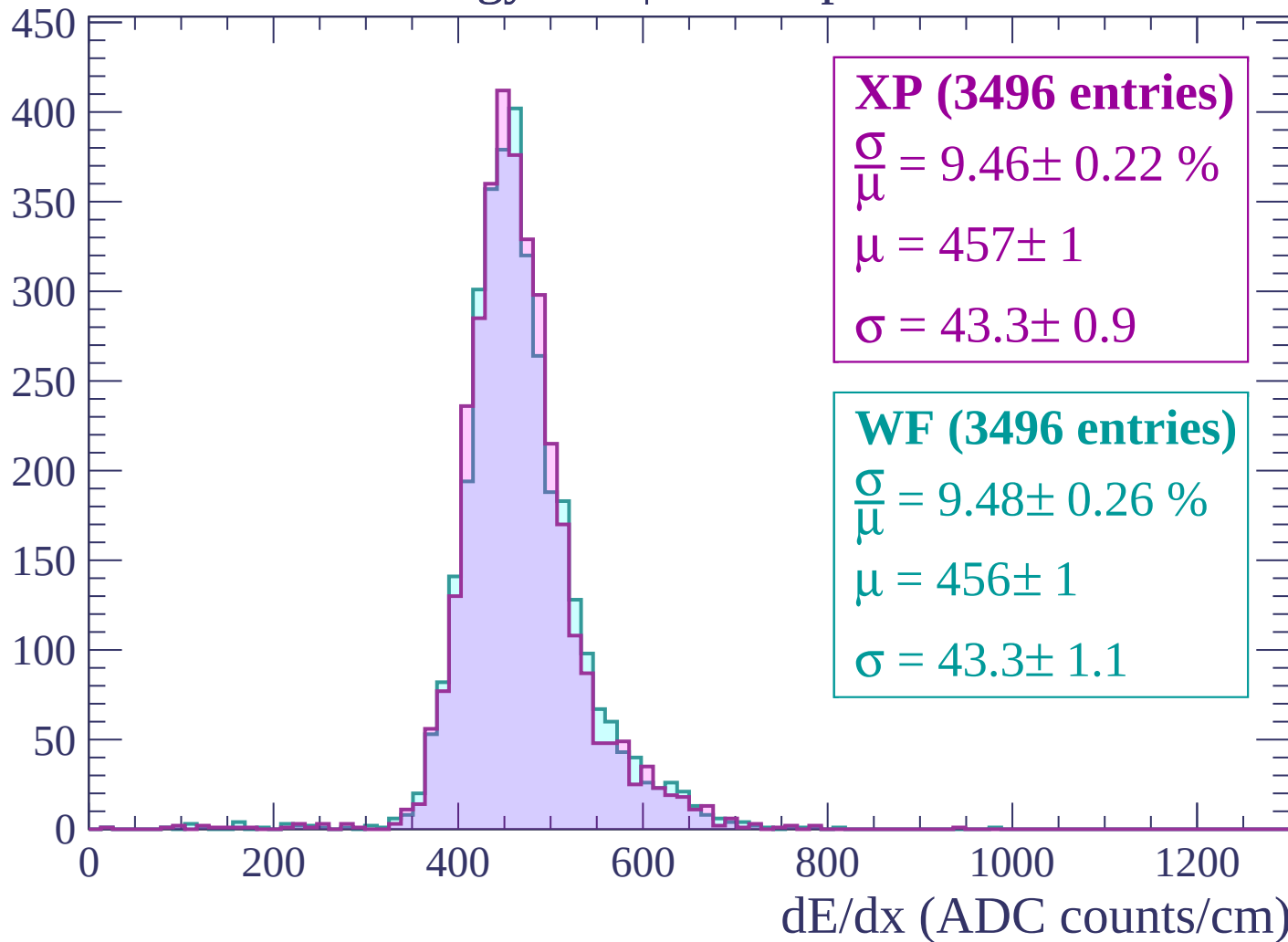
$$\mu = 417 \pm 1$$

$$\sigma = 37.4 \pm 0.6$$

dE/dx (ADC counts/cm)

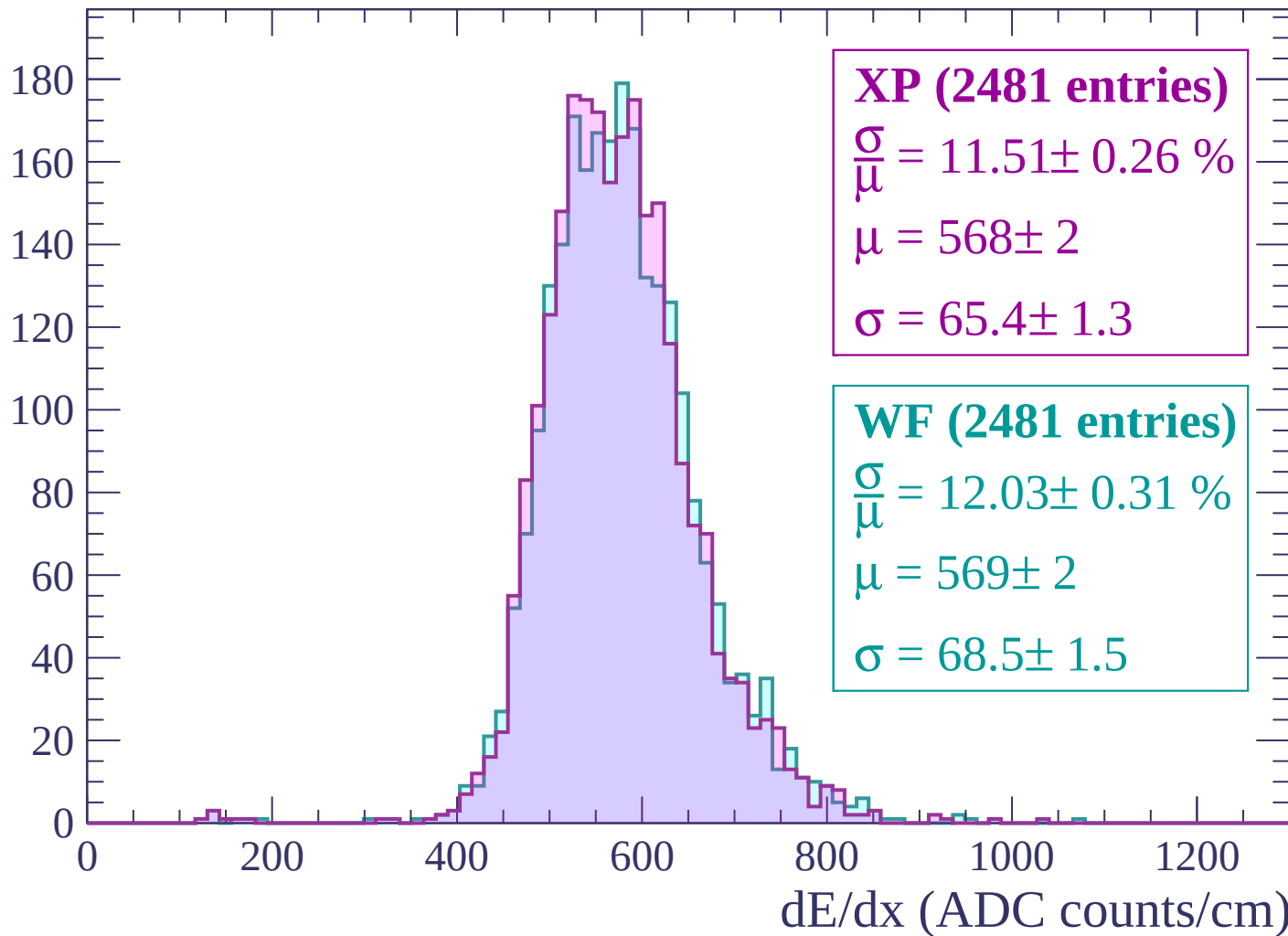
Energy loss | -180 < p < -120

Count



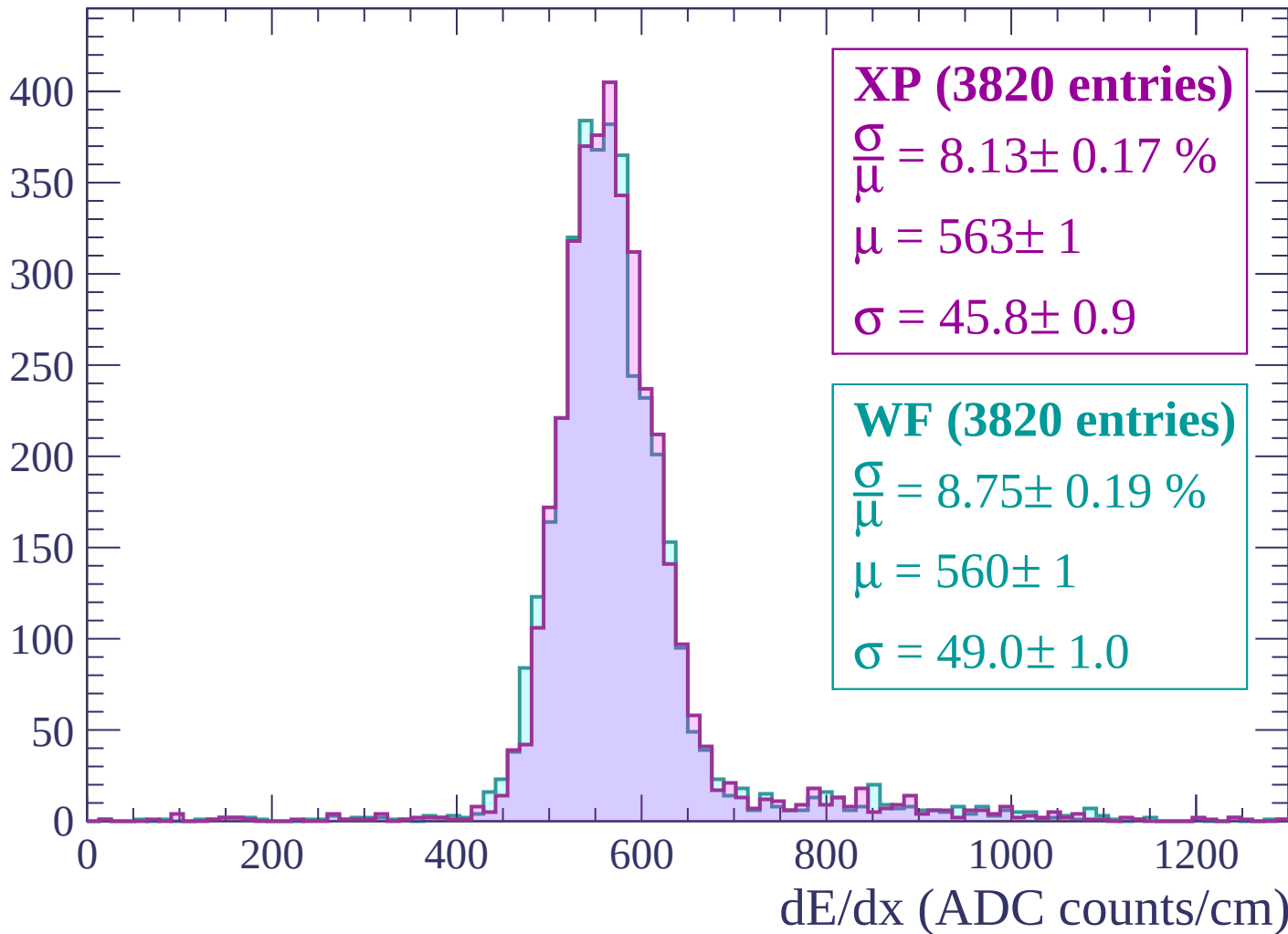
Energy loss | $-120 < p < -60$

Count



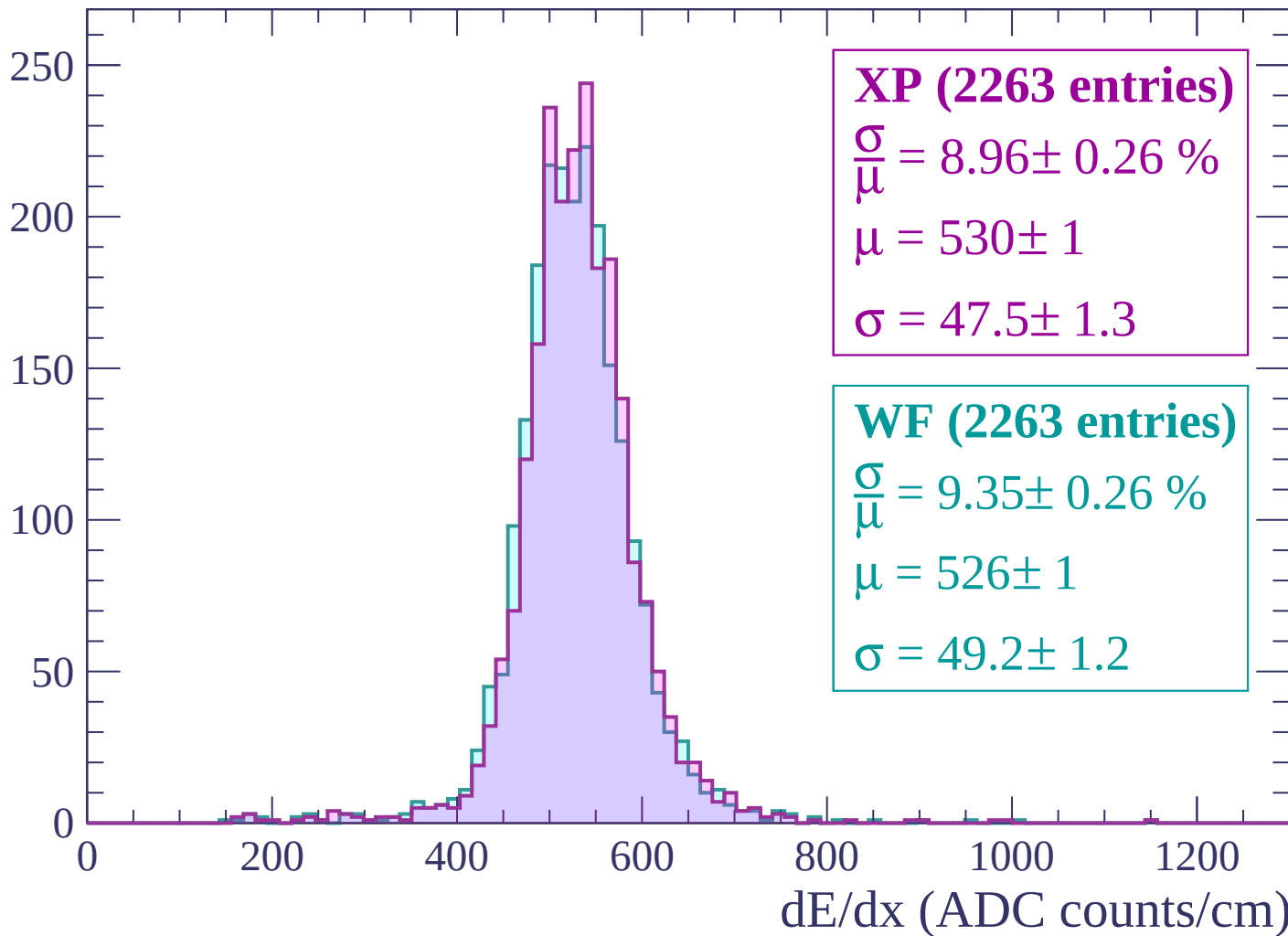
Energy loss | $-60 < p < 0$

Count



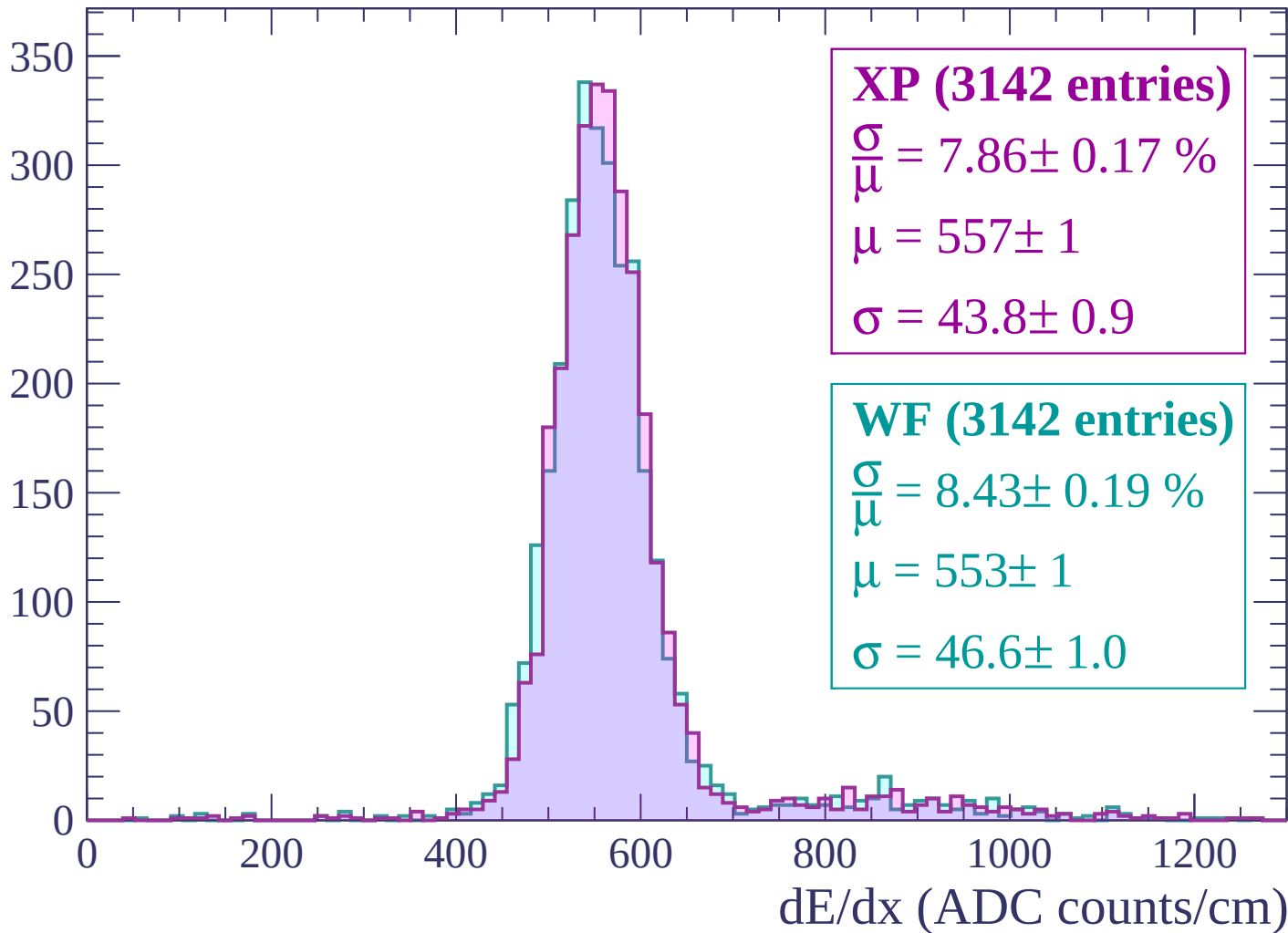
Energy loss | $0 < p < 60$

Count



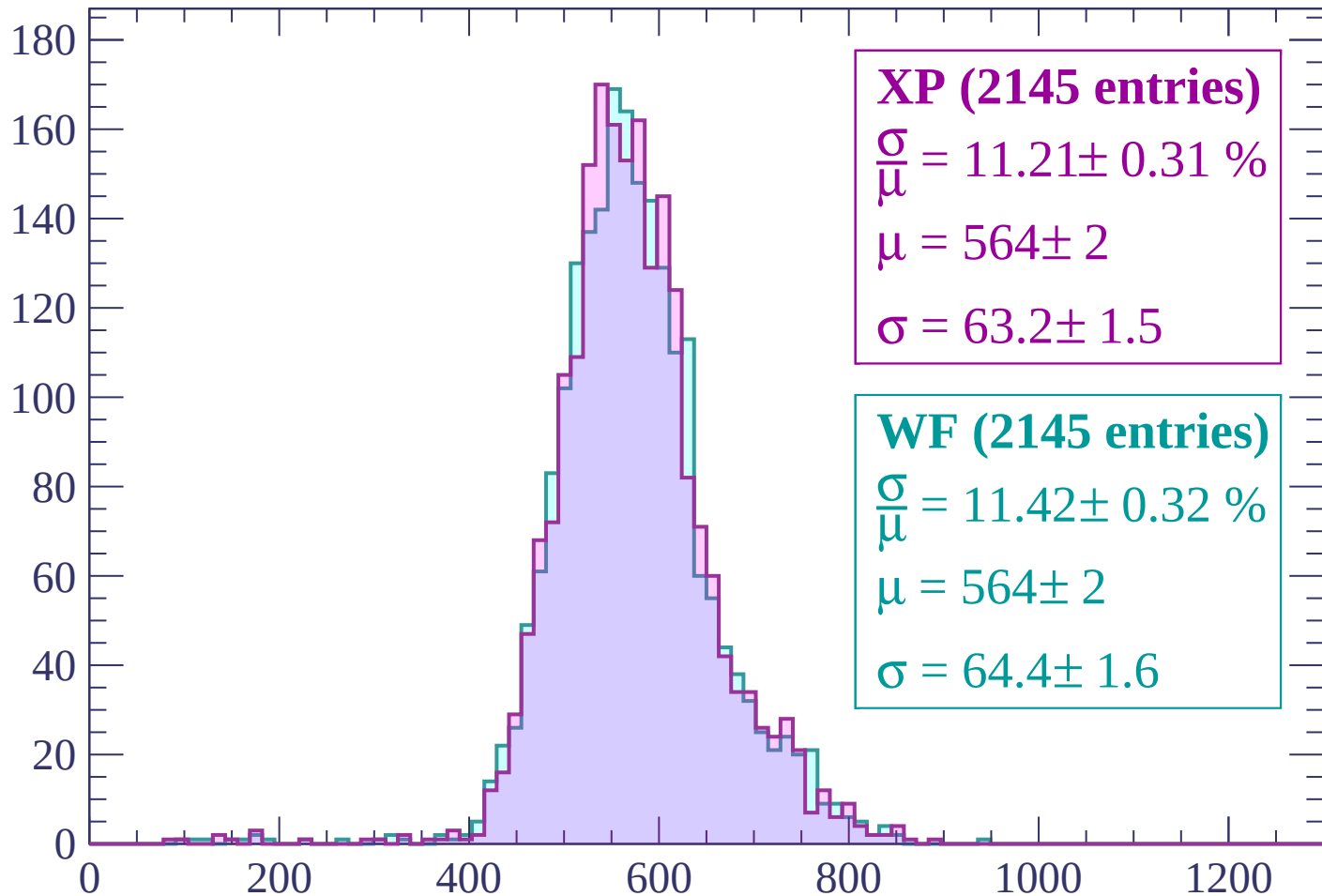
Energy loss | $60 < p < 120$

Count



Energy loss | $120 < p < 180$

Count



XP (2145 entries)

$$\frac{\sigma}{\mu} = 11.21 \pm 0.31 \%$$

$$\mu = 564 \pm 2$$

$$\sigma = 63.2 \pm 1.5$$

WF (2145 entries)

$$\frac{\sigma}{\mu} = 11.42 \pm 0.32 \%$$

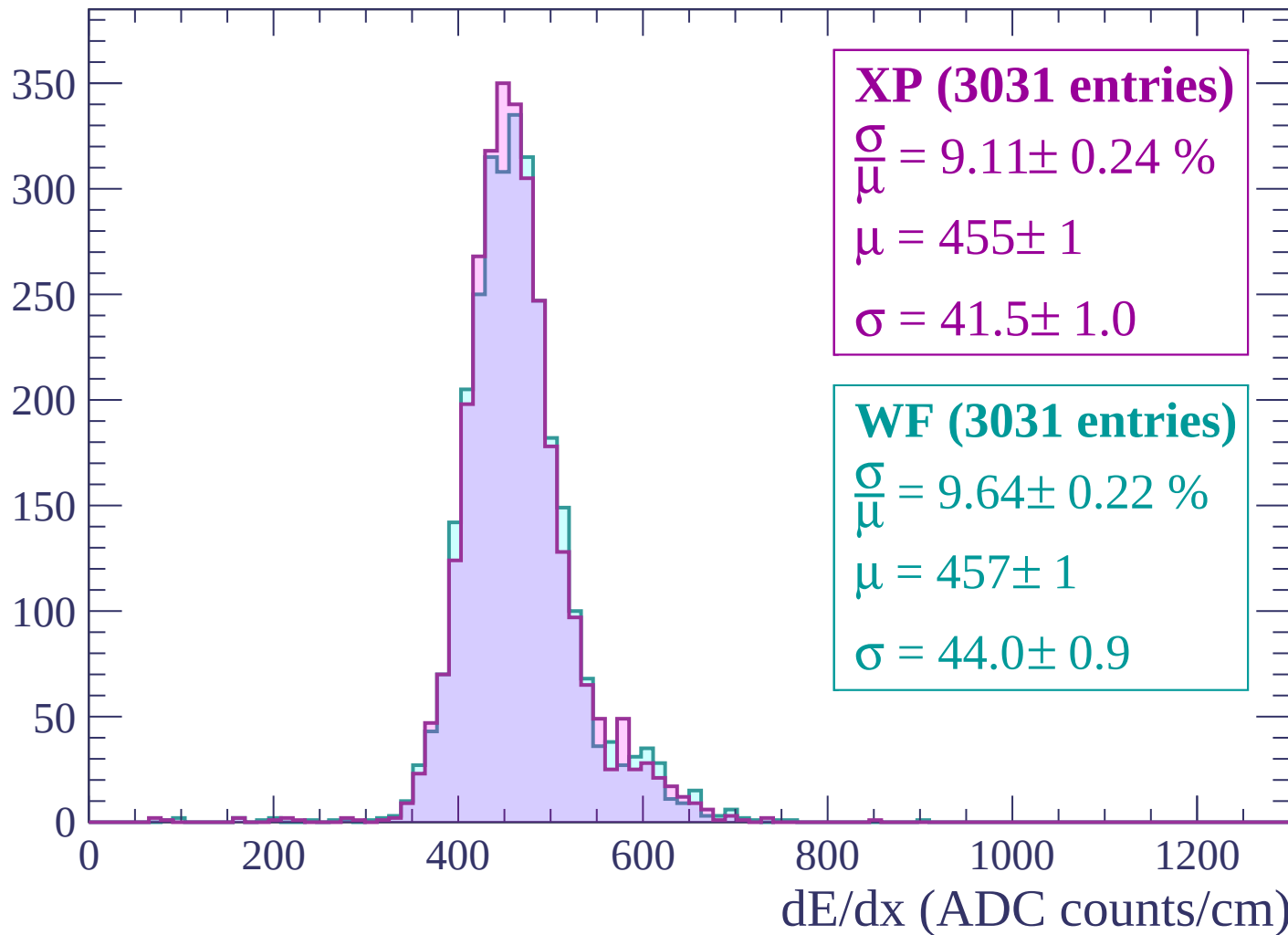
$$\mu = 564 \pm 2$$

$$\sigma = 64.4 \pm 1.6$$

dE/dx (ADC counts/cm)

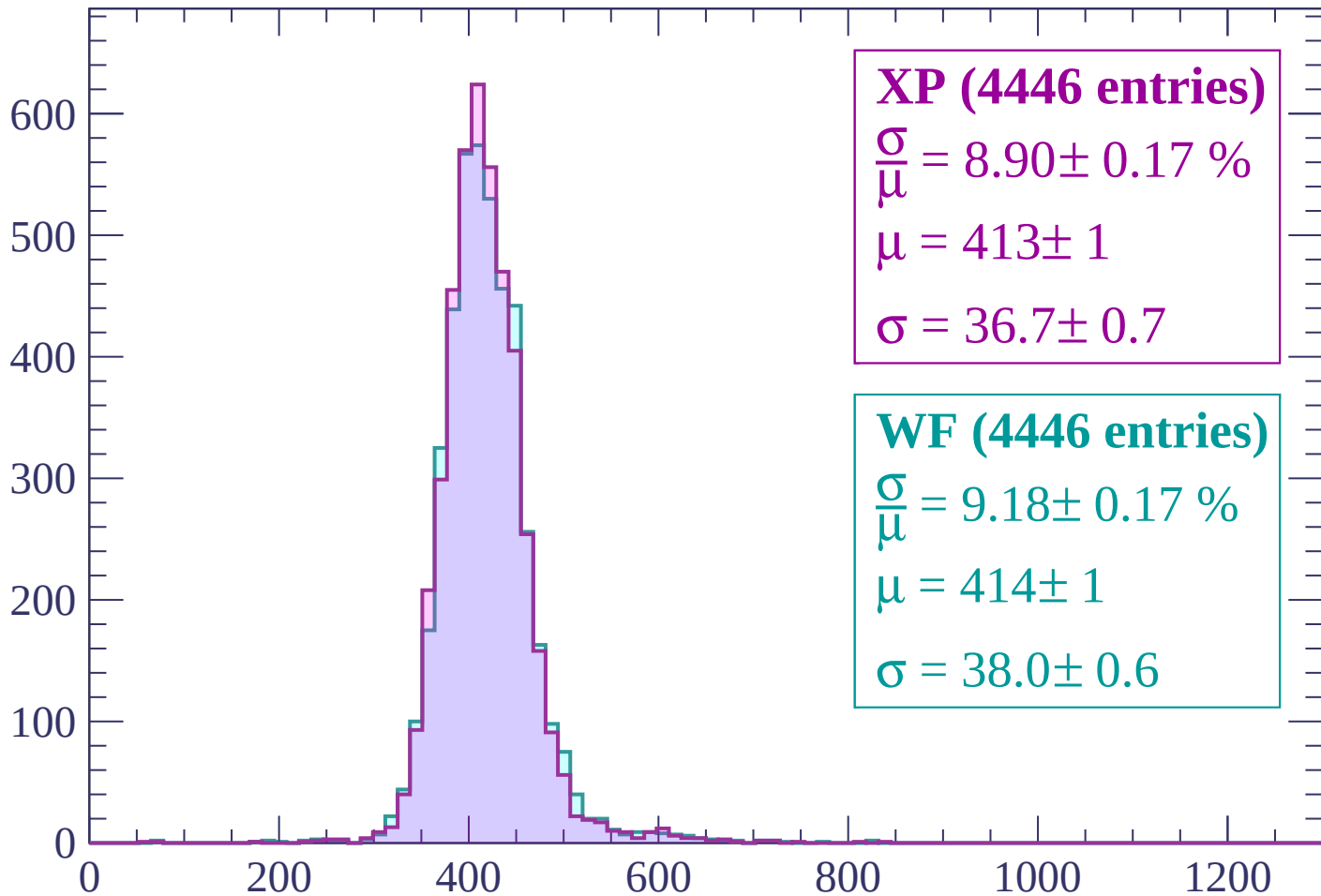
Energy loss | $180 < p < 240$

Count



Energy loss | $240 < p < 300$

Count



XP (4446 entries)

$$\frac{\sigma}{\mu} = 8.90 \pm 0.17 \%$$

$$\mu = 413 \pm 1$$

$$\sigma = 36.7 \pm 0.7$$

WF (4446 entries)

$$\frac{\sigma}{\mu} = 9.18 \pm 0.17 \%$$

$$\mu = 414 \pm 1$$

$$\sigma = 38.0 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $300 < p < 360$

Count

1000

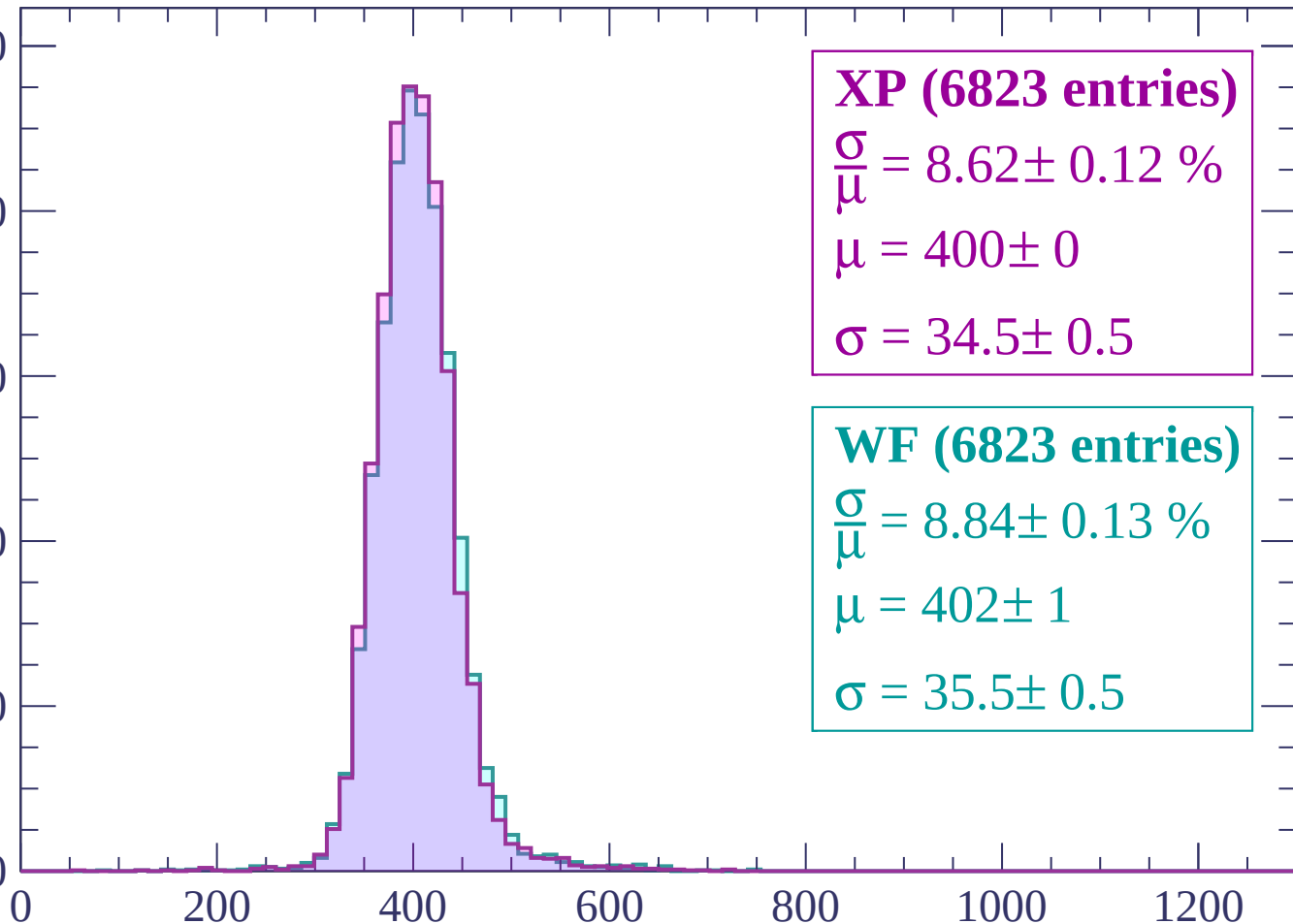
800

600

400

200

0



XP (6823 entries)

$\frac{\sigma}{\mu} = 8.62 \pm 0.12 \%$

$\mu = 400 \pm 0$

$\sigma = 34.5 \pm 0.5$

WF (6823 entries)

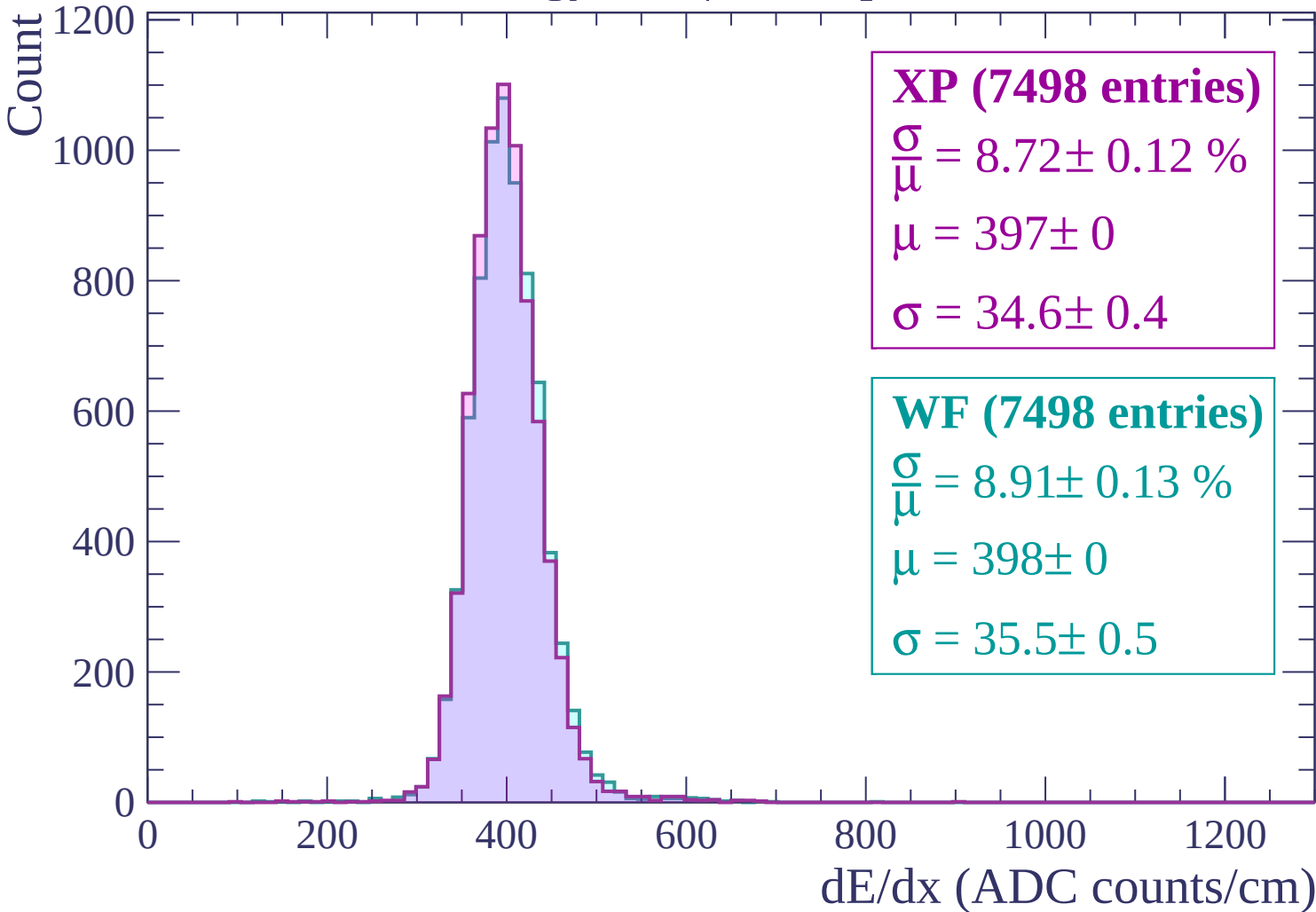
$\frac{\sigma}{\mu} = 8.84 \pm 0.13 \%$

$\mu = 402 \pm 1$

$\sigma = 35.5 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $360 < p < 420$



Energy loss | $420 < p < 480$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (7660 entries)

$\frac{\sigma}{\mu} = 8.60 \pm 0.12 \%$

$\mu = 398 \pm 0$

$\sigma = 34.3 \pm 0.4$

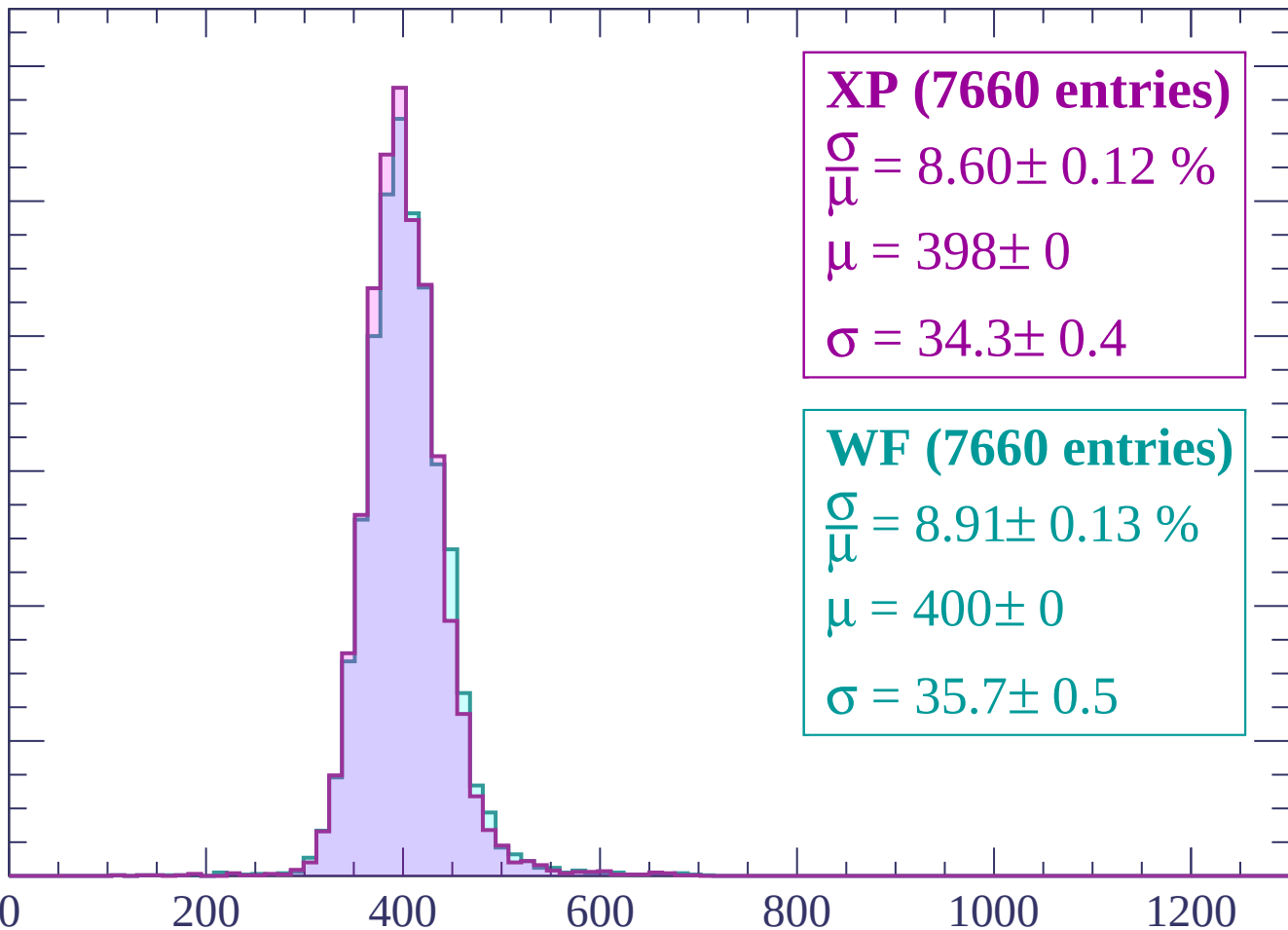
WF (7660 entries)

$\frac{\sigma}{\mu} = 8.91 \pm 0.13 \%$

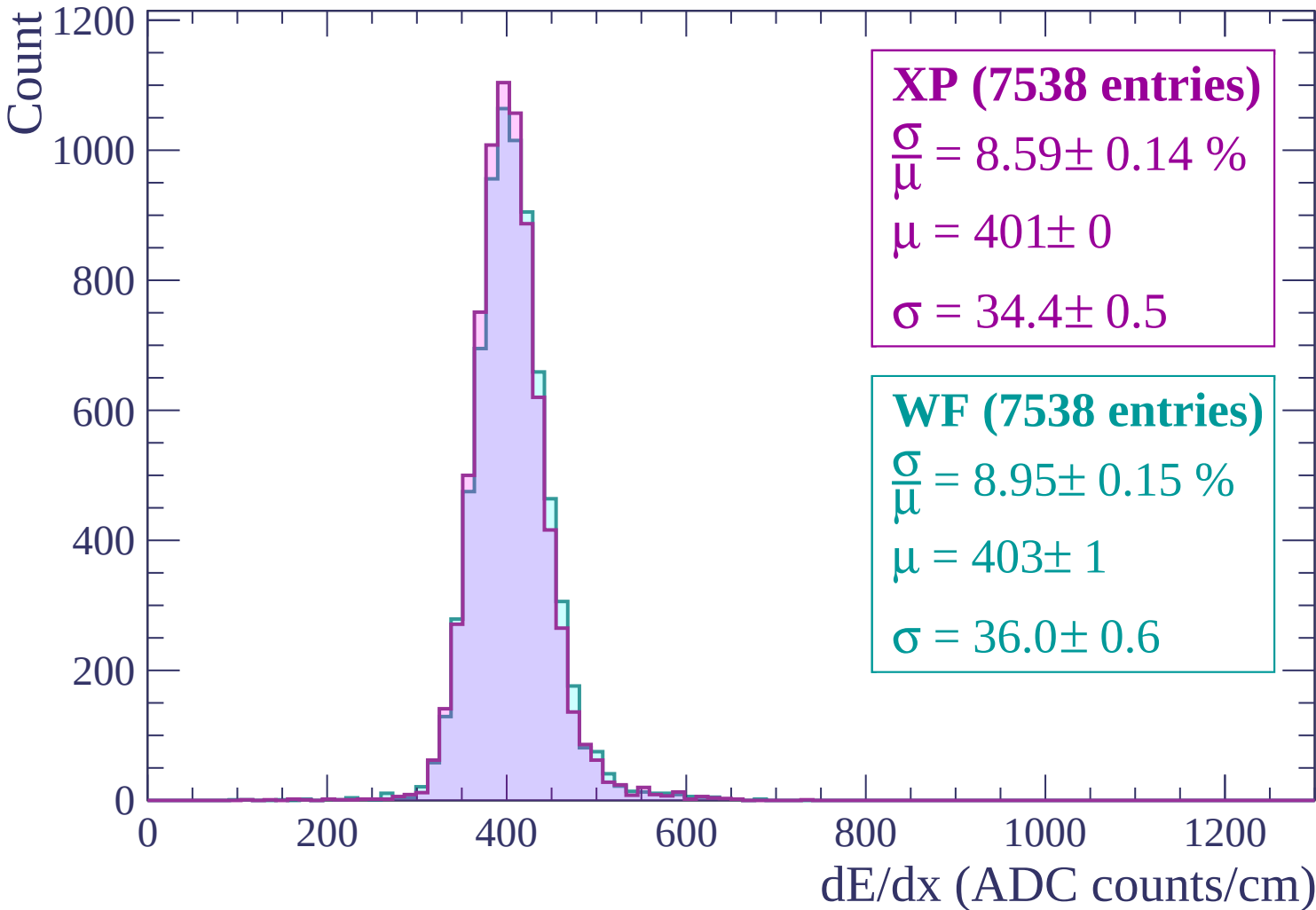
$\mu = 400 \pm 0$

$\sigma = 35.7 \pm 0.5$

dE/dx (ADC counts/cm)

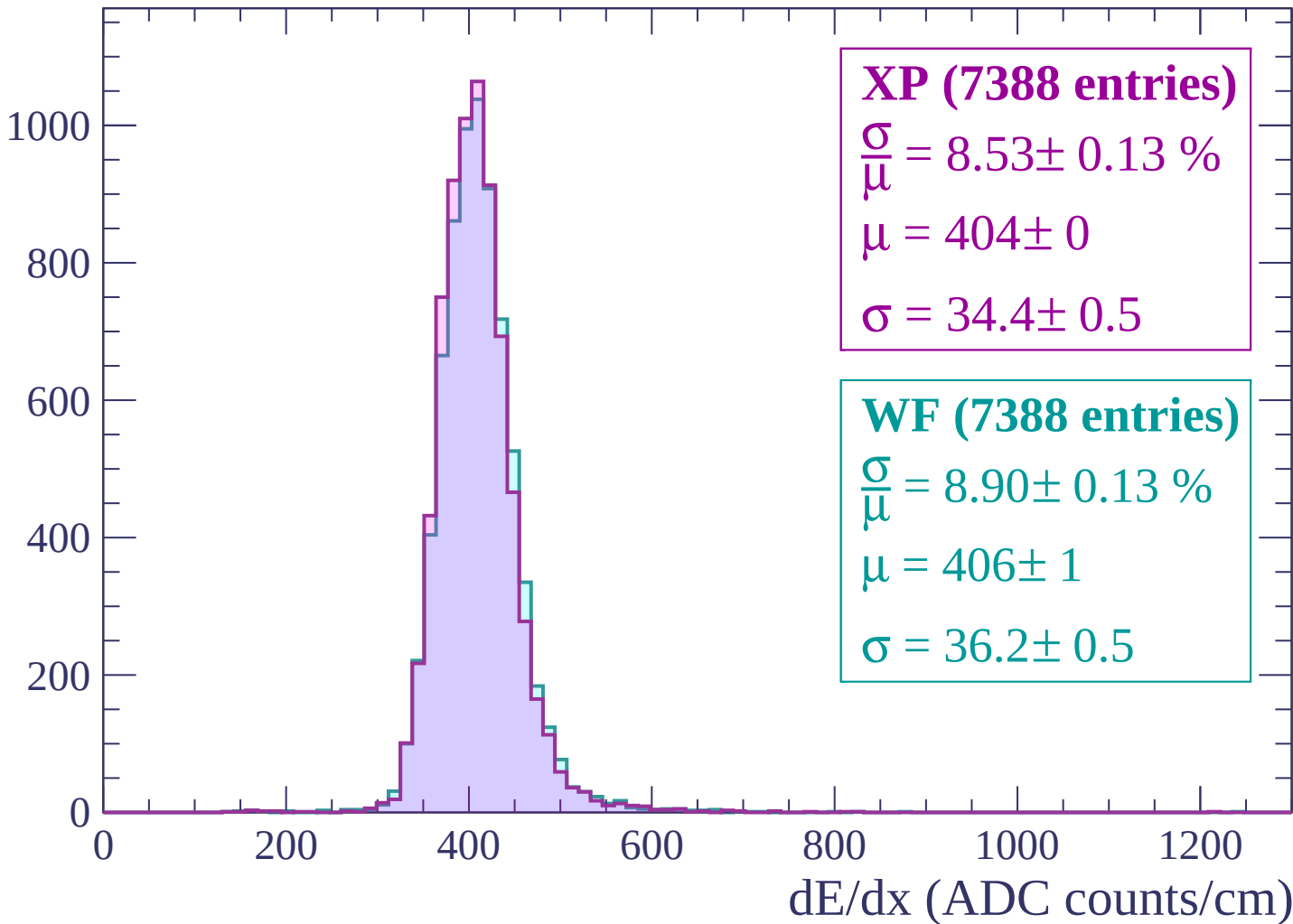


Energy loss | $480 < p < 540$



Energy loss | $540 < p < 600$

Count



Energy loss | $600 < p < 660$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (7197 entries)

$\frac{\sigma}{\mu} = 8.72 \pm 0.13 \%$

$\mu = 407 \pm 0$

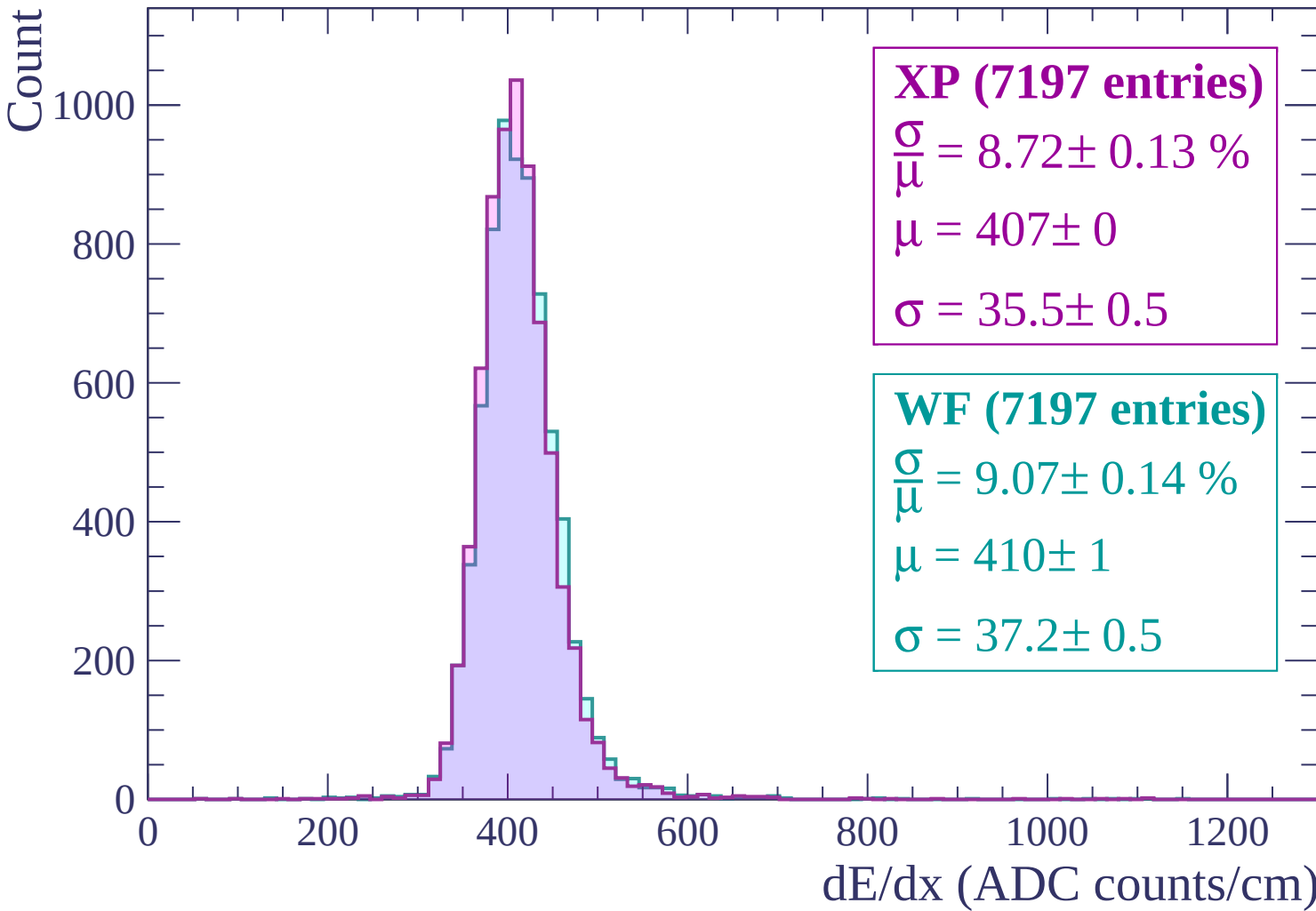
$\sigma = 35.5 \pm 0.5$

WF (7197 entries)

$\frac{\sigma}{\mu} = 9.07 \pm 0.14 \%$

$\mu = 410 \pm 1$

$\sigma = 37.2 \pm 0.5$



Energy loss | $660 < p < 720$

Count

1000

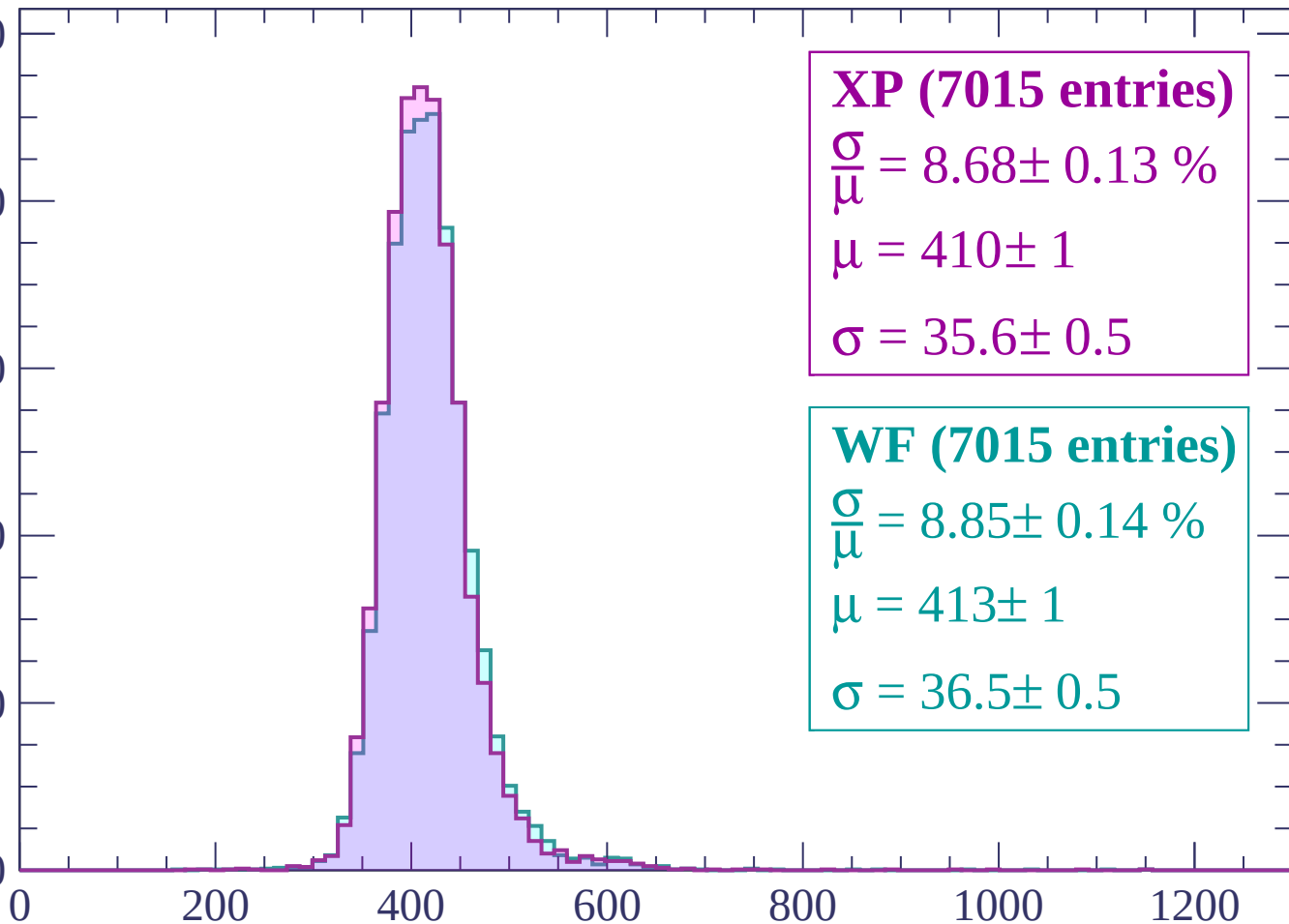
800

600

400

200

0



XP (7015 entries)

$\frac{\sigma}{\mu} = 8.68 \pm 0.13 \%$

$\mu = 410 \pm 1$

$\sigma = 35.6 \pm 0.5$

WF (7015 entries)

$\frac{\sigma}{\mu} = 8.85 \pm 0.14 \%$

$\mu = 413 \pm 1$

$\sigma = 36.5 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $720 < p < 780$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (7055 entries)

$\frac{\sigma}{\mu} = 8.63 \pm 0.13 \%$

$\mu = 414 \pm 1$

$\sigma = 35.7 \pm 0.5$

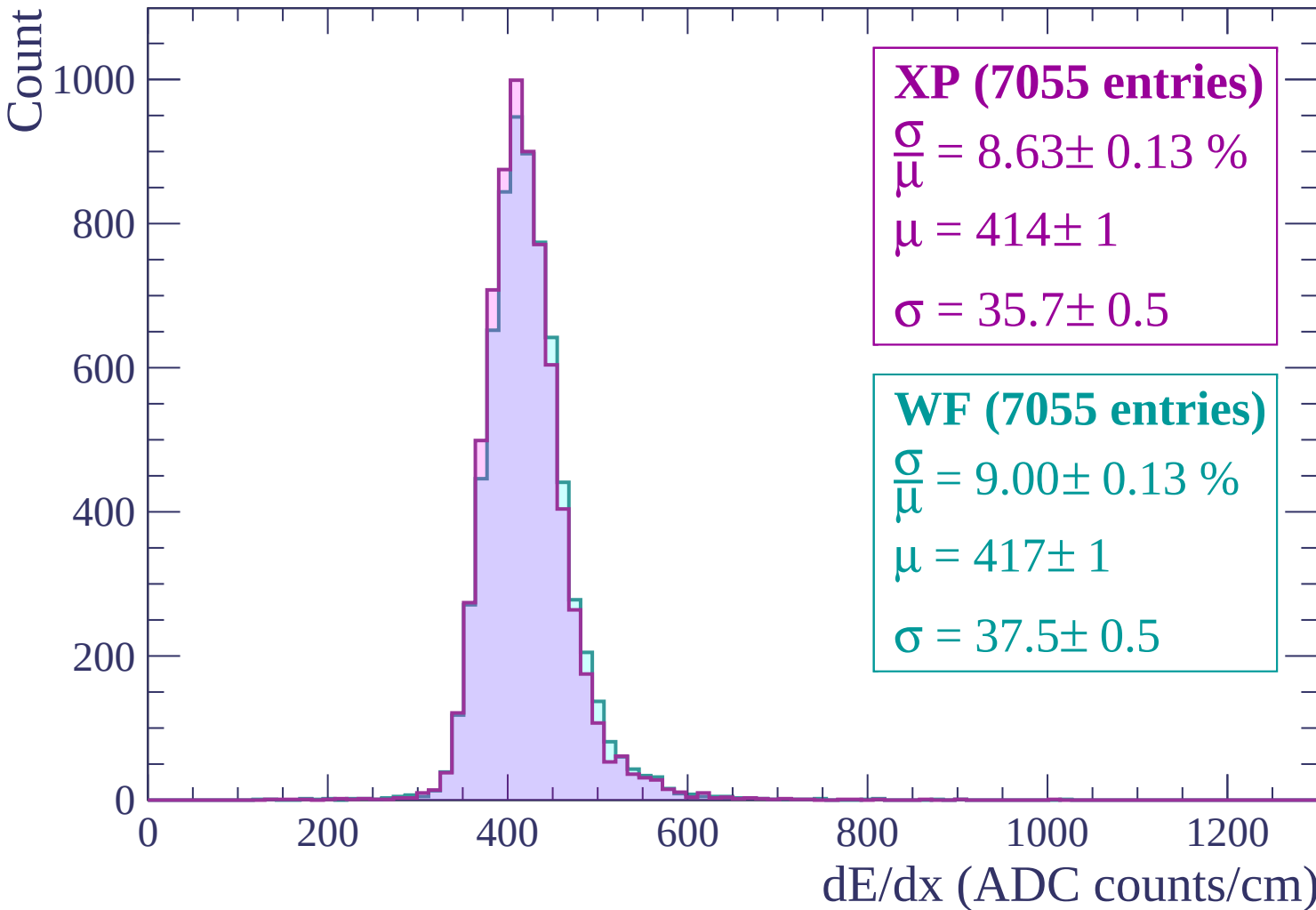
WF (7055 entries)

$\frac{\sigma}{\mu} = 9.00 \pm 0.13 \%$

$\mu = 417 \pm 1$

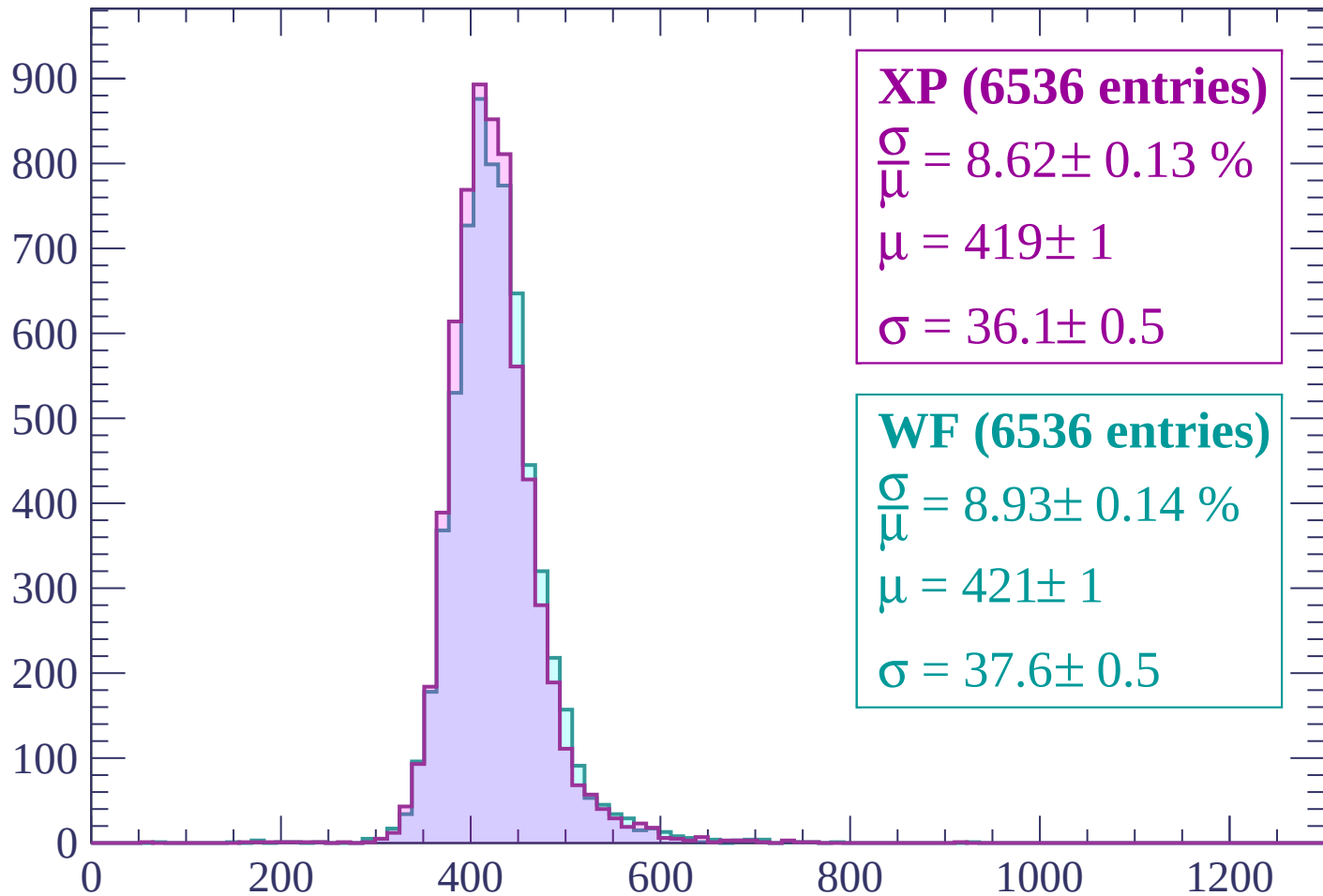
$\sigma = 37.5 \pm 0.5$

dE/dx (ADC counts/cm)



Energy loss | $780 < p < 840$

Count



XP (6536 entries)

$$\frac{\sigma}{\mu} = 8.62 \pm 0.13 \%$$

$$\mu = 419 \pm 1$$

$$\sigma = 36.1 \pm 0.5$$

WF (6536 entries)

$$\frac{\sigma}{\mu} = 8.93 \pm 0.14 \%$$

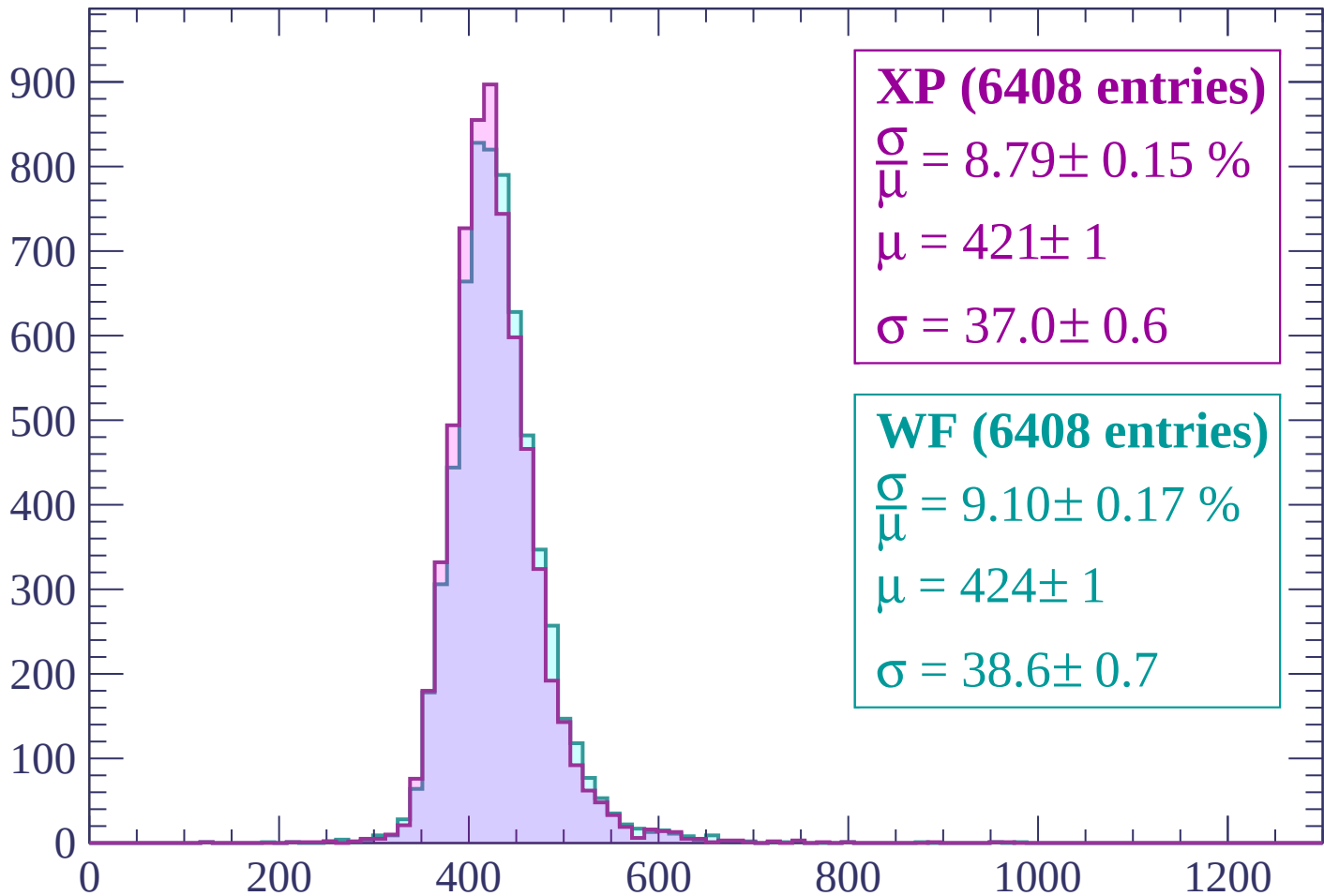
$$\mu = 421 \pm 1$$

$$\sigma = 37.6 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $840 < p < 900$

Count



XP (6408 entries)

$\frac{\sigma}{\mu} = 8.79 \pm 0.15 \%$

$\mu = 421 \pm 1$

$\sigma = 37.0 \pm 0.6$

WF (6408 entries)

$\frac{\sigma}{\mu} = 9.10 \pm 0.17 \%$

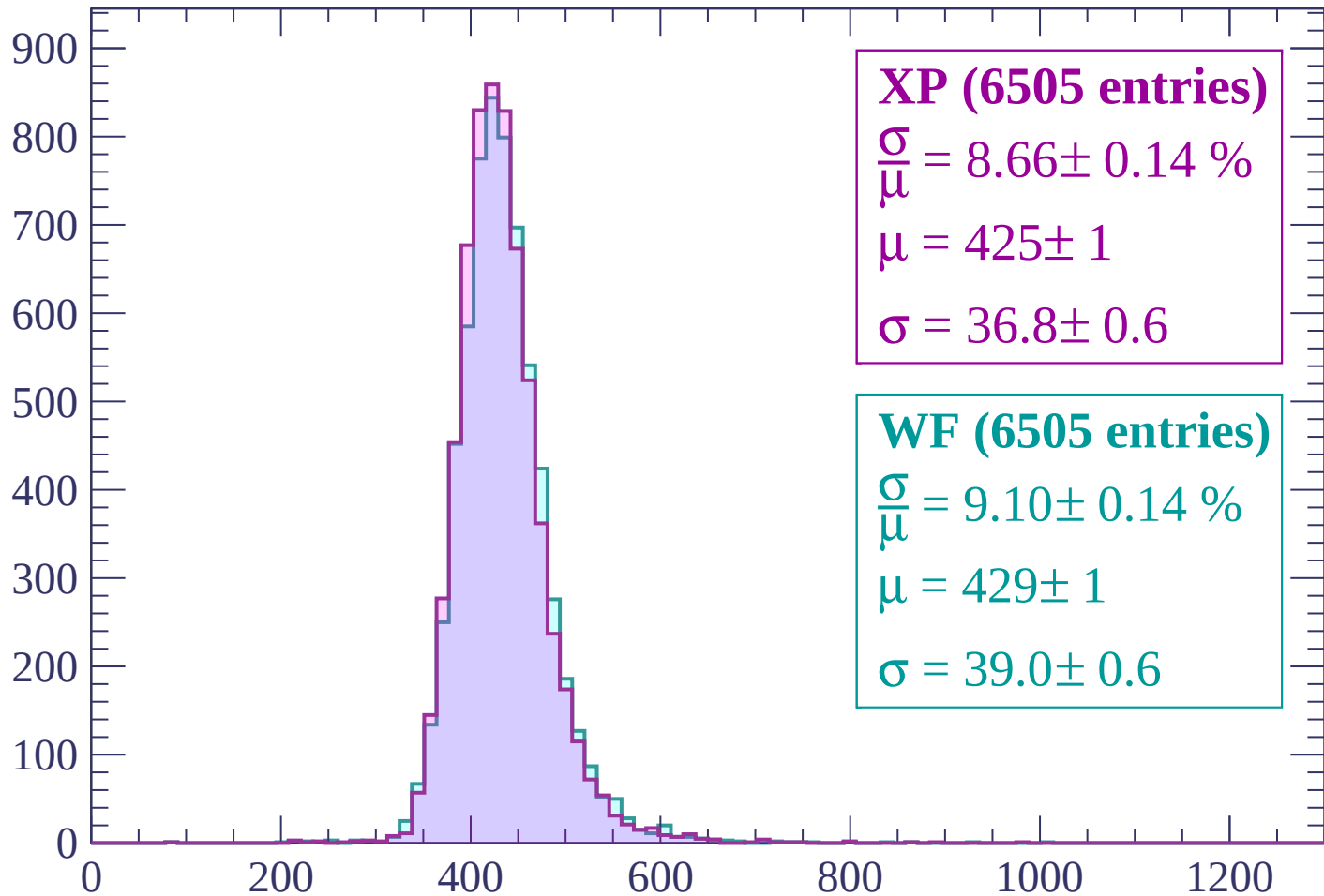
$\mu = 424 \pm 1$

$\sigma = 38.6 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $900 < p < 960$

Count



XP (6505 entries)

$$\frac{\sigma}{\mu} = 8.66 \pm 0.14 \%$$

$$\mu = 425 \pm 1$$

$$\sigma = 36.8 \pm 0.6$$

WF (6505 entries)

$$\frac{\sigma}{\mu} = 9.10 \pm 0.14 \%$$

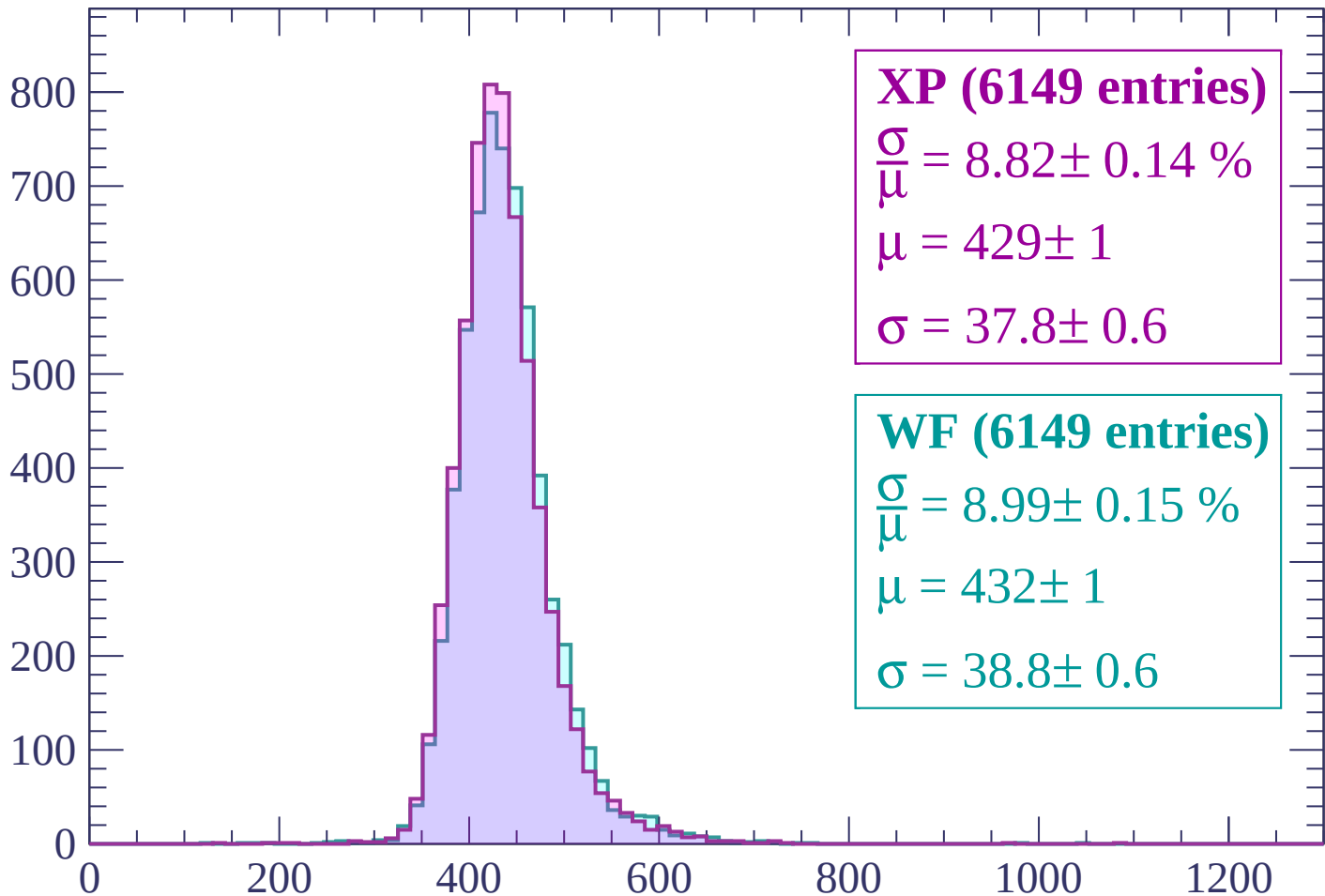
$$\mu = 429 \pm 1$$

$$\sigma = 39.0 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1020$

Count



XP (6149 entries)

$\frac{\sigma}{\mu} = 8.82 \pm 0.14 \%$

$\mu = 429 \pm 1$

$\sigma = 37.8 \pm 0.6$

WF (6149 entries)

$\frac{\sigma}{\mu} = 8.99 \pm 0.15 \%$

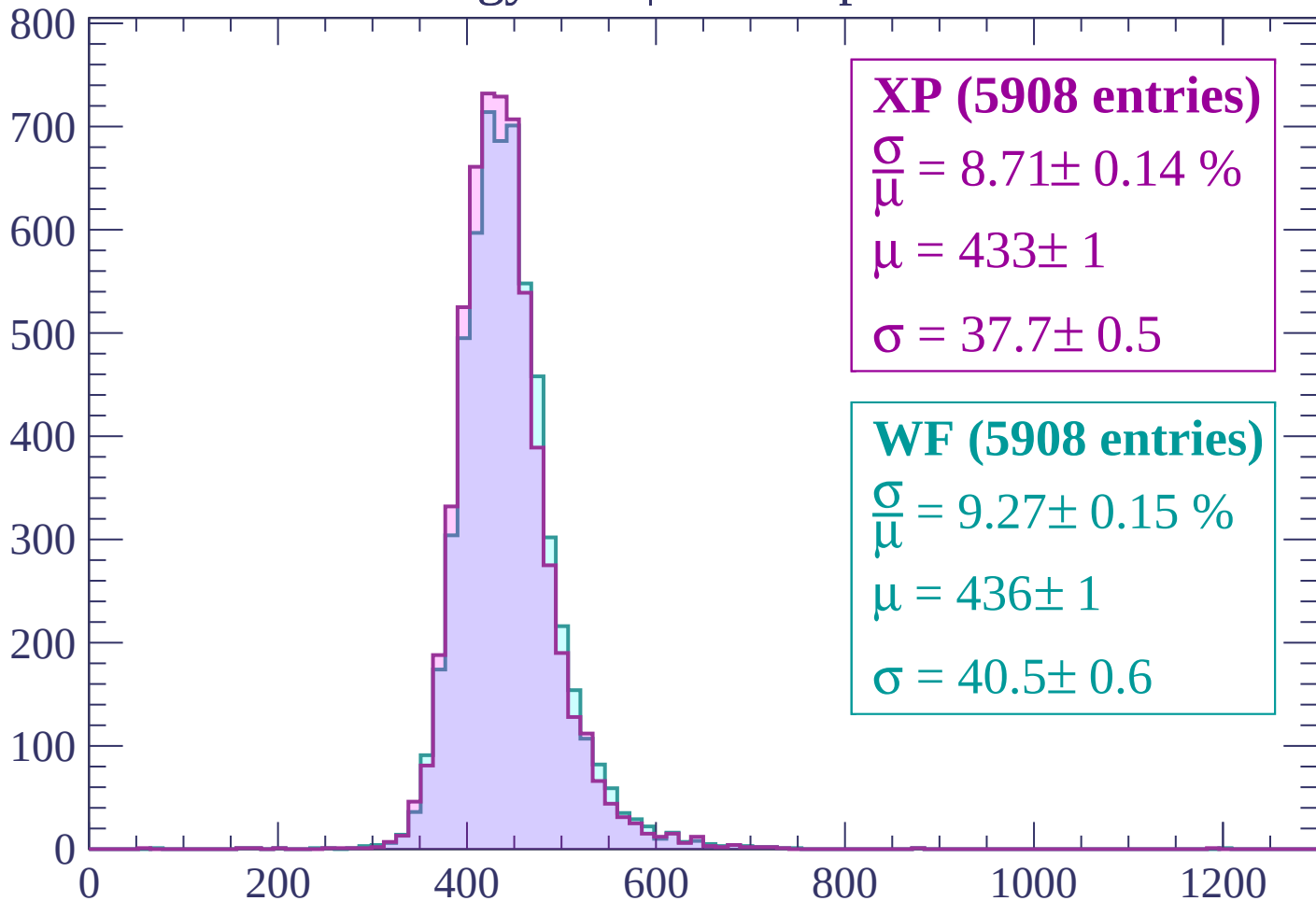
$\mu = 432 \pm 1$

$\sigma = 38.8 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $1020 < p < 1080$

Count



XP (5908 entries)

$$\frac{\sigma}{\mu} = 8.71 \pm 0.14 \%$$

$$\mu = 433 \pm 1$$

$$\sigma = 37.7 \pm 0.5$$

WF (5908 entries)

$$\frac{\sigma}{\mu} = 9.27 \pm 0.15 \%$$

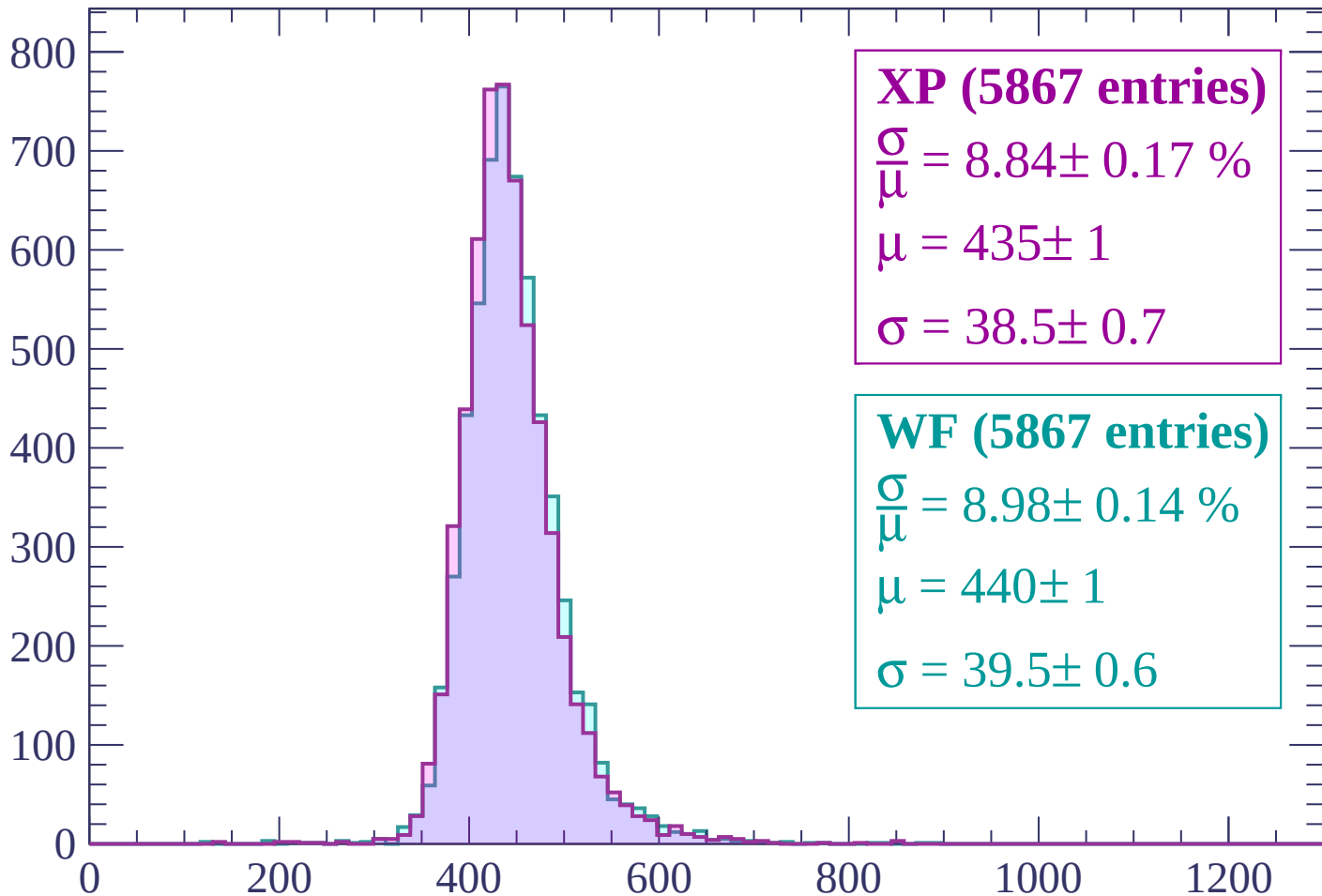
$$\mu = 436 \pm 1$$

$$\sigma = 40.5 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $1080 < p < 1140$

Count



XP (5867 entries)

$\frac{\sigma}{\mu} = 8.84 \pm 0.17 \%$

$\mu = 435 \pm 1$

$\sigma = 38.5 \pm 0.7$

WF (5867 entries)

$\frac{\sigma}{\mu} = 8.98 \pm 0.14 \%$

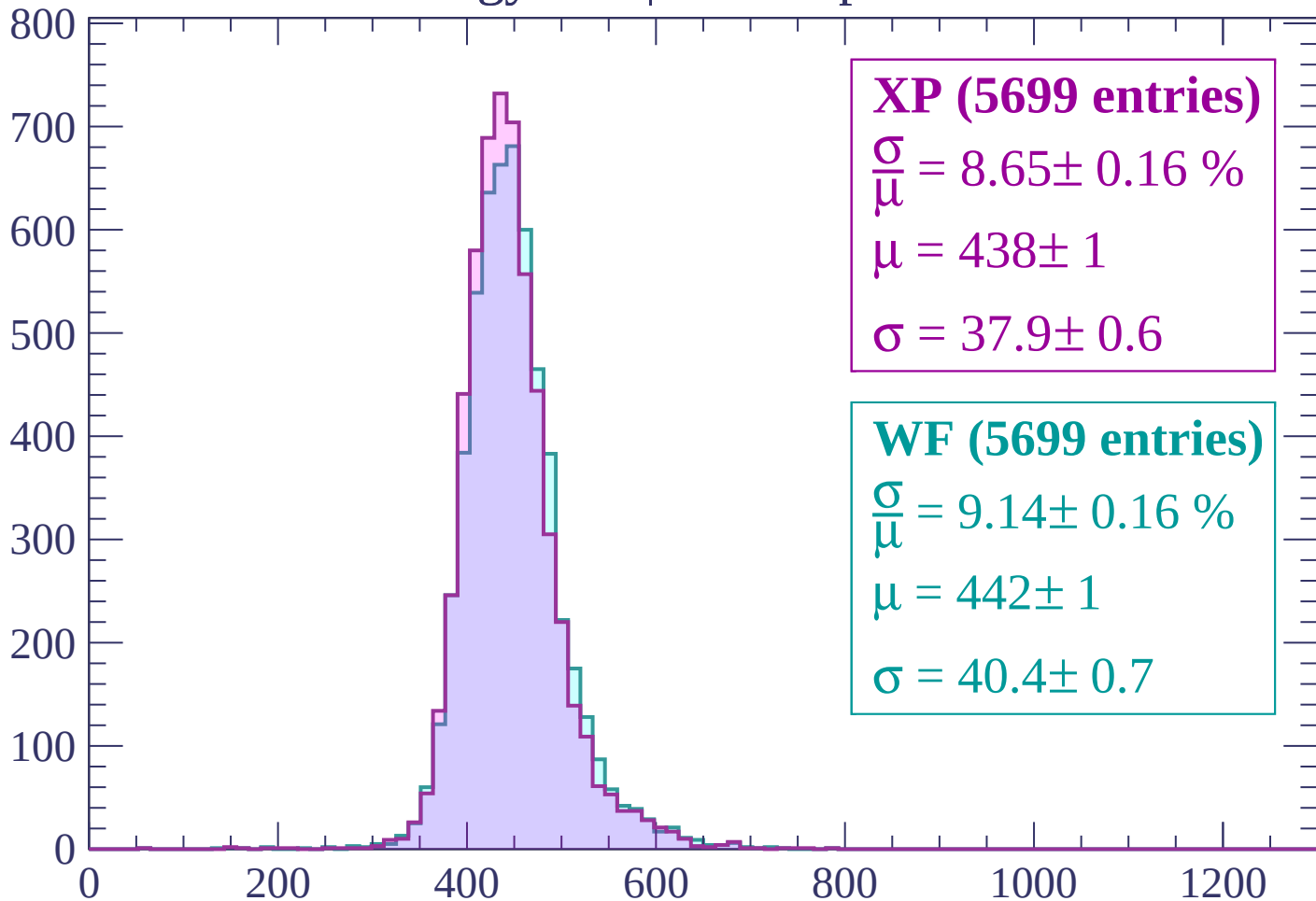
$\mu = 440 \pm 1$

$\sigma = 39.5 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $1140 < p < 1200$

Count



XP (5699 entries)

$$\frac{\sigma}{\mu} = 8.65 \pm 0.16 \%$$

$$\mu = 438 \pm 1$$

$$\sigma = 37.9 \pm 0.6$$

WF (5699 entries)

$$\frac{\sigma}{\mu} = 9.14 \pm 0.16 \%$$

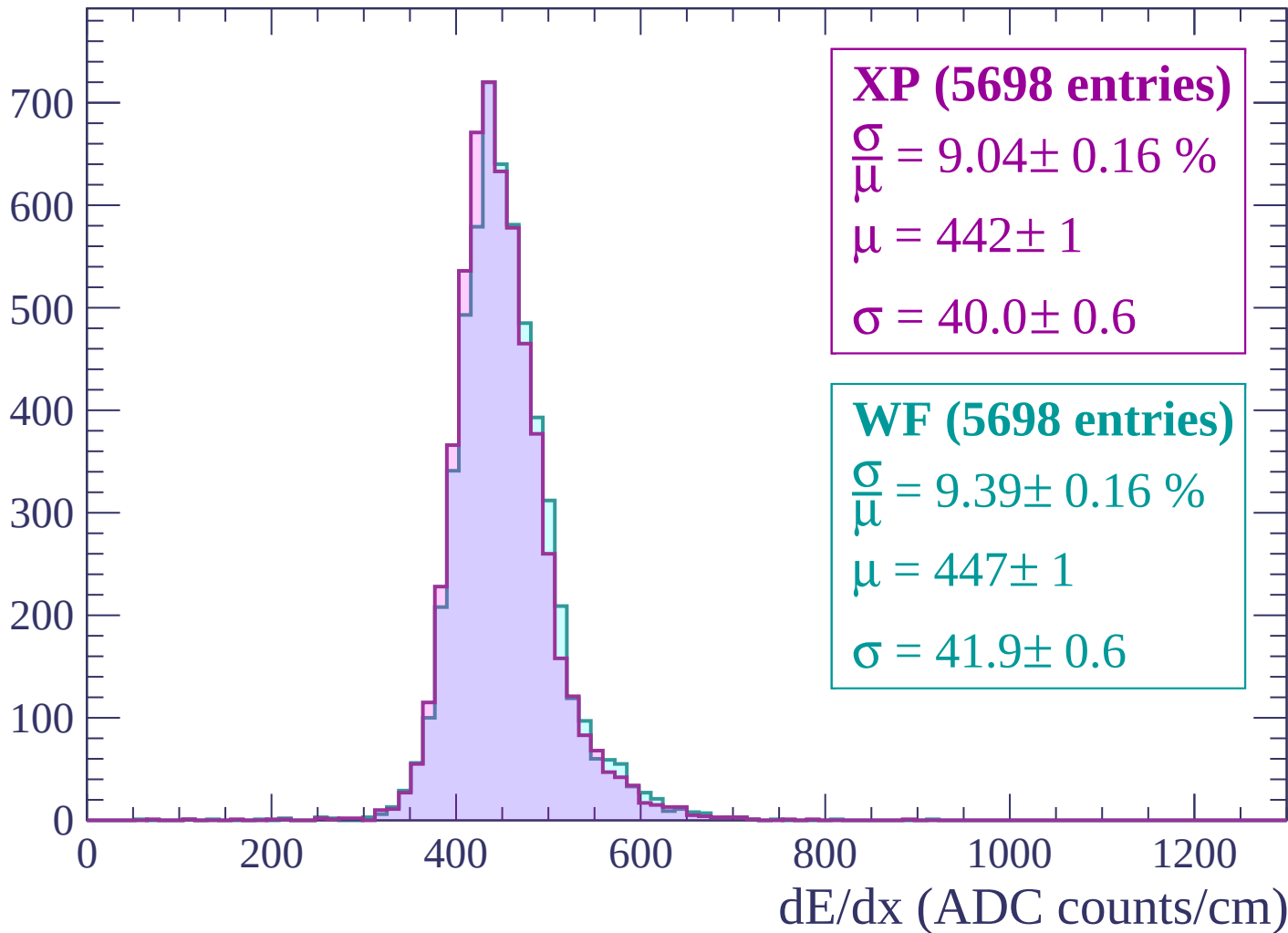
$$\mu = 442 \pm 1$$

$$\sigma = 40.4 \pm 0.7$$

dE/dx (ADC counts/cm)

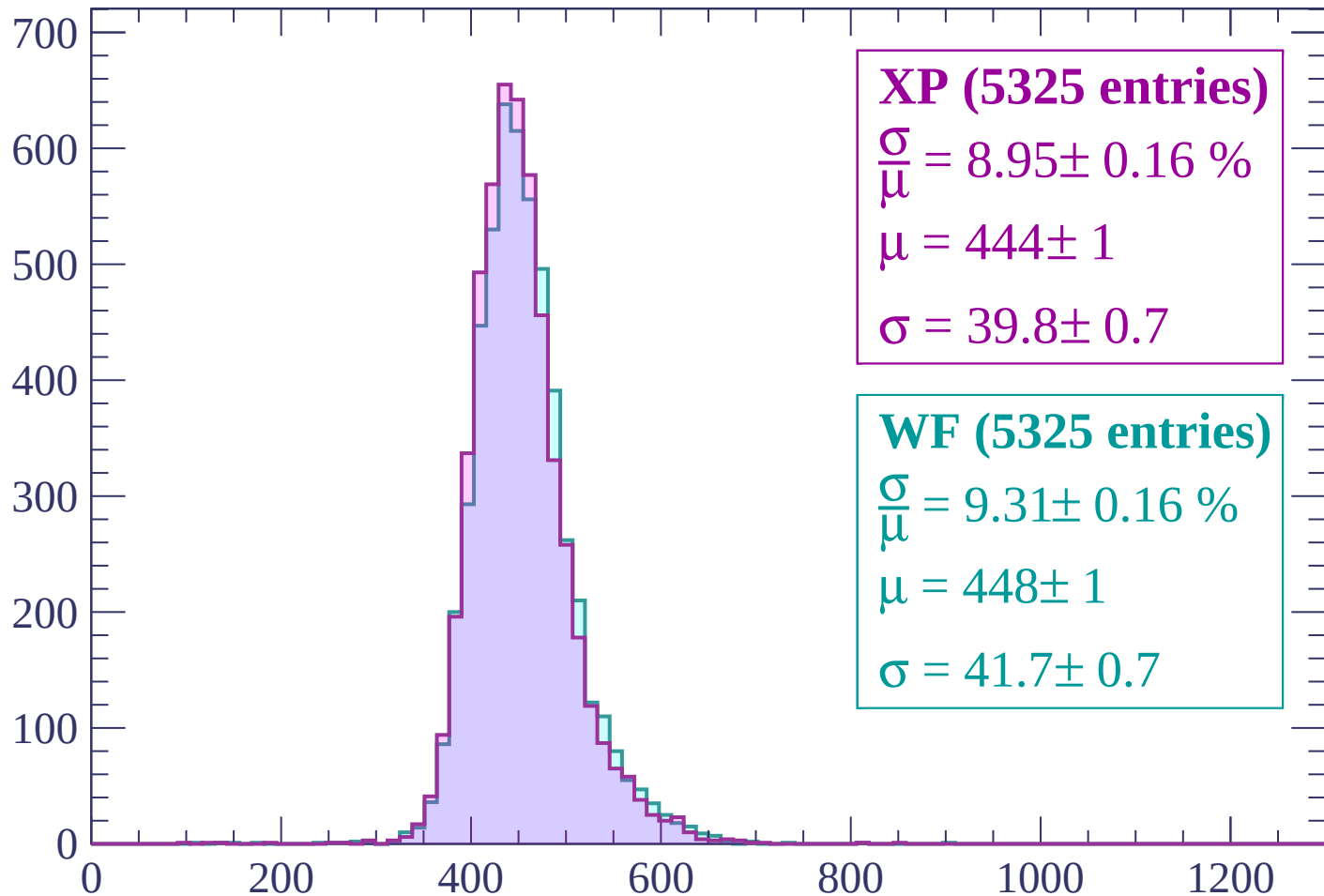
Energy loss | $1200 < p < 1260$

Count



Energy loss | $1260 < p < 1320$

Count



XP (5325 entries)

$\frac{\sigma}{\mu} = 8.95 \pm 0.16 \%$

$\mu = 444 \pm 1$

$\sigma = 39.8 \pm 0.7$

WF (5325 entries)

$\frac{\sigma}{\mu} = 9.31 \pm 0.16 \%$

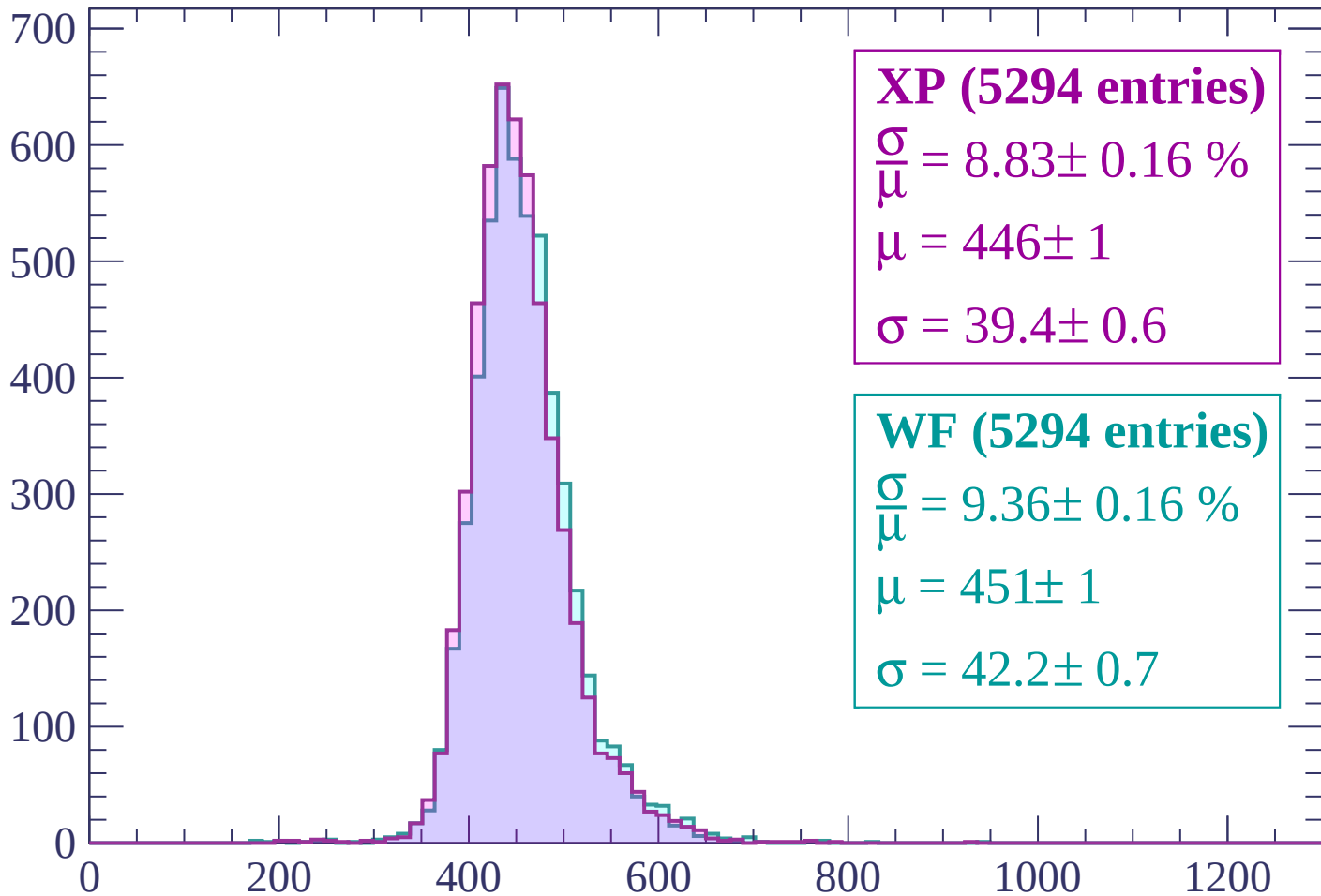
$\mu = 448 \pm 1$

$\sigma = 41.7 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $1320 < p < 1380$

Count



XP (5294 entries)

$$\frac{\sigma}{\mu} = 8.83 \pm 0.16 \%$$

$$\mu = 446 \pm 1$$

$$\sigma = 39.4 \pm 0.6$$

WF (5294 entries)

$$\frac{\sigma}{\mu} = 9.36 \pm 0.16 \%$$

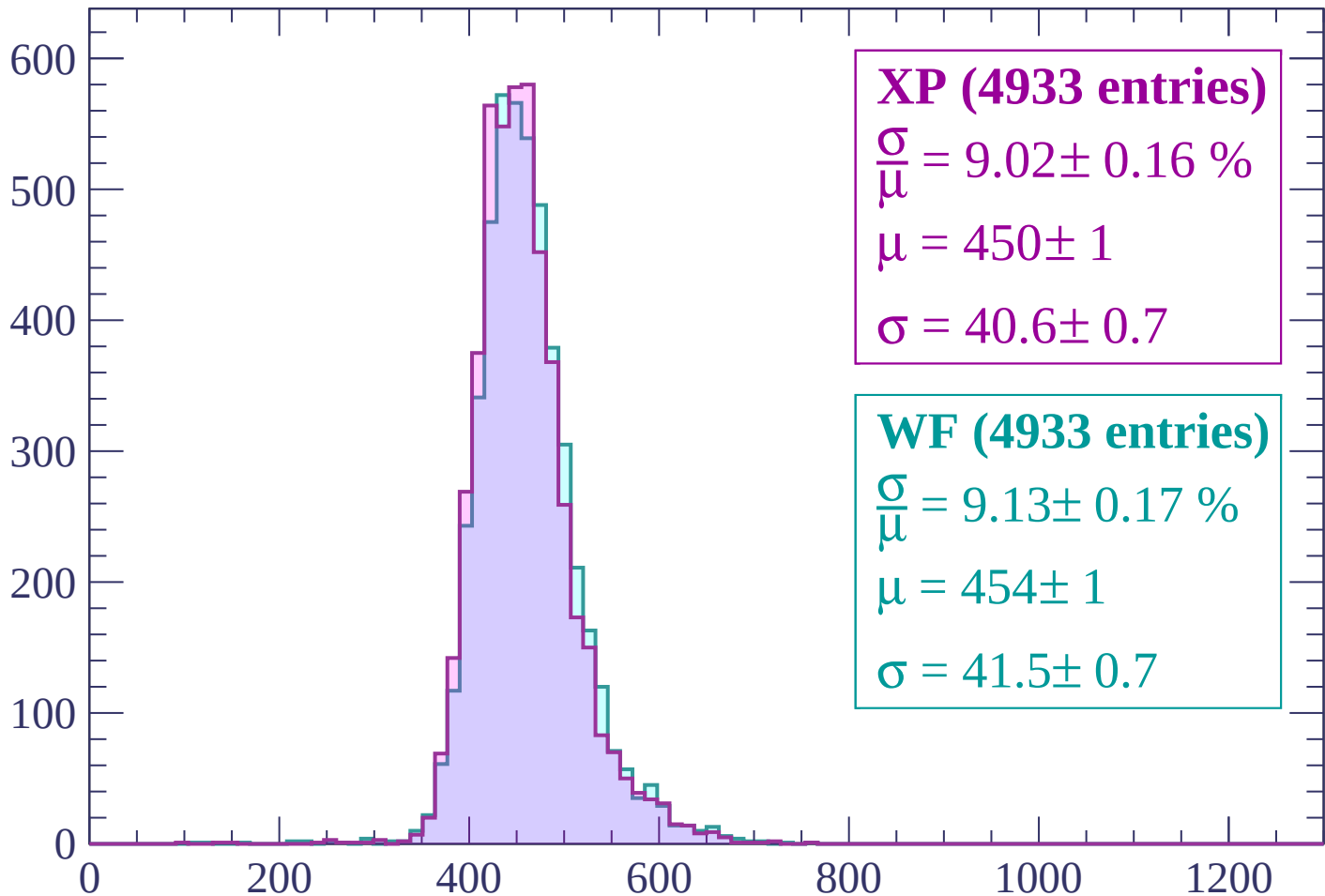
$$\mu = 451 \pm 1$$

$$\sigma = 42.2 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1380 < p < 1440$

Count



XP (4933 entries)

$\frac{\sigma}{\mu} = 9.02 \pm 0.16 \%$

$\mu = 450 \pm 1$

$\sigma = 40.6 \pm 0.7$

WF (4933 entries)

$\frac{\sigma}{\mu} = 9.13 \pm 0.17 \%$

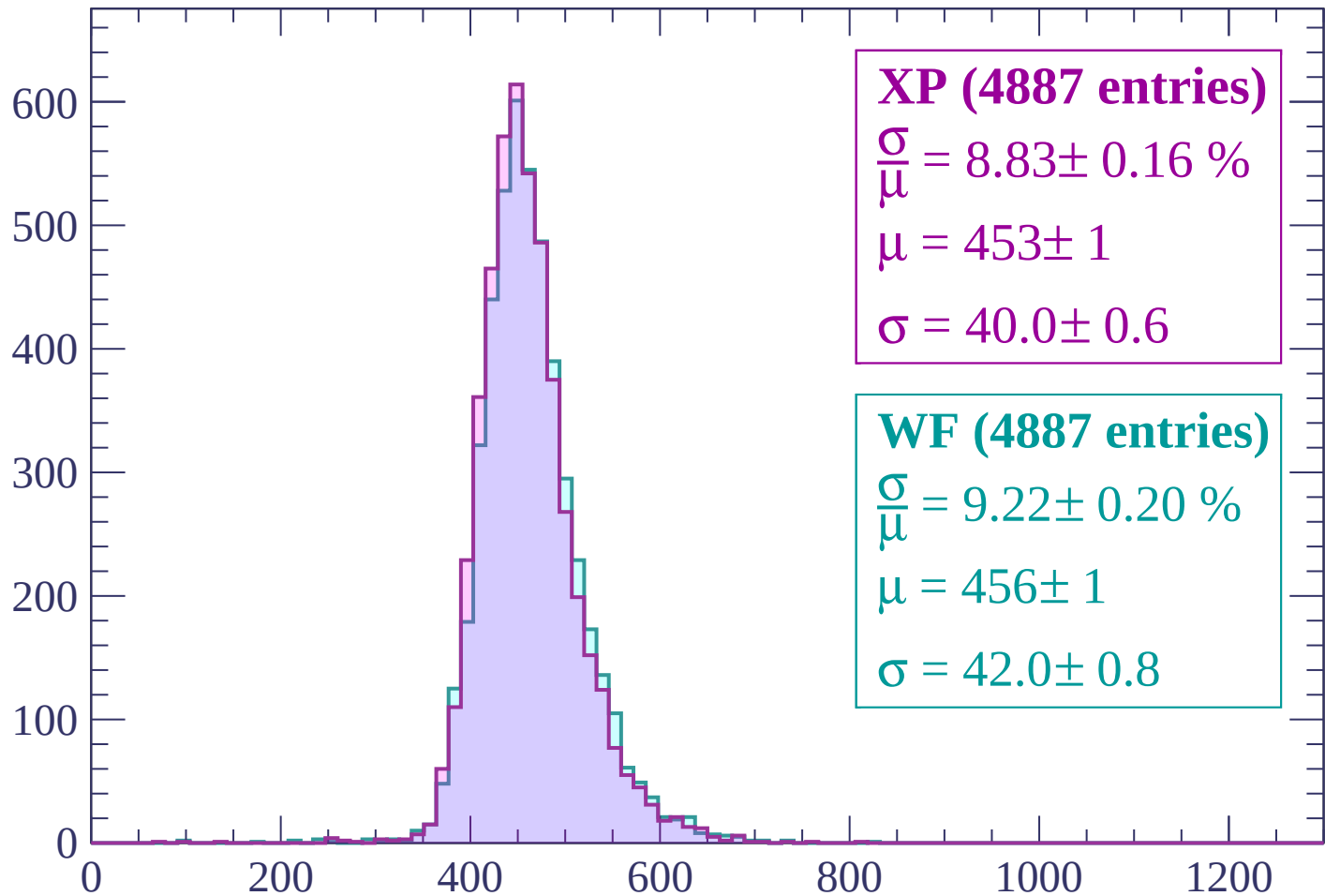
$\mu = 454 \pm 1$

$\sigma = 41.5 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $1440 < p < 1500$

Count



XP (4887 entries)

$$\frac{\sigma}{\mu} = 8.83 \pm 0.16 \%$$

$$\mu = 453 \pm 1$$

$$\sigma = 40.0 \pm 0.6$$

WF (4887 entries)

$$\frac{\sigma}{\mu} = 9.22 \pm 0.20 \%$$

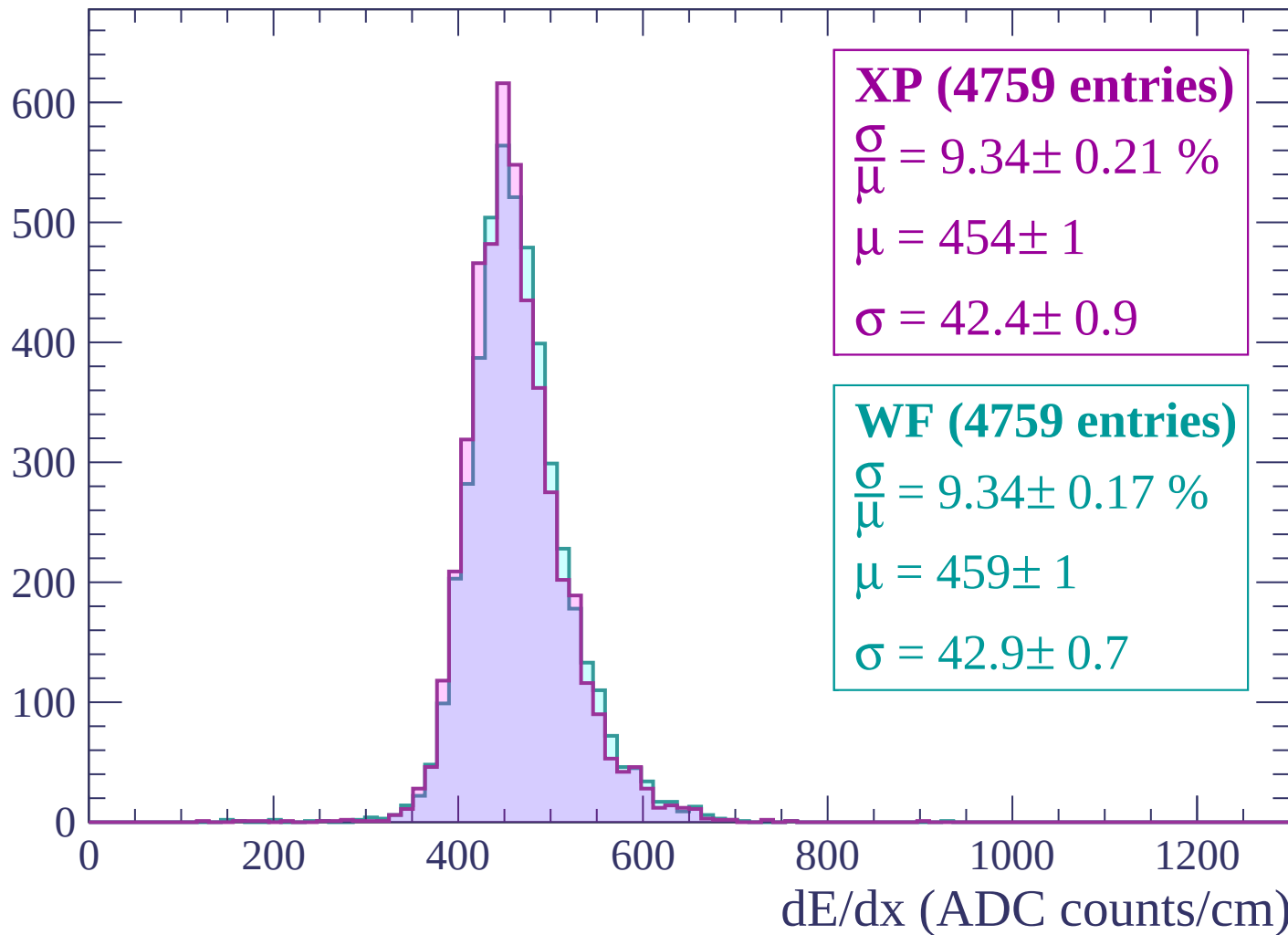
$$\mu = 456 \pm 1$$

$$\sigma = 42.0 \pm 0.8$$

dE/dx (ADC counts/cm)

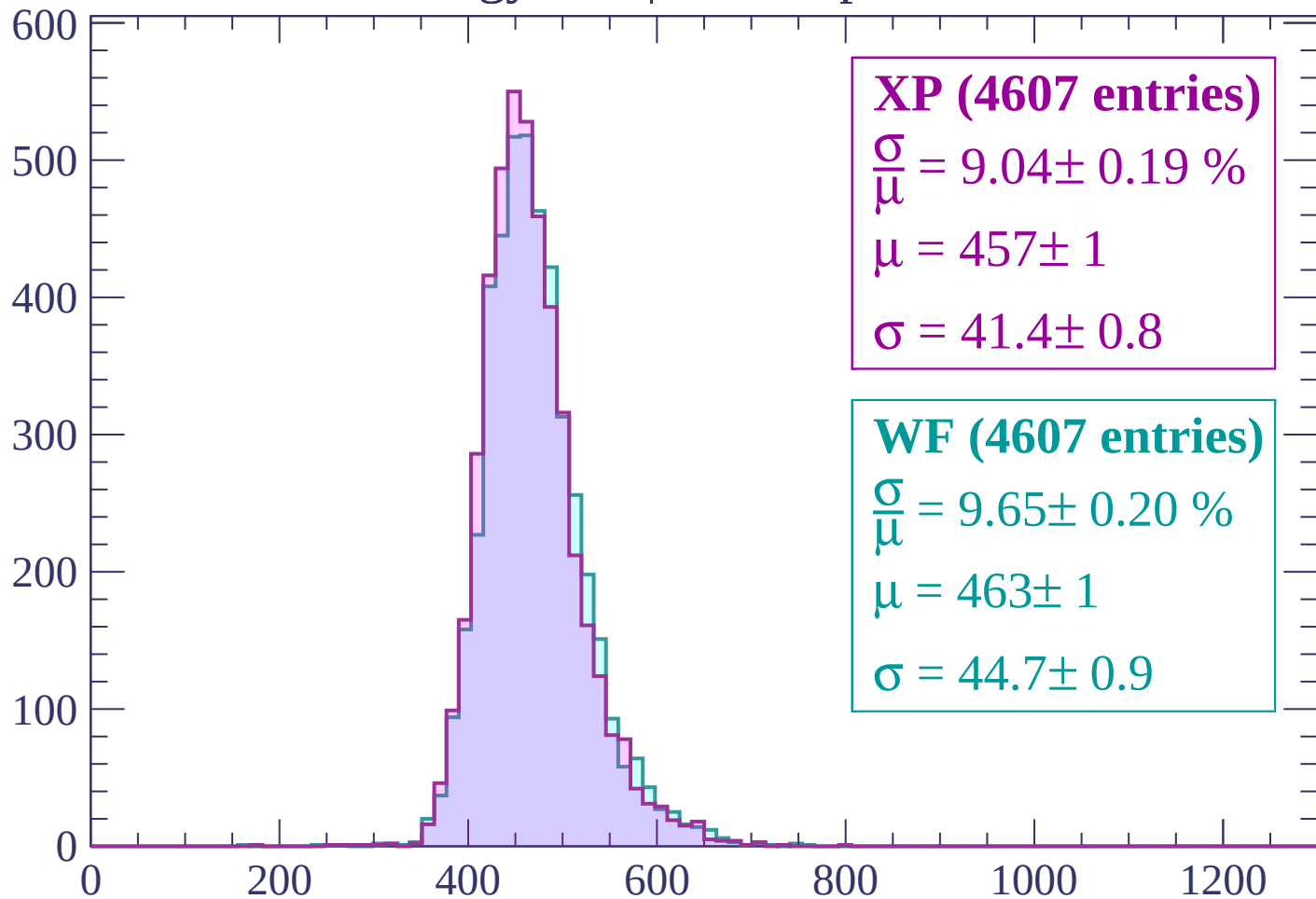
Energy loss | $1500 < p < 1560$

Count



Energy loss | $1560 < p < 1620$

Count



XP (4607 entries)

$\frac{\sigma}{\mu} = 9.04 \pm 0.19 \%$

$\mu = 457 \pm 1$

$\sigma = 41.4 \pm 0.8$

WF (4607 entries)

$\frac{\sigma}{\mu} = 9.65 \pm 0.20 \%$

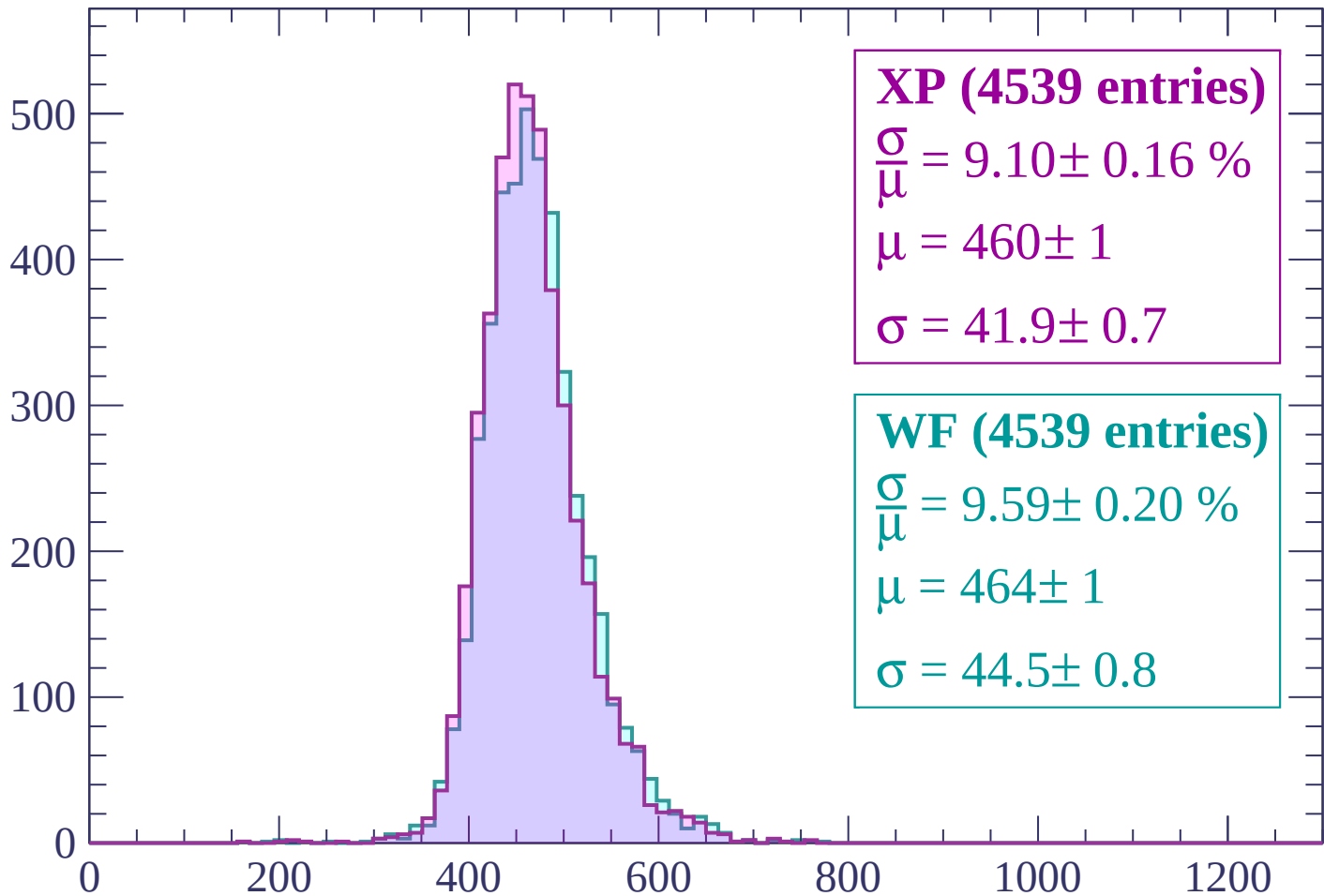
$\mu = 463 \pm 1$

$\sigma = 44.7 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss | $1620 < p < 1680$

Count



XP (4539 entries)

$$\frac{\sigma}{\mu} = 9.10 \pm 0.16 \%$$

$$\mu = 460 \pm 1$$

$$\sigma = 41.9 \pm 0.7$$

WF (4539 entries)

$$\frac{\sigma}{\mu} = 9.59 \pm 0.20 \%$$

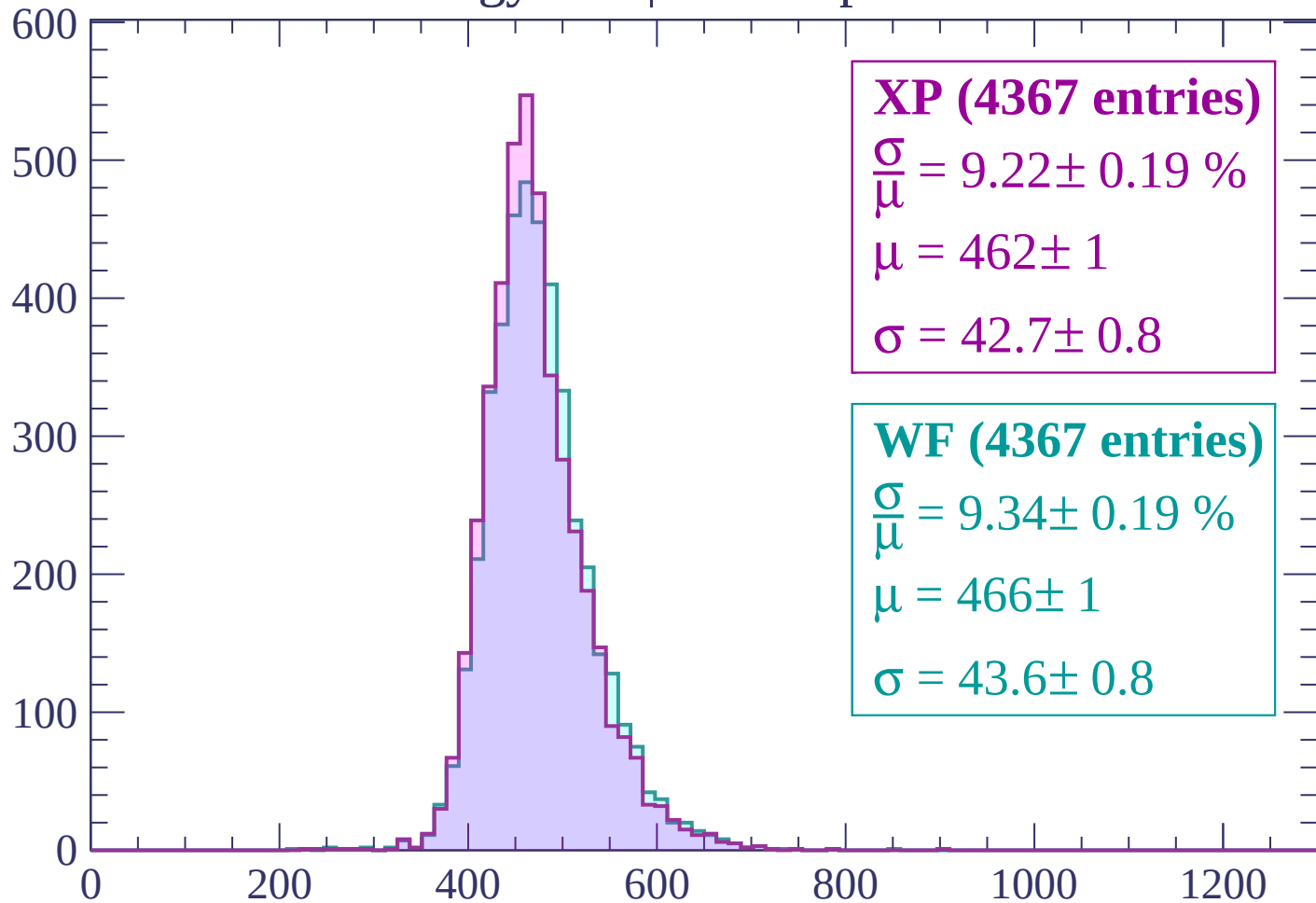
$$\mu = 464 \pm 1$$

$$\sigma = 44.5 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1680 < p < 1740$

Count



XP (4367 entries)

$$\frac{\sigma}{\mu} = 9.22 \pm 0.19 \%$$

$$\mu = 462 \pm 1$$

$$\sigma = 42.7 \pm 0.8$$

WF (4367 entries)

$$\frac{\sigma}{\mu} = 9.34 \pm 0.19 \%$$

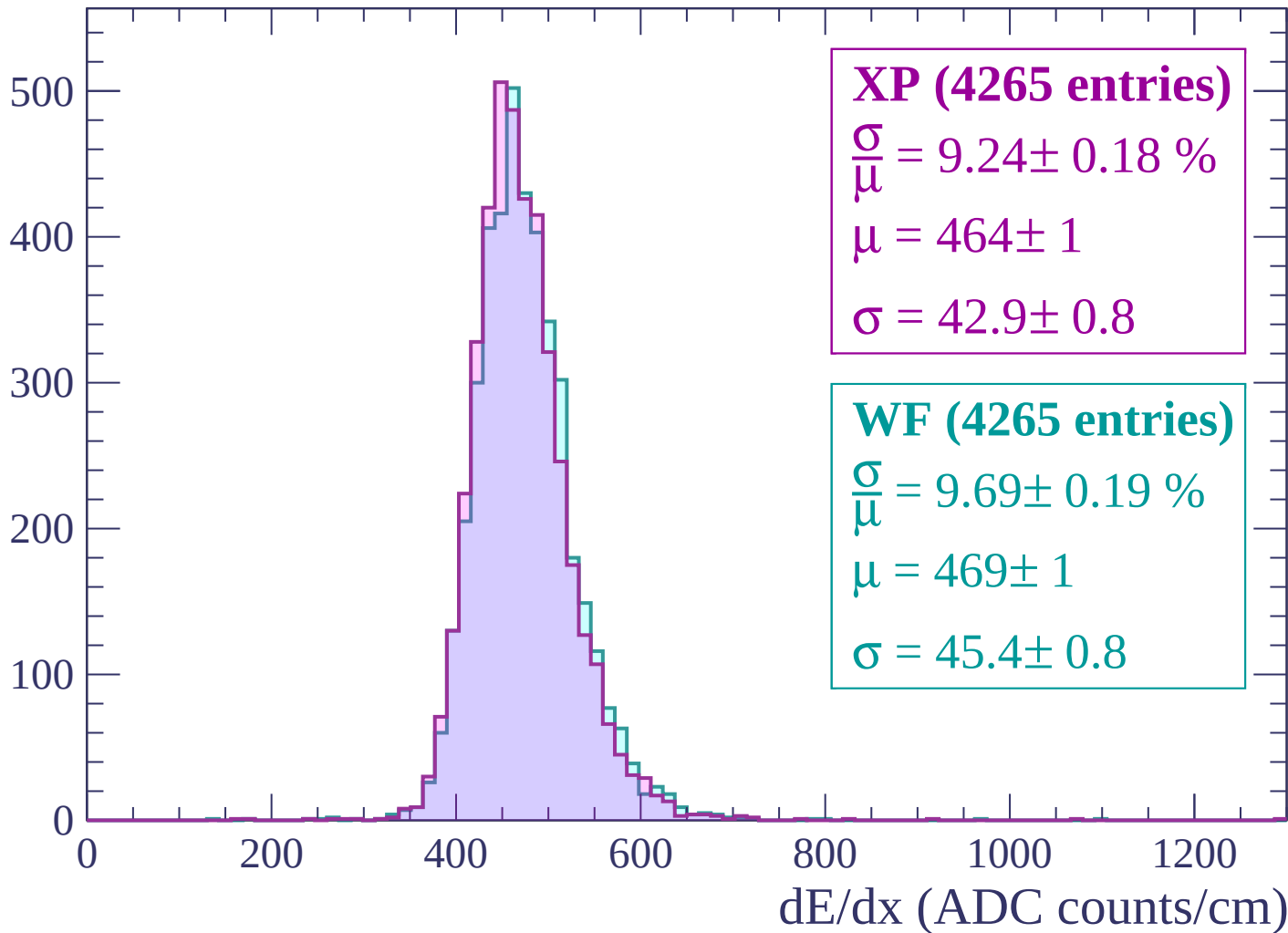
$$\mu = 466 \pm 1$$

$$\sigma = 43.6 \pm 0.8$$

dE/dx (ADC counts/cm)

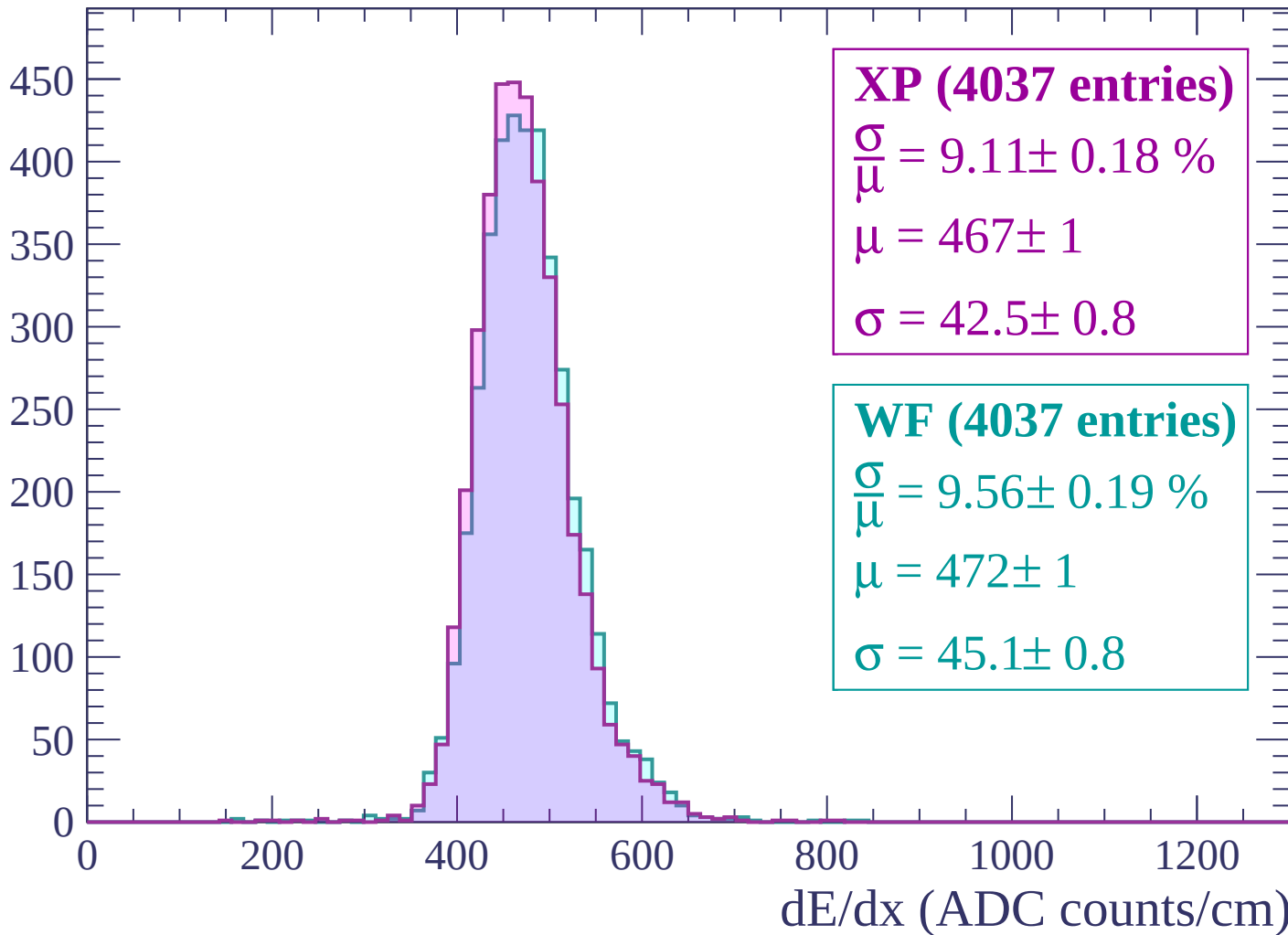
Energy loss | $1740 < p < 1800$

Count



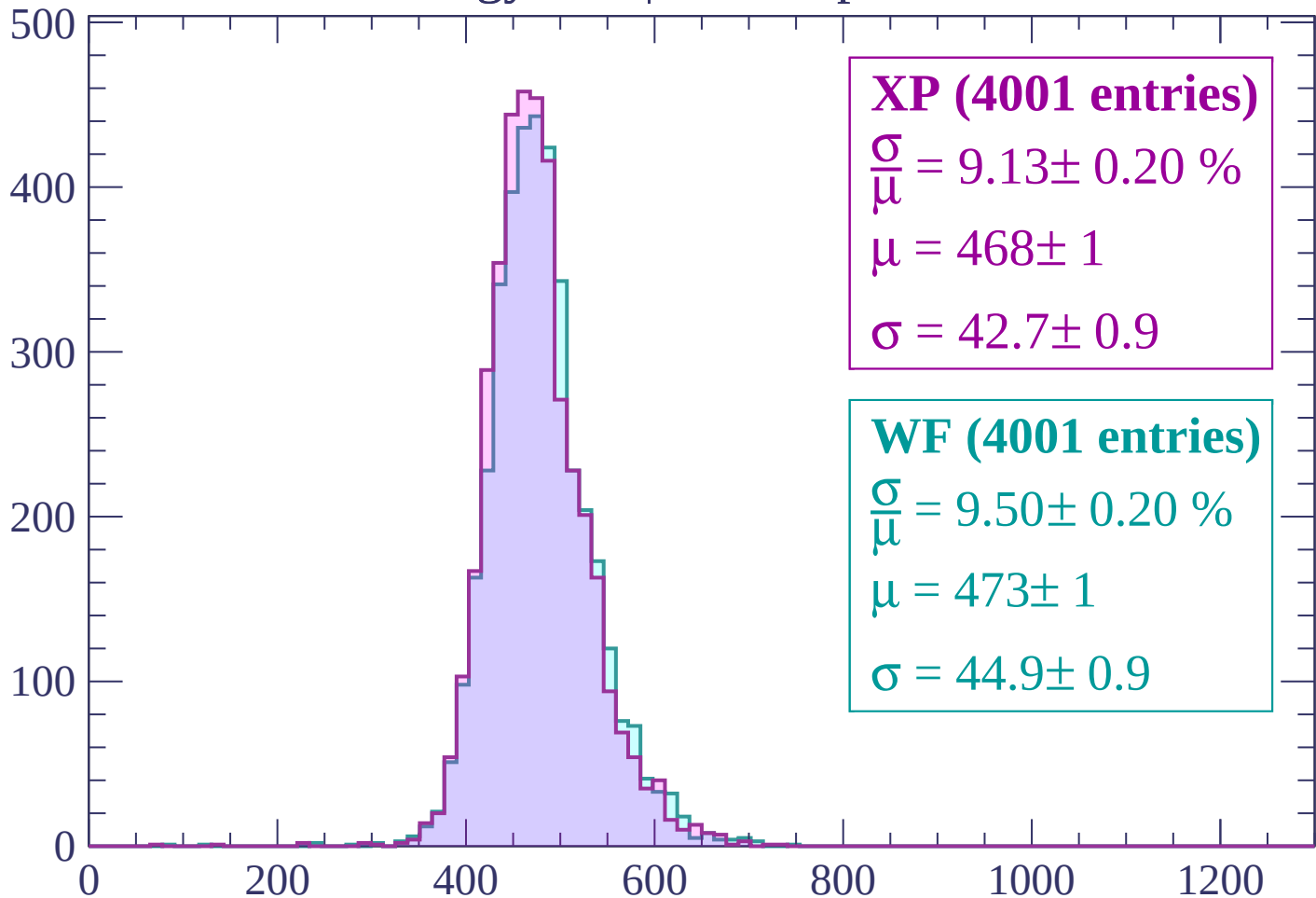
Energy loss | $1800 < p < 1860$

Count



Energy loss | $1860 < p < 1920$

Count



XP (4001 entries)

$\frac{\sigma}{\mu} = 9.13 \pm 0.20 \%$

$\mu = 468 \pm 1$

$\sigma = 42.7 \pm 0.9$

WF (4001 entries)

$\frac{\sigma}{\mu} = 9.50 \pm 0.20 \%$

$\mu = 473 \pm 1$

$\sigma = 44.9 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss | $1920 < p < 1980$

Count

500

400

300

200

100

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (3907 entries)

$\frac{\sigma}{\mu} = 9.52 \pm 0.20 \%$

$\mu = 471 \pm 1$

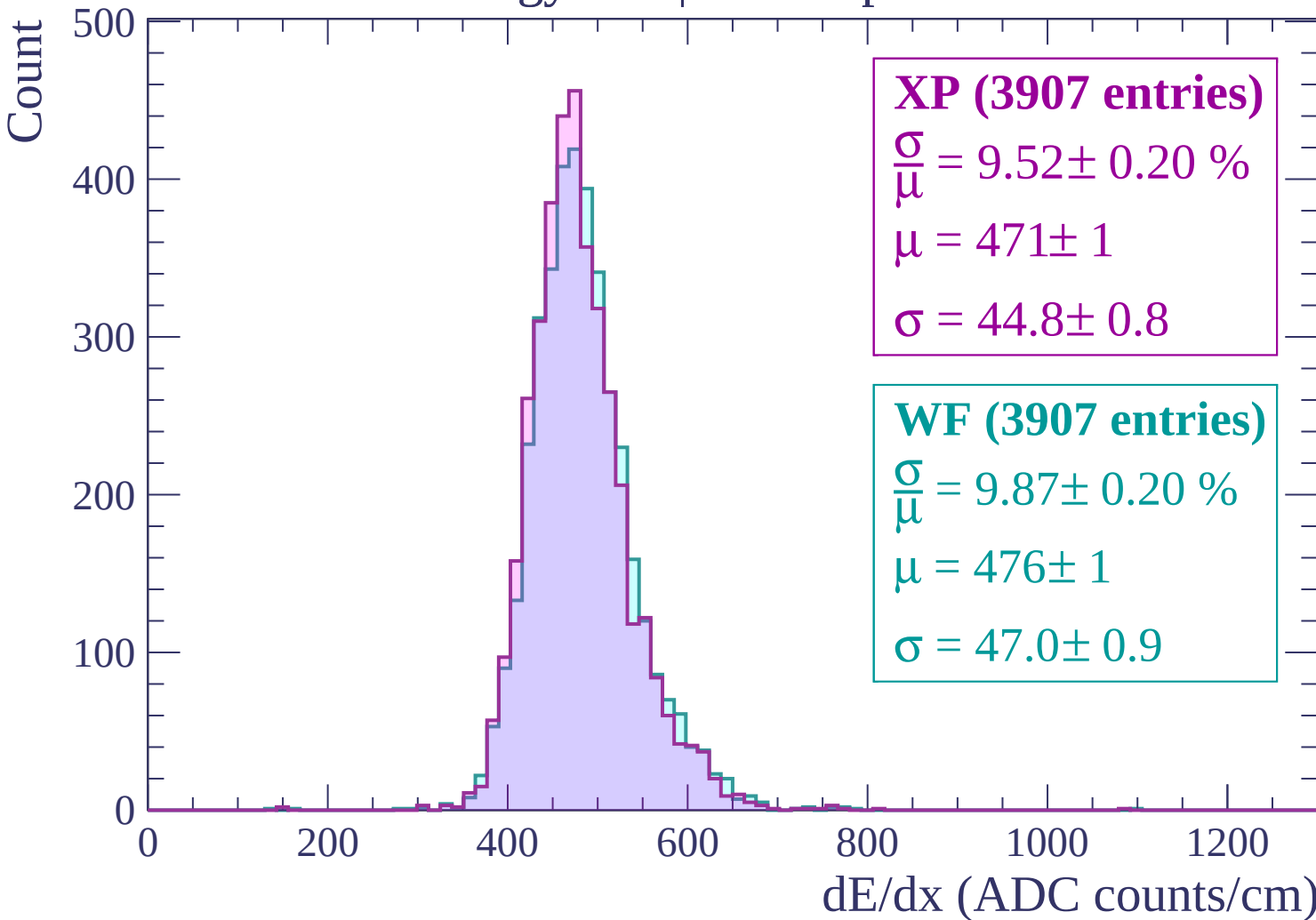
$\sigma = 44.8 \pm 0.8$

WF (3907 entries)

$\frac{\sigma}{\mu} = 9.87 \pm 0.20 \%$

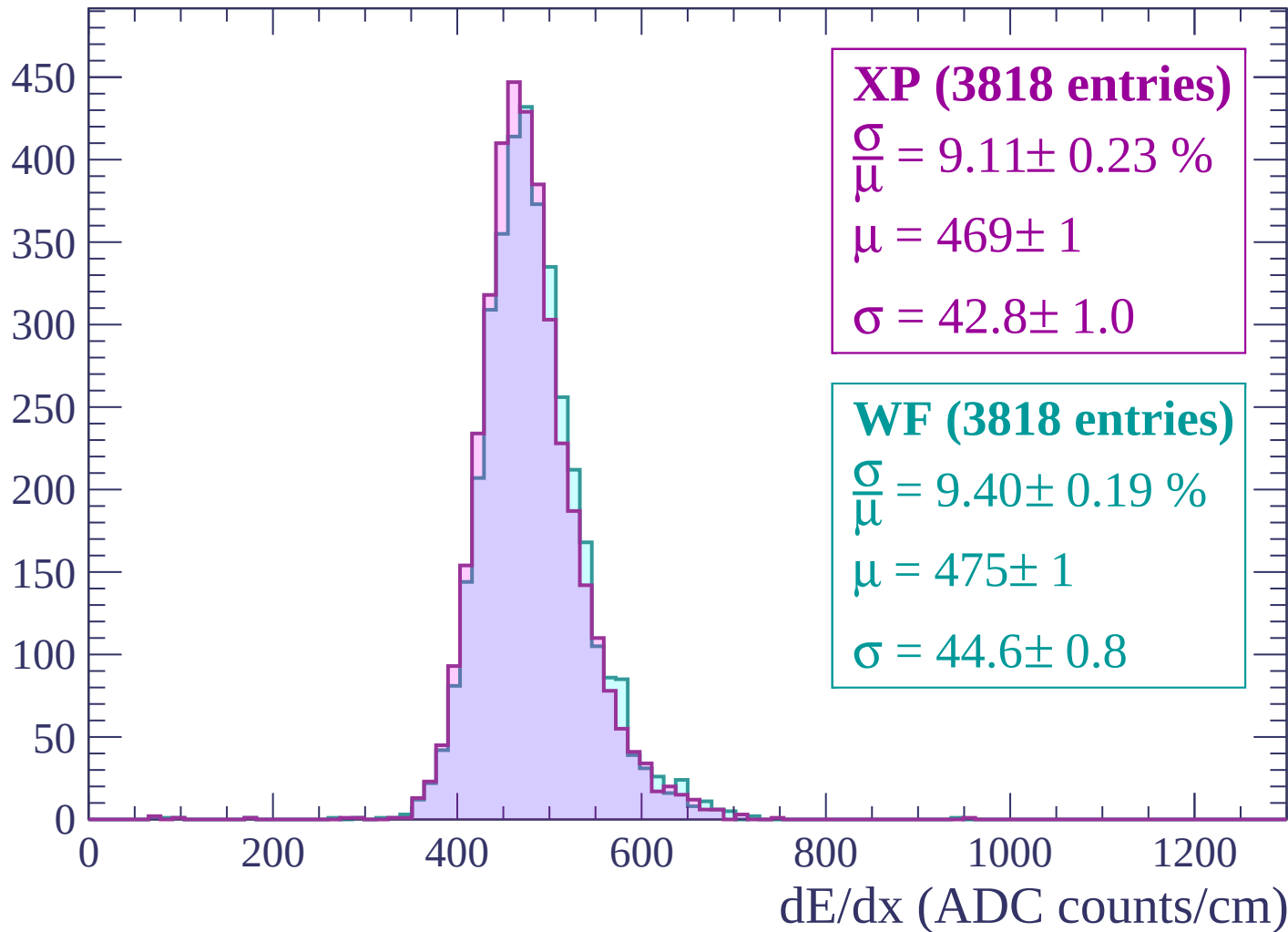
$\mu = 476 \pm 1$

$\sigma = 47.0 \pm 0.9$



Energy loss | $1980 < p < 2040$

Count



Energy loss | $2040 < p < 2100$

Count

400
350
300
250
200
150
100
50
0

XP (3689 entries)

$\frac{\sigma}{\mu} = 9.62 \pm 0.21 \%$

$\mu = 473 \pm 1$

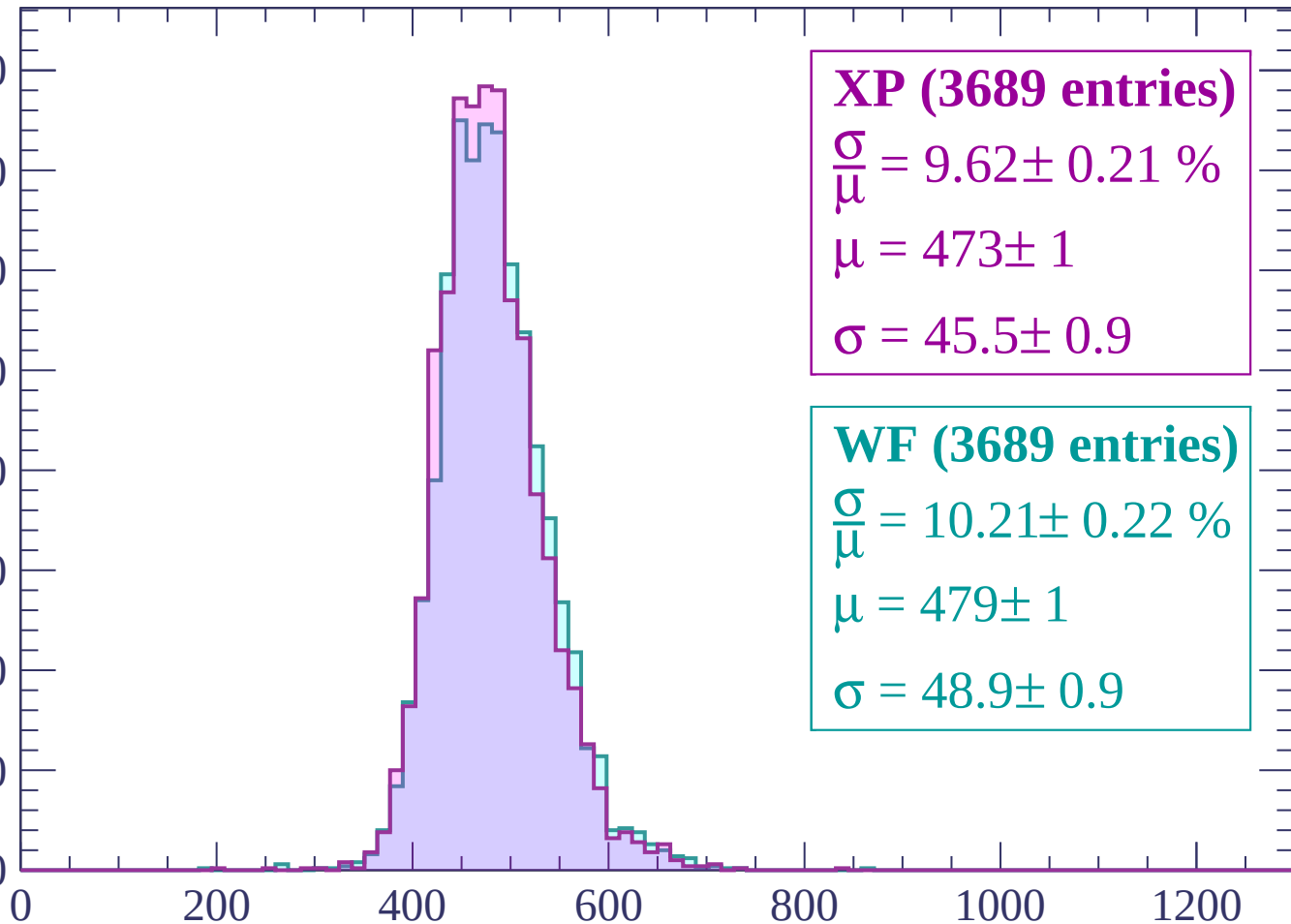
$\sigma = 45.5 \pm 0.9$

WF (3689 entries)

$\frac{\sigma}{\mu} = 10.21 \pm 0.22 \%$

$\mu = 479 \pm 1$

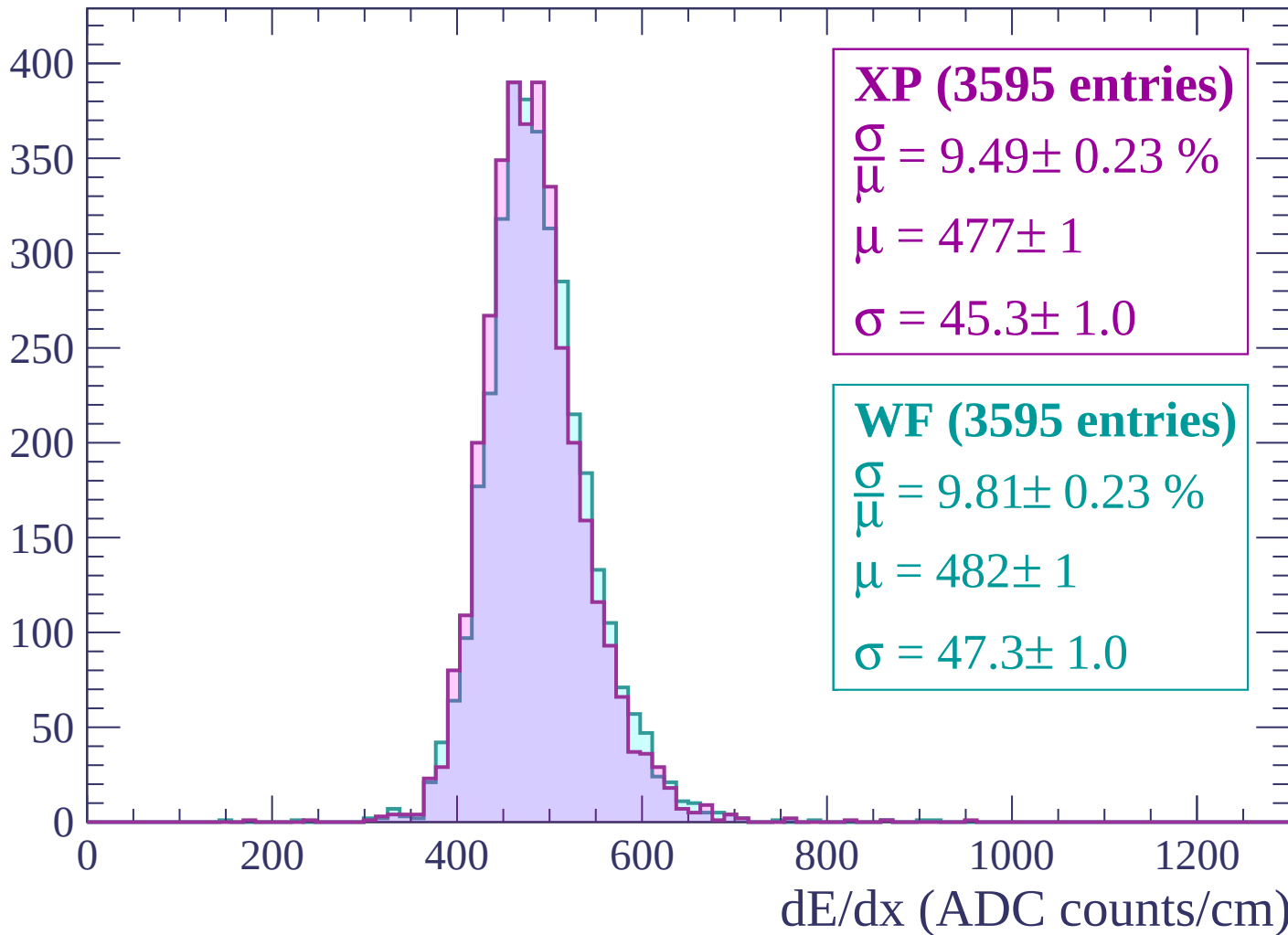
$\sigma = 48.9 \pm 0.9$



dE/dx (ADC counts/cm)

Energy loss | $2100 < p < 2160$

Count



Energy loss | $2160 < p < 2220$

Count

400
350
300
250
200
150
100
50
0

XP (3466 entries)

$\frac{\sigma}{\mu} = 9.39 \pm 0.23 \%$

$\mu = 477 \pm 1$

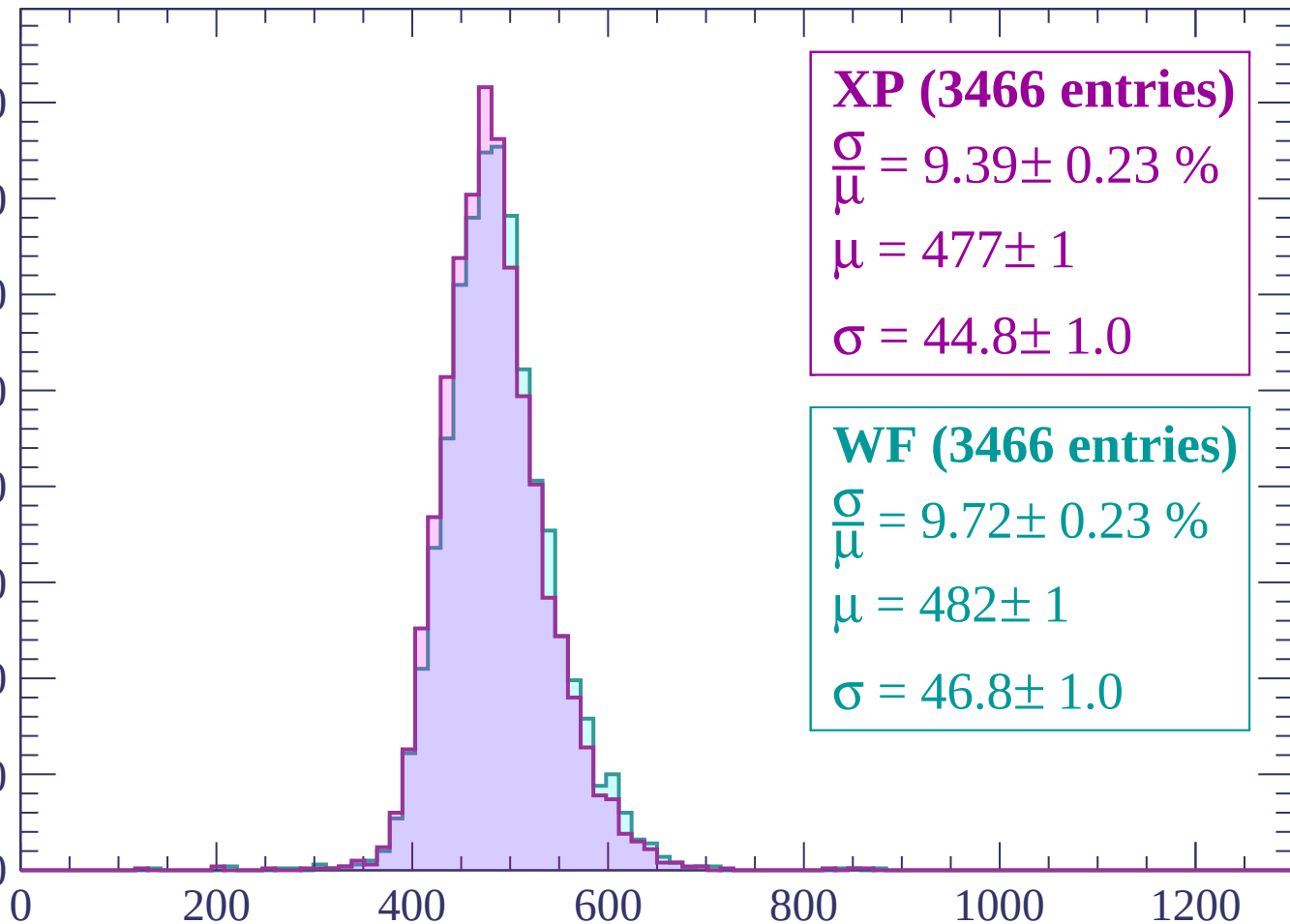
$\sigma = 44.8 \pm 1.0$

WF (3466 entries)

$\frac{\sigma}{\mu} = 9.72 \pm 0.23 \%$

$\mu = 482 \pm 1$

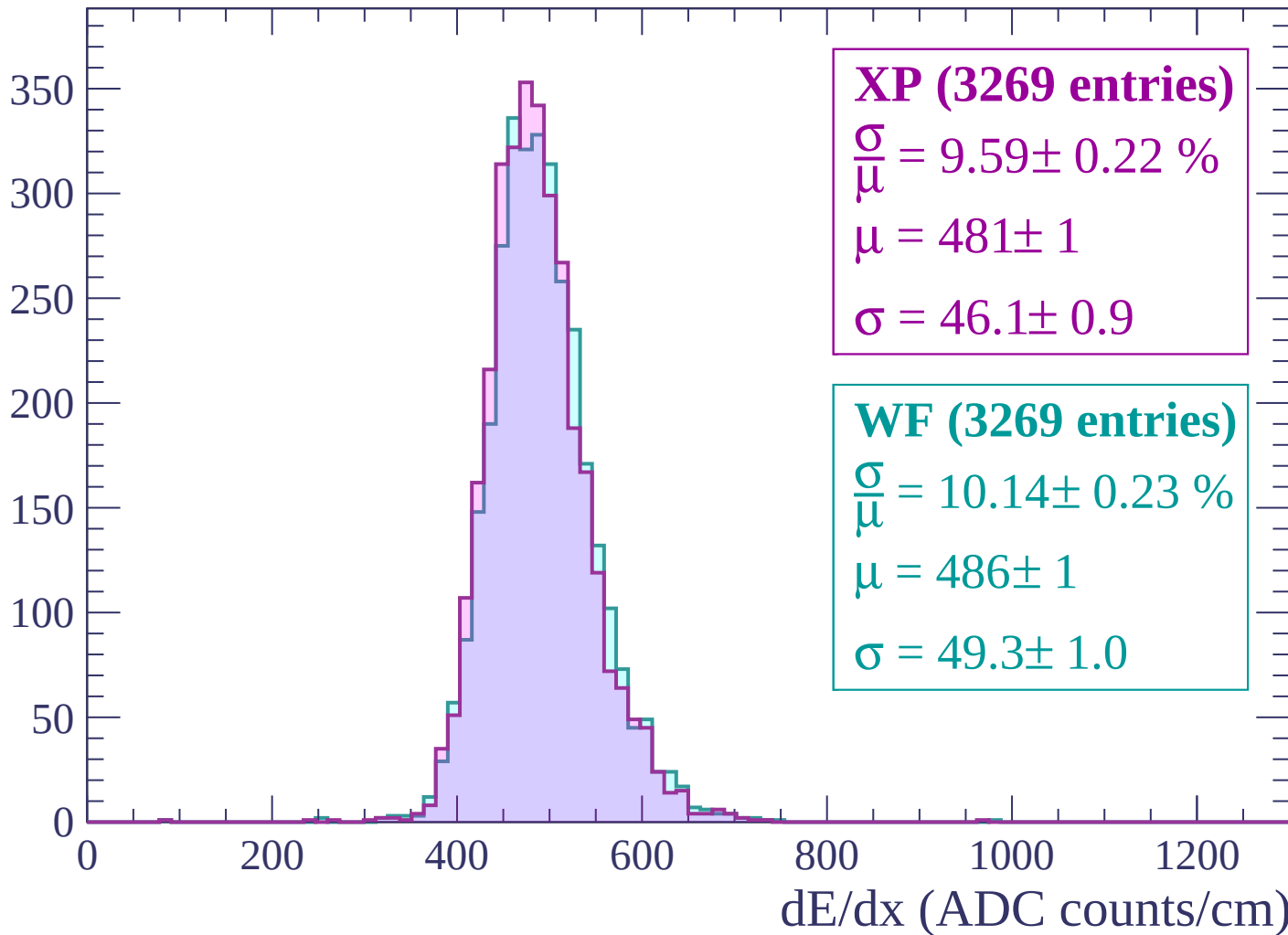
$\sigma = 46.8 \pm 1.0$



dE/dx (ADC counts/cm)

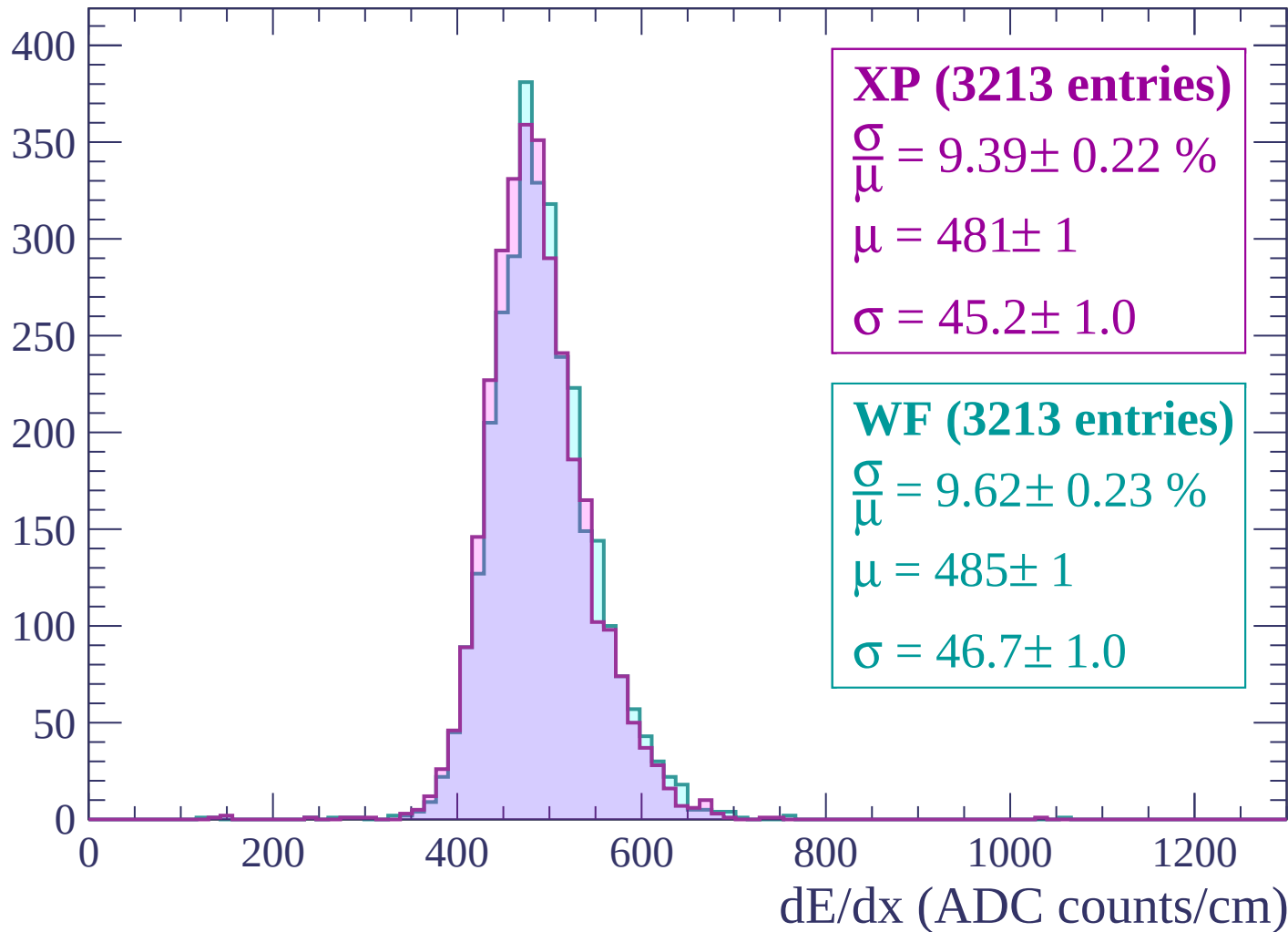
Energy loss | $2220 < p < 2280$

Count



Energy loss | $2280 < p < 2340$

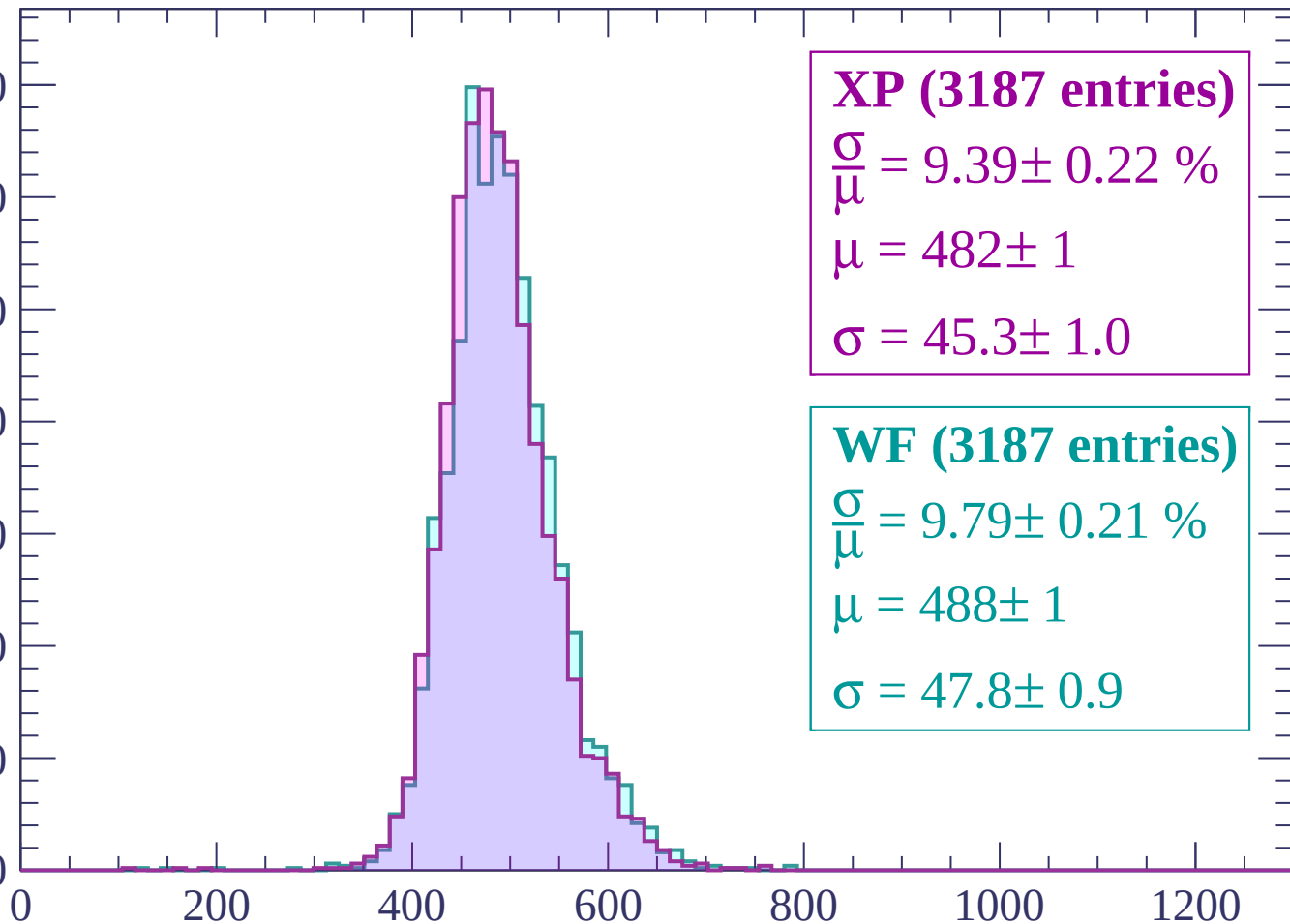
Count



Energy loss | $2340 < p < 2400$

Count

350
300
250
200
150
100
50
0



XP (3187 entries)

$\frac{\sigma}{\mu} = 9.39 \pm 0.22 \%$

$\mu = 482 \pm 1$

$\sigma = 45.3 \pm 1.0$

WF (3187 entries)

$\frac{\sigma}{\mu} = 9.79 \pm 0.21 \%$

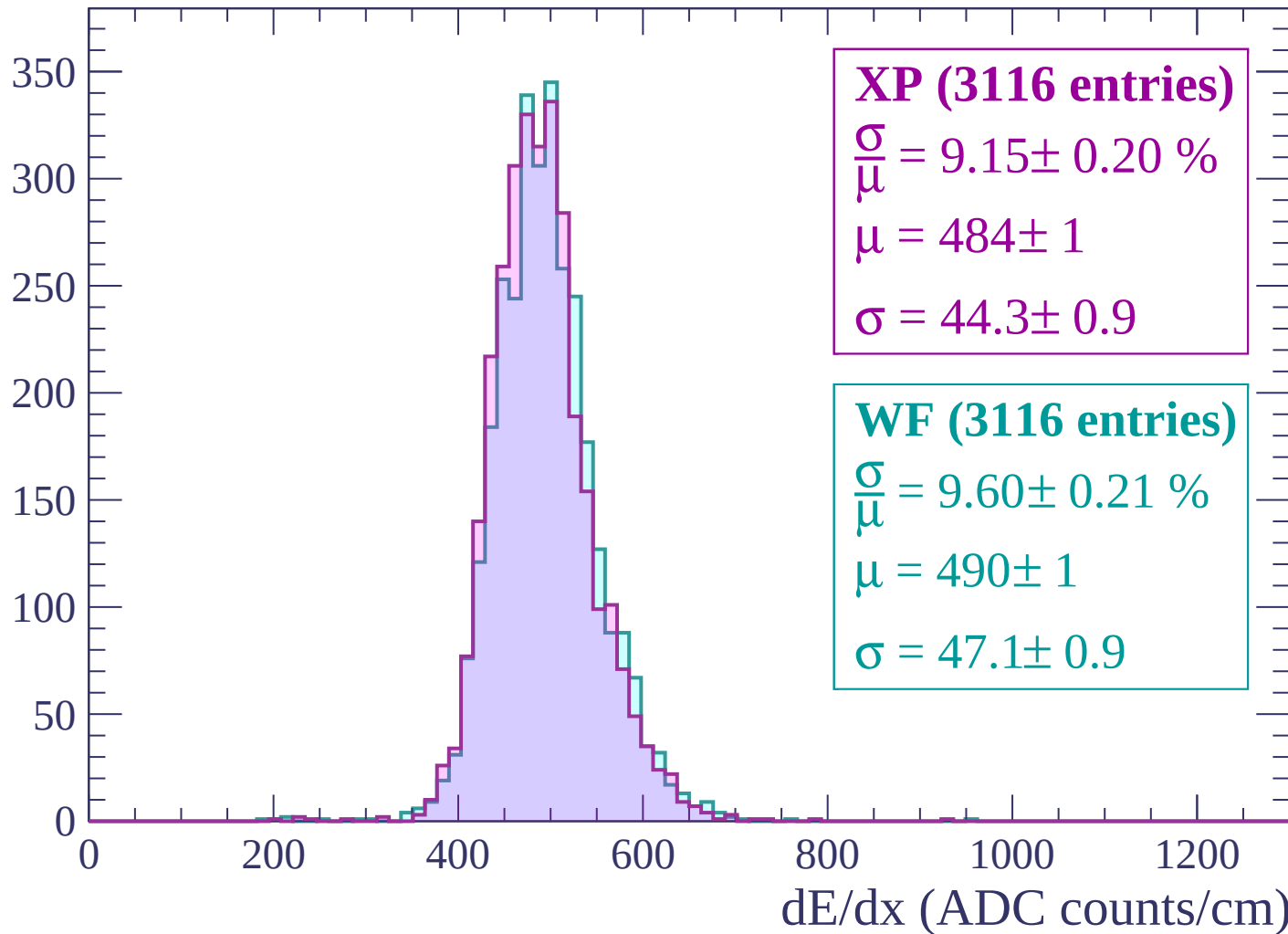
$\mu = 488 \pm 1$

$\sigma = 47.8 \pm 0.9$

dE/dx (ADC counts/cm)

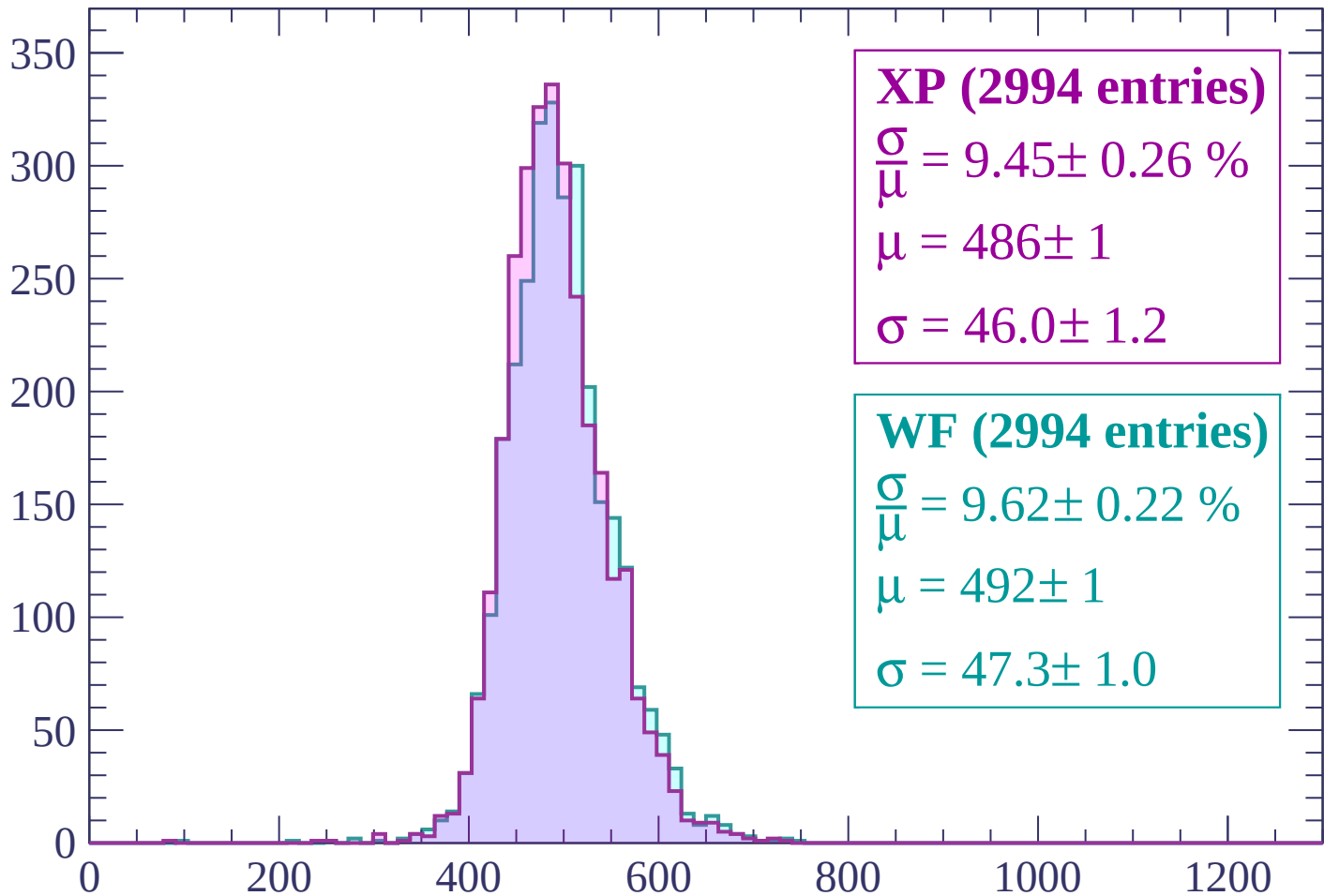
Energy loss | $2400 < p < 2460$

Count



Energy loss | $2460 < p < 2520$

Count



XP (2994 entries)

$$\frac{\sigma}{\mu} = 9.45 \pm 0.26 \%$$

$$\mu = 486 \pm 1$$

$$\sigma = 46.0 \pm 1.2$$

WF (2994 entries)

$$\frac{\sigma}{\mu} = 9.62 \pm 0.22 \%$$

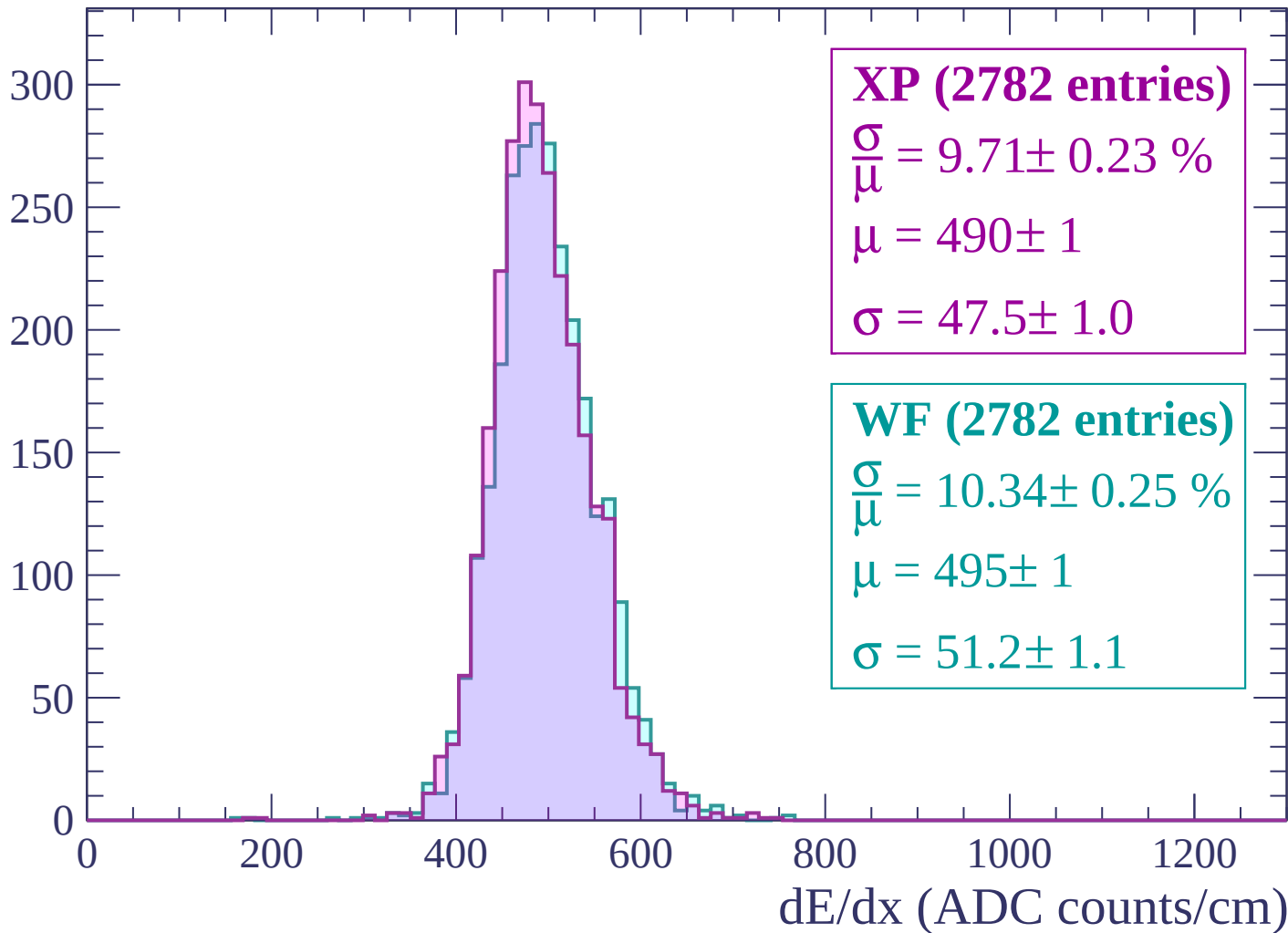
$$\mu = 492 \pm 1$$

$$\sigma = 47.3 \pm 1.0$$

dE/dx (ADC counts/cm)

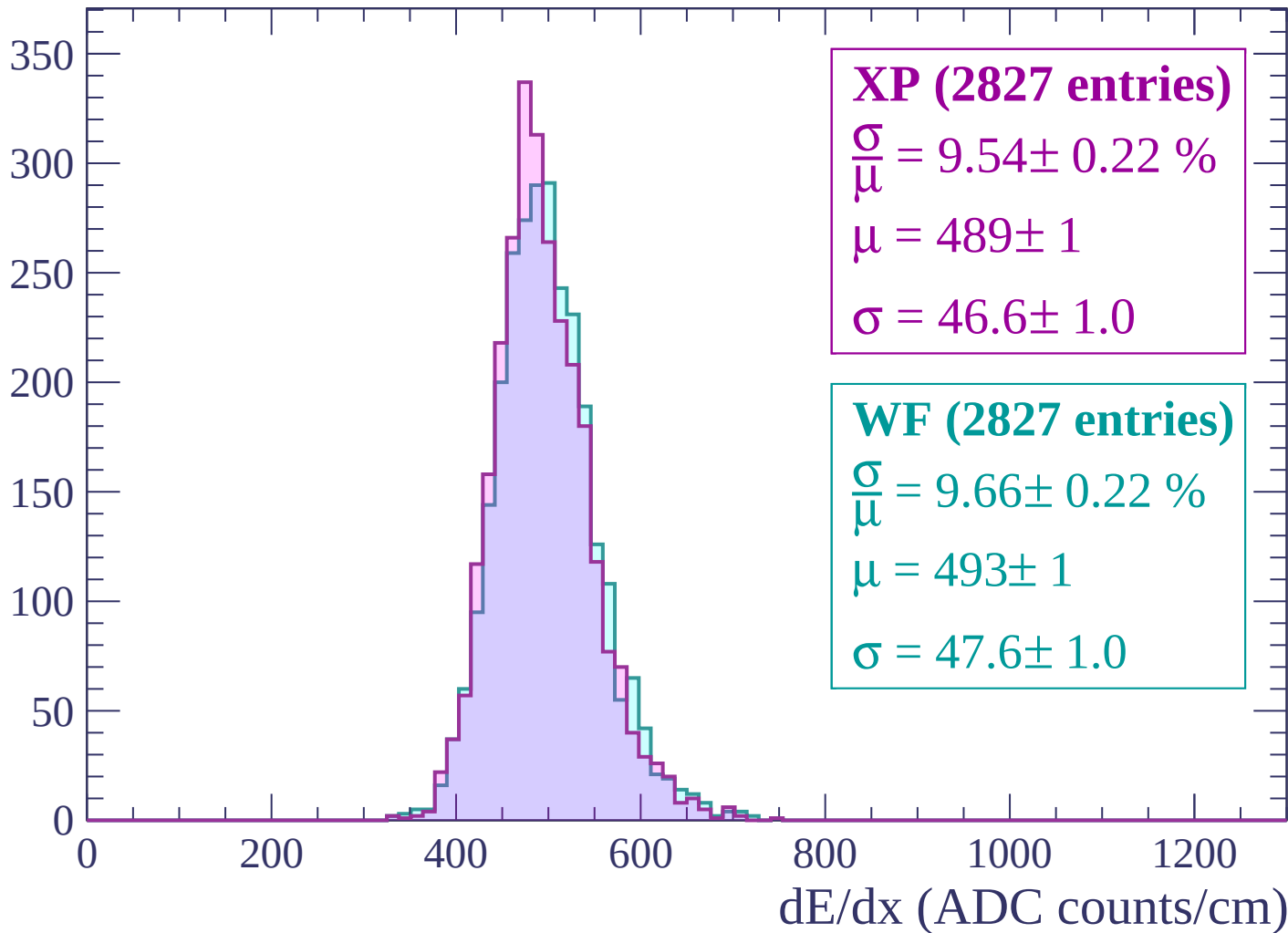
Energy loss | $2520 < p < 2580$

Count



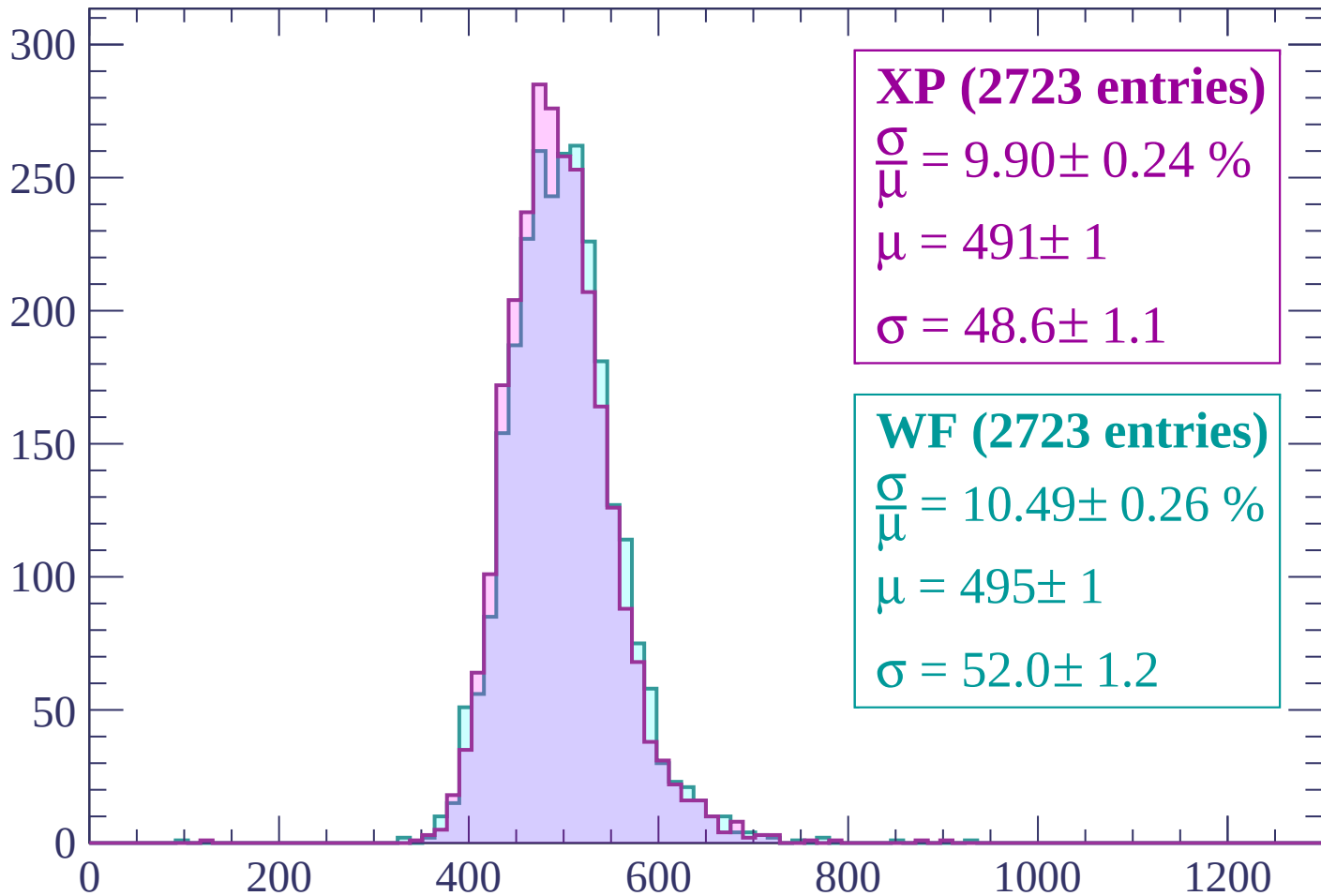
Energy loss | $2580 < p < 2640$

Count



Energy loss | $2640 < p < 2700$

Count



XP (2723 entries)

$\frac{\sigma}{\mu} = 9.90 \pm 0.24 \%$

$\mu = 491 \pm 1$

$\sigma = 48.6 \pm 1.1$

WF (2723 entries)

$\frac{\sigma}{\mu} = 10.49 \pm 0.26 \%$

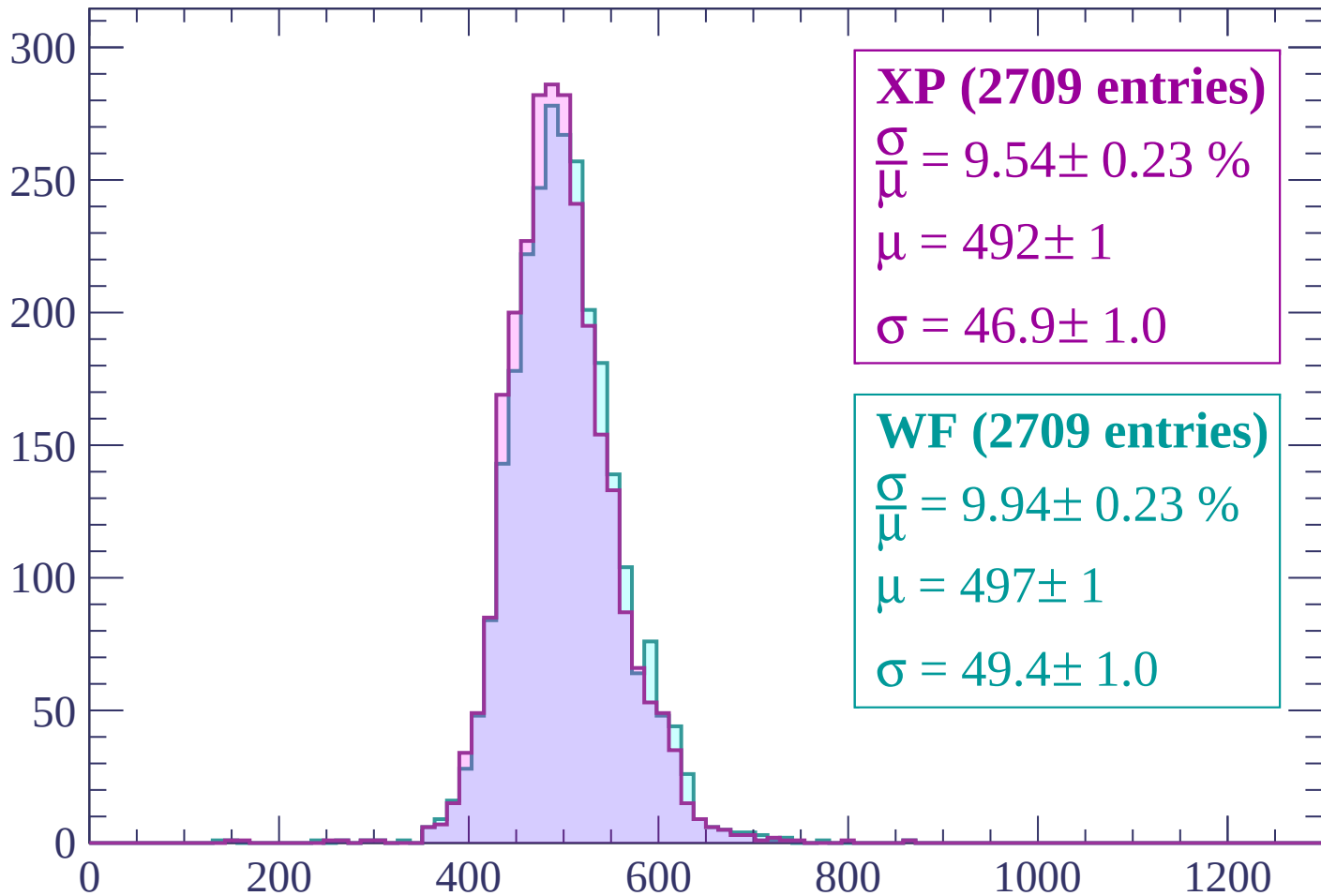
$\mu = 495 \pm 1$

$\sigma = 52.0 \pm 1.2$

dE/dx (ADC counts/cm)

Energy loss | $2700 < p < 2760$

Count



XP (2709 entries)

$\frac{\sigma}{\mu} = 9.54 \pm 0.23 \%$

$\mu = 492 \pm 1$

$\sigma = 46.9 \pm 1.0$

WF (2709 entries)

$\frac{\sigma}{\mu} = 9.94 \pm 0.23 \%$

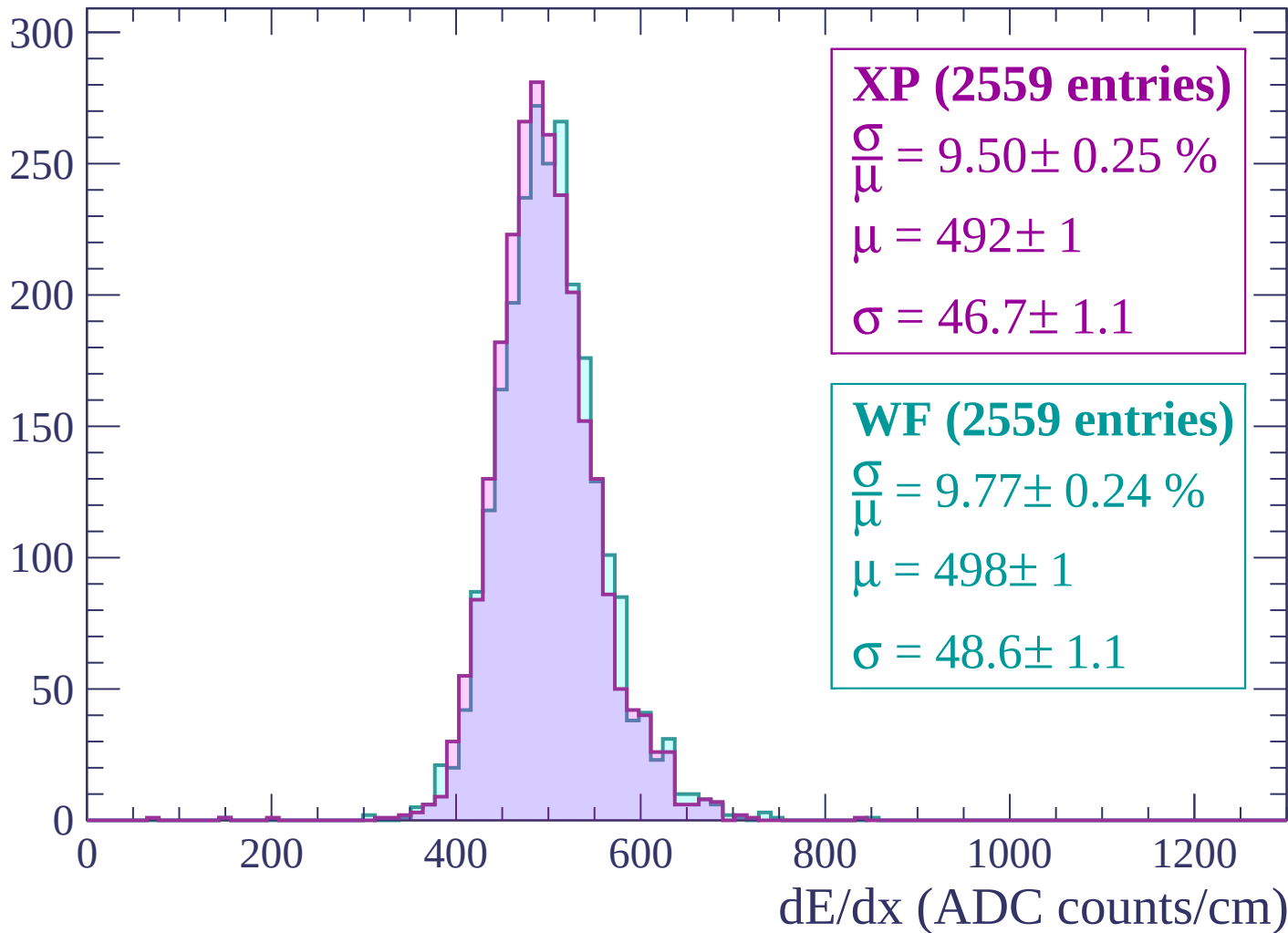
$\mu = 497 \pm 1$

$\sigma = 49.4 \pm 1.0$

dE/dx (ADC counts/cm)

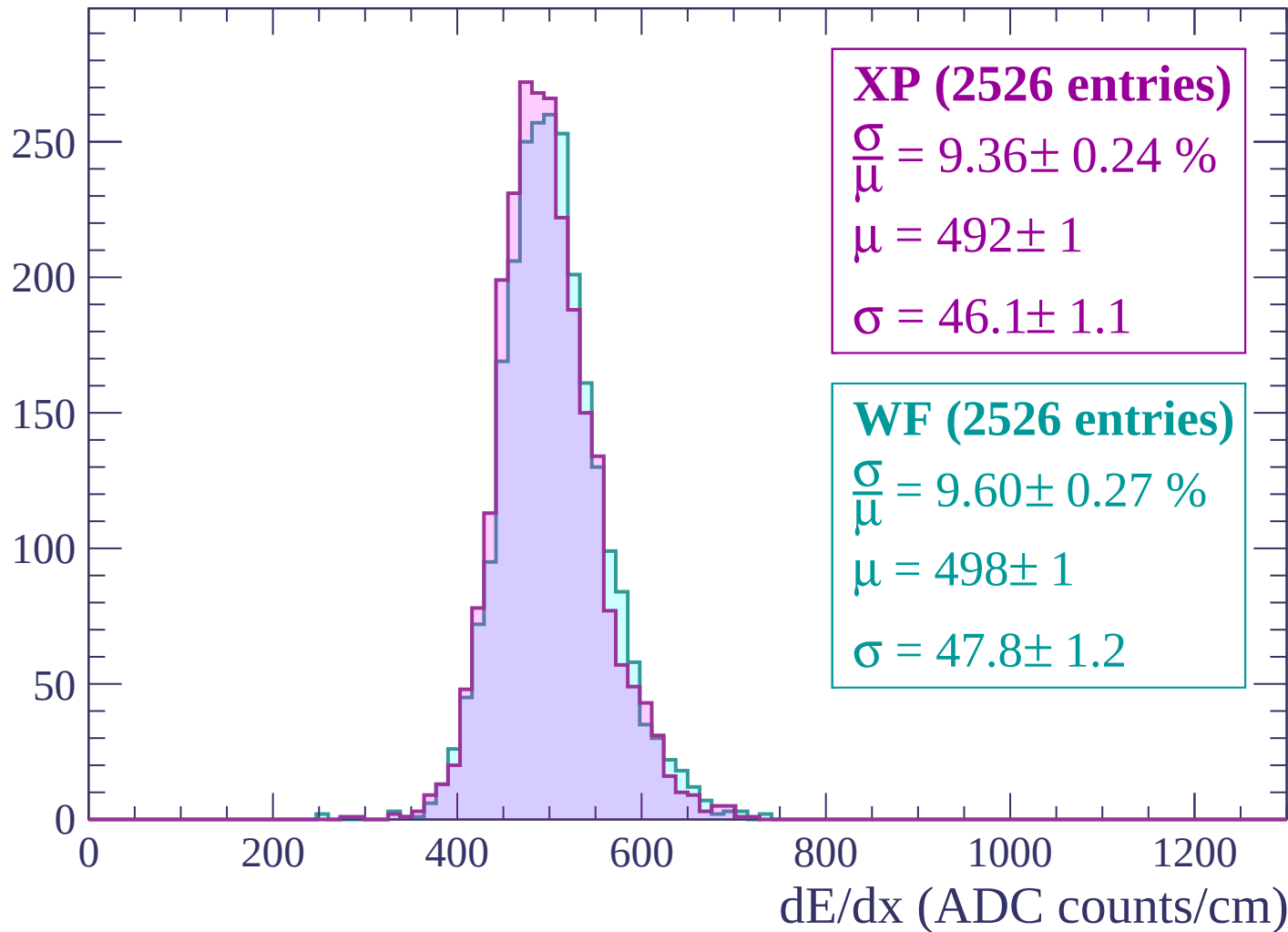
Energy loss | $2760 < p < 2820$

Count



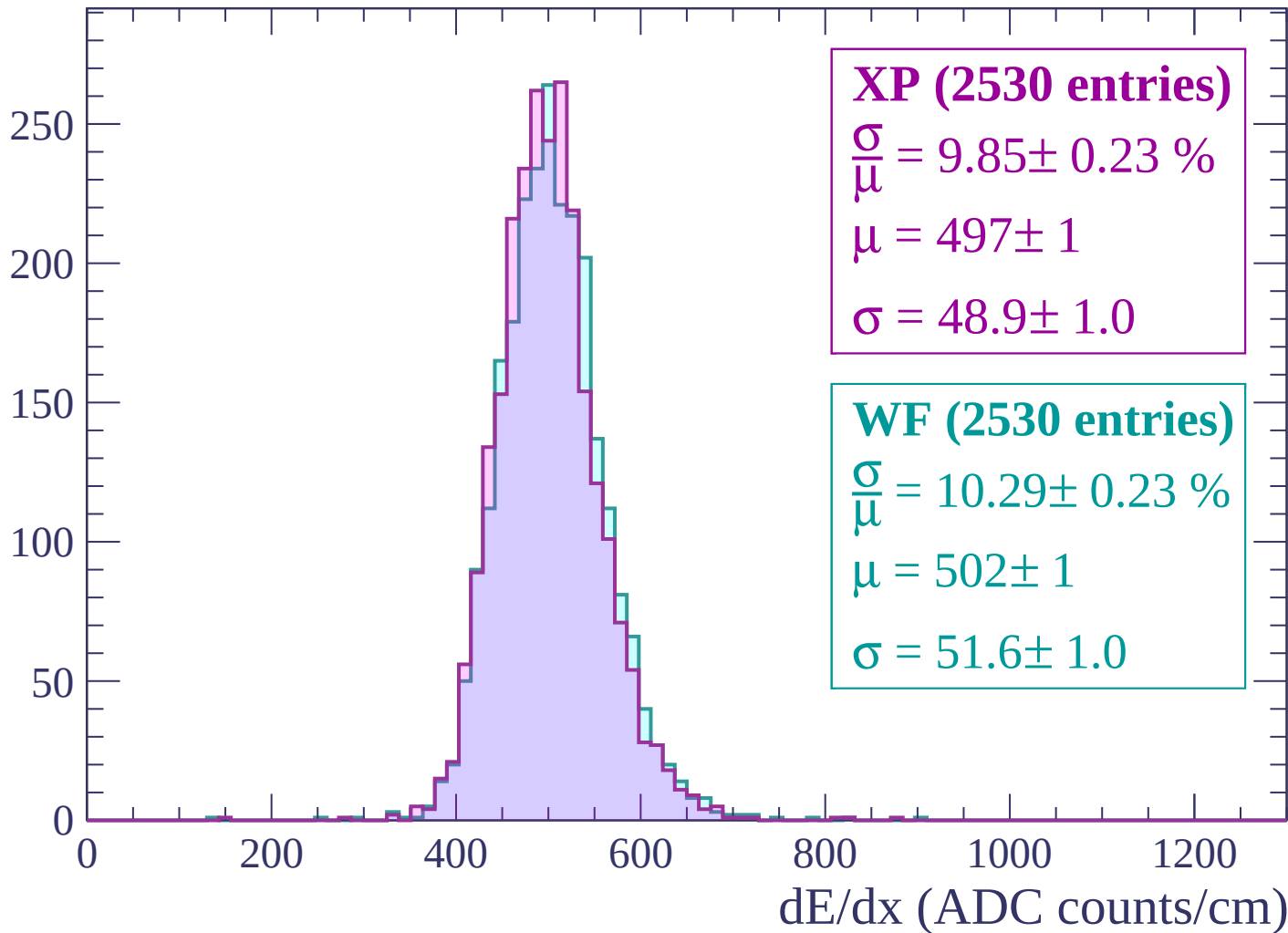
Energy loss | $2820 < p < 2880$

Count



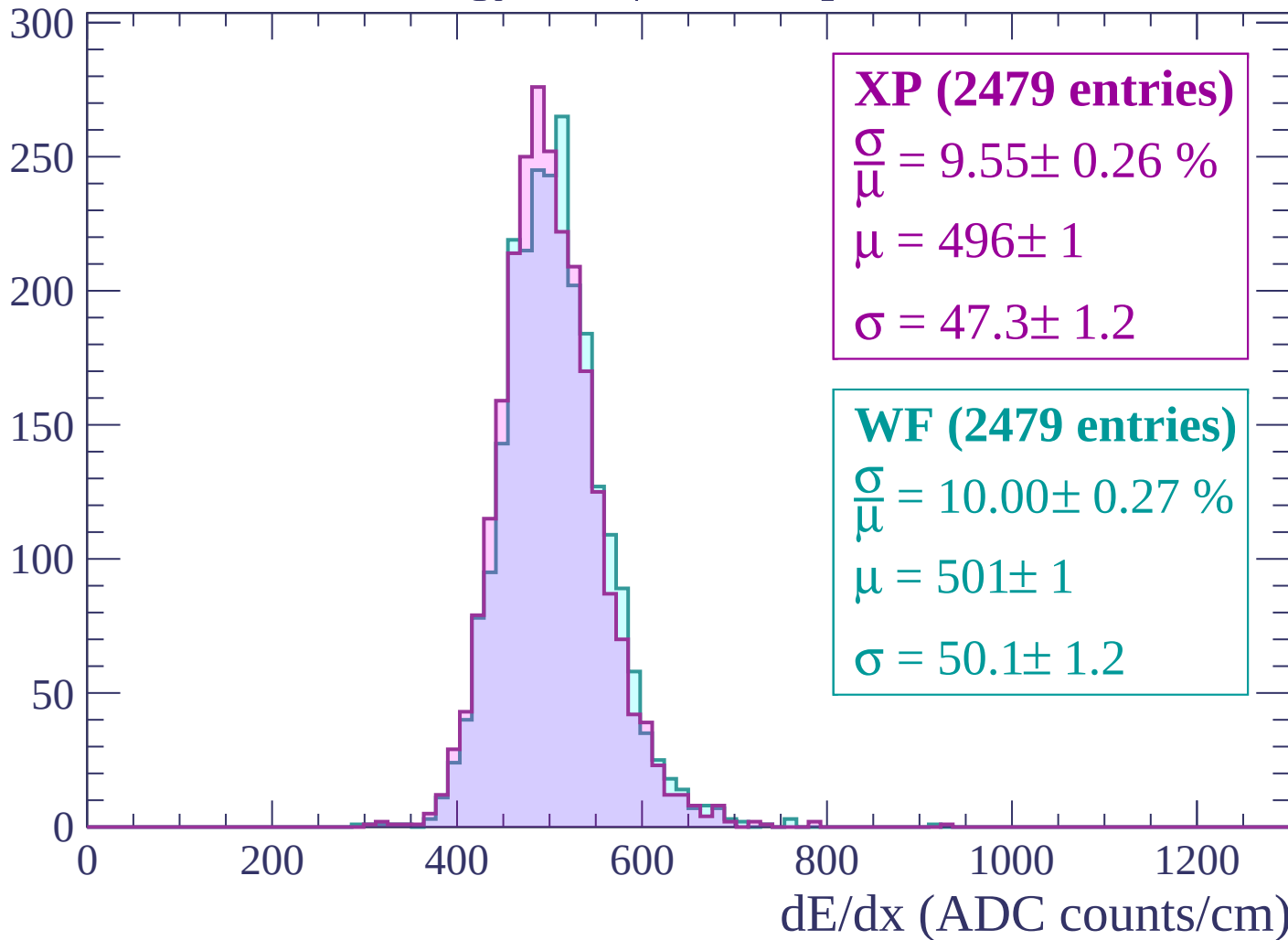
Energy loss | $2880 < p < 2940$

Count



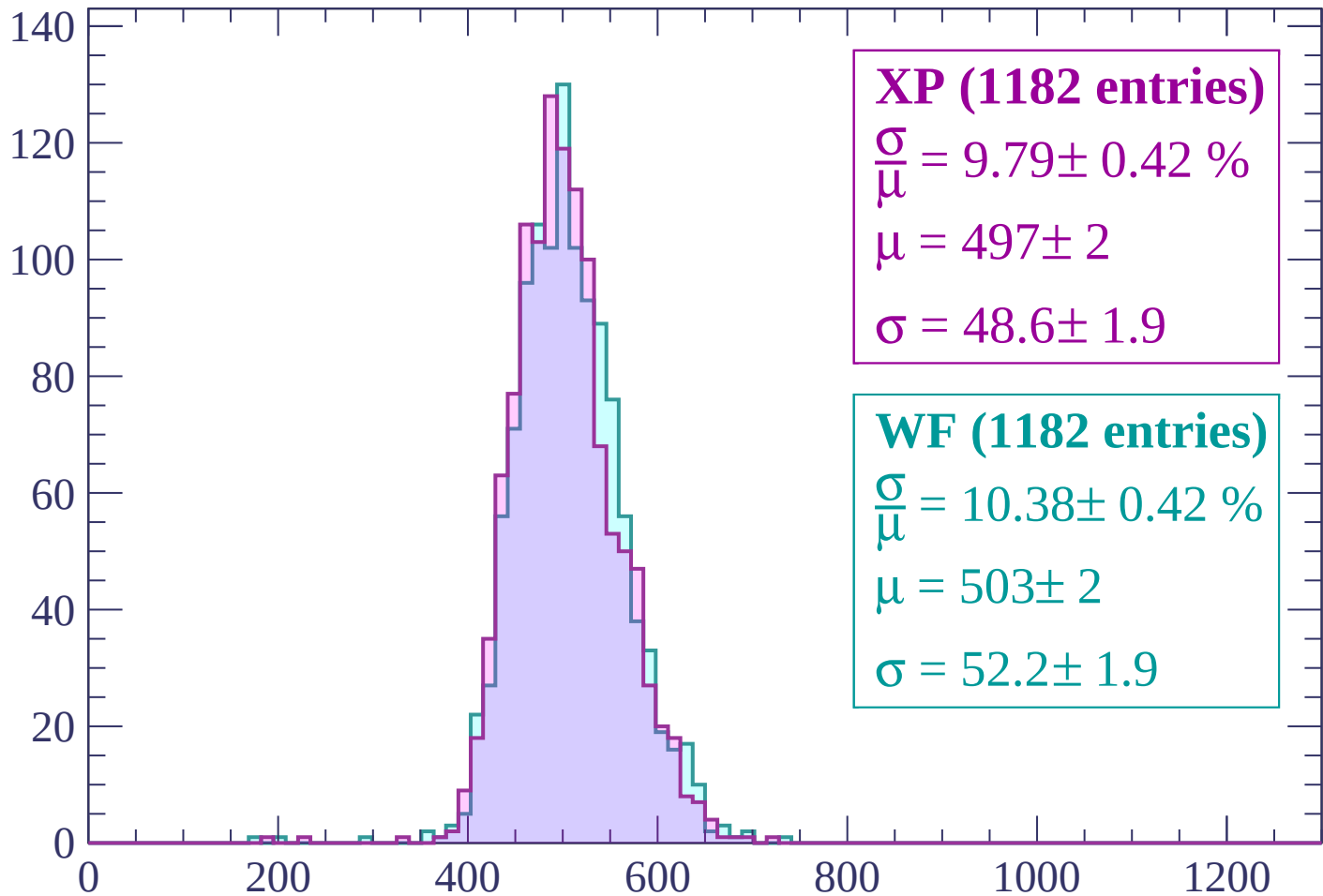
Energy loss | $2940 < p < 3000$

Count



Energy loss | $3000 < p < 3060$

Count



XP (1182 entries)

$$\frac{\sigma}{\mu} = 9.79 \pm 0.42 \%$$

$$\mu = 497 \pm 2$$

$$\sigma = 48.6 \pm 1.9$$

WF (1182 entries)

$$\frac{\sigma}{\mu} = 10.38 \pm 0.42 \%$$

$$\mu = 503 \pm 2$$

$$\sigma = 52.2 \pm 1.9$$

dE/dx (ADC counts/cm)

